

Springer Handbooks of Computational Statistics

Cathy Yi-Hsuan Chen
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Handbook of Blockchain Analytics



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Handbook of Blockchain Analytics

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About This Book

A blockchain is a robust technology that ensures secure, transparent, and trustworthy recording of data, making it suitable for a wide range of applications. However, as any new technology it comes with risks and chances at different levels. Understanding and balancing opportunities and downside risks requires skillful management based on reliable data and algorithms, hence on a *prima facie* scientific investigation of blockchains. Obviously, a scientific approach is not equivalent to simply invoking a portfolio of empirical studies of various features since this modern technology requires domain adapted analysis techniques. This handbook of blockchain analytics is offering the collective wisdom of pioneering and leading scientists on how to properly analyze this new and advancing technology. The following summary captures the essence of each part and chapter, providing a comprehensive overview of the topics covered here.

The handbook is divided into different parts reflecting the domain-specific analysis techniques for different application fields that cover digital finance, security, supply chains among others. Part I provides an introduction to blockchains and presents fundamental concepts including its structure, operation, and potential applications. It lays the groundwork for more detailed discussions in subsequent parts of the book. The next two contributions cover digital assets in blockchains and explore Non-Fungible Tokens (NFTs) and their integration with visualization techniques (VizTech). It also highlights the unique attributes of NFTs and their impact on digital art and ownership. “Markets require liquidity,” the topic of the cryptocurrency liquidity forecasting paper, providing insights crucial for traders and investors.

Part III introduces “Cryptoeconomics” where first the application of Metcalfe’s Law to cryptocurrencies is analyzed on how the value of a crypto network increases with its user base. Next, an empirical study on forecasting volatility in cryptocurrency prices is presented. It is particularly accounting for sudden market jumps, offering a more accurate predictive model. The blockchain analytics part is based on process mining for blockchain analytics by examining applications through a process lens, more precisely it investigates how process mining techniques can be applied to improve understanding and optimization of blockchain processes.

Another section sees blockchains meeting network analytics, that explores the intersection of blockchain technology and network analytics, focusing on heuristic methods for location analysis and fraud detection.

A standard field of blockchains is Digital Finance, which is the topic of the next part. It contains the chain of thoughts on “Blockchain Transaction Decision: Through the Lens of Mempool,” a contribution on decision-making processes for blockchain transactions, particularly within the Mempool where transactions are queued before being added to the blockchain. Part VI contains modern applications with the following papers.

The ongoing theme of “Blockchain Business Development” discusses strategies and opportunities for using blockchain in business development, highlighting case studies and potential markets. “The Implications of DLT and Blockchain: Legal and Financial Aspects,” in which the legal and financial implications of Distributed Ledger Technology (DLT) is discussed. “Applications of Blockchain in Healthcare” explores how blockchain can be used in healthcare to improve data security, patient privacy, and operational efficiency. “A Survey on Blockchain-Based Federated Learning Frameworks,” surveys the use of blockchain in federated learning frameworks, which allow for collective machine learning while preserving data privacy. “Blockchains and Cybersecurity” examines the role of blockchain technology in enhancing cybersecurity, addressing how it can protect against various cyber threats.

Part VII covers the blockchain ecosystem with articles on “The Use of Blockchain Technology in the Compliance Function” shows how blockchain can be utilized to improve compliance functions, ensuring regulatory adherence and reducing compliance costs. “Who Cares About Decentralized Finance? Evidence from DeFi Attacks” analyzes the impact of attacks on decentralized finance (DeFi) systems, providing evidence and insights into the vulnerabilities and resilience of DeFi platforms. “Imagining a Democratic, Affordable Future of Foundation Models: A Decentralized Avenue” envisions a future where foundational models for AI and machine learning are collective, making these technologies more accessible and democratic. Finally, “The Development of Blockchain Platform for Maritime Supply Chain” discusses the development and application of blockchain platforms in the maritime supply chain, focusing on improving transparency, efficiency, and traceability.

Living in the Information Age, the power of data and correct statistical analysis has never been more prevalent. Academics and practitioners always require an accurate application of quantitative methods. We have therefore put presentations of this handbook on a (evolving blockchain) network, Quantinar (quantinar.com), to support quantitative analytics knowledge paired with code in the form of Quantlets (quantlet.com) or software snippets. The integration of blockchain technology makes Quantinar a decentralized autonomous organization (DAO) that ensures fully transparent and reproducible scientific research.

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Part I

Introduction to Blockchain

Understanding Blockchain Technology



Min-Bin Lin, Daniel Traian Pele, and Rui Ren

If you don't think about today, today will bite you hard tomorrow.

Abstract Blockchain technology has gained great attention from academics and practitioners, and its implementation has been considered as a crucial component for the development of digital economy and finance. From a technical perspective, it is a secured database managed by a network of participants who follows a certain protocol to maintain a distributed ledger, ultimately substituting a centralized system through a consensus algorithm. Over the past 5 to 10 years, the industry has witnessed significant expansion in blockchain applications, particularly in the area of digital assets. More recent practices are in peer-to-peer lending, e-governance, healthcare, and supply chain management. These applications capitalize on the inherent features of blockchain, including decentralization and security, which are widely touted as key selling points. The technology, however, as any newly introduced instrument for today's economy, has borne risks and chances due to the disequilibrium between the hype and hope. Yet, blockchain technology has not been fully comprehended. In this chapter, we are not only focusing on cryptocurrencies, smart contracts, and non-fungible tokens but also its evolution and functionality.

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1 Genesis of Blockchain

In recent years, blockchain and other decentralized technologies have received substantial attention from the public. In particular, with the rapid rise in popularity of DeFi (decentralized finance) markets and the expanding array of cryptocurrencies (cryptos), there is a growing emphasis on researching the economics of these novel markets and their foundational technology [19]. Yet, comprehension of blockchain technology remains constrained to both economic and information infrastructure perspectives. Doubt persists among the public regarding its practical application, often relegating it to speculative assets. This chapter aims to elucidate the technology and apply various perspectives to enhance understanding.

Given that individual coin/token ownership is recorded in a distributed ledger, blockchain technology enables a network of anonymous users to transact directly through a consensus decision-making protocol, thereby eliminating the need for a traditional trusted intermediary. The emergence of such technology in the form of currency has encroached on the development of other fields such as healthcare management, supply chain management, e-governance, Internet of Things (IoT), cybersecurity, social media, and banking and financial services [3, 28]. This shift is significant because it introduces decentralized and tamper-resistant database systems that could potentially revolutionize various industries by providing greater transparency, security, and efficiency in transactions and data management.

The evolution of blockchain technology has surpassed the initial vision outlined by Nakamoto [16] as merely a digital medium of exchange. For instance, tokenizing assets and information on a blockchain infrastructure extends the potential for various applications by enabling efficient and transparent transfer, ownership, and management. However, the necessity and realistic application of tokenization depend on factors such as the type of assets or information being tokenized, the regulatory environment, scalability, and user adoption. It is essential to avoid blindly adopting the technology without a clear understanding of its benefits and limitations.

As researchers explore the intricate economics of these markets and the broader implications, we anticipate ongoing innovation and integration spanning various sectors. This trajectory promises to usher in a more decentralized and interconnected digital economy. In this chapter, we introduce the fundamental concept of blockchain technology and relevant extensions and explore potential research approaches.

2 Blockchain in the Intersection

In the following, we introduce blockchain technology as the intersection of three domains—peer-to-peer network, cryptography, and game theory in Fig. 1. A peer-to-peer network facilitates anonymous user connections, with cryptography ensuring secure information distribution among participants. Through consensus mecha-

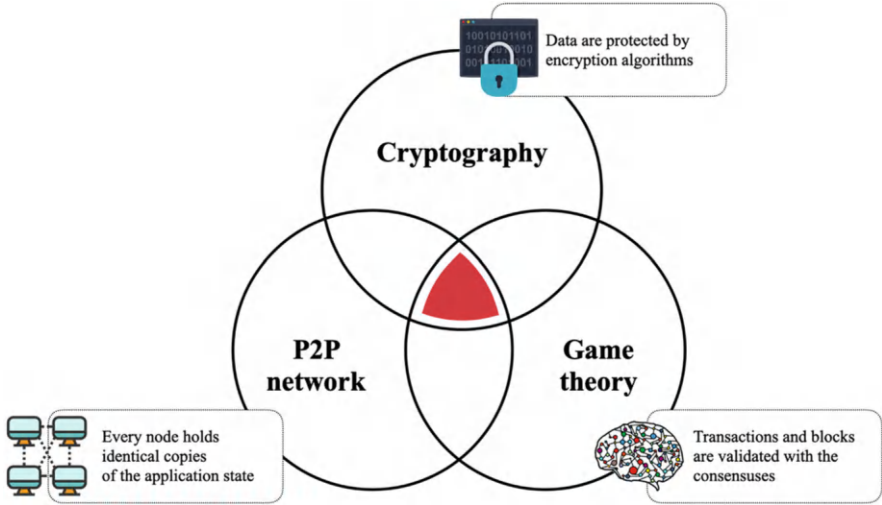


Fig. 1 Blockchain technology and its knowledge intersection

nisms, users reach agreements on verified information updates to the blockchain, a process highly intertwined with game theory.

2.1 Peer-to-Peer Network

Blockchain technology serves as a quintessential illustration of distributed ledger technology (DLT), celebrated for its decentralized architecture. This architecture eliminates dependence on central authorities, bypassing intermediaries, and mitigating single points of failure, such as those inherent in the traditional banking system, where the ledger is controlled and verified solely by banks. In this decentralized setup, every user upholds a replicated ledger and validates its authenticity with other nodes, thereby safeguarding the accuracy of the data. This framework epitomizes the peer-to-peer network concept, fostering a collaborative and trust less environment.

Based on the level of user anonymity within the blockchain network, we can classify the peer-to-peer network as either an anonymous network (public blockchain) or a transparent network (private blockchain). Public blockchains are hosted on public servers and are open for anyone to join and participate in, while private blockchains operate within privately owned infrastructure and are restricted to certain authorized participants. Figure 2 categorizes blockchains into four sections according to the ownership of data infrastructure and permission levels. We can differentiate blockchains based on their levels of permission and categorize them as permissioned and permissionless. A permissionless system grants every user the freedom to join and manage the ledger autonomously; however, within a private

		Read-write-commit permission granted	
Ownership of data infrastructure	Public	Permissioned	Permissionless
		Medium scalability • Anyone can join and read • Only authorized and known participants can write and commit	Low scalability • Anyone can join, read, write and commit
	Private	Very high scalability	High scalability
		• Only authorized participants can join and read • Only the network operator can write and commit	• Only authorized participants can join, read and write

Fig. 2 Ownership and permission management in blockchains. Source: Mckinsey and Company

server setting, participation is regulated. On the other hand, a permissioned system regulates users’ rights for read, write, and commit access. Because of the network’s scale and the permission management design, the collective decisions within the peer-to-peer network can vary in timing and scalability outcomes. For instance, a permissioned system operating within a private infrastructure may resemble a centralized system, with decisions made by a select few users, resulting in considerably high scalability.

Cryptos that their distributed ledgers are accessible to anyone like Bitcoin (BTC) and Ethereum are the examples for public blockchains and being permissionless. These crypto platforms adhere to the classic principles of blockchain technology, as outlined by Nakamoto [16] in his seminal work. However, reaching agreement on ledger updates (consensus) can be difficult in anonymous peer-to-peer networks that underpin these blockchains.

The Byzantine Generals Problem (BGP) vividly exemplifies the daunting challenge posed by faulty actors or an environment lacking in mutual trust among users [12]. In this scenario, generals commanding segments of a Byzantine army face the critical task of coordinating their decisions regarding attack or retreat via messengers. The inclusion of disloyal generals significantly heightens the risk of misinformation, thereby jeopardizing the success of the entire military operation. Essentially, they need a critical mass of concurring votes to achieve consensus or a protocol to assist them in efficiently and effectively coordinating their actions. In the context of blockchains, this protocol relies on both the consensus algorithm and cryptography for information encryption. Next, we will introduce how information is secured within the blockchain and how one can verify whether the information originates from an honest user or a faulty one.

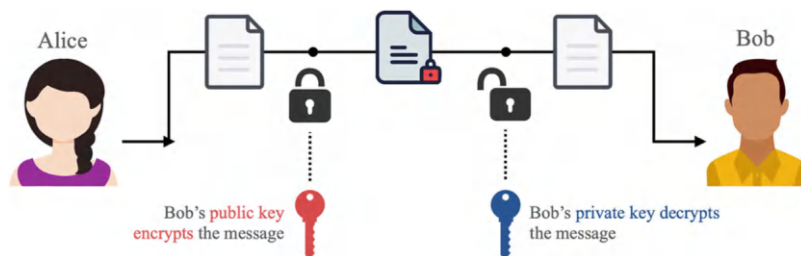


Fig. 3 Asymmetric encryption

2.2 Cryptography

Cryptography acts as the key for accessing and verifying information within a blockchain. It includes the algorithms of concealing or encoding information, ensuring that only the intended recipient can decipher it securely. These algorithms are instrumental in protecting sensitive data like bank cards, computer passwords, and e-commerce transactions from unauthorized access. For example, a cipher is one of the simplest algorithms that converts plaintext into ciphertext, typically by replacing each letter in the plaintext with another letter or symbol according to a predetermined mapping. Next, we will provide a brief overview of the cryptography concepts commonly employed in blockchains.

Asymmetric Encryption It is also known as public-key cryptography, which uses a pair of keys for encryption and decryption: a public key and a private key, commonly seen in the operation of crypto wallets. The public key is widely distributed and can be freely shared, while the private key is closely guarded by the owner. Data encrypted with the public key can only be deciphered by its corresponding private key, and conversely, for example, in Fig. 3, Alice encrypts the message using Bob's public key, which is accessible to all network participants. Upon receiving the encrypted message, Bob can use his private key to decrypt and see its contents. This concept is analogous to the idea of email, where everyone may know the email address, but only the individual with the password can access and review the contents.

Hash Algorithm A hash algorithm is a cryptographic function that operates in one direction, taking an input (message) and outputting a fixed-size string of characters, called as a hash (value). This output, typically comprising a sequence of numbers and letters, is unique to the input data. The one-way function exhibits the following properties for a hash: irreversibility (rendering it computationally infeasible to reverse the hashing process), collision resistance (making it challenging to find two distinct messages that produce the same hash value), high avalanche effect (having input changes result in significantly different hash outputs), and determinism (ensuring consistent hash value generation for identical input data).

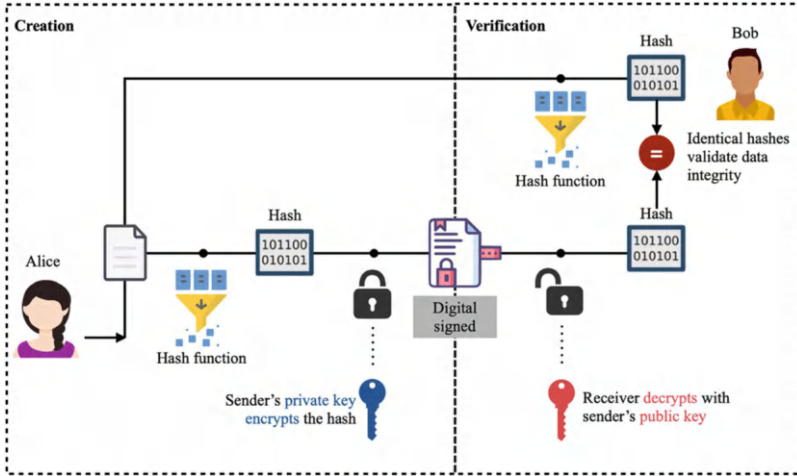


Fig. 4 Digital signature

Common hash algorithm designs include the Merkle-Damgård hash construction, which generates a hash tree structure by linking hashes over hashes, and the sponge construction, which iteratively absorbs input data into a fixed-size internal state and releases hash values in a controlled manner [26]. Furthermore, while hash algorithms are employed to secure data, in proof-of-work (PoW) consensus mechanisms (i.e., used by BTC), validators must locate a specific hash value that meets certain criteria (i.e., start with the correct amount of zeroes) to validate transactions.

Digital Signature Beyond decrypting the message, how can Bob verify that the sender is indeed Alice? This involves confirming the digital signature accompanying the message to ensure its origin from Alice and integrity during transmission, see Fig. 4.

When Alice wants to send a message to Bob, she generates a digital signature using her private key. Upon receiving the message, Bob can use Alice's public key (which is freely available) to verify the digital signature by decrypting it and comparing the resulting hash with a hash generated from the received message. If the two hashes match, it confirms the authenticity and integrity of the message, indicating that it is indeed from Alice and has not been altered in transit. This verification process relies on the fact that only Alice's private key could have generated a signature that corresponds to her public key.

2.3 Game Theory

Once we have established secure and verifiable one-on-one communication of data, the next step is to conclude transactions through a collective agreement, akin to

resolving the BGP in peer-to-peer networks. This concept is related to game theory, where a group of agents strives to achieve an agreement on a shared value for their collective output through cooperative or competitive games. In blockchain, the rules of games are represented by consensus algorithms. These algorithms provide a protocol for the network of users to follow, determining which list of transactions (block) should be appended to the chain, thus processing these transactions.

The common consensus algorithms include PoW, where miners (validators) compete to solve complex mathematical puzzles (i.e., a specific hash value) to validate transactions and add new blocks to the chain. This process demands substantial computational power and energy consumption. Once a miner finds a valid solution, it undergoes verification by other nodes in the network. If accepted, the new block is appended to the blockchain, and the miner receives cryptos as a reward. Even cryptos like BTC, Litecoin (LTC) still use PoW, the consensus mechanism is highly criticized for its energy consumption and scalability issues, leading to the development of alternative consensus methods.

Proof of stake (PoS), used by cryptos like Ethereum, shifts away from heavy computational power reliance for transaction verification. Instead, validators create new blocks and verify transactions based on the amount of cryptos they hold and are willing to stake as collateral, effectively granting them voting power. This does raise concerns about potential influence by a dominant minority. However, due to the network's substantial scale, the likelihood of a monopoly scenario is significantly reduced.

Numerous other consensus algorithms are currently undergoing continuous improvements and upgrades, each tailored to specific properties and scenarios where they excel. After traversing the three domains, we can establish a basic blockchain transaction system. In this system, Alice broadcasts the information (including the message, digital signature, and public key), which validators verify and append to the blockchain alongside other transactions using a consensus algorithm. Subsequently, Bob receives the transaction.

3 Blockchain Trilemma

The consensus algorithm and other design elements within a blockchain can vary depending on their specific purposes, resulting in different types of cryptos or tokens. The blockchain trilemma, also known as the scalability trilemma, is a concept proposed by Buterin [2] suggesting a trade-off exists among three crucial properties: decentralization, scalability, and security.

Decentralization This refers to the distribution of control across multiple nodes (users) in the peer-to-peer network. A highly decentralized blockchain consists of numerous independent nodes, rendering it more resilient against censorship and single points of failure. However, it necessitates considerable time for data transition and updates.

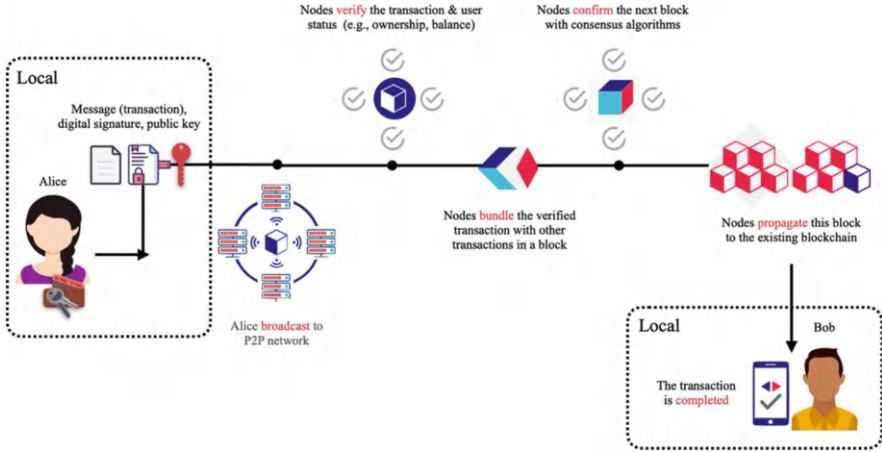


Fig. 5 A simplified blockchain system

Scalability Scalability relates to the ability of a blockchain to handle an increasing number of transactions or users without sacrificing performance. A scalable blockchain can efficiently process transactions as the user base or transaction volume expands.

Security Security is the protection level of a blockchain against various threats, including hacking, fraud, and network attacks. A secure blockchain ensures that transactions are immutable and resistant to tampering.

It is challenging for a blockchain system to simultaneously achieve high levels of all three. Improving one property often comes at the expense of another. For instance, in PoW, although the difficulty for miners to find the hash meeting the criteria is high, the time required for generating a block can be prolonged. This results in the blockchain having a higher level of security but lower scalability. Blockchain developers and researchers are constantly seeking solutions that achieve a balance among these three properties, aiming to mitigate the inherent trade-offs in the scalability trilemma. They explore various consensus mechanisms, network architectures, and protocol enhancements to address this challenge.

4 Blockchain Beyond Cryptos

Often, the public misunderstands and considers cryptocurrency to be synonymous with blockchain, overlooking other technology-driven tokens such as NFTs and utility tokens. These tokens serve various functions, including representing tangible or intangible assets and acting as a medium for energy trading, among other uses.

4.1 *Crypto*

Since the emergence of BTC in 2008, the crypto ecosystem has experienced a surge in market capitalization, reaching the global level around \$2.55 trillion in April 2024 according to the CoinMarketCap. The crypto market is characterized by its high volatility, frequent changes in market structure, and the emergence and disappearance of cryptos on a daily basis.

Unlike fiat currencies issued by governments, cryptos operate on decentralized networks based on blockchain technology. Accordingly, decentralization is a key attribute of cryptos, as they are not controlled by any central authority like a government or financial institution. Another important feature is pseudonymity, meaning that transactions are recorded on a public ledger using cryptographic techniques to secure them while maintaining the privacy of the parties involved.

CRIX, a crypto market index developed by Trimborn and Härdle [27], offers the first scientifically backed proxy to the crypto market analogous to S&P 500 measuring the forward-oriented risk in the crypto market and was acquired by Royalton Partners in 2021. The index allows to trace the evolution of diverse and fast changing crypto market with a small number of constituents. The CRIX index is being calculated by S&P Global since November 2021, is renamed as Royalton CRIX Crypto Index, and is published on the [website](#).

4.2 *Utility Token*

It is a form of crypto token specifically crafted to raise funds for project or platform development, aiming to function as a payment medium for services or products provided within the project or platform ecosystem. Utility tokens generally do not confer ownership rights and are not structured as investment instruments. Instead, their purpose is to grant access to services or products, such as unlocking premium features, covering transaction fees, or engaging in governance decisions. These tokens are often created and distributed through initial coin offerings (ICOs) or token sales, where investors purchase the tokens in exchange for existing cryptos like BTC or Ethereum. For instance, Golem (GLM) is a token that enables individuals to lease out their available computing power. The ICP token (Internet Computer Protocol token) serves as a multifaceted utility within the network, facilitating governance through voting rights, rewarding positive contributions, and facilitating transactions by acting as a means of fee payment.

4.3 *Smart Contract*

A smart contract (SC) is a self-executing piece of software, algorithms, code, or processes with predetermined terms of the agreement without any intermediary's

involvement for calculations and various problem-solving operations [29]. It has a wide range of applications due to its ability to automate and streamline processes while ensuring transparency and security, such as in supply chain management, decentralized finance protocols, real estate, and governance.

SCs enable the automation of complex processes and the creation of new types of decentralized applications (DApps). The Ethereum blockchain is one of the most popular platforms for deploying SCs, although other blockchains like Binance Smart Chain, Solana, and others also support SC functionality. Key characteristics of SCs include:

- **Autonomy:** SCs are performed automatically by the network and reduce the need for a third party to manage transactions.
- **Immutability:** SCs cannot be altered or edited once deployed, ensuring the integrity of the agreement.
- **Transparency:** SCs enable the visibility of contracts to all parties involved on the blockchain.
- **Decentralization:** An SC is replicated and distributed by all the nodes in the decentralized network, indicating there is no single point of failure and no central authority controlling the contract execution.
- **Efficiency:** By eliminating the need for intermediaries, SCs lead to reduced costs and increased efficiency.
- **Security:** SCs leverage data encryption and are tamper-proof, rendering them a highly secure alternative to paper-based contracts.

4.4 Non-fungible Token

Non-fungible tokens, commonly known as NFTs, are digital tokens existing on blockchain networks, each symbolizing a unique asset such as artwork, digital content, or media. Driven by blockchain technology and SC, NFTs inherit their functional features, including network decentralization, privacy and encryption, data immutability, and contract programmability. The NFT digital art market is increasingly recognized as an emerging avenue for creators of all levels, from professionals to amateurs, to engage with underserved populations. Lin et al. [14] propose a robust price index to depict the NFT art market and offer a gauge for market volatility.

NFT standards like Ethereum Request for Comment (ERC)-721 and ERC-1155 are employed for the creation and management of NFTs on the Ethereum blockchain. These standards establish the guidelines and protocols for the creation, ownership, and exchange of NFTs, enabling developers to create applications and marketplaces based on NFTs. ERC-721 is generally recognized as the top choice for creating NFTs due to its unique identifiers, indivisibility into smaller units, and extensive support from NFT marketplaces. ERC-1155 is another Ethereum token standard that allows for the creation of both fungible and non-fungible tokens

within the same contract. It is more flexible than ERC-721 because it supports batch transfers and reduces gas costs for transactions involving multiple tokens.

5 Blockchain Analytics

Much research has been conducted on blockchain technology and cryptos. In the following section, we showcase some perspectives and tools for analyzing them.

5.1 Network Analysis

As blockchain technology is powered by a network structure, in this section, we provide a network analysis framework. The financial risk meter (FRM) method is designed to identify individual risk characteristics and indicate systemic risk in a network topology. FRM extends the conditional value at risk (CoVaR) that accounts for two financial institutions' dependence structure changes across time and varied market conditions [1]. FRM allows for numerous nodes to be at risk and involves network dynamics combined with the analysis of tail behavior. FRM is based on the tail-event-driven NETwork (TENET) risk ideas in [8], quantile lasso regression [15], or expectile version as detailed in [23].

A linear regression for log return series $X_{j,t}^k$ is given by

$$X_{j,t}^k = \alpha_j^k + A_{j,t}^k \beta_j^k + \varepsilon_{j,t}^k,$$

with N crypto assets and m macroeconomic variables, $j \in \{1, 2, \dots, N\}$, $A_{j,t}^k = [X_{-j,t}^k, M_{t-1}^k]$ representing a $p = N + m - 1$ dimensional vector of covariates, T denoting the total number of observations, $t \in \{1, \dots, T\}$, k denoting the index of rolling windows, s representing the window size, $k \in \{1, \dots, T - s + 1\}$, and the vector β_j^k collecting p parameters. The estimated coefficients are obtained by minimizing the quantile/expectile Lasso loss function, i.e., $\min_{\alpha_j^k, \beta_j^k} \left\{ \frac{1}{s} \sum_{t=k}^{s+k-1} \rho_\tau \left(X_{j,t}^k - \alpha_j^k - A_{j,t}^k \beta_j^k \right) + \lambda_j^k \left\| \beta_j^k \right\|_1 \right\}$ with the corresponding asymmetric loss function ρ_τ at level τ , and L_1 -norm penalization parameter λ_j^k , which measures the change of tail events in terms of the change of risk driver influence. The evolution of averaged λ_j^k represents the variation of the systemic tail risks. Thereby, we establish the FRM index, denoted as $\text{FRM}_k = \frac{1}{N} \sum_{j=1}^N \lambda_j^k$, to quantify joint tail events.

The quantile/expectile lasso regression coefficients can be arranged into an adjacency matrix $A = \{\beta_{j,i}^k\}$ where $\beta_{j,j}^k = 0$ for every considered day. The adjacency matrix representation, in turn, allows us to consider the interaction between the selected cryptos in a network. Figure 6 visualizes the interdependencies in a directed graph with a focus on BTC in the crypto market from [22]. Blue

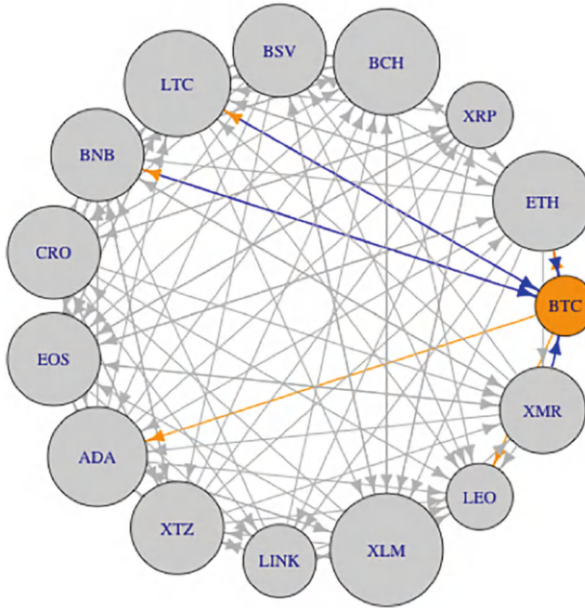


Fig. 6 Interdependencies among cryptos with a focus on BTC

lines represent coins influencing BTC, while yellow lines indicate cryptos affected by BTC. Node sizes reflect total degree. For live information, please refer to [FRM@Crypto](#), where one can find a video displaying their dynamic spillover effects.

FRM indices and dynamic networks are calculated for multiple channels including FRM@Crypto, FRM@ Europe, FRM@China, FRM@Americas, and FRM@Romania on the [FRM website](#). These risk indices and their interconnections serve as crucial tools for communicating risks to the public, understanding how risks evolve over time, and supporting decision-making.

5.2 Time-Series Methods for Cryptos

Cryptos have garnered significant attention in financial markets, leading researchers to explore various modeling techniques to predict their prices. Classical time-series models have traditionally been utilized in financial forecasting, and their application to cryptos presents both challenges and opportunities. The scarcity of reliable data sources and the unique characteristics of crypto markets, such as high volatility and limited historical data, can pose challenges for modeling using classical time-series methods [4].

One common classical time-series model used in crypto price prediction is the generalized autoregressive conditional heteroscedasticity (GARCH) model. This model incorporates conditional heteroscedasticity in the residuals to capture the non-Gaussian dynamics of crypto returns. The GARCH(1,1) model is typically formulated as follows:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

where σ_t^2 is the variance forecast, ϵ_{t-1} is the lagged error term, and ω , α , and β are parameters to be estimated. By accounting for volatility clustering and leverage effects, GARCH models can provide insights into the risk and uncertainty associated with crypto price movements [11].

While classical time-series models like ARIMA lack the probabilistic interpretation offered by machine learning techniques, they remain valuable for capturing the underlying trends and patterns in crypto price data [10]. The ARIMA(p,d,q) model is expressed as

$$\left(1 - \sum_{i=1}^p \phi_i L^i\right) (1 - L)^d X_t = \left(1 + \sum_{j=1}^q \theta_j L^j\right) \epsilon_t,$$

where L is the lag operator, d is the order of differencing, p and q are the orders of the autoregressive and moving average parts, respectively, and ϕ_i and θ_j are parameters.

Additionally, models like GARCH have been applied to analyze conditional heteroskedasticity in crypto-asset returns, shedding light on the volatility dynamics of cryptos [25]. Moreover, stability analysis of cryptos using classical time-series models has been explored to understand the persistence and predictability of crypto price trends. Researchers have assessed the stability and predictability of crypto time-series data by measuring parameters like the largest Lyapunov exponent and approximate entropy [18].

Crypto price prediction has attracted significant attention in recent years, with researchers exploring various models and techniques to forecast the volatile nature of crypto markets. Deep learning methods, particularly recurrent neural networks (RNN) and long short-term memory (LSTM) models, have emerged as promising tools for predicting crypto prices [6, 20]. These models leverage the sequential nature of time-series data to capture patterns and trends in crypto prices [7].

In addition to deep learning approaches, some studies have delved into regime switching prediction models that consider the illiquid and highly volatile nature of cryptos. By incorporating econometrics and statistical models, researchers have aimed to better understand and predict crypto price volatility [5]. Furthermore, sentiment analysis and market sentiment have been integrated into prediction models to enhance forecasting accuracy [9].

Researchers have also explored the use of reinforcement learning to improve crypto price prediction performance, particularly in dealing with non-stationary

time-series data [24]. By incorporating trend classification and deep learning techniques, these models aim to capture the dynamic nature of crypto prices. Moreover, the integration of IoT concepts, cloud computing, and data parallelism has been proposed to enhance real-time crypto price prediction models [21].

Various studies have highlighted the importance of utilizing machine learning algorithms, such as XGBoost, LightGBM, and linear models, to predict the prices of popular cryptos like BTC, Dogecoin, and Ethereum [13]. These models leverage historical price data and technical indicators to forecast future price movements. Additionally, the application of optimum convolutional neural networks (CNN) has shown promising results in predicting crypto prices with high accuracy [17].

6 Conclusion

Similar to the development of other technologies throughout history, the journey often starts with hype and doubts, eventually integrating seamlessly into our daily lives. The evolution of blockchain technology represents not only a technological advancement but also heralds the emergence of a transformative economic paradigm. Since BTC's emergence in 2008, the crypto market has experienced exponential growth, boasting over 10,000 different cryptos in circulation today. This surge has prompted numerous startups and corporations to pivot their business models to incorporate blockchain technology and digital assets. Governments have begun to take notice of its development, engaging in discussions regarding the implementation and potential of governmental digital currencies, i.e., Central Bank Digital Currency (CBDC). This chapter provides an overview of the fundamental concepts of blockchain technology, along with a selection of methodologies that offer different perspectives for evaluating this technology. It serves as a simple guide for readers to explore this handbook.

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Part II

Digital Assets in Blockchains

NFTs and VizTech



Bingling Wang, Min-Bin Lin, and Wolfgang Karl Härdle

Abstract Non-fungible tokens, or NFTs, have been in the focus of emerging digital asset transactions, have gained a lot of attention of art investors, and are among the fastest growing Web 3.0 and crypto products. The popularity of NFTs has started with a generative art collection—CryptoPunks from Larva Lab in 2017, and the total market capitalization has been over US\$ 41 billion since 2021.

We present a short history sketch of this NFT art phenomenon and investigate the properties of CryptoPunks via their flattened pixel vector. This leads to very high-dimensional data, which we represent via modern VizTech, in order to detect clusters and similarities among the digital assets. Employing the UMAP technique, we are able to attach to and patterns and elements. The Uniform Manifold Approximation and Projection (UMAP) is explained in detail and compared with other localizing techniques like t-SNE. The two-dimensional projections of UMAP disclose clusters among these CryptoPunks which one can associate with prices and traits.

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1 Introduction

Non-fungible tokens (NFTs) have emerged as a new form of digital asset whose proof of ownership is recorded on the Ethereum blockchain through smart contracts. Since the beginning of 2021, NFTs have garnered significant interest from investors, and the craze shows no signs of slowing down. NFTs are versatile, enabling trading of various digital assets, including artworks, videos, audios, texts, tickets, domain names, and even virtual lands.

NFTs are a realization of smart contracts by which they enable programmatically executed transactions within a blockchain [11]. Minted through smart contracts, the ownership of NFTs can be transferred to other accounts.

We examined one of the most well-known NFT collections, CryptoPunks, which is representative of this emerging digital art market. Other similar collectable art works include Bored Ape Yacht Club and Hashmasks. These NFTs not only hold artistic value but also possess social attributes. People often use them as avatars on social media platforms for community identification and to flaunt their ownership. CryptoPunks is an algorithm generating a set of 10,000 NFTs as 24×24 pixel art images of punky-looking figures, including male, females, and rarer types such as apes, zombies, and aliens. As one of the pioneers of applying ERC-20 standard on Ethereum blockchain, CryptoPunks is still one of the collections with the highest market cap, with prices ranging from hundreds of thousands to millions of dollars.

CryptoPunks is characterized by images with similar styles, components, and structures. To gain insight into the high-dimensional space of the image data, we used several visualization techniques. A modern VizTech tool is Uniform Manifold Approximation and Projection (UMAP). UMAP is capable of projecting and visualizing data in low-dimensional space, providing a better understanding of the relationships between high-dimensional data points. UMAP falls in the class of nonlinear dimensionality reduction techniques, which include LLE (Local Linear Embedding) [6], Isomap [7], Diffusion Maps [1], and t-distributed Stochastic Neighborhood Embedding (t-SNE) [8, 9]. The linear dimensionality reduction methods include PCA (principal component analysis) [5] and multidimensional scaling (MDS) [2].

2 Curse of Dimensionality

Visualization of unstructured data in high dimensions is a challenging task for the simple reason that our human visual perception system is concentrating on 2–3 dimensional structures [3]. The most important element of any Visualization Technique (VizTech from here on) is that “close” neighbors of data in high dimension should also be “close” neighbors in the two-dimensional plane that the human retina perceives from a flat computer screen. On purpose we use here the “”

quote symbols to indicate that any VizTech will depend on the relation of geometries used in both the high- and low-dimensional space.

Any geometry defining neighborhoods of data will use some sort of ball or neighborhood to define closeness. Unfortunately the number of data points in high dimensions is so sparsely distributed and that the unit ball is almost empty. This fact is also called the curse of dimensionality that we meet here again by flattening the 24×24 pixel images of, say, CryptoPunks. This results in a collection of $n = 10000$ vectors of size $d = 576$. This dimensionality gets pushed up further since each pixel is represented by a three-dimensional RGB color code. So in fact we are looking at $n = 10000$ digital art CryptoPunks that are flattened to $d = 1728$ vectors. This leads to serious projection issues, as we explain below.

Why is the HD (high dim) space so sparse? The following thoughts will make this evident. Indeed, assume to sample data points that are uniformly distributed in a unit cube $(-1, 1)^d$. One easily calculates the volume of the unit ball in $d = 2$ dimensions as π , i.e., the proportion of the unit circle in two dimensions relative to the unit cube volume of $\pi/4 = 0.79$. Hence we can count on average only on 79 points lying in the neighborhood defined by the unit ball. The general formula for the unit ball in d dimensions is

$$V_d = \frac{\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2} + 1)},$$

where

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt,$$

$$\Gamma(u) = (u - 1)!$$

Hence for $d = 5$ dim we have only 16 points in the unit ball, and in $d = 15$ dim one would need $n = 100000$ data points to guarantee 1 data point inside the unit ball. Feel free to calculate this ratio for $d = 1728$, based on $n = 10000$ CryptoPunk images... We may conclude here that the higher dimensions we try to understand the emptier are the neighborhoods we like to describe. A glimpse at $\Gamma(1728/2 + 1)$ shows that this equals $\Gamma(\frac{1728}{2} + 1) = \Gamma(865) = 864! \approx (n/e)^n \sqrt{2\pi n} \approx (318)^{864} \cdot 74$, $n = 864$.

3 Visualization Techniques

In the following, we showcase six state-of-the-art modern visualization techniques and apply them directly on image data.

3.1 *Multidimensional Scaling*

Multidimensional Scaling (MDS) is one of the most classical linear visualization techniques which projects high-dimensional data to a low-dimensional space aiming at maximizing the pairwise dissimilarities between data points. Thus, for an original dataset X , its low-dimensional representation Y can be estimated by minimizing the disagreement between an input distance matrix D calculated from the original data and the output embedding. The stress function is written as

$$\Phi(Y) = \left(\frac{\sum_{i,j} (b_{ij} - Y_i^\top Y_j)^2}{\sum_{i,j} b_{ij}^2} \right)^{1/2},$$

where $i, j = \{1, 2, \dots, N\}$, and Y_i and Y_j are m -dimensional vectors. b_{ij} are the elements of the centered distance matrix B , where $B = -\frac{1}{2}CD^{(2)}C$ is the squared distance matrix centered by $C = I_N - \frac{1}{N}J_N$. J_N is an $N \times N$ matrix of all ones. Thus, the embedded output Y can be derived by solving $B = YY^\top$ via applying spectral decomposition on matrix B , obtaining $Y = E_m \Lambda_m^{1/2}$, where E_m is the matrix of top m eigenvectors and Λ_m is the diagonal matrix of the largest m eigenvalues of B . Both focusing on maximizing the dissimilarities among the data, MDS is equivalent to PCA when the distance measure is Euclidean, resulting in B to be the variance-covariance matrix of X . But when we talk about MDS, we are actually talking about a series of variants, including different ways to form distance matrices and nonmetric MDS algorithms. In general, MDS uses the distance matrix and PCA works on the original data [4].

3.2 *Isomap*

A nonlinear visualization technique that inherits the advantages of PCA and MDS in having a global optimal solution and asymptotically converges to the true geometry for certain class of manifold is so called Isomap. By only incorporating a distance measure of geodesic distances, it tackles the problem that has always been inherent in linear dimensional reduction techniques—learning the high-dimensional manifold structures.

The geodesic distances can be calculated by summing up edge weights along the shortest path between two data points. This requires to build a neighborhood graph on the data and renders a distance measure capturing local neighborhood structures. Thus the three steps of Isomap algorithm is designed as follows: First, define a graph G for X by connecting neighboring points with edges of weight $d_X(i, j)$. Neighborhood of X_i is defined either as all the points in a unit ball centered X_i with radius of ϵ or by taking K -nearest neighbors. Second, calculate the geodesic distance between X_i and X_j by $\min\{d_G(i, j), d_G(i, k) + d_G(k, j)\}$,

$k = 1, 2, \dots, K$, where d_G is defined as the pairwise edge weight and is set to be infinity when the pairs are not connected. Third, similar to PCA and MDS, form an embedding Y in a low-dimensional space where the coordinates are the top d eigenvectors according to the largest d eigenvalues of the centered geodesic distance matrix.

PCA and MDS are good at learning global relationships in a Euclidean structure by either maximizing global variance (PCA) or maximizing pairwise distances (MDS) from high-dimensional space. But in terms of high-dimensional nonlinear structure, Isomaps are proved to be capable of detecting true degrees of freedom of the “handwriting number” and the “faces” data.

3.3 Locally Linear Embedding

Another manifold-based nonlinear dimensional reduction technique is Locally Linear Embedding (LLE), which intends to restore global nonlinear relationship in a lower dimensional space by applying only local linear fits. In this manner, LLEs avoid calculating the pairwise distances among data points which can lead to a very sparse matrix and simultaneously save computing time and space. The algorithm consists of a two-step estimation procedure. Suppose there is a D -dimensional dataset X , for each point $X_i, i = \{1, 2, \dots, N\}$, (1) estimating an $N \times K$ weight matrix W presenting the local structure of X_i regarding its K -nearest neighbors, with a loss function

$$e(W) = \sum_i |X_i - \sum_j W_{ij} X_j|^2$$

$$s.t. \sum_j W_{ij} = 1,$$

which enables X_i to be reconstructed only by the linear relationship with its neighbors, and (2) estimating a low-dimensional representation Y maximizing the retaining of the global manifold structure of X . With W_{ij} being estimated from the first step, the second step can be estimated by solving a quadratic form with eigenvalue decomposition, as the cost function being formed as

$$\Phi(Y) = \sum_i |Y_i - \sum_j W_{ij} Y_j|^2.$$

These two steps formulate the problem in a locally linear fashion, but resulting in a nonlinear embedding that preserves both local and global geometric structures. It has been shown to more effectively preserve global geometric relationships on the ‘Swiss roll’ dataset compared to traditional techniques like PCA and MDS. Course and codes are found in [Quantinar.com](https://www.quantinar.com) .

3.4 *t*-SNE

t-SNE, compared to the abovementioned nonlinear visualization techniques, is the most widely used method and produces stable and better performance on various types of data in the sense of interpretability. Compared to techniques developed before, such as LLE and Stochastic Neighbor Embedding (SNE) which mainly focus on preserving the local similarities in low-dimensional space, t-SNE is able to restore the smaller pairwise distances as much as possible yet ensure gaps between clusters. As assumed before, for HD data points X_i , X_j , and their LD mapping Y_i , Y_j , $i, j = \{1, 2, \dots, N\}$, define HD and LD pairwise similarity as conditional probability $p_{j|i}$ and $q_{j|i}$, respectively,

$$p_{j|i} = \frac{\exp(-\|X_i - X_j\|^2/2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|X_i - X_k\|^2/2\sigma_i^2)};$$

$$q_{j|i} = \frac{(1 + \|\mathbf{Y}_i - \mathbf{Y}_j\|^2)^{-1}}{\sum_{k \neq i} (1 + \|\mathbf{Y}_i - \mathbf{Y}_k\|^2)^{-1}}.$$

This measures the probability of X_j or Y_j being in the neighborhood centered at X_i or Y_i , assuming neighborhoods are Gaussian distributed with certain variance in HD space and t-distributed with degree of freedom in LD space. Different from SNE, using a heavy-tailed distribution instead of Gaussian to describe the LD neighborhood is aiming at keeping dissimilar points far apart, thus solving the “crowding problem.” Now define the pairwise joint probability distribution

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2N};$$

$$q_{ij} = \frac{(1 + \|Y_i - Y_j\|^2)^{-1}}{\sum_k \sum_{l \neq k} (1 + \|Y_k - Y_l\|^2)^{-1}}.$$

The LD mapping can be calculated by minimizing the mismatch between HD and LD local structures described by joint distribution P and Q , with the cost function:

$$KL(P \parallel Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}.$$

The optimization of the cost function can be obtained by gradient descent. Unlike SNE, t-SNE restores global structure better as the gradient of t-SNE approaches zero when high-dimensional (HD) and low-dimensional (LD) distances both are small or large. However, t-SNE has a weakness where the increase in HD distance may not correspond to an equal rate increase in LD distance (with the gradient still near zero), resulting in inadequate cluster concentration. This can cause dissimilar clusters to be positioned too closely, meanwhile similar clusters to be not close enough, leading to randomness in cluster location in LD space. Course and codes are found in [Quantinar.com](https://quantinar.com) .

3.5 UMAP

UMAP (Uniform Manifold Approximation and Projection) can solve the aforementioned problem. It has been designed to preserve HD (High Dim) local and global structure. t-SNE tries to do the same, but UMAP is often better at displaying global structure than t-SNE [10] on nearly all kinds of data. The basic idea is to construct a topological representation of HD data and then to optimize the cross-entropy between HD and LD (Low Dim) topological representations.

Let us demonstrate this via a simple sine curve in two dimensions with irregularly distributed points and an open cover (via balls!). The basic idea is to convert overlapping and intersecting balls into simplices of dimensions 0, 1, and 2 here. Converting these simplices into edge weights and using some local metrics leads to an optimization procedure that aims at finding an LD representation that is “close” to the corresponding HD one. The fundamental idea is to compare vectors of 1-simplices, i.e., the probabilities of being close or edge weights between 0-simplices (data points). These are Bernoulli random variables, since an edge either exists or not. This brings up classical Kullback-Leibler information theory and naturally leads to cross-entropy as the appropriate measure. Hence the task to “minimize cross-entropy” leads to maximum “closeness.” The loss function of UMAP reads

$$\sum_{e \in E} w_h(e) * \log \left\{ \frac{w_h(e)}{w_l(e)} \right\} + \{1 - w_h(e)\} * \log \left\{ \frac{1 - w_h(e)}{1 - w_l(e)} \right\},$$

where here E is the set of edges, $w_h(e)$ is the edge weight in HD, and $w_l(e)$ is the edge weight in LD.

This leads to a gradient which only approaches zero when small and large HD distances are well restored in LD space, and the penalty is large enough when the HD and LD distances do not match. This creates sufficient gaps between clusters, resulting in a better-organized embedding that is easier to interpret. Course and codes are found in [Quantinar.com](https://quantinar.com) .

4 Application on CryptoPunks

4.1 Data

Our image data are collected from [UNPKG](https://unpkg.com), where they were originally stored on by their official creator larvalabs.com. Now the image and attributes data are fully embedded in the smart contract and stored on blockchain. The whole collection contains 10000 images, each in the size of 24×24 pixels with transparent background.

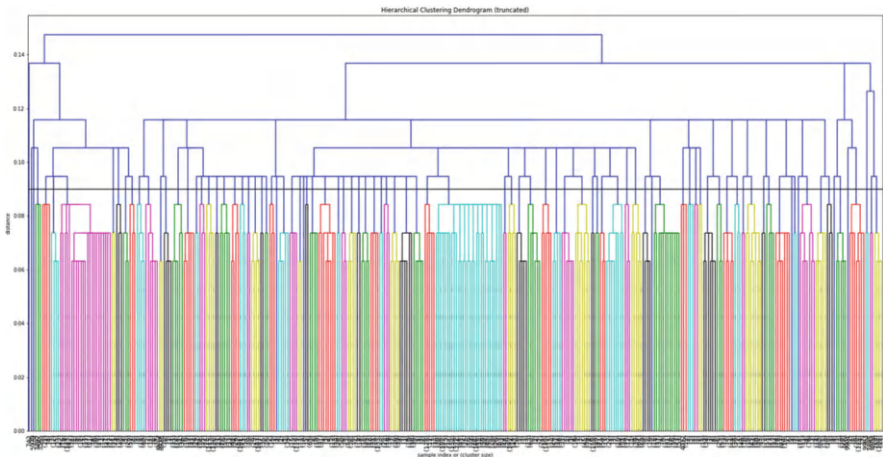


Fig. 2 Truncated hierarchical clustering dendrogram of CryptoPunks

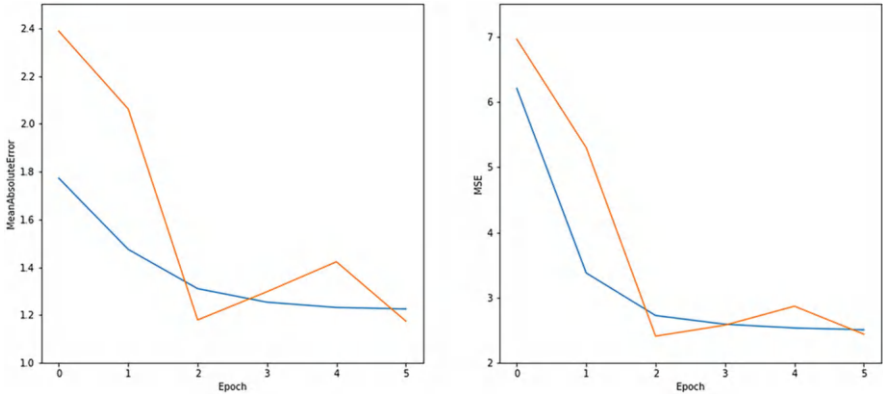


Fig. 3 Metrics of CNN regressor on 24×24 images. Sample loss and validation loss are marked in blue and orange colors, respectively

another. Figure 2 shows the truncated dendrogram of hierarchical clustering with complete linkage and Hamming distance, where distance threshold of 0.09 has been marked with a vertical line. This results in 93 clusters which will be used to evaluate the Viztech results (Figs. 3, 4, 5, 6, and 7).

4.3 Visualization Results

Figures 8, 9, 10, 11, and 12 show the results of the aforementioned five visualization techniques applied to the flattened raw image data. Each image originally loaded as

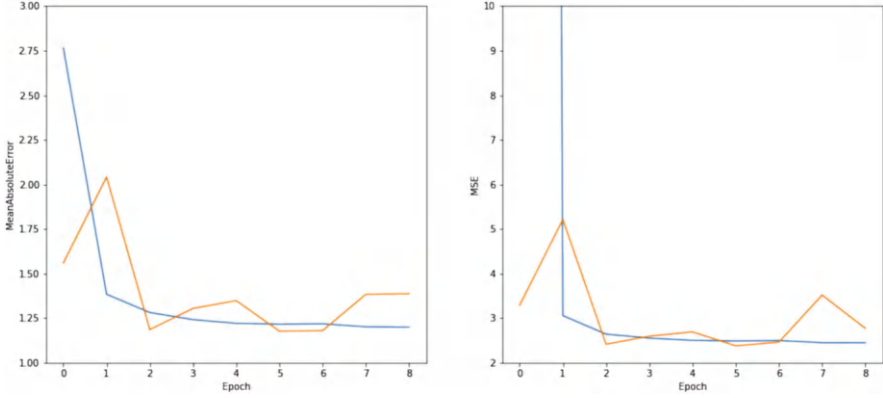


Fig. 4 Metrics of CNN regressor on 240×240 images. Sample loss and validation loss are marked in blue and orange colors, respectively

$24 \times 24 \times 3$ RGB data is flattened into a vector of 1,728 dimensions, and then all 4964 CryptoPunks with a historical sale price are mapped to two-dimensional space. Subplots of the 2-dim embedding from top to bottom and left to right are marked with different attributes, such as the original punk faces, the color-coded sale prices, color-coded 5 Punk types, and 93 clusters as defined in Sect. 4.1.

From the top left figure of each VizTech results, it can be observed that most of the visualization techniques can detect local structures from the very high dimensional image data and present them in an LD mapping quite well even without any preprocessing. It manifests in the fact that the similarities in the color blocks of the images are well preserved. For example, from the results of MDS (Fig. 8), we can roughly observe three clusters with gaps among them, representing female, male with darker skin colors, and male with lighter skin colors, but the clusters did not separate smaller details, such as men and women with the same hair color well.

From MDS to UMAP, it can also be found that there is more and more emphasis on keeping global structures or balancing the local and global structures of the data. In Fig. 8, although it can be found that the 2-Dim mapping is regularly arranged from the center to the outside according to the skin color and hair color, indicating the true degree of freedom embedded in the data found by MDS, the gap between the data did not separate the Punks well with respect to the skin and hair color. In Isomap and LLE we have more visible clusters, but similar concern still remains. t-SNE has a much better representation in searching balance between keeping local and global structures, forming groups with similar attributes and gaps between them. But as mentioned before, due to the unbalanced loss function on keeping local and global distances, t-SNE still has the issue of presenting the location of similar groups. For example, the groups with yellow hair are arranged on two sides of the map, far away from each other, while UMAP keeps the yellow haired groups close to each other in

Fig. 12. Furthermore, UMAP results in more concentrated groups by keeping wider gaps between groups.

The colored mark of the bottom two subplots helps us further evaluate the embedding by comparing the mapping to the similarity of text descriptions. It can be shown that UMAP and LLE can set apart the major groups of male and female (marked yellow and orange in the bottom left figure) better, while UMAP and MDS have a comparably better performance on keeping minor groups of apes, zombies, and aliens closer (marked pink, gray, and red, respectively). From the color marks of the bottom right subplots, we can conclude that none of the methods are visualizing the image from the same perspective as the text attributes; thus no obvious color clusters can be observed. This may due to the different logic of defining similarities. But Isomap, t-SNE, and UMAP can still form several groups similar to the hierarchical clusters.

As for the price marked as color ranges shown in the top right subplots, no clusters or regular pattern can be found, which means no obvious link between embedding and the price.

In conclusion, UMAP performs best in visualizing and interpreting the local and global structure of the raw images of CryptoPunks.

5 CNN with VizTech

Based on the VizTech results of the previous chapter, VizTech is applied with convolutional neural network (CNN) to the images of CryptoPunks and a vector of sale price, aiming at further investigating links between the images and the prices.

5.1 CNN on CryptoPunks

A regression CNN has been built based on the VGG16 model from [Keras](#), with 4964 CryptoPunk images as input and log price as output. This model contains five convolutional blocks, each with a max pooling layer and “ReLU” as activation function of the convolution layer, two fully connected layers with “SeLU” as activation function, as well as a last layer with sigmoid function and dropout rate of 0.2.

The model is designed for an input of the original image size (24, 24, 3) and an enlarged image size (240, 240, 3) to compare the impact of high and low resolution on the results. To ensure that both the training and test sets contain samples representing a wide range of attributes, the training set consists of 4,183 observations randomly selected from all 93 hierarchical clusters. Test set contains

788 observations over 88 clusters. The size ratio of training set and test set is about 5 : 1, and the validation set is randomly drawn from the training set in the size of 847 observations containing samples from all 93 groups. The model performances are evaluated by Mean Absolute Error (MAE) and Mean Squared Error (MSE). The training loss of model of 24×24 and 240×240 pixels input is shown in Figs. 3 and 4, respectively. Although the training loss converges slower, input with better resolution will not lead to a better performance. So we take the first model which provides an out of sample MSE of 2.76 and MAE of 1.35. Filters of the first and seventh CNN layers are plotted in Figs. 6 and 7. We can see that filters in the first layer put weights on larger features such as skin, eyes, and hair. Filters in the seventh layer put more emphasis on minor details, such as the pipe and other small areas. This enables CNN to catch attributes from the image and combine them by giving them different weights.

5.2 Visualizing CNN Layers with UMAP

Results are further investigated by applying UMAP on the output of the dense layer before the output layer. VizTech helps us to visualize and analyze the pricing mechanism fitting by the regression model.

From Fig. 5 it can be observed that overall the gaps between groups are smaller compared to UMAP result on raw data, even with a smaller neighborhood, which means CNN found more link between groups. There are several components which may link to the price. From the left to the right of the embedding, the color contrast changes from low to high, cool hair colors are mostly located in the lower middle, light skin tones are mostly located in the upper right, and similar hair colors are often grouped together. We can also find that males and females are not clearly separated, but that rarer Punk types form clusters, such as aliens. Accessory and price groups still arranged quite random on the mapping, but groups seem more concentrated. This also indicates that CryptoPunk prices are not related to individual attributes, but rather the combination of them. In conclusion, the pricing mechanism can be learned by CNN from images, and VizTech forms different visualization which helps interpreting the model.

Appendix

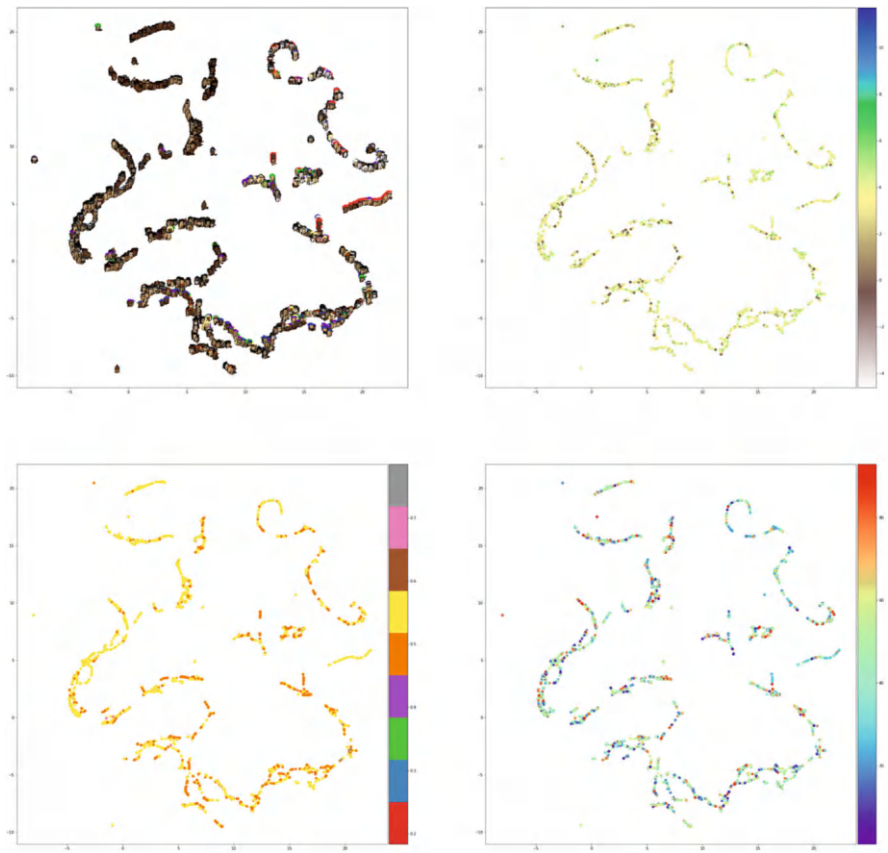


Fig. 5 The layer before output layer in UMAP



Fig. 6 Filters in the first CNN layer

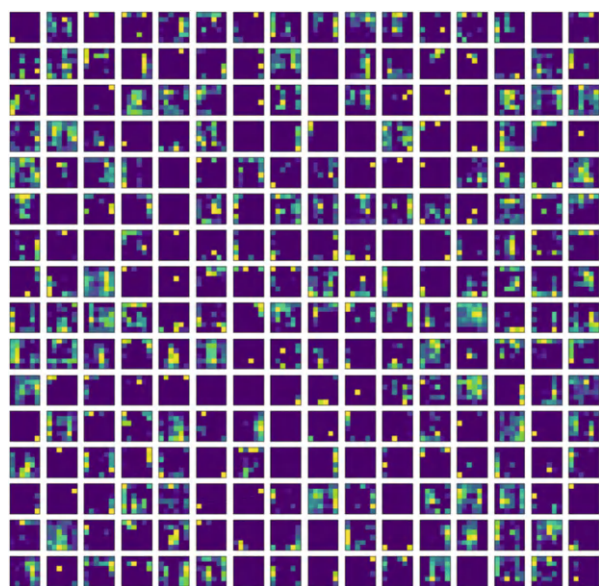


Fig. 7 Filters in the seventh CNN layer

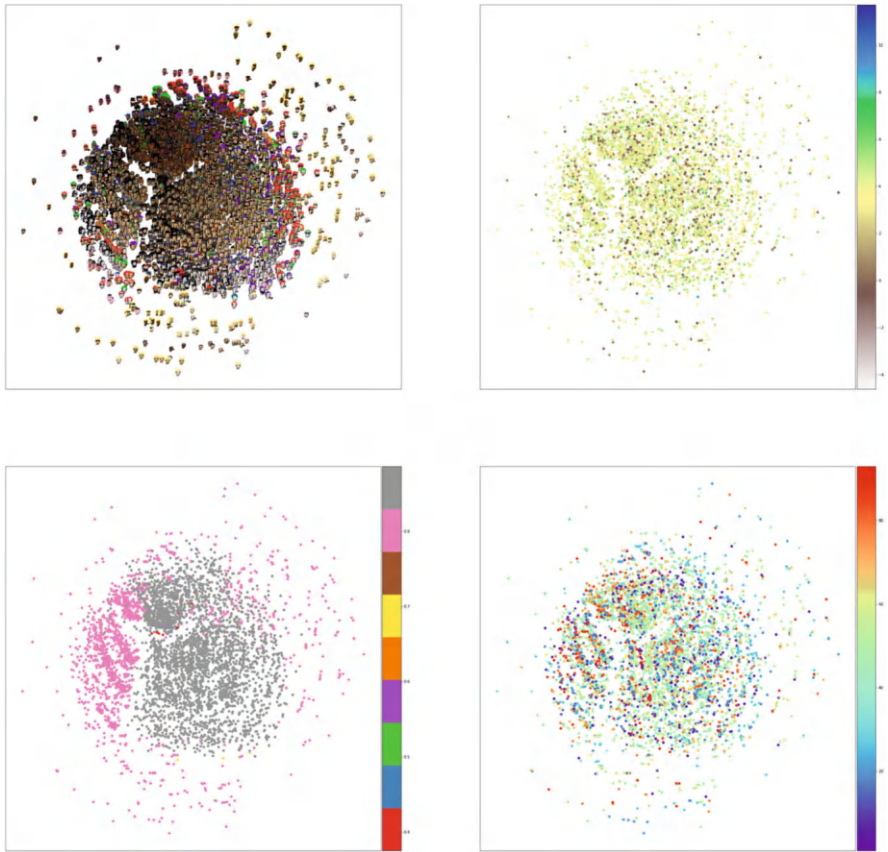


Fig. 8 MDS visualization result of CryptoPunks 

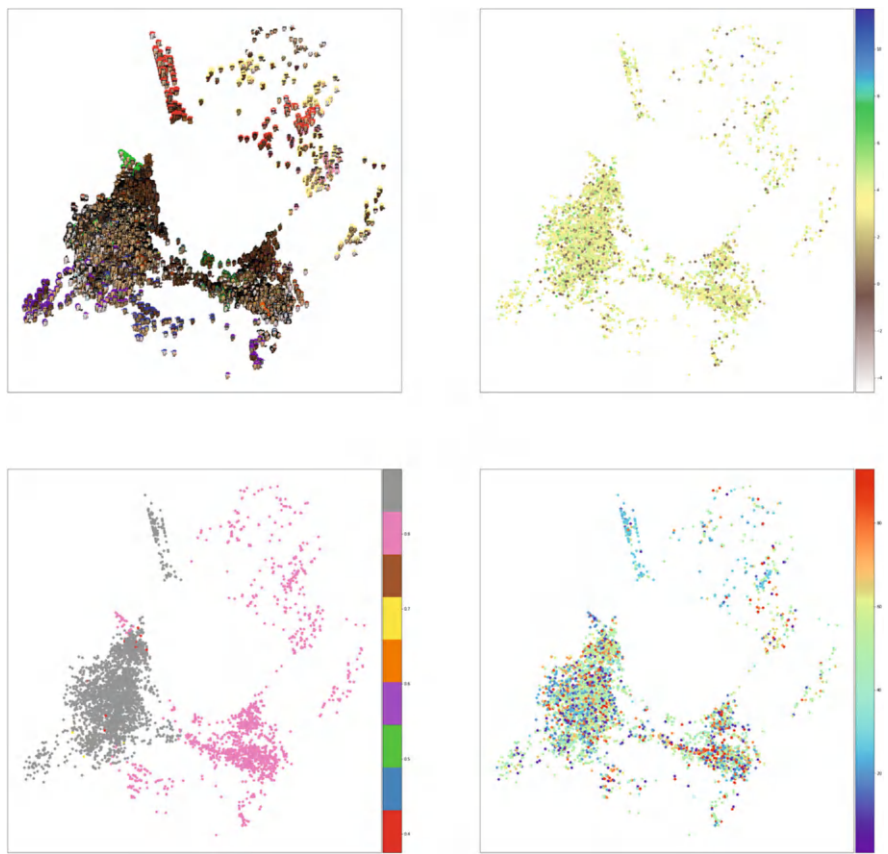


Fig. 9 Isomap visualization result of CryptoPunks 

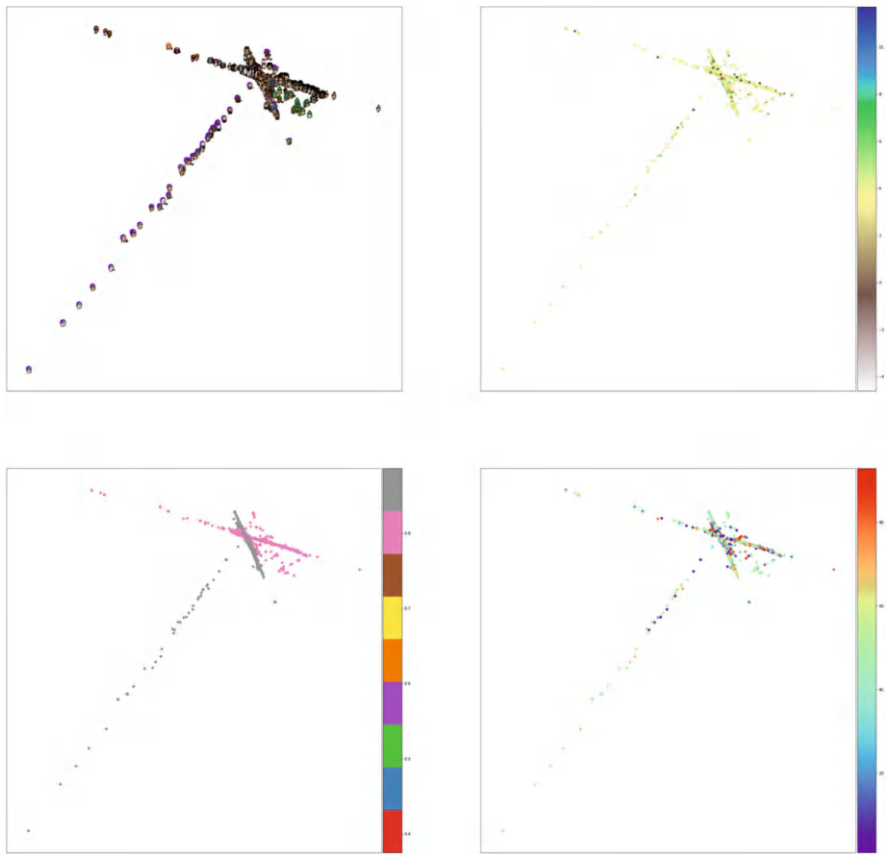


Fig. 10 LLE visualization result of CryptoPunks



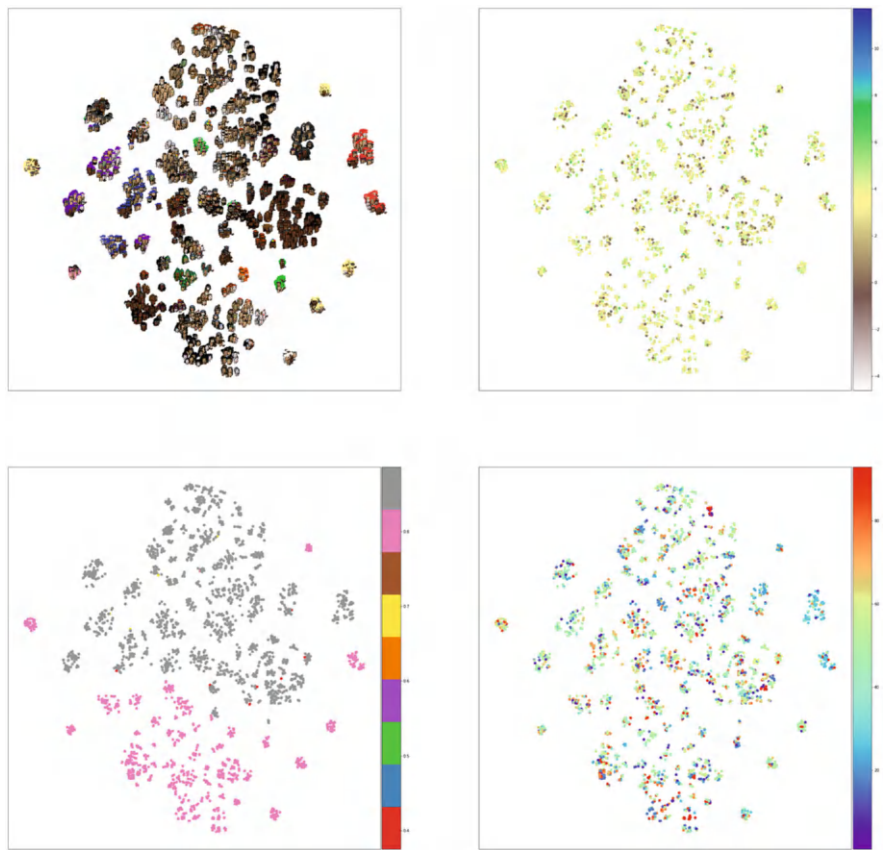


Fig. 11 t-SNE visualization result of CryptoPunks 

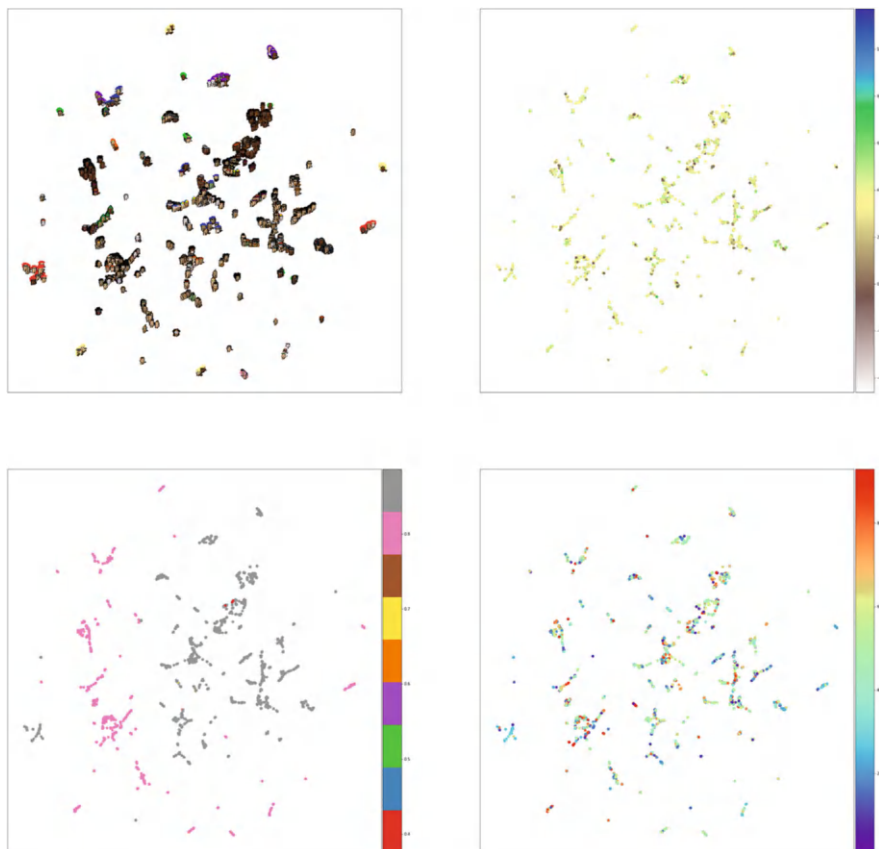


Fig. 12 UMAP visualization result of CryptoPunks 

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Cryptocurrency Liquidity Forecasting



Ilyas Agakishiev, Wolfgang Karl Härdle, and Daniel Traian Pele

Abstract Market makers provide liquidity to small investors. To optimally provide liquidity and deal with potential demand spikes, it is important to know the liquidity required in advance. This chapter describes an algorithm that attempts to predict the liquidity for a 4-hour period using previous transaction data, return data, sentiment data, and macroeconomic data. One of the challenges to consider is the daily and weekly seasonality of order data. The main algorithm is based on LSTM, with data being preprocessed using a wavelet-based low-pass filter. As alternatives, SARIMAX and TBATS algorithms are considered, as they can also deal with seasonality. Experiments have shown that while all three algorithms capture seasonality well, the LSTM-based algorithm outperforms the other two

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overall, mostly by dealing slightly better with unusual situations. In addition to that, an ensemble model was built that combines the three approaches using a ridge regression, boosting the performance further. Data for experiments was provided by Nuri, a market maker in Berlin, Germany.

1 Introduction

The cryptocurrency market is still relatively new and contains unpredictable surprises. As Bitcoin and other assets constantly switch between growth and decline, the market attracts more and more investors, both new and experienced. High-profile banks, such as Bank of America and Goldman Sachs, allow cryptocurrency purchases to some degree. Standard and Poors, at the time of writing, is planning to offer ETF shares of CRIX, a cryptocurrency index [14, 27], while Blackrock is planning to offer a Bitcoin ETF [18]. For the liquidity providers of cryptocurrencies, this means that there is, in general, more demand for Bitcoin than there is supply, but there could also be sudden movement in the opposite direction. The FTX incident has provided us with an example of how important liquidity is at a cryptocurrency exchange. To be optimally prepared for any scenario, a prediction algorithm is required.

While there is a plethora of papers aiming at predicting the returns or volatility of assets, including cryptocurrencies, this is, to our knowledge, the first paper that attempts to predict the demand and supply of a digital asset for a liquidity provider. This is important, as having too little liquidity may prevent some orders from executing, thus missing out on fees, while too much liquidity results in unnecessary opportunity costs. We also provide the regulators with a tool to assess the conditions in the cryptocurrency market. The tools described here may assist them in formulating relevant policies to stabilize and regulate the market.

There are several existing models useful for such an analysis. ARIMA [1] is the classical tool for time series analysis. It has received several extensions, such as VARIMA [3] for multiple time series, SARIMA [24] for time series with a seasonality, and SARIMAX, which includes external factors. Other methods that account for seasonality include Facebook prophet [26] and TBATS [17]. Lastly, machine learning-based methods, like LightGBM [16] or LSTM [12], have also seen success. For volatility forecasting, GARCH [5] models and their extensions (such as EGARCH [20], GJR-CARCH [10], and more) are applied instead.

In a related study [8], the objective was to forecast market liquidity in both crypto- and fiat currencies. The researchers defined market liquidity using the logarithmic rates of bid-ask spreads. The findings indicated that the K-nearest neighbor (KNN) approach outperformed ARMA and GARCH models in predicting the logarithmic rates of bid-ask spreads for both crypto- and fiat currencies.

Preprocessing techniques allow to retain useful information while discarding the rest. They can also transform the provided information, such that models learn faster from that information. The fast Fourier transform [11] represents the data

in a frequency domain. Wavelets [13] expand on the idea, by allowing localized decomposition.

During the research, we faced many challenges, but the biggest one was the seasonality of data. As the goal is to predict demand for Bitcoin for the next 4-hour period, there will be more activity at some times than others. Nuri, a blockchain banking service, operated exclusively in Germany, so naturally there was more activity in the day of Central European Time (CET). In addition, there is more activity on workdays than on weekends. This and other unique features of the dataset will be addressed when designing the approximation method for market activity.

The remaining chapters are as follows. In Chap. 2, the provided data is described and analyzed, and additional features are synthesized to form a data universe for modeling and forecasting. Next, in Chap. 3, the methodology is described. This includes preprocessing using wavelets and predictive models. Then, in Chap. 4, the evaluation is conducted on the dataset, the performance is measured, and the results are interpreted. Finally, suggestions for future research are stated.

The code for the chapter is written in Python and published in Quantlet, while an overview of the chapter is presented in Quantinar.

2 Dataset

2.1 Original Data

The dataset consists of confidential data provided by Nuri and data from public sources. The currency pair to investigate is BTC/EUR. Nuri provided the prices and the transaction order history. The time span is March 2019 to January 2021.

Now, the order data will be evaluated. In the given dataset, every single transaction is recorded, and there may be multiple transactions each second. As the data is very granular, it will be consolidated in 4-hour time periods in order to reduce noise. The independent (target) variable for this research will be the difference between the number of Bitcoins sold to Nuri and the number of Bitcoins sold by Nuri in a time period, as stated below:

$$o_t = \sum_{\tau=t-1}^t V_s - \sum_{\tau=t-1}^t V_b, \quad (1)$$

where o_t is the target variable, V_s are Bitcoins sold to Nuri in a transaction, and V_b are Bitcoins sold by Nuri in a transaction. Thus, the transaction volumes are accumulated for each 4-hour period.

The metric is shown below in a histogram (Fig. 1):

It can be observed that, both in the norm and in the extremes, Nuri sells more Bitcoins than it buys most of the time. This will be important when interpreting the final results.

Fig. 1 Histogram for the independent (target) variable [CryptoVolumeForecasting-Data_visualization](#)

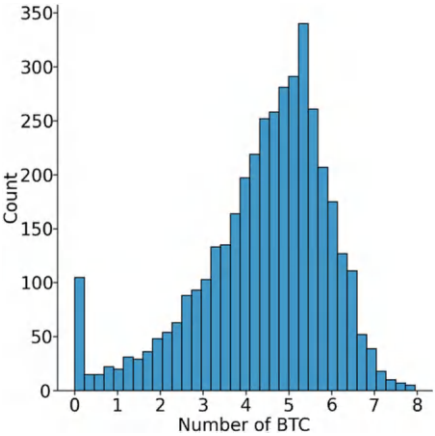
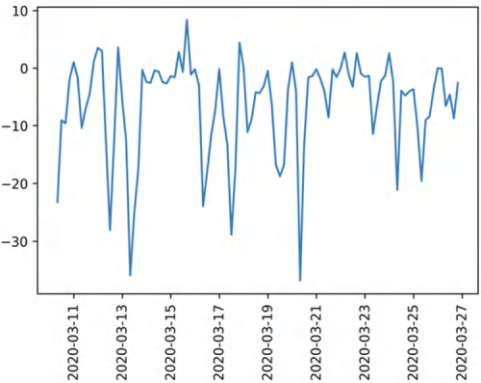


Fig. 2 Snapshot of trading volume. Activity is much lower at night and on weekends [CryptoVolumeForecasting-Data_visualization](#)



Besides the noticeable skewness, another property of the dataset is the seasonality. In fact, there are two types of seasonality: daily and weekly. They can be observed in the following data samples (Fig. 2):

This seasonality is easy to explain. As Nuri operates exclusively in Germany, the trades are usually done during daytime of the CET time zone and during weekdays, even though Nuri operates around the clock. The potential effect of algorithmic trading on the predictions seems to be negligible. This is consistent with other research on this topic [21].

To further elaborate on the seasonality, box plots are considered for every 4-hour interval (Fig. 3):

One can observe that most activity is in the morning hours. This is also where the skew toward Bitcoin demand is the highest. More descriptive plots can be found in Appendix A.

Next, the Bitcoin prices are observed. During most of 2019 and 2020, the price fluctuated between 5000 and 10000 dollars, before initiating a substantial rise from the end of 2020 onward (Fig. 4).

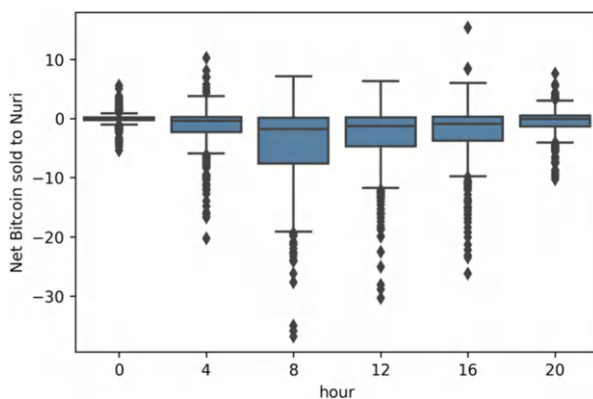


Fig. 3 Snapshot of trading volume. Activity is much lower at night and on weekends [CryptoVolumeForecasting-Data_visualization](#)

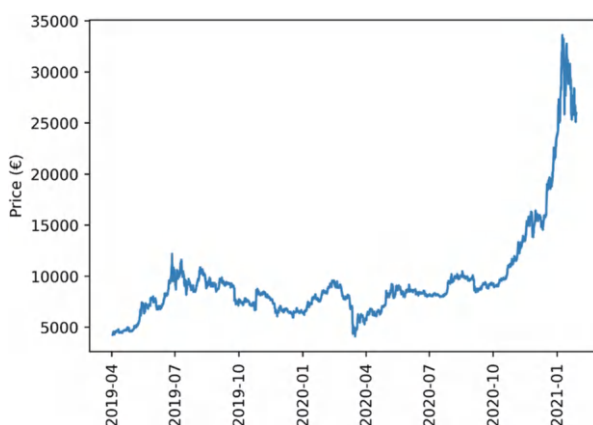
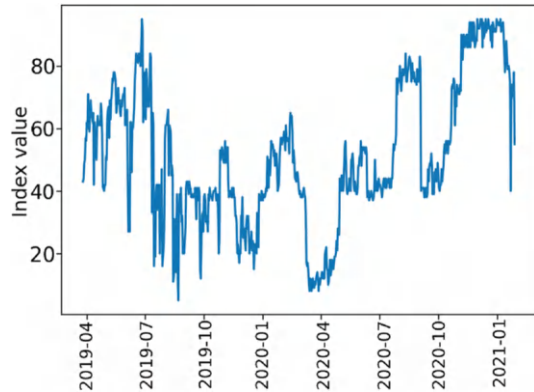


Fig. 4 BTC/EUR price [CryptoVolumeForecasting-Data_visualization](#)

The price contains useful information that can be extracted for use to predict volume. Empirical observations show that volume is highest, when there are larger price fluctuations. A sharp increase in price may attract nonprofessional investors, thus increasing volume in that time period. Such a behavior would also be predictable, making price a source of useful indicators for the current objectives.

Other data used for prediction includes:

- Fear-and-Greed Index (alternative.me)—an index measuring investor sentiment
 - It consists of these factors:
 - * Volatility (25%)
 - * Market volume (25%)

Fig. 5 Fear-and-Greed Index

- * Social media sentiment (15%)
- * Surveys (15%)
- * Coin dominance (10%)
- * Google Trends (10%)
- It ranges from 0 to 100.
- In the dataset, it is binned in the following way:
 - * Extreme Fear (0–25)
 - * Fear (26–45)
 - * Neutral (46–54)—not included in dataset due to redundancy
 - * Greed (55–75)
 - * Extreme Greed (76–100)
- BTC/EUR daily volume (CoinGecko using API)
- EUR/USD minute data (HistData.com)
- EURIBOR (Banque de France, ECB using Quandl API) includes EONIA, 1 week, 1 month, 3 months, 6 months, and 12 months EURIBOR (Fig. 5, 6, 7, and 8)

These parameters are picked because they are often used for return forecasting or volatility forecasting [6]. As volume is heavily dependent on returns and volatility, it makes sense to use the same parameters for volume prediction.

The Fear-and-Greed Index and volume from CoinGecko aid the evaluation of general investor behavior, while EUR/USD and EURIBOR are useful macroeconomic indicators.

It is important to note that volume and EURIBOR were not available for each 4-hour-interval, as it is only provided on a daily basis. In that case, the last available value is used instead.

Fig. 6 BTC volume
(CoinGecko)

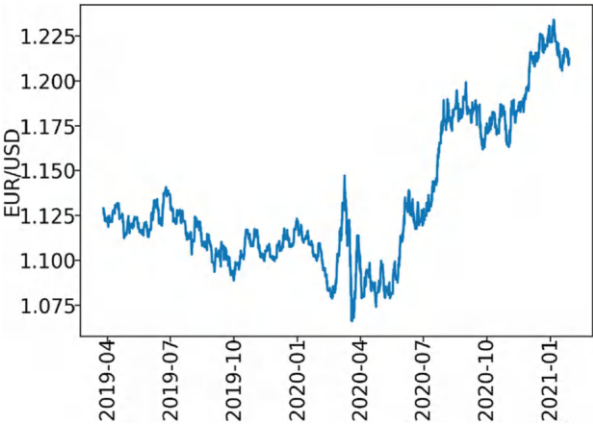
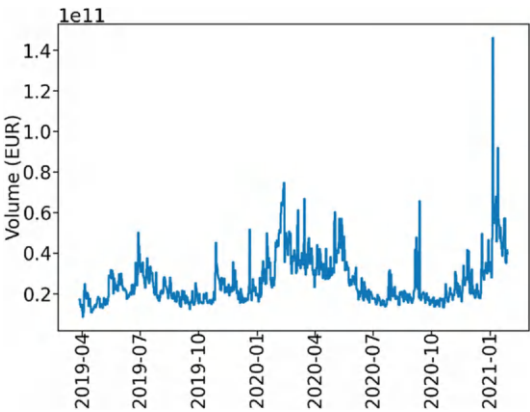


Fig. 7 EUR/USD exchange rate

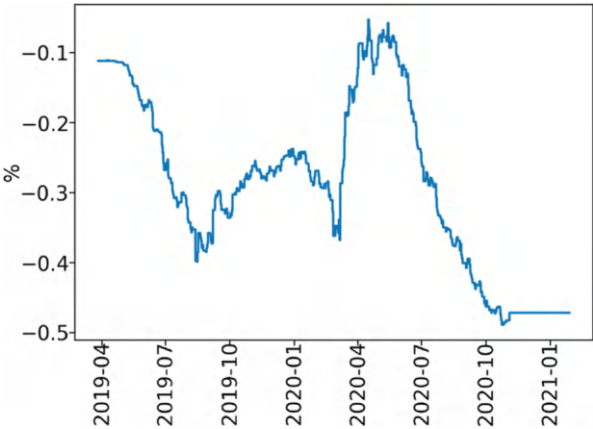


Fig. 8 12-month EURIBOR [CryptoVolumeForecasting-Data_visualization](#)

2.2 Feature Engineering

With the required data assembled, certain features can be derived from it:

- Hour of the day determines whether it is 0 am, 4 am, 8 am, 12 pm, 4 pm, or 8 pm and is used to account for daily seasonality.
- Holiday/weekend determines whether it is currently a holiday/weekend or workday. Empirical observations seem to suggest that, like weekends, holidays have generally less activity than workdays [21]. Nuri is a German company, so German holidays (as stated in the “holidays” Python package) were considered.
- Realized volatility of BTC/EUR

$$\sigma_t = \sqrt{\sum_{i=t-29}^t r_i^2},$$

where r_i is the return at time point i .

- OHLC data for BTC/EUR—Open, High, Low, and Close prices for each interval. Both volatility and fluctuating OHLC data offer clues to potentially high trading activity.
- Trading indicators based on OHLC and volume (available from Python package “ta”). They include momentum indicators, volume indicators, volatility indicators, and trend indicators. All of them may be applied to trading volume forecasting.

3 Methodology

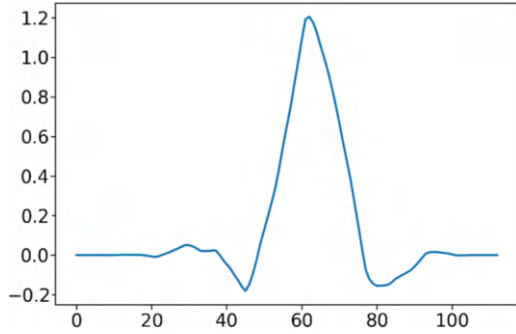
3.1 Preprocessing

First, the data is appropriately scaled. The target variable, trading volume, receives standard scaling, while the Bitcoin order size is log-scaled, and the other variables are Min-Max scaled (between 0 and 1).

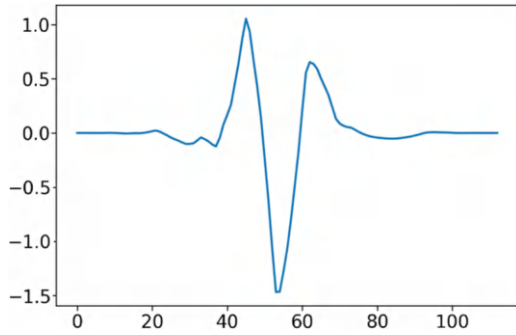
Next, a low-pass filter is applied to all non-dummy variables. For this method, a Symlet wavelet is used for decomposition [13]. Empirically, this improves stability of the approximation procedure. Due to the averaging, the transformation is smoother, compared to the Haar wavelet. Alongside the Daubechies wavelet, the Symlet wavelet is the most popular wavelet and often yields better forecasting performance in financial applications or other instances, which require denoising [25].

Local signal decomposition:

$$X_{a,b} = \int x(t) \phi\left(\frac{t-b}{a}\right) dt,$$

Fig. 9 ϕ —father wavelet**Fig. 10** ψ —mother wavelet

 CryptoVolumeForecasting-Symlet



where $\phi(t)$ is the mother Symlet wavelet. For order N and father Symlet φ , the following properties hold:

$$\text{supp } \varphi \subseteq [0, 2N - 1]$$

$$\text{supp } \phi \subseteq [-N + 1, N]$$

$$\int x^l \phi(x) dx = 0, \quad l = 0, \dots, N - 1.$$

For our application, a Symlet of order 4 is used, and the highest two layers are eliminated.

Figures 9 and 10 show a visualization of the Symlet.

3.2 LSTM

Long Short-Term Memory (LSTM) is a neural network architecture designed to process sequential data, such as time series. It is an evolution from RNNs (Recurrent Neural Networks), aimed at fixing some of RNN flaws. Unlike RNN, LSTM has the

ability to retain information for a longer time span, while removing irrelevant past information, thanks to the gate system. The LSTM has one more gate with the forget gate and introduces the memory cell C for controlled error flow. These innovations mitigate the vanishing gradient problem.

An LSTM cell has three gates:

- Input gate protects data in the cell from irrelevant inputs

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \quad (2)$$

$$C_t^* = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c). \quad (3)$$

- Forget gate protects data in the cell from irrelevant data in previous cells

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f) \quad (4)$$

$$C_t = f_t C_{t-1} + i_t C_t^*. \quad (5)$$

- Output gate protects data in other cells from irrelevant data in the current cell.

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \quad (6)$$

$$h_t = o_t \tanh(C_t). \quad (7)$$

W_{yz} is a trained weighted matrix that is propagated from gate y to gate z . x is the input, h is the propagated value from the previous time step, and b is a constant. An illustration can be found in Figure 11. σ is the sigmoid function:

$$\sigma(x) = \frac{e^x}{e^x + 1}.$$

Since LSTM allows to retain useful information for extended time periods, it should be a good fit for the seasonal data.

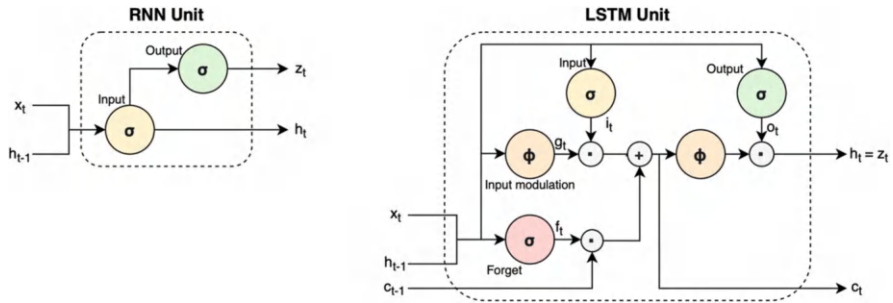


Fig. 11 Illustration of RNN and LSTM unit

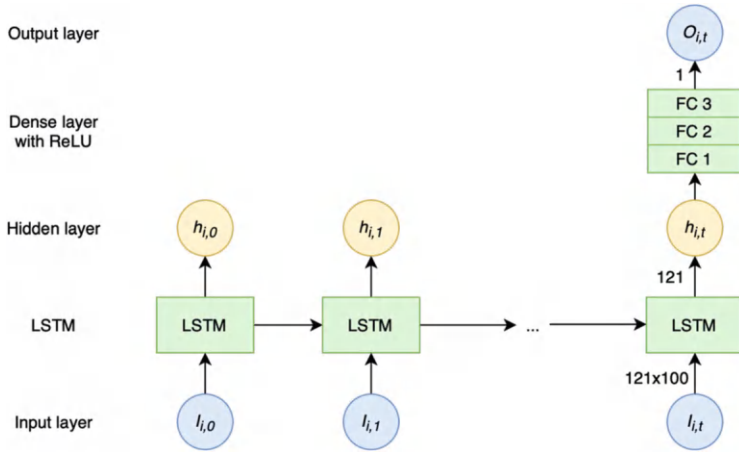


Fig. 12 LSTM architecture

The complete architecture of the predictor includes the LSTM layer and three dense layers. The activation function for the hidden layers is the Rectifier Linear Unit (ReLU), as it is efficient and mitigates the vanishing gradient problem. The output layer lacks an activation function to allow any output. Each batch has 100 time units. This setup worked best in tests based on validation set metrics. As the metrics did not change a lot between different time frames, we can conclude that, without significant structural breaks, the results are consistent and replicable (Fig. 12).

The output is a prediction of volume (in BTC), with the sign determining the direction of the trades, as shown in (1). The mean absolute error (MAE) is the loss function.

4 Experiment

4.1 Experiment Setup

The dataset is split into the subsets shown in the Table 1. The cryptocurrency market is very dynamic, which possibly affects the prediction quality over time. To investigate the performance for consistency, four validation sets are used, in order to reflect various possible scenarios.

For the LSTM, the following parameters are used, as this yielded consistent results for the validation sets, showing that there is no overfitting (Table 2):

For benchmarks, two simple models are used: SARIMAX and TBATS. Just like LSTM, these models are capable of capturing seasonality. Unlike LSTM, they only use returns as input. For SARIMAX, external inputs are used to cover the weekly

Table 1 Dataset overview

Set	Start date	End date	Data points
Training set	2019-03-26 00:00:00	2020-08-14 20:00:00	3204
Validation set 1	2020-08-15 00:00:00	2020-09-17 04:00:00	200
Validation set 2	2020-09-17 08:00:00	2020-10-20 12:00:00	200
Validation set 3	2020-10-20 16:00:00	2020-11-22 20:00:00	200
Validation set 4	2020-11-23 00:00:00	2020-12-26 04:00:00	200
Test set	2020-12-26 08:00:00	2021-01-28 12:00:00	200

Table 2 LSTM parameters.
Learning rate relates to the
Adam optimizer

Parameter	Value
Input length	100
Epochs	30
Dense layers	3
Dropout	0.25
Optimizer	Adam
Learning rate	0.001

seasonality, in addition to the daily one. TBATS is a derivative of exponential smoothing for time series, which accounts for seasonality [17]. Additionally, an even simpler method, which predicts the value of the previous time period, is used as a benchmark (referred to as “BENCHMARK” in the result tables).

4.2 Experiment Results

The results with various different metrics can be found below (Tables 3 and 4).

It can be observed that all three methods are superior to the benchmark. In addition, LSTM beats the two seasonal autoregressive models, thus showing that the supplementary data had an impact on the performance. LSTM outperformed in all validation sets and the test set, meaning that the performance is consistent and replicable for many different scenarios (Fig. 13).

The resulting data implies that seasonality (regularly bought Bitcoin) is captured, but the magnitude is often not as high as the true values. On the other hand it is correctly predicted that weekends have low general activity. Most of the time, the prediction is not lagging behind the true values. On the other hand, the selloffs that occur occasionally are not captured in the predictions. They seem to happen too rarely, and the variables available in the dataset do not contain the information required for a better prediction. Another remark is the slight skew in the predictions, which lead to small underestimations, as seen in Figure 14. The reason could be the use of mean absolute error instead of mean squared error, but the overall performance with MAE as a loss function has been better in empirical tests.

Table 3 Experiment error metrics. MSE—mean square error, MAE—mean absolute error

Set	Val. set 1	Val. set 2	Val. set 3	Val. set 4	Test set
LSTM MSE	11.63	9.07	15.18	18.2	25.49
SARIMAX MSE	13.6	10.62	18.9	22.37	27.51
TBATS MSE	14.16	10.81	17.17	21.94	27.18
BENCHMARK MSE	23.53	18.95	43.33	49.08	54.63
LSTM MAE	2.38	2.15	2.68	2.92	3.49
SARIMAX MAE	2.62	2.48	3.09	3.28	3.89
TBATS MAE	2.65	2.5	3.01	3.27	3.88
BENCHMARK MAE	3.58	3.24	4.63	4.77	5.39

Table 4 R^2 values for LSTM and reference models

Set	SARIMAX R^2	TBATS R^2	LSTM R^2
Validation set 1	0.35	0.33	0.46
Validation set 2	0.41	0.41	0.53
Validation set 3	0.35	0.40	0.49
Validation set 4	0.39	0.4	0.51
Test set	0.38	0.39	0.46

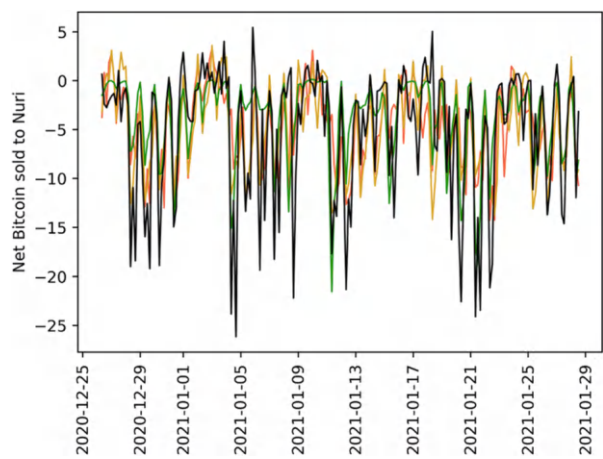
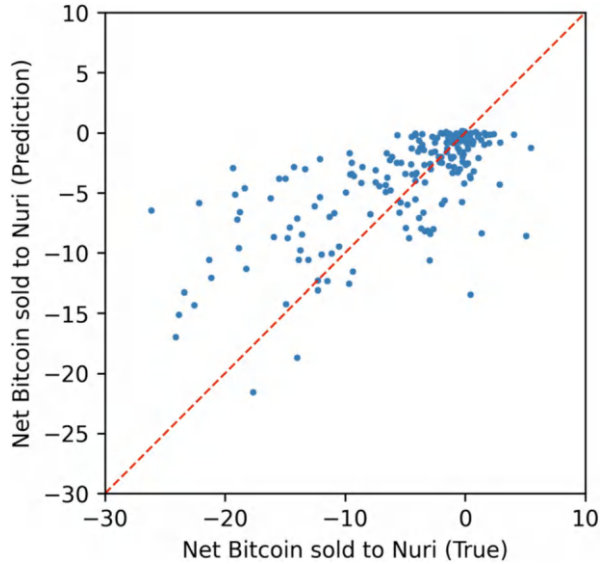


Fig. 13 Net Bitcoin sales (black) and their predictions using LSTM (green), TBATS (yellow), and SARIMAX (red) for test set (2020-12-26 08:00:00—2021-01-28 12:00:00) [CryptoVolume-Forecasting-Validation_test_plot](#)

Overall, the performance of the models is fairly consistent and does not change significantly over time despite some changes in structure, as can also be observed in Appendix B. However, the data considered here did not include a bear market—where there could be a sudden swing toward selling the coins. To account for these situations, more data may be required.

Fig. 14 Scatterplot for test set (LSTM and actual values)
[CryptoVolumeForecasting-Data_visualization](#)



5 Discussion and Further Research

To improve accuracy of the model, it could be beneficial to add sentiment data, as cryptocurrency trading is still very sentiment-driven [7, 15, 19].

When analyzing the user transaction data, it was noticed that there are mostly two types of investors:

- Informed investors—investors doing systematic buy or sell trades over multiple days
- Buy-and-hold investors—investors who buy Bitcoin once and have no other activity afterward

Agent-based modeling may allow to understand the market from a different angle. A model with two types of investors could be calibrated. It could also include frequency of new traders appearing and existing traders leaving. Such a model could be used to simulate certain scenarios and better understand the market [23].

Another direction to follow would be the application of reinforcement learning to determine the required amount of liquidity to hold [22]. The input would be the same, and LSTM could be the processing neural network. The output would consist of two numbers—the amount of Bitcoin and Euro to hold. The loss function could reward smaller holdings of Bitcoin and Euro, but issue a penalty, if the demand cannot be covered with the available funds.

The LSTM model works only as long as the structure is relatively stable. When the structure changes, there may not be sufficient data to adapt to the new situation. Change point detection may help detect those changes and take action as soon as possible [2].

Other potential direction of future research would be market microstructure [9], to understand how order book dynamics, bid-ask spreads, or trading mechanisms influence liquidity provision, trading behavior, and market efficiency.

At the same time, it is worth investigating how changes in liquidity levels impact price volatility and vice versa, by examining the feedback loops between liquidity and volatility of cryptocurrencies [4].

6 Conclusion

In this chapter, Bitcoin volume data of a local market maker was analyzed, with an attempt to predict the volume and direction for a 4-hour time period. The study found that during the time period analyzed, there was a consistent net demand for purchasing Bitcoin, exhibiting both intraday and intra-week seasonality.

To make predictions, we compared the performance of different models, including statistical models such as SARIMAX and TBATS, as well as a Long Short-Term Memory (LSTM) network, using MSE as an evaluation criterion.

The results of the analysis showed that the LSTM network outperformed the statistical models in terms of overall predictive accuracy. This suggests that the LSTM model was able to capture the complex temporal dependencies present in the Bitcoin volume data more effectively than the statistical models.

However, it is important to note that even with the best-performing LSTM network, the study found that occasional sudden jumps in Bitcoin selling were not predictable by any model. These unpredictable events are likely to occur in general and cannot be reliably anticipated. Therefore, the chapter emphasizes the importance of implementing appropriate risk mitigation measures to address such cases.

In conclusion, the study demonstrated that analyzing Bitcoin volume data using machine learning techniques, specifically the LSTM network, can provide valuable insights into predicting transaction volume and direction. While the LSTM model showed superior performance compared to statistical models, it is crucial to acknowledge the inherent unpredictability of sudden jumps in Bitcoin selling and take appropriate measures to manage risk in such instances.

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A Data overview supplements

(See Figs. 15, 16, 17)

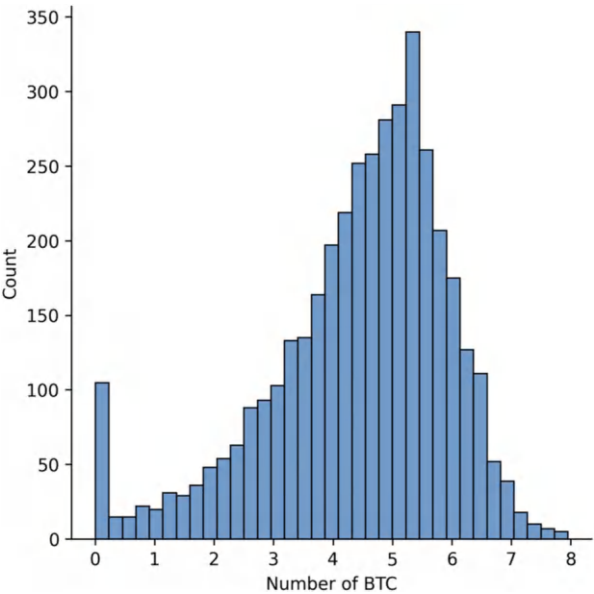
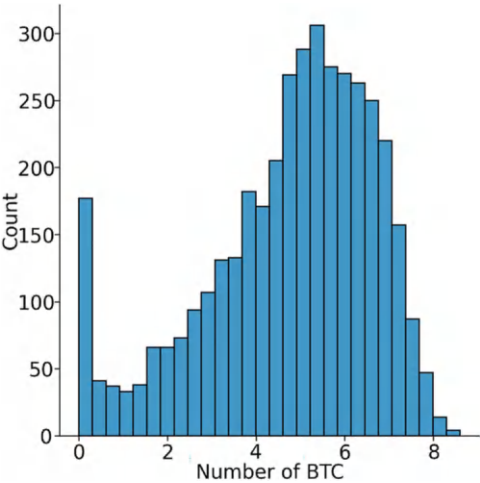


Fig. 15 Histogram of BTC bought for every 4-hour time period

Fig. 16 Histogram of BTC sold for every 4-hour time period



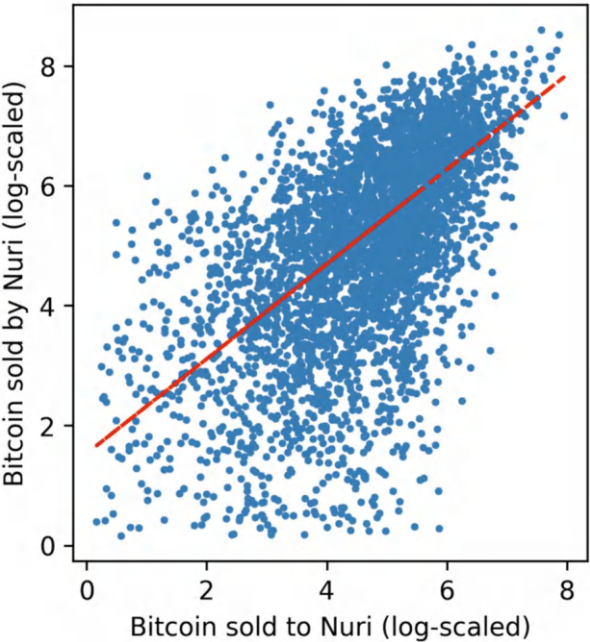


Fig. 17 Scatterplot of BTC bought vs BTC sold (with regression line)

B Results Supplements

(See Figs. 18, 19, 20, and 21)

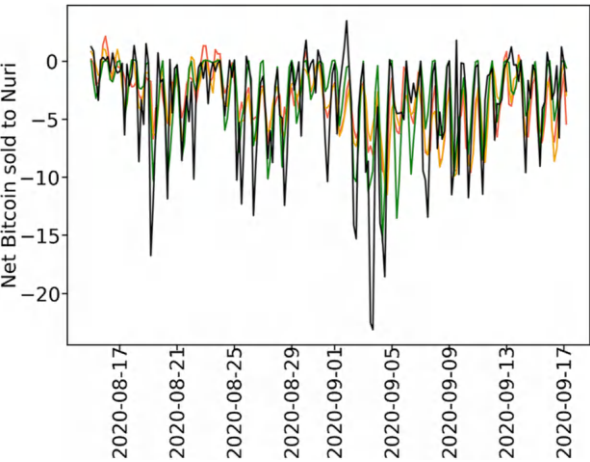


Fig. 18 Net Bitcoin sales and their predictions for validation set (2020-08-15 00:00:00—2020-09-17 04:00:00)

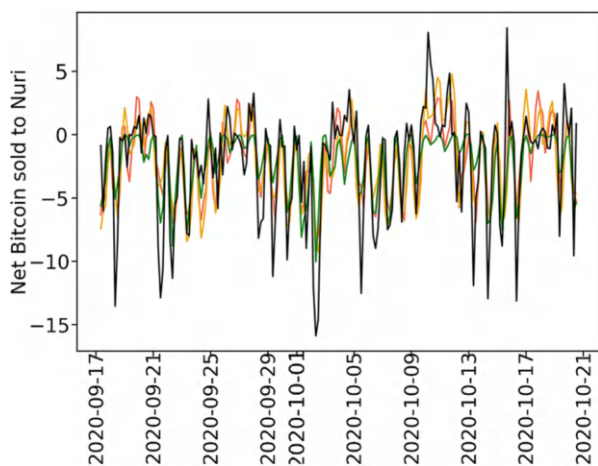


Fig. 19 Net Bitcoin sales and their predictions for validation set (2020-09-17 08:00:00—2020-10-20 12:00:00)

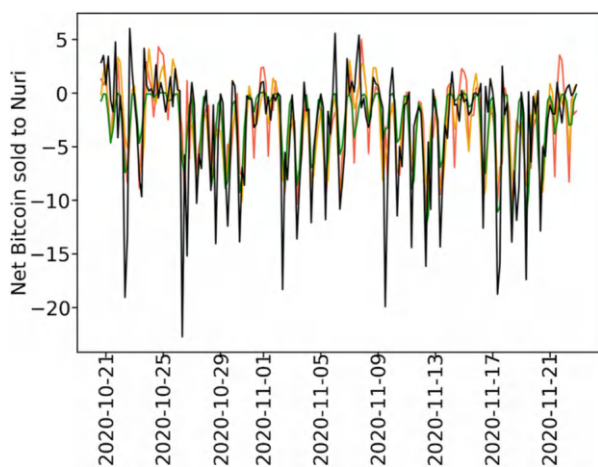


Fig. 20 Net Bitcoin sales and their predictions for validation set (2020-10-20 16:00:00—2020-11-22 20:00:00)

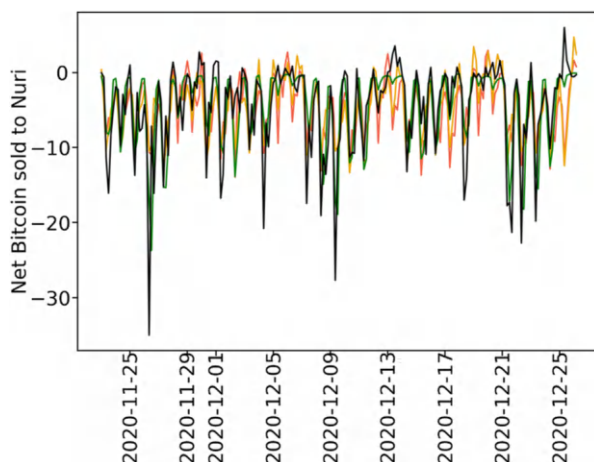


Fig. 21 Net Bitcoin sales and their predictions for validation set (2020-11-23 00:00:00—2021-01-28 12:00:00)

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Part III

Cryptoeconomics

Cryptocurrency Market Analysis: Insights from Metcalfe's Law and Log-Periodic Power Laws



Alexandra Ioana Conda, Andrei-Theodor Ginavar, Daniel Traian Pele,
and Miruna Mazurencu-Marinescu-Pele

Abstract In this paper, we investigate the statistical properties of Bitcoin and the CRIX index using two layers of analysis: Metcalfe's law and bubble behaviour through LPPL modelling. The results show that, in the medium to long run, Metcalfe's law (which states that the value of a network is proportional to the square of the number of connected users) is valid for evaluating cryptocurrencies. However, in the short run, the validity of Metcalfe's law for Bitcoin is questionable. The DS LPPLS methodology was used to capture the behaviour of Bitcoin exchange rates during an endogenous bubble and to predict the most probable time of regime switching. The main conclusion of this paper is that while Metcalfe's law may be valid in the long run, its validity in the short run, across various data regimes, is highly debatable.

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1 Introduction

After 2008, when the pseudonymous Satoshi Nakamoto developed Bitcoin, a surge in other cryptocurrencies began, rooted in blockchain technology. By mid-March 2024, the total market capitalisation of cryptocurrencies had reached approximately 2.72 trillion USD, highlighting their significant appeal within the global asset market.

This emerging class of assets has attracted attention not only from traders but also from market regulators and academics. Recent warnings from the European Supervisory Authorities (ESAs), which include the European Banking Authority (EBA), the European Securities and Markets Authority (ESMA), and the European Insurance and Occupational Pensions Authority (EIOPA), emphasise the high risks associated with crypto-assets. In a joint warning issued in March 2022, the ESAs highlighted that many crypto-assets are highly speculative and not suitable for most retail consumers. They warned of the real possibility of losing all invested money and cautioned against misleading advertisements, particularly those involving social media influencers [1, 2].

From an academic perspective, a plethora of papers have explored cryptocurrencies, particularly focusing on their statistical properties and risk modelling. The research landscape now includes more sophisticated analyses of cryptocurrency market behaviours. Recent studies utilise advanced multivariate time-series models and machine learning to predict log-return prices of cryptocurrencies, showing significant enhancements in prediction accuracy over traditional models [3].

The valuation of cryptocurrencies often employs Metcalfe's law, which posits that the value of a network is proportional to the square of its users. Recent findings confirm the complex dynamics between user growth and price, suggesting intricate interactions rather than straightforward correlations [4]. This underscores the importance of network effects in cryptocurrency valuations.

Furthermore, the study of herding behaviour in the cryptocurrency market has shown that market uncertainty is closely linked with the likelihood of crashes, indicating that smaller cryptocurrencies tend to follow the trends set by the largest ones, signalling potential market inefficiencies [5].

Log-Periodic Power Law (LPPL) models, traditionally used in stock market analyses, have been adapted to study cryptocurrency markets. These models have been employed to anticipate the timing of market crashes by analysing the trading prices and identifying super-exponential growth indicative of bubble dynamics [6].

In this study, we focus on employing two major statistical methods to explore the behaviours of the cryptocurrency market, aiming to update and expand upon the findings of Pele et al. [4]. First, we apply a generalised Metcalfe's law to Bitcoin to understand the interrelations among its price, market capitalisation, network user numbers, and transaction volume. Second, we use the DS LPPLS methodology to model the dynamics of bubbles in Bitcoin and a major cryptocurrency index, the CRIX, highlighting the value of log-periodic power laws in predicting regime shifts.

Considering the findings from the literature, our contribution to the research on the statistical analysis of cryptocurrencies is mostly empirical.

The most important contribution of this paper is that in the long term, Metcalfe's law tends to be valid for Bitcoin; however, in the short term, its validity is questionable. The rest of the paper is organised as follows: Sect. 2 outlines the methodology; Sect. 3 presents the dataset and the empirical results; and Sect. 4 concludes.

2 Methodology

As previously mentioned, the methodology used in this paper has two layers. First, we investigate the validity of Metcalfe's law for the most popular cryptocurrency, Bitcoin, showing the potential for herding behaviour. Subsequently, we apply the DS LPPLS methodology to identify the bubble regime in Bitcoin prices and in the evolution of the Cryptocurrency Index (CRIX).

The reason for applying these two different techniques lies in the following argumentation: speculative behaviour, which may be caused by herding, can be tested using LPPL models. At the same time, herding behaviour may be derived from a modified version of Metcalfe's law, where the price increase serves as an incentive for more investors to join the market, driven by high expected returns.

2.1 Metcalfe's Law

Originally stated by Robert Metcalfe in the 1980s, Metcalfe's Law suggests that the value of a network is proportional to the square of the number of its connected users. This principle has been verified in various studies focusing on different network types. Zhang et al. [7] confirmed its applicability to social media platforms like Facebook and Tencent, a significant Chinese network. Further research by Madureira et al. [8], Van Hove [9, 10], and Metcalfe [11] demonstrated the law's relevance to internet networks, indicating consistent patterns across diverse digital ecosystems.

In the domain of cryptocurrencies, Peterson [12] applied Metcalfe's Law to analyse Bitcoin's pricing dynamics, particularly examining the interplay between supply (number of Bitcoins) and demand (number of Bitcoin wallets). Additionally, Van Vliet [13] extended this model to include the logistic diffusion of innovation, further affirming Metcalfe's Law in the evolution of Bitcoin. These studies highlight the robust applicability of Metcalfe's Law across both traditional internet and emerging financial networks, offering a valuable framework for understanding the growth and value dynamics of interconnected systems.

In this study, we are using as reference the Metcalfe's law version from Wheatley et al. [14]:

$$C_t = e^{b_0} u_t^{b_1} \quad (1)$$

where:

- C_t is the Bitcoin's market capitalisation at time t .
- u_t is the Bitcoin's number of unique addresses at time t .

If Metcalfe's law is valid for Bitcoin, then the coefficient $b_1 = 2$; we are testing the Eq. (1) over the entire sample and by using a rolling window approach.

In addition to the classical form of Metcalfe's law, we are testing the hypothesis that the Bitcoin's price itself is driven by the Bitcoin's network size, showing potential for some herding behaviour.

2.2 Log-Periodic Power Laws (LPPL)

According to the field theory [15], an imitative process can be described through its hazard rate $h(t) : \frac{dh}{dt} = Ch^d$, where $C > 0$, and $d + 1 > 1$ is the average number of interactions between investors. Then $h(t) = \left(\frac{h_0}{t_c - t}\right)^a$, with $a = \frac{1}{d-1}$ and t_c being the critical time, so the price dynamics prior to the crash should be $\log \frac{p(t)}{p(0)} = k \int_{t_0}^t h(u) du$.

As the crash probability should be compensated by larger price changes, prior to the stock market crash [16], the hazard rate could be expressed via the Ising model: $h(t) \approx B_0(t_c - t)^{-a} + B_1(t_c - t)^{-a} \cos [\omega \ln (t_c - t) + \psi']$.

Thus, the trading price before the crash follows a log-periodic power law [17]:

$$E [\log p(t)] = A + B(t_c - t)^b \left\{ 1 + C \cos [\omega \ln (t_c - t) + \phi] \right\}, \quad (2)$$

where $p(t)$ is the price at moment t , t_c is the critical time (the most probable moment of the crash), and $b, B_0, B_1, \omega, \phi$ are the parameters of the model which give its log-periodic feature. To have a proper specification of the model, there are several constraints applied to the parameters: $A > 0, B < 0, C \neq 0, |C| < 1, 0 < b < 1, \omega \in (0, \infty)$ and $\phi \in [0, 2\pi]$.

We are using the LPPL model to test the propensity for herding behaviour in the case of Bitcoin and the CRIX Index. We use an alternative formulation for the LPPL model which simplifies the calibration of the parameters used by Filomonov et al. [18]:

$$\log E [p(t)] = A + B(t_c - t)^m + C_1(t_c - t)^m \cos (\omega \log (t_c - t)) + C_2(t_c - t)^m \sin (\omega \log (t_c - t)), \quad (3)$$

by changing the parameters C and ϕ into: $C_1 = C \cos (\phi), C_2 = C \sin (\phi)$.

To diagnose a bubble, we use the LPPLS Confidence Indicator [19], which is defined as the fraction of fitting windows in which the LPPLS calibrations satisfy the specified filter conditions. A value close to 1 means that most of the fitted windows met the filtering conditions, thus indicating a high degree of confidence in a bubble. The filtering conditions are based on evidence from previous studies on bubbles [20].

3 Empirical Results

3.1 Dataset

The dataset presented in Table 1 consists of daily cryptocurrency data (price, market capitalisation, and exchange volume).¹ One market index was also used for the analysis: Cryptocurrency Index² as a reference for the cryptocurrencies market [21]. For data analysis, the programming language Python was used.

3.2 Metcalfe’s Law for Bitcoin

To evaluate the applicability of Metcalfe’s law to cryptocurrencies, we focus on the most well-known and widely traded cryptocurrency, Bitcoin, primarily due to the availability of transaction and network data.

As stated in the original formulation of Metcalfe’s law, the value of a network should be proportional to the square of the number of network users. However, in the case of cryptocurrencies, the actual number of users is unknown, and we need to use a proxy for it, namely the number of unique addresses. Unique addresses in the Bitcoin ecosystem are payment addresses with a non-zero balance. This metric

Table 1 Description of the dataset

No.	Symbol	Cryptocurrency/Index	Number of daily observations	Start date	End date
1	BTC	Bitcoin	3500	2014-09-17	2024-04-16
2	CRIX	CRyptocurrency IndeX	2010	2017-09-06	2023-03-07 ^a

^aThe latest data publicly available for CRIX

¹ Blockchain research centre: <https://blockchain-research-center.com/>.

² <http://thecrix.de>. The CRyptocurrency IndeX is a benchmark for the cryptocurrencies market, being real-time computed by the Ladislaus von Bortkiewicz Chair of Statistics at Humboldt University Berlin, Germany.

can be used as a proxy for the number of network users although we cannot assume that the number of users is equal to the number of unique addresses.

The number of unique addresses is not constant over time. When fees are high, investors may leave their cryptocurrencies in multiple addresses, as consolidating into a single address would incur a high cost. Conversely, when fees are low, investors may consolidate their funds into a single address. As the Bitcoin network grows, the number of unique addresses will also increase. However, when the market is declining, fewer unique addresses are in use due to a decrease in the number of transactions, as seen in Fig. 1.

3.2.1 Metcalfe's Law for the Entire Data Sample

In this section, we are estimating the generalised Metcalfe's law, which is a log-linearisation of Eq. (1) [22].

$$\log C_t = b_0 + b_1 \log u_t + b_2 \log v_t + \varepsilon_t, \quad (4)$$

where:

- C_t is the Bitcoin's market capitalisation at time t .
- u_t is the number of unique Bitcoin addresses used at time t .
- v_t is the Bitcoin's volume at time t .

The estimation results for Eq. (4) are shown in Table 2, using daily data for the period 2015-01-01–2024-01-08 (Fig. 2).

The model has a high explanatory power ($R^2_{\text{adj}} = 0.817$), supporting the validity of Metcalfe's law for Bitcoin.

The log-linear relationship between Bitcoin's market capitalisation, the number of unique Bitcoin addresses, and the Volume used is illustrated in Fig. 3.

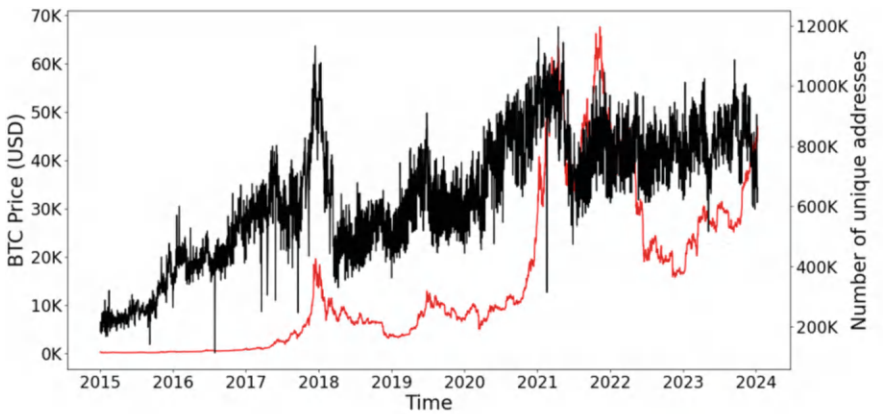
From the validity of Metcalfe's law for Bitcoin one can infer the existence of a possible herding effect: as an increase of the number of users is reflected in an increase of the market capitalisation, this may be explained by the fact that there is a mimetic effect among users, making the price to have an ascending trend.

One insight into this direction can be found by estimating the generalised Metcalfe's law for the Bitcoin's price:

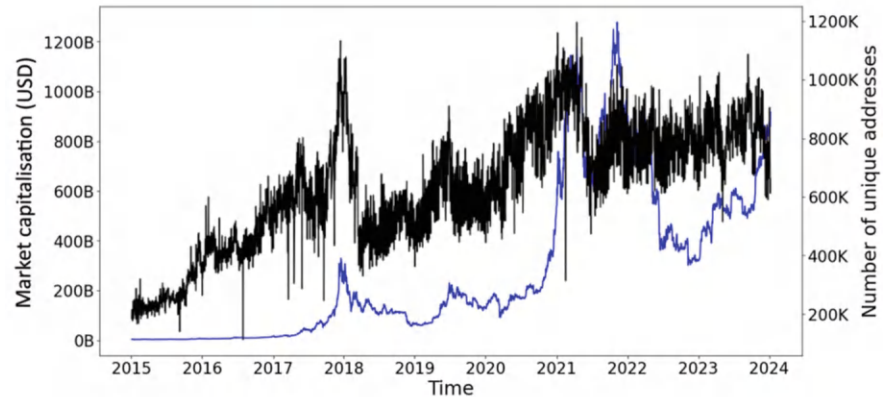
$$\log P_t = b_0 + b_1 \log u_t + b_2 \log v_t + \varepsilon_t \quad (5)$$

The estimation results of Eq. (5) are shown in Table 3.

The results of the estimation show that there is strong log-linear relationship between the Bitcoin's market price and the number of unique addresses, as a proxy for the number of Bitcoin's network users (see also Fig. 4); moreover, the price increase may be a direct effect of the increasing network size, through a possible mimetic behaviour.



(a) BTC price (USD); Number of unique addresses 



(b) Market capitalisation (USD), Number of unique addresses 

Fig. 1 Price, market capitalisation, and number of unique BTC addresses. (a) BTC price (USD); Number of unique addresses  . (b) Market capitalisation (USD), Number of unique addresses 

3.2.2 Metcalfe’s Law on Rolling Windows

The validity of Metcalfe’s law for Bitcoin’s market capitalisation is questionable: as shown below, due to the different regimes, one will obtain very different parameter estimates by fitting the model on different sub-windows of the data.

For w the length of a rolling window, we estimated the following model:

Table 2 Estimation results for Eq. (4)

Parameter	Estimated value
b_0	$-14.007^{***}(0.580)$
b_1	$2.281^{***}(0.056)$
b_2	$0.403^{***}(0.011)$
R_{adj}^2	0.817^{***}

Note: *** denotes statistical significance at 99% confidence level; standard errors in (). The sample covers the period 2015-01-01–2024-01-08

$$\log C_t = b_{0k} + b_{1k} \log u_t + b_{2k} \log v_t + \varepsilon_t, \quad (6)$$

where $t \in \{k + 1, \dots, k + w\}$, $k \in \{0, \dots, T - w + 1\}$ and T is the number of observations in the sample. The results are presented in the succession of graphs from Fig. 5, which depicts the values of the Adjusted R-squared for various rolling windows; there are periods when the explanatory power of the model is close to, or higher than 80%, but there also situations when the number of network users has no explanatory power on the market capitalisation of Bitcoin.

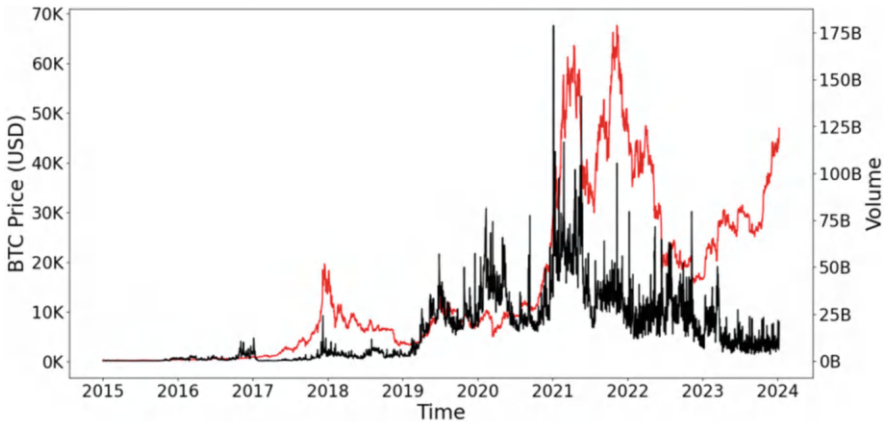
Figure 5 shows the estimated values of the slope coefficient from the Eq. (6), which, according to the classical formulation of Metcalfe’s law, should be equal to 2. However, there is a huge volatility in the evolution of this coefficient, its average values being significantly lower than 1, for the rolling windows of 60, 90, and 250 days and significantly lower than 2 for the 500 days rolling window (Fig. 6).

It can be noticed that the R -squared of 500-days rolling window is very close to zero around 2021 and the slope of the Eq. (6) for 500-days rolling window is – 1 around 2021. During 2020 and 2021, the relationship between the two variables from the Eq. (6) was not linear, as seen in Fig. 7.

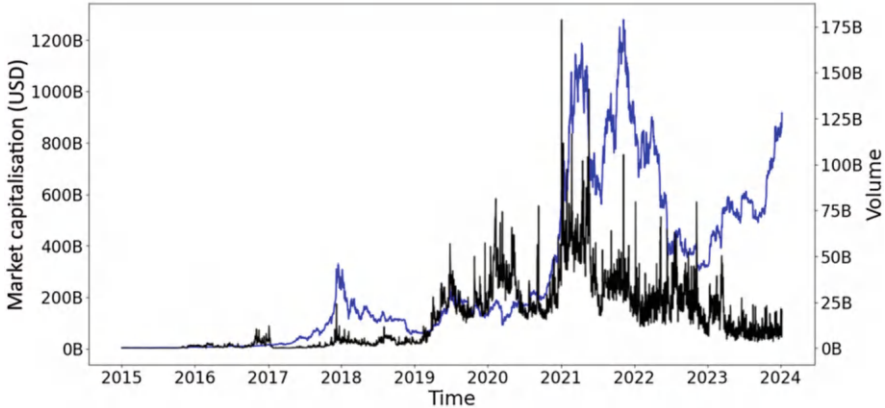
Overall, this analysis shows that there is a clear pattern of inconsistency over time, questioning the validity of Metcalfe’s law for Bitcoin, when considering different sub-windows of the data.

3.3 LPPL Models

The strong short-term relationship between the number of users and Bitcoin’s market capitalisation may be explained by investors exhibiting herd behaviour, driven by rising prices or external factors. The super-exponential price growth resulting from such behaviour can be modelled using the LPPL approach. The Confidence Indicator provides information regarding the most probable time of a bubble burst. To capture the bubble regime and estimate the most probable time of a crash, we used the LPPLS package in Python.



(a) BTC price (USD); Volume



(b) Market capitalisation (USD), Volume

Fig. 2 Price, market capitalisation, and volume. (a) BTC price (USD); Volume . (b) Market capitalisation (USD), Volume

3.3.1 Numerical Results for Bitcoin

In the case of Bitcoin, the first case of regime switching was recorded in December 2017, the exchange rate hitting a local maximum on December 16th, 2017. In addition to our previous analysis, here we have a second case where a bubble can be seen, which takes place around early 2021. The sample for fitting the LPPL model in the case of Bitcoin for predicting these phase transitions was 2014-09-17–2024-

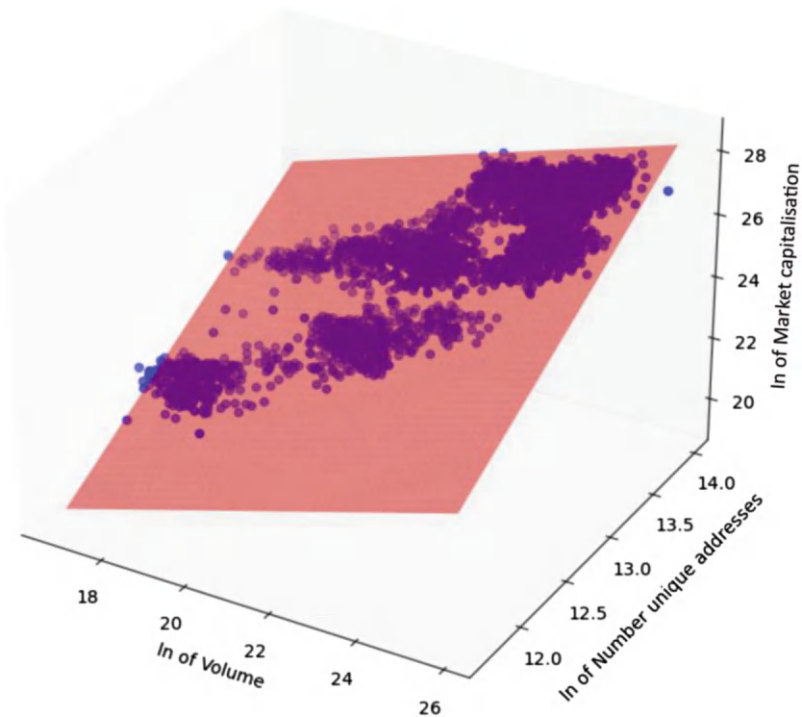


Fig. 3 Generalised Metcalfe’s law for Bitcoin’s market capitalisation, for the entire data sample



Table 3 Estimation results for Eq. (5)

Parameter	Estimated value
b_0	$-28.5960^{***}(0.560)$
b_1	$2.1635^{***}(0.054)$
b_2	$0.3801^{***}(0.010)$
R^2_{adj}	0.811

Note: *** denotes statistical significance at 99% confidence level; standard errors in (). The sample covers the period 2015-01-01–2024-01-08

04-16 (3500 daily observations). Using a shrinking window, going from 120 to 10 days, several estimations are made for each day in the period of analysis.

The best results of the LPPL models, in terms of proximity to the peak price and strength of the bubble indicator, are given in Table 4.

The model with the largest value for bubble indicator anticipated on December 8th, 2017, an imminent crash for the respective day, as seen in Fig. 8. The other two selected models offer predictions for 2021, more specifically January 9th, 2021, and January 8th, 2021.

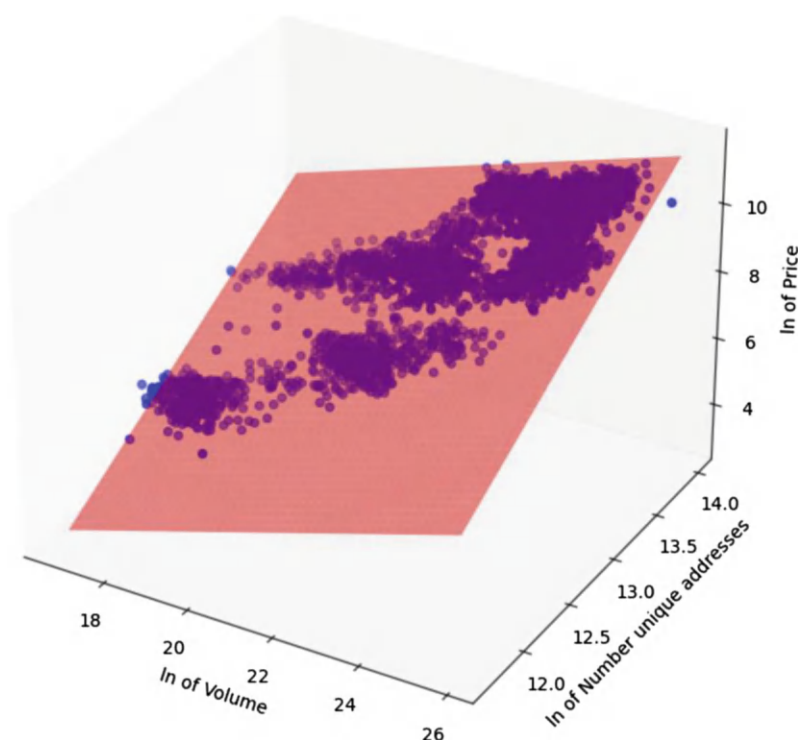


Fig. 4 Generalised Metcalfe's law for Bitcoin's price



As seen in Fig. 8, there is a large concentration of bubble signals between 2015 and the crash of 2018. More specifically, the bubble indicator closest to the crash estimates a critical time corresponding to December 14, 2017. The local maximum was recorded around December 16, while the model with the strongest indicator estimates a critical time on December 8. Although the model is relatively close to the actual crash date, it slightly underestimates the timing, failing to capture the true peak since a first dip in Bitcoin's price occurred around December 8. The price hike leading to this dip was super-exponential and faster than the rise leading to the actual peak. While Bitcoin's price experienced a short rebound during this period, it began declining after January 14, 2018, and did not return to previous levels until roughly 3 years later.

Additionally, our current analysis identifies another period of super-exponential price growth around late 2020 to early 2021, with Bitcoin's price dropping by more than 30% in January 2021. For this early 2021 crash, the model estimates several critical times leading up to the peak registered on January 9, with the closest estimation suggesting a critical time on that exact date. Although the price eventually recovered, the entire year of 2021 was marked by high volatility.

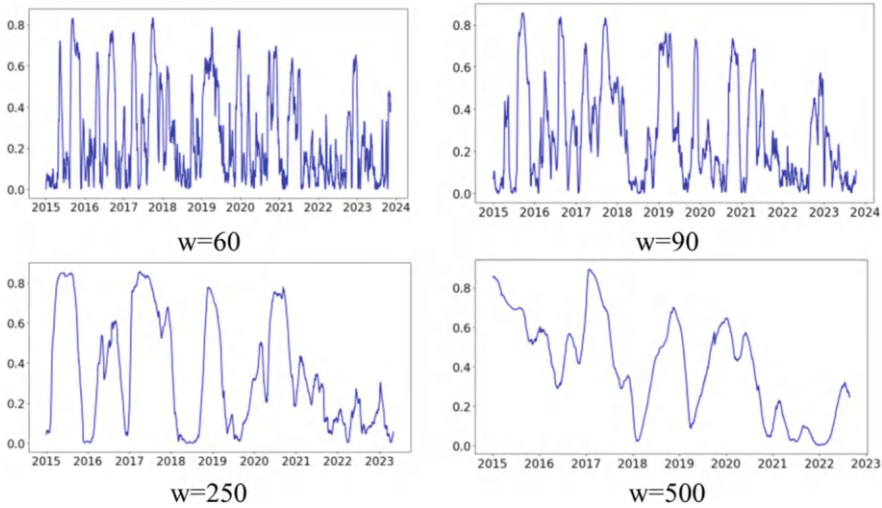


Fig. 5 Adjusted R -squared for Metcalfe’s law (Eq. 6), estimated on rolling windows of 60, 90, 250, 500 trading days

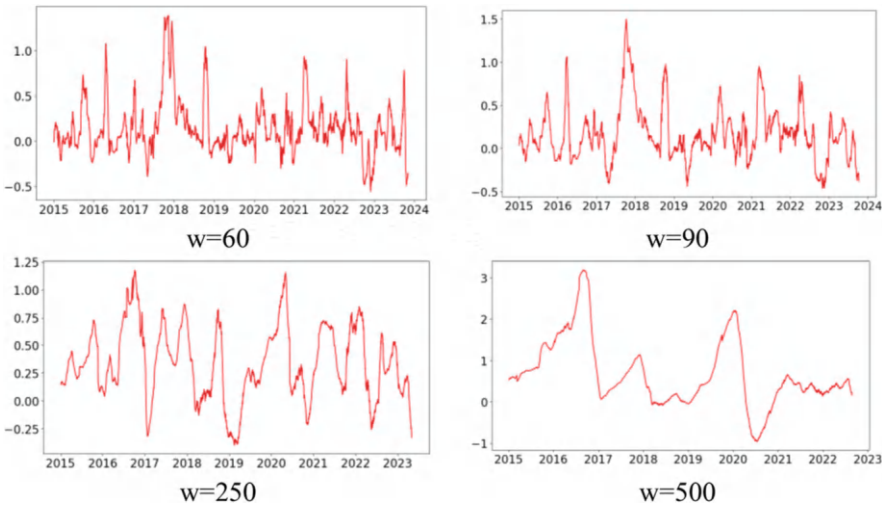


Fig. 6 The b_1 coefficient for Metcalfe’s law (Eq. 6), estimated on rolling windows of 60, 90, 250, 500 trading days

Furthermore, in April 2024, there is a cluster of positive indicators. However, this may be a false positive, as the price growth leading up to the peak was not as extreme as in previous crashes, and the cause of this price drop is believed to be a strengthening of the US Dollar.

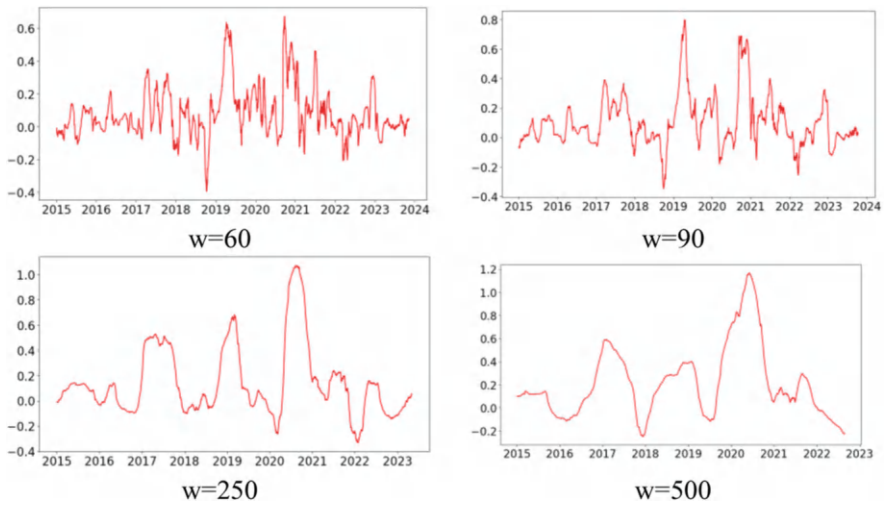


Fig. 7 The b_2 coefficient for Metcalfe’s law (Eq. 6), estimated on rolling windows of 60, 90, 250, 500 trading days

Table 4 The best fits for Bitcoin’s LPPL model

	Model 1—2017/18 Crash	Model 2—2021 Crash	Model 3—2021 Crash
Bubble signal registered on:	2017-12-08	2021-01-08	2021-01-07
A	9.569	11.009	10.666
B	−0.102	−0.399	−0.142
C1	0.005	−0.009	−0.013
C2	−0.0004	−0.017	0.001
b	0.580	0.301	0.510
ω	10.974	5.992	9.541
Window start date	2015-01-15	2015-01-15	2015-01-15
Window end date	2024-04-16	2024-04-16	2024-04-16
Estimated date of crash	2017-12-08	2021-01-09	2021-01-08

3.4 Numerical Results for the CRIX Index

Since Bitcoin is part of the CRIX index, the bubble periods are similar. For 2018, CRIX has a local maximum on the 1st of January 2018, and in the case of the 2021 crash, the price has a very similar progression. The initial sample for fitting the LPPLS model in the case of the CRIX index was 2017-08-06–2023-03-08 (2010 daily observations). The shrinking window procedure used above for Bitcoin was

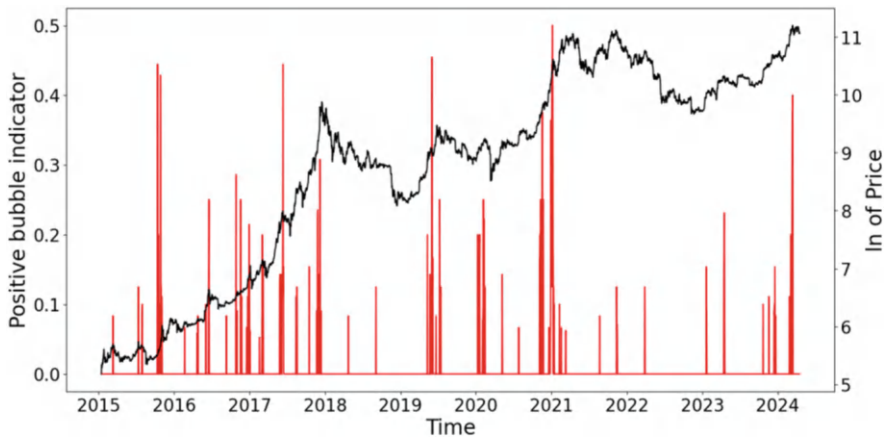


Fig. 8 Positive bubble indicator for BTC

Table 5 The best fit for CRIX’s LPPL model

	Model 1—2017/18 Crash	Model 2—2021 Crash	Model 3—2021 Crash
Bubble signal registered on:	2017-12-14	2021-01-05	2021-01-06
A	8.381	16.268	8.723
B	−0.180	−7.382	−0.127
C1	−0.005	−0.023	0.010
C2	0.001	0.019	0.003
b	0.522	0.030	0.505
ω	7.291	9.074	10.957
Window start date	2017-10-11	2017-10-11	2017-10-11
Window end date	2023-03-08	2023-03-08	2023-03-08
Estimated date of crash	2017-12-16	2021-01-08	2021-01-06

also used for the CRIX index although with a shorter starting window of 35 days, selected after several trials.

The best results of the LPPL models, in terms of strength of the bubble indicator and proximity to the peak value, are given in Table 5. There is a cluster of strong positive bubble indicators leading to a first dip in the index value after the December maximum, on the 17th.

The best fit for the CRIX index during this crash, with the positive bubble indicator with a value of 1, estimates a critical time on December 16th 2017, 1 day before the date of the maximum. Compared to the previous results, the model fails to capture the actual local maxima (5th of January), possibly because of the price drop seen after December 17.

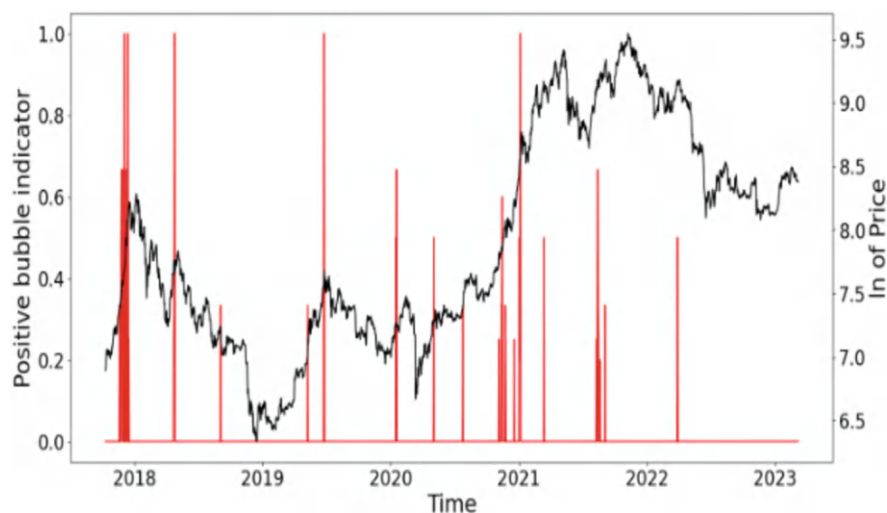


Fig. 9 Positive Bubble indicator for the CRIX index 

As for January 2021, the bubble indicator registers values of 0.5 and 0.2 nine, respectively, 8 days before the local maximum. Then, on the following 2 days it registers values of 1, estimating an imminent crash within the following 5 days, with the best estimation anticipating a crash on the 8th of January 2021, the actual date of the local maximum (Fig. 9).

4 Conclusions

Our paper deals with a new class of assets, cryptocurrencies, from the point of view of their statistical properties and behaviour.

While the findings of Pele et al. [4] are still valid and available, by analysing Bitcoin-related data, we prove, in the medium to long run, the validity of Metcalfe's law for the evaluation of cryptocurrencies; however, in the short run, the validity of Metcalfe's law for Bitcoin is debatable.

This analysis shows that there is a clear pattern of inconsistency over time, questioning the validity of the Metcalfe's law for Bitcoin, when considering different sub-windows of the data.

As there is a strong correlation between the size of the network and the market price of cryptocurrencies, this may indicate mimetic behaviour among investors. These investors enter the market driven by high expected currency rates, potentially leading the market into a super-exponential bubble regime. LPPLS models can be useful in estimating the most probable time of regime switching for an endogenous cryptocurrency bubble. By analysing the behaviour of Bitcoin's price and the

CRIX index, we have demonstrated that LPPL models can be an effective tool in recognising and mapping the behaviour of a developing bubble. This validates the predictive power of LPPLS models in detecting imitative behaviour among investors in the cryptocurrency market, providing valuable insights both theoretically and practically.

At the same time, this validation of LPPLS models also questions the universal validity of Metcalfe's law for Bitcoin when modelling the relationship between price and the number of network users.

The main conclusion of the paper is that while Metcalfe's law may be valid in the long run, its validity in the short run, across various data regimes, is debatable.

Furthermore, the findings from the analysis of Bitcoin and the CRIX index using Metcalfe's law and LPPL models have significant implications for both regulatory frameworks and practical applications. For regulators, these insights can enhance risk management and improve market surveillance. For investors and market participants, understanding the long-term applicability of Metcalfe's law and the predictive power of LPPL models can lead to more informed investment strategies and better risk management practices. Additionally, the development of advanced analytical tools based on these findings can support more robust market analysis and forecasting, benefiting the entire financial ecosystem involved in cryptocurrencies.

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Realized Cryptocurrency Volatility Forecasting with Jumps: An Empirical Study



Meng-Jou Lu, Weiyu Kuo, Junjie Hu, and Wolfgang Karl Härdle

Abstract In this chapter, we examine the realized volatility of highly volatile cryptocurrencies by employing the threshold realized variance method to distinguish the jump component from the continuous process. Despite the fact that the jump process does not enhance explanatory power, we still observe that jumps convey valuable information. Specifically, we find that the one-day lagged threshold jump component has a significant positive impact on future realized volatility, indicating a likelihood of increased volatility in the cryptocurrency market following jumps occurring one day before. However, the unthresholded jump component exhibits a significant negative correlation with future realized volatility.

JEL Classification: C01, C58, E42, E51, G10, K24, K42, L86, O31

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1 Introduction

In this chapter, we study realized volatility processes of two of the most liquid cryptocurrencies, Bitcoin and Ethereum, traded in some major exchanges. A lot of investors or speculators are attracted by the emerging market of cryptocurrency for the price of Bitcoin was soaring from 900 U.S. dollars to nearly 20,000 U.S. dollars in 2017 only within one year and fell back to around 5,000 U.S. dollars 2 months after the peak. As of today, the Bitcoin price is again at high level of near 70,000 U.S. dollars. We aim to provide an overall study on the risk characteristics and investigate if the realized volatility can be forecasted based on its dynamics and separated jump process.

1.1 *Why Cryptocurrency?*

Blockchain is viewed as one of the revolutionary technologies that can possibly change our lives in the future from all aspects. The blockchain functions as an immutable ledger comprising a sequential chain of records, each cryptographically linked to the preceding one. Its security and reliability stem from a decentralized network of nodes, governed by computational algorithms, ensuring consensus and preventing unauthorized alterations to the recorded data.

In this context, decentralized, encrypted systems operate autonomously, offering a viable solution for sectors reliant on trust. It is foreseeable that blockchain technology will experience widespread adoption across various industries as its potential for enhancing security and trustworthiness becomes increasingly recognized.

Blockchain technology relies on the presence of cryptocurrencies as an essential component for its operation. Cryptocurrencies serve as the incentive mechanism driving individuals to invest in computational power, thereby competing to validate transactions and maintain the integrity of the distributed ledger system. Indeed, from Satoshi Nakamoto's seminal paper in 2008 introducing Bitcoin and blockchain technology, it is evident that blockchain is intricately linked with associated cryptocurrencies. This symbiotic relationship is grounded in both economic principles and practical outcomes. Increased competition for validating transactions on the blockchain enhances its security, leading to greater investments and energy consumption. Consequently, this heightened security translates into higher valuations for cryptocurrencies.

The significance of cryptocurrencies parallels that of blockchain technology. Despite criticisms from business leaders labeling Bitcoin and other cryptocurrencies as scams or asset bubbles destined to burst, notable developments have countered such assertions. Notably, in 2017, the Chicago Mercantile Exchange (CME) introduced Bitcoin futures, underscoring growing institutional interest. Moreover, academia's attention has been drawn to this domain, evidenced by a proliferation of research papers and the establishment of cryptocurrency index benchmarks [17],

further legitimizing their role in the financial landscape. Should individuals embrace blockchain as an integral component of our future, acceptance of cryptocurrencies as a legitimate means of payment becomes imperative. Consequently, if cryptocurrencies integrate into our currency system, the establishment of a financial market becomes justified. Notably, blockchain technology has exerted a direct and profound impact on the financial sector, catalyzing the emergence of a rapidly growing cryptocurrencies financial market characterized by escalating trading activity and an expanding array of digital coins.

1.2 Realized Volatility and Timing

In finance, volatility is the center of derivatives pricing, portfolio optimization, and risk management [11], and it can also carry jumps on price which can be potentially more informative. Numerous literature has been discussed the properties of the model-free estimate, namely realized volatility, found that it is highly right-skewed, logarithmic normal distributed, and strongly temporal dependent [2]. Unlike asset returns, the realized volatility is more likely to be predictable due to the long-memory property. Interestingly, when we looked into the realized volatility process, we find that the cryptocurrencies are not only more volatile than other assets with “fat tails” but also evolve with much more jumps comparing with other assets, and we use a more reliable threshold method to separate the jump process which will be elaborated in latter sections.

One major interest of volatility in finance is the economic value that it brings. Discussion on volatility risk premium (VRP) has lasted for a quite long time, papers have found that timing strategy using the intradaily constructed realized volatility gains more than static portfolio [11, 12], and further research confirms such finding across different types of realized variance estimators [16].

For cryptocurrencies realized volatility, recent paper [7] studies the long-short-term realized volatility of Bitcoin and finds that, in the long term, it is negatively impacted by S&P 500 volatility, however, positively correlated to Baltic dry index volatility, and some other papers also discussed realized volatility focuses more on forecasting side [14].

2 Threshold Realized Jumps Detection Model

In this section, we will discuss the method that we employ to estimate realized volatility and jumps in our empirical part and describe our motivation to those methods.

Under the continuous-time jump diffusion process framework, for logarithmic asset price $p(t)$, we have

$$dp(t) = \mu(t)dt + \sigma(t)dW(t) + \kappa(t)dq(t), 0 \leq t \leq T, \quad (1)$$

where μ is the continuous process with local variation bounded, σ is a càdlàg process, and W is the Brownian motion process. The third term on the right-hand side is the jump process including two parameters q and k , q counts the number of jumps with time-varying intensity, and κ denotes the size of corresponding jumps.

Naturally, logarithmic return at time $t + j\Delta$ ($j = 1, 2, 3, \dots$) is defined as

$$r_{t+j\Delta} \stackrel{\text{def}}{=} p(t + j\Delta) - p(t + (j-1)\Delta). \quad (2)$$

Realized variance, hereafter denoted as RV , is the cumulative squared log returns over time period $[t, t+h]$, denoted as $RV_{t,t+h}$. Only in this section, we omit t for simplicity and still have the meaning clearly, i.e., $RV_{t+h} = RV_{t,t+h}$. We will discuss further about different periods of RV in Sect. 3.1

$$RV_{t+h}(\Delta) \stackrel{\text{def}}{=} \sum_{j=1}^{h/\Delta} r_{t+j\Delta}^2. \quad (3)$$

To calculate $RV_{t+h}(\Delta)$, following the above definition, we partition time period $[t, t+h]$ into N intervals evenly and then the sampling time interval $\Delta = h/N$, for example, we have 288 observations each day, and then $\Delta = 1/288$. As the quadratic variation of $p(t)$ process on that time period, hereafter QV , can be expressed as

$$QV_{t+h} = \text{p-lim}_{\Delta \rightarrow 0} \sum_{j=1}^{h/\Delta} r_{t+j\Delta}^2, \quad (4)$$

QV measures the variation of $p(t)$ that comes from two sources, one driven by the càdlàg process and one caused by the jump process

$$QV_{t+h} = \int_t^{t+h} \sigma^2(s)ds + \sum_{t < s \leq t+h} \kappa^2(s). \quad (5)$$

Then we employ the famous relation that RV converges in probability to QV as Δ goes to 0

$$RV_{t+h}(\Delta) \xrightarrow{P} QV_{t+h} \quad (6)$$

and

$$RV_{t+h}(\Delta) \xrightarrow{P} \underbrace{\int_t^{t+h} \sigma^2(s)ds}_{IV_{t+h}} + \underbrace{\sum_{t < s \leq t+h} \kappa^2(s)}_{J_{t+h}}. \quad (7)$$

Hence, the RV consists of two components. One is a continuous component, integrated variance (IV), and the other one is jump component (J). Barndorff-Nielsen and Shephard [3] provide us with a way to separate those two components and grant us the chance to investigate the information contained in the jump component. Key for the separation is a model-free and consistent estimator called “Bipower Variation” that measures the continuous process, hereafter BPV . We start with the general definition of “Multipower Variation,” hereafter MPV , for time period $[t, t + h]$

$$MPV_{t+h}(\Delta, \eta_1, \dots, \eta_m) \stackrel{\text{def}}{=} \left(\prod_{k=1}^m \mu_{\eta_k}^{-1} \right) \cdot \Delta^{1-\frac{1}{2}(\eta_1+\dots+\eta_m)} \cdot \sum_{j=m}^{h/\Delta} \prod_{k=1}^m |r_{t+(j-k+1)\Delta}|^{\eta_k}. \quad (8)$$

For any $\eta_k \geq 0, k = 1, 2, \dots$, where

$$\mu_{\eta_k} = 2^{\eta_k/2} \frac{\Gamma\{(\eta_k + 1)/2\}}{\Gamma(\frac{1}{2})},$$

the convergence property is given by

$$MPV_{t+h}(\Delta, \eta_1, \dots, \eta_m) \xrightarrow{P} \int_t^{t+h} \sigma^{\eta_1+\dots+\eta_m}(s) ds. \quad (9)$$

For simplicity, to calculate BPV , we specify $m = 2$ and $\eta_1 = \eta_2 = 1$, so that

$$BPV_{t+h}(\Delta) = \mu_1^{-2} \sum_{j=2}^{h/\Delta} |r_{t+j\Delta}| \cdot |r_{t+(j-1)\Delta}|. \quad (10)$$

BPV converges in probability to IV term in Eq. (7) as Δ goes to 0. Intuitively, BPV is robust to an infrequent jump process as it smooths the variation process by cumulating the adjacent log returns. Then, we can separate the J component by taking the difference of RV and BPV . In order to guarantee that J is nonnegative, we truncate the negative values of difference to 0

$$RV_{t+1}(\Delta) - BPV_{t+1}(\Delta) \xrightarrow{P} \sum_{t < s \leq t+1} \kappa^2(s) \quad (11)$$

$$J_{t+1} \stackrel{\text{def}}{=} \max \{RV_{t+1}(\Delta) - BPV_{t+1}(\Delta), 0\}.$$

BPV is effective in the sense of smoothing variation process when size and quantity in jump process are relatively small. In other words, BPV is biased to large and consecutive jumps which can be a major problem for investigating cryptocurrencies. As shown in Fig. 1, we are able to find consecutive jumps occur frequently in our studying asset.

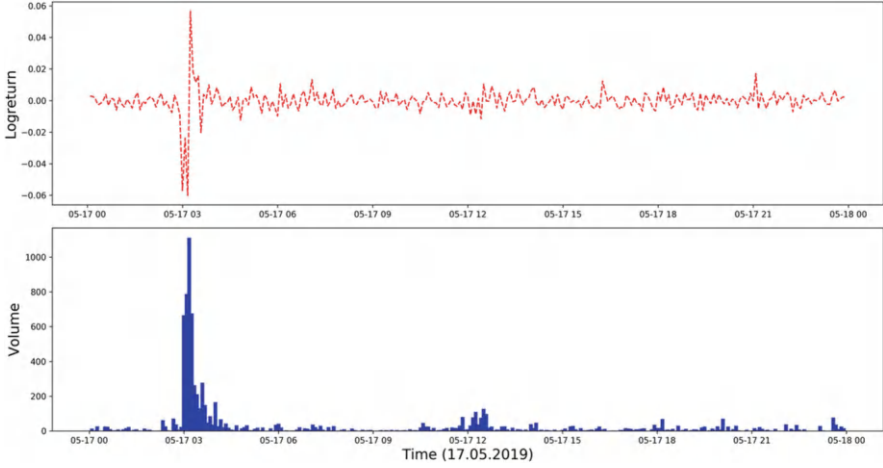


Fig. 1 A sample of the intraday behavior of 5-min freq BTC on 10th Mar. 2017. Upper panel shows the log-return process, and the bottom panel is corresponding trading volume

To alleviate this bias, we are motivated to implement the threshold variation model provided by Corsi et al. [9]. The main idea from [15] provides threshold estimators to measure multipower of integrated variance. The main idea is thresholding log return when it exceeds a certain level θ and the estimator still holds the convergence property,

$$\begin{aligned}
 TMPV_{t+h}(\Delta, \eta_1, \dots, \eta_m) &= \left(\prod_{k=1}^m \mu_{\eta_k}^{-1} \right) \cdot \Delta^{1-\frac{1}{2}(\eta_1+\dots+\eta_m)} \\
 &\cdot \sum_{j=m}^{h/\Delta} \prod_{k=1}^m |r_{t+(j-k+1)\Delta}|^{\eta_k} \\
 &\cdot \mathbf{I}\left\{|r_{t+(j-k+1)\Delta}|^2 \leq \theta_{t+(j-k+1)\Delta}\right\}.
 \end{aligned} \tag{12}$$

The threshold value θ_t is varying along with local variance $\widehat{V}_t$ for given constant coefficient,

$$\theta_t = c_\theta^2 \cdot \widehat{V}_t. \tag{13}$$

Constant c_θ makes certain impact on $TMPV$ via adjusting the threshold values, as c_θ goes larger the estimator has less truncated values, and hence, $TMPV$ goes to MPV . In our main empirical results, we follow the literature and choose $c_\theta = 3$, while we will also investigate how this coefficient affects our results. Furthermore, we employ a nonparametric method from [10] to estimate realized local variance σ^2

$$\widehat{V}_t^{[n]} = \frac{\sum_{i=-l, i \neq -1, 0, 1}^l K\left(\frac{i}{l}\right) \cdot r_{t+i}^2 \cdot \mathbf{I}\{r_{t+i}^2 \leq c_\theta^2 \cdot \widehat{V}_{t+i}^{[n-1]}\}}{\sum_{i=-l, i \neq -1, 0, 1}^l K\left(\frac{i}{l}\right) \cdot \mathbf{I}\{r_{t+i}^2 \leq c_\theta^2 \cdot \widehat{V}_{t+i}^{[n-1]}\}}, n = 1, 2, 3, \dots, \quad (14)$$

where K is a Gaussian kernel, and here we choose bandwidth value $l = 25$ and drop out the adjacent two values of t along with the value at t . To avoid using future information, $\widehat{V}_t$ is estimated within period $[t, t+h]$, for example, $h = 1\text{-day}$ means that we compute $\widehat{V}_t$ within one day iteratively until converged, and consequently, we lose 50 observations each day. In our case, we have 288 observations for each day, so losing 50 of them would not create too much bias. However, this may be problematic for those assets that only trade for about 7 hours (about 84 samples per day), and dropping 50 out of 84 observations could create huge information loss.

However, truncating the large returns may also “kill” some of the price changes that are not big enough to be jumps which then will create bias. We implement the corrected version of $TMPV$ that relieves such “double-sword” bias [9]. Essentially, instead of eliminating those points that have square returns r^2 larger than certain threshold value θ , this method replaces the η -th power log return $|r|^\eta$ with $r^e(\theta, \eta)$ which is the expected value under assumption that $r_t \sim N(0, \sigma^2)$. The expectation value is given by

$$\begin{aligned} r^e(\theta, \eta) &= \mathbb{E} \left\{ |r|^\eta \mid r^2 > \theta \right\} \\ &= \frac{(2\sigma^2)^{\frac{\eta}{2}}}{2\sqrt{\pi}\Phi\left(-\frac{\sqrt{\theta}}{\sigma}\right)} \cdot \Gamma\left(\frac{\eta+1}{2}, \frac{\theta}{2\sigma^2}\right). \end{aligned} \quad (15)$$

After plugging in relation at Eq. (13), σ^2 is approximated by $\widehat{V}$, and we have

$$r^e(\theta, \eta) = \frac{1}{2\sqrt{\pi}\Phi(-c_\theta)} \cdot \left(\frac{2\theta}{c_\theta^2}\right)^{\frac{\eta}{2}} \cdot \Gamma\left(\frac{\eta+1}{2}, \frac{c_\theta^2}{2}\right), \quad (16)$$

where Φ is the cumulative density function of standard normal distribution and Γ is the upper incomplete gamma function, defined as follows:

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{s^2}{2}} ds, \quad \Gamma(\alpha, x) = \int_x^{+\infty} s^{\alpha-1} e^{-s} ds.$$

The corrected estimator of $TMPV$ is defined as

$$\begin{aligned} TMPV_{t+h}(\Delta, \eta_1, \dots, \eta_m) &\stackrel{\text{def}}{=} \left(\prod_{k=1}^m \mu_{\eta_k}^{-1} \right) \\ &\cdot \Delta^{1-\frac{1}{2}(\eta_1+\dots+\eta_m)} \cdot \sum_{j=m}^{h/\Delta} \prod_{k=1}^m C_{\eta_k} \left(r_{t+(j-k+1)\Delta}, \theta_{t+(j-k+1)\Delta} \right), \end{aligned} \quad (17)$$

where $C_\eta(r, \theta)$ is defined using Eq. (16)

$$C_\eta(r, \theta) = \begin{cases} |x|^\eta, & r^2 \leq \theta \\ r^e(\theta, \eta), & r^2 > \theta. \end{cases} \quad (18)$$

In this chapter, we use two special cases of corrected $TMPV$ to estimate $\int_t^{t+h} \sigma^2(s)ds$ and $\int_t^{t+h} \sigma^4(s)ds$. For the former, we specify $m = 2, \eta_1 = \eta_2 = 1$ ($TBPV$), and for the latter, we specify $m = 3, \eta_1 = \eta_2 = \eta_3 = \frac{4}{3}$ ($TPPV$), definitions for time period $[t, t + h]$ as follows, respectively,

$$TBPV_{t+h}(\Delta) = \mu_1^{-2} \cdot \sum_{j=2}^{h/\Delta} C_1(r_{t+j\Delta}, \theta_{t+j\Delta}) C_1(r_{t+(j-1)\Delta}, \theta_{t+(j-1)\Delta}). \quad (19)$$

$$\begin{aligned} TTPV_{t+h}(\Delta) &= \mu_{\frac{4}{3}}^{-3} \cdot \Delta^{-1} \cdot \sum_{j=3}^{h/\Delta} C_{\frac{4}{3}}(r_{t+j\Delta}, \theta_{t+j\Delta}) C_{\frac{4}{3}}(r_{t+(j-1)\Delta}, \theta_{t+(j-1)\Delta}) \\ &\quad \times C_{\frac{4}{3}}(r_{t+(j-2)\Delta}, \theta_{t+(j-2)\Delta}). \end{aligned} \quad (20)$$

Equation (21) is the test for threshold jumps, hereafter denoted as t - z , provided by Corsi et al. [9] which is based on the test from [4], and we use them for testing threshold jumps and jumps, respectively.

$$t\text{-}z = \frac{\{RV_{t+h}(\Delta) - TBPV_{t+h}(\Delta)\} RV_{t+h}^{-1}(\Delta)}{\sqrt{\Delta \cdot \zeta \cdot \max \left\{ 1, \frac{TTPV_{t+h}(\Delta)}{\{TBPV_{t+h}(\Delta)\}^2} \right\}}}, \quad (21)$$

where $\zeta = \frac{\pi^2}{4} + \pi - 5$. Under a series of assumptions, for the null hypothesis that no jump exists, t - z converges to standard normal distribution as Δ goes to 0

$$t\text{-}z \xrightarrow{\mathcal{L}} N(0, 1).$$

Now, we can calculate the threshold realized jumps, hereafter TJ , taking account only the significant jumps by t - z test. Likewise, we also use the z test to retain only the significant jumps J in Eq. (11)

$$TJ_{t+h}(\Delta) = \max \{RV_{t+h}(\Delta) - BPV_{t+h}(\Delta), 0\} \cdot \mathbf{I}_{\{t\text{-}z_{t+h} > \Phi_\alpha\}}. \quad (22)$$

To make sure that $RV = C + J$ always holds, the continuous part TC is defined by TJ

$$TC_{t+h}(\Delta) = RV_{t+h}(\Delta) - TJ_{t+h}(\Delta) \quad (23)$$

as well as $C_{t+h}(\Delta) = RV_{t+h}(\Delta) - J_{t+h}(\Delta)$.

3 Empirical Results

Our empirical results use the two of the most liquid cryptocurrencies that also have the largest market capitals, Bitcoin and Ethereum. Many of the literature have discussed the sampling frequency issues [2, 13], in this chapter, we mainly focus on the 5-min frequency sample which is the most used one over all the previous researches, while we will also compare key results from other different sampling frequencies to ensure the robustness of our findings.

Cryptocurrency markets are trading 24 hours and 7 days, and we will have 288 samples within each day. Hence, we are exempted from over-night effect and gain more information from markets.

3.1 Data Description

Both of our coins, Bitcoin (BTC) and Ethereum (ETH), are being traded on multiple exchanges. Although here we assume that no arbitrage available between different exchanges, still we employ two sources of data for comparison, one is from a private data source called DYOS¹ and the other one is an online free provider.²

For the first data source, the price consists of three exchanges, including Poloniex, Bittrex, and Bitfinex. To be distinguishable, we denote BTC and ETH from DYOS as BTC-D and ETH-D. For the second source, they offer us data from different exchanges, however, most of them are hourly frequency, and thus we only take data from an exchange named Gemini which provides 1-min high-frequency data and subsample the dataset to 5-min high frequency. We denote Bitcoin and Ethereum as BTC-G and ETH-G, respectively.

All our empirical findings are based on the annualized unconditional daily realized variance ($RV_{t+1}(\Delta)$), which naturally leads other derived variables to be also annualized. Based on annualized daily RV , we normalize realized variance for longer horizon defined by Eq. (24), i.e., $RV_{t,t+h}(\Delta)$, $h > 1$, by averaging over all annualized realized variance h -days.³ Of course, one can also use Eq. (3) with larger h to yield equivalent RV . Our argument is that the averaging method incorporates information from longer horizons and ensures that RVs are on the same scale, thereby facilitating consistent comparisons across horizons in our empirical analysis.

$$RV_{\tau_1, \tau_2}(\Delta) = \frac{1}{\tau_2 - \tau_1 + 1} \sum_{\tau=\tau_1}^{\tau_2} RV_{t+\tau}. \quad (24)$$

¹ <https://dynamicsoft.io/>.

² <https://www.cryptodatadownload.com/>.

³ $RV_{t,t+1} = RV_{t+1}$ by definition.

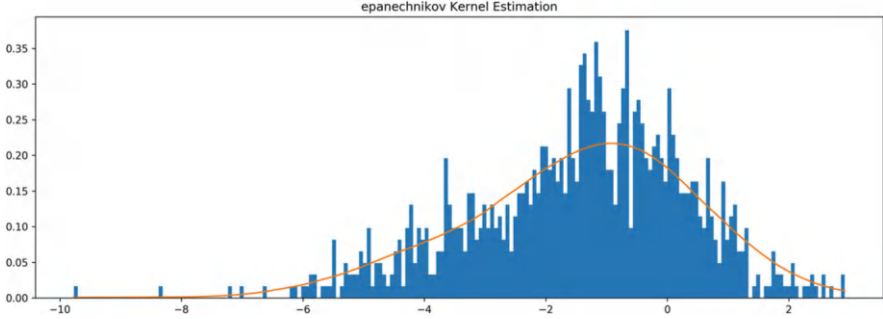


Fig. 2 Kernel density distribution of annualized unconditional daily logarithmic realized BTC-G variance ($\log(RV)$), $c_\theta = 3$

3.2 Realized Volatility and Jump Process

From Fig. 2, we can see the distribution of unconditional realized variance, similar to [2] and [6], RV is log normally distributed, and from Table 1, we can see the skewness and excessive kurtosis of $\log(RV)$ are all close to 0.

According to the $ACF(n)$ coefficients in Tables 1, 2, 3, and 4, consistent with results from the existent literatures, the $\log(RV)$ processes of all cryptos have clearly strong serial dependency structure comparing with other variables, which is a well-known property of the memory of RV .

From the continuous components defined by the difference between RV and significant jump components as shown in Tables 1, 2, 3, and 4, we can see the long-memory property from them. In addition, the logarithmic form of the continuous component is close to normal distribution.

Under the confidence level $\alpha = 99.99\%$, we can find totally 862 significant TJ out of 965 days (around 89%) for BTC-G (Table 1), and this finding is strongly against the empirical results on U.S. stocks documented on which is averagely 8.3% ($\alpha = 99.9\%$) days with jumps out of 1256 trading days across six most liquid stocks of S&P500 [9]. Similarly, there are around 70% of all days that are detected with jumps in terms of J , and this also contradicts results from which use BPV method.⁴ From Table 1, we can also see both J and TJ are extremely leptokurtic and positive skewed. All those findings suggest that crypto price processes are characterized by a series of jumps with large sizes, and in the meanwhile, extreme jumps are commonly seen.

By comparing J and TJ , we can also see that the threshold method has a stronger detection ability in terms of the number of jumps and size of jumps averagely. Especially, we can see TJ has a much larger value at 95 percentile and maximum,

⁴ In [1], under $\alpha = 99.99\%$, they detect 8.3%, 5.1%, and 7.6% of all days that have jumps on DM/\$, S&P 500 and U.S. T-Bond, respectively.

Table 1 Summary statistics on BTC-G, $\alpha = 99.99\%$, 5-min frequency, $C_\theta = 3$

	RV		$\log(RV)$		C		$\log(C)$		TC		$\log(TC)$		J_α		$\log(J_\alpha + 1)$		TJ_α		$\log(TJ_\alpha + 1)$	
Count	965		965		963		963		958		958		678		965		862		965	
Mean	0.81		-1.44		0.70		-2.08		0.51		-2.62		0.16		0.08		0.33		0.20	
Std	1.68		1.78		1.53		2.37		1.22		2.64		0.52		0.17		0.88		0.29	
Min	0.00		-9.77		0.00		-14.93		0.00		-15.64		0.00		0.00		0.00		0.00	
5%	0.01		-4.75		0.00		-6.29		0.00		-7.40		0.00		0.00		0.01		0.00	
50%	0.29		-1.24		0.20		-1.62		0.13		-2.06		0.07		0.03		0.14		0.11	
95%	2.99		1.10		2.93		1.08		2.02		0.70		0.45		0.30		1.12		0.69	
Max	18.59		2.92		18.28		2.91		18.28		2.91		10.90		2.48		15.38		2.80	
Skewness	5.59		-0.54		5.23		-0.89		6.60		-0.93		14.83		6.54		10.59		3.63	
Kurtosis	41.21		0.45		37.46		1.32		65.15		1.02		275.28		65.37		149.12		19.72	
ac(1)	0.49		0.79		0.57		0.76		0.50		0.77		0.12		0.26		0.15		0.30	
ac(7)	0.18		0.64		0.22		0.65		0.20		0.69		0.07		0.18		0.06		0.13	
ac(30)	0.11		0.42		0.16		0.49		0.17		0.52		0.01		0.03		-0.00		0.01	
ac(100)	0.05		0.21		0.07		0.29		0.08		0.30		-0.01		-0.04		-0.00		0.01	

Table 2 Summary statistics on ETH-G, $\alpha = 99.99\%$, 5-min frequency, $C_\theta = 3$

	RV	$\log(RV)$	C	$\log(C)$	TC	$\log(TC)$	J_α	$\log(J_\alpha + 1)$	TJ_α	$\log(TJ_\alpha + 1)$
Count	836	836	806	806	782	782	482	836	699	836
Mean	3.11	0.16	2.04	-0.59	1.33	-1.24	1.97	0.38	2.22	0.58
Std	8.68	1.29	4.93	1.83	3.15	2.21	7.25	0.62	7.94	0.71
Min	0.01	-4.25	0.00	-7.73	0.00	-9.51	0.01	0.00	0.01	0.00
5%	0.17	-1.79	0.02	-3.95	0.00	-5.68	0.05	0.00	0.08	0.00
50%	1.06	0.06	0.68	-0.39	0.45	-0.80	0.48	0.09	0.57	0.34
95%	11.72	2.46	7.88	2.06	5.59	1.72	7.31	1.64	8.59	2.10
Max	181.48	5.20	62.57	4.14	49.57	3.90	138.01	4.93	165.08	5.11
Skewness	12.27	0.34	6.23	-0.75	7.16	-0.99	14.37	2.59	13.93	2.17
Kurtosis	221.29	0.58	50.61	1.42	80.14	1.13	256.88	8.33	260.51	5.70
ac(1)	0.38	0.63	0.45	0.59	0.46	0.51	0.26	0.69	0.29	0.54
ac(7)	0.04	0.26	0.08	0.34	0.11	0.42	0.05	0.46	0.04	0.32
ac(30)	0.01	0.02	-0.01	0.17	0.03	0.26	0.05	0.37	0.05	0.21
ac(100)	-0.00	-0.09	0.02	0.07	0.05	0.13	0.01	0.20	0.00	0.10

Table 3 Summary statistics on BTC-D, $\alpha = 99.99\%$, 5-min frequency, $C_\theta = 3$

	RV		$\log(RV)$		C		$\log(C)$		TC		$\log(TC)$		J_α		$\log(J_\alpha + 1)$		TJ_α		$\log(TJ_\alpha + 1)$	
Count	913		913		913		913		912		912		352		913		706		913	
Mean	1.54		-0.12		1.33		-0.35		1.02		-0.66		0.55		0.13		0.68		0.3	
Std	2.66		0.96		2.26		1.09		1.73		1.43		1.28		0.28		1.95		0.38	
Min	0.10		-2.33		0.00		-6.96		0.00		-20.47		0.02		0.00		0.04		0.00	
5%	0.23		-1.48		0.16		-1.81		0.11		-2.22		0.06		0.00		0.07		0.00	
50%	0.77		-0.26		0.63		-0.46		0.49		-0.71		0.25		0.00		0.29		0.18	
95%	5.09		1.63		4.48		1.50		3.55		1.27		1.82		0.64		2.51		1.11	
Max	49.04		3.89		29.77		3.39		22.94		3.13		19.27		3.01		45.54		3.84	
Skewness	8.39		0.62		5.63		-0.18		5.6		-5.40		10.05		3.92		17.95		2.64	
Kurtosis	119.31		0.29		46.09		3.04		45.74		71.76		131.95		22.69		400.79		11.62	
ac(1)	0.40		0.71		0.48		0.70		0.52		0.55		0.35		0.7		0.16		0.45	
ac(7)	0.11		0.42		0.12		0.38		0.12		0.31		0.27		0.62		0.10		0.36	
ac(30)	0.06		0.21		0.05		0.16		0.06		0.14		0.18		0.45		0.09		0.23	
ac(100)	0.01		0.00		0.02		0.04		0.03		0.03		0.06		0.21		0.02		0.09	

Table 4 Summary statistics on ETH-D, $\alpha = 99.99\%$, 5-min frequency, $C_\theta = 3$

	RV	$\log(RV)$	C	$\log(C)$	TC	$\log(TC)$	J_α	$\log(J_\alpha + 1)$	TJ_α	$\log(TJ_\alpha + 1)$
Count	914	914	908	908	904	904	428	914	739	914
Mean	4.47	0.72	3.10	0.32	2.17	-0.04	2.97	0.37	2.87	0.64
Std	10.99	1.09	5.94	1.67	3.84	2.02	10.25	0.68	10.14	0.76
Min	0.00	-6.31	0.00	-19.23	0.00	-32.10	0.00	0.00	0.00	0.00
5%	0.47	-0.76	0.28	-1.27	0.2	-1.61	0.12	0.00	0.16	0.00
50%	1.80	0.59	1.39	0.33	1.02	0.02	0.67	0.00	0.73	0.44
95%	17.48	2.86	11.54	2.45	7.97	2.08	15.62	1.83	11.85	2.23
Max	193.42	5.26	71.36	4.27	47.03	3.85	168.53	5.13	173.51	5.16
Skewness	9.60	0.53	5.81	-5.22	5.14	-8.43	11.34	2.86	10.99	2.28
Kurtosis	131.71	2.57	45.59	52.05	36.41	112.44	165.64	9.52	156.92	6.32
ac(1)	0.53	0.73	0.54	0.39	0.46	0.40	0.40	0.88	0.47	0.69
ac(7)	0.21	0.42	0.08	0.17	0.05	0.19	0.26	0.80	0.27	0.59
ac(30)	0.16	0.20	0.09	0.02	0.1	0.06	0.14	0.62	0.16	0.43
ac(100)	0.02	0.04	0.02	-0.01	0.02	0.00	0.02	0.27	0.02	0.18

and this may suggest that some of the large size jumps or consecutive jumps are missed in J for the reasons we stated above by using Fig. 1. Unlike RV process, the jump processes are not characterized by significant series autocorrelation.

3.3 In-Sample Forecasting

In this subsection, we focus on building and analyzing the in-sample forecast of RV . By comparing results between different models, we will discuss questions including how does separating jumps improve our forecasting power on RV , what role does the separated jump play, and whether the threshold method makes significant differences. Please note that here in this subsection, all the explanatory variables are relied on $\alpha = 0.9999$ and $c_\theta = 3$.

There is no doubt that the long-memory dependency structure of the variance process is critically important in forecasting. A series of literature uses different ARCH, ARMA, and stochastic volatility models to capture this property. However, we would rather study the improvement of forecast accuracy from different models than absolute forecast accuracy, hence we follow many of the literature on realized volatility forecasting, and instead of using complicated long-memory models, we use HAR- RV models. Although the HAR model does not take account of the whole long-memory decay process, with very simple structure, it can still capture the key characteristics. To ensure this point, we also implement exponential-weighted moving average component and analyze the comparison.

There are three types of dependent variables that we will use in our forecasting models, square form, square-root form, and logarithmic form. For each forecasting model, related variables are in the same form. All variables are square form by default, for example, RV means realized variance, and we define $\widetilde{RV}_{t-h,t} = \{RV_{t-h,t}, RV_{t-h,t}^{1/2}, \log(RV_{t-h,t})\}$, $\widetilde{C}_{t-h,t} = \{C_{t-h,t}, C_{t-h,t}^{1/2}, \log(C_{t-h,t})\}$, and $\widetilde{J}_{t-h,t} = \{J_{t-h,t}, J_{t-h,t}^{1/2}, \log(J_{t-h,t} + 1)\}$.

To simplify the notation in regression equations, we predefine the following matrix by combining the variables defined above

$$X_{RV} = (\widetilde{RV}_{t-1,t}, \widetilde{RV}_{t-7,t}, \widetilde{RV}_{t-30,t})$$

$$X_C = (\widetilde{C}_{t-1,t}, \widetilde{C}_{t-7,t}, \widetilde{C}_{t-30,t})$$

$$X_{TC} = (\widetilde{TC}_{t-1,t}, \widetilde{TC}_{t-7,t}, \widetilde{TC}_{t-30,t})$$

$$X_J = (\widetilde{J}_{t-1,t}, \widetilde{J}_{t-7,t}, \widetilde{J}_{t-30,t})$$

$$X_{TJ} = (\widetilde{TJ}_{t-1,t}, \widetilde{TJ}_{t-7,t}, \widetilde{TJ}_{t-30,t})$$

$$\beta_1 = (\tilde{\beta}_D, \tilde{\beta}_W, \tilde{\beta}_M)$$

$$\beta_2 = (\tilde{\beta}_{JD}, \tilde{\beta}_{JW}, \tilde{\beta}_{JM})$$

, $X_{CS} = \{X_C, X_{TC}\}$, and $X_{JS} = \{X_J, X_{TJ}\}$.

3.3.1 HAR Model

Equation (25) is heterogeneous autoregression model, namely HAR, [8], that takes the multiperiod lagged $RV_{t-h,t}$ as explanatory variables to forecast normalized realized variance of different horizons $RV_{t,t+h}$. We focus on daily, weekly, and monthly horizons. Due to the fact that the cryptocurrency markets are trading 24 hours, 7 days, we take $h = 1, 7, 30$ accordingly. When forecast horizon $h > 1$, to adjust standard errors of estimates caused by serial correlations of the error term, we use Newey-West covariance matrix estimator with 7, 14, and 60 lags for daily, weekly, and monthly forecast horizon, respectively.

$$\widetilde{RV}_{t,t+h} = \alpha + X_{RV}\beta_1 + \varepsilon_{t,t+h}. \quad (25)$$

3.3.2 HAR-RV(T)J Models

To answer the question that if separating jump component improves forecasting power, we use *HAR-RV(T)J* model [1] which is adding lagged jump terms to HAR model. In particular, here we add daily, weekly, and monthly lagged jump terms.

$$\widetilde{RV}_{t,t+h} = \alpha + X_{RV}\beta_1 + X_{JS}\beta_2 + \varepsilon_{t,t+h}. \quad (26)$$

We will plug both $X_{JS} = X_J$ (for *HAR-RVJ*) and $X_{JS} = X_{TJ}$ (for *HAR-RVTJ*) into the right side of the regression to test if the jump component detected from *TBPV* method has advantages over the one from the *BPV* method.

3.3.3 HAR-(T)CJ Models

Based on the *HAR* model, apart from adding jump terms into *HAR* model as explanatory variables, we also decompose the lagged *RV* into jump component and continuous component as the following:

$$\widetilde{RV}_{t,t+h} = \alpha + X_{CS}\beta_1 + X_{JS}\beta_2 + \varepsilon_{t,t+h}. \quad (27)$$

For *HAR-CJ* model, $X_{CS} = X_C$, $X_{JS} = X_J$, and for threshold version, *HAR-TJ* model, $X_{CS} = X_{TC}$, $X_{JS} = X_{TJ}$.

3.3.4 *HAR-Exp-RV-(T)(C)J* Models

The models above involve stepwise change which may omit information between each weekly and monthly time points. To avoid that, we also implement a smoothing method called exponential-weighted moving average on the past *RV* and continuous components as the following:

$$\begin{aligned} RV_{t-h,t}^{Exp} &= \frac{\sum_{k=0}^t (1-\alpha)^k \cdot RV_{t-k}}{\sum_{k=0}^h (1-\alpha)^k} \\ &= \frac{\sum_{k=0}^t \left(1 - \frac{1}{1+CoM(t-h,t)}\right)^k \cdot RV_{t-k}}{\sum_{k=0}^h \left(1 - \frac{1}{1+CoM(t-h,t)}\right)^k}, \end{aligned} \quad (28)$$

where α is the smoothing factor which is defined as $\alpha = \frac{1}{1+CoM(t-h,t)}$. For each given *CoM*, we compute the corresponding α and yield exponential smoothed *RV*. The *CoM*($t-h, t$) is the center of mass for EWMA, and for time period $[t-h, t]$, we take *CoM*($t-h, t$) = $h/2$.

Likewise, we can also have the exponentially smoothed version of continuous components, $\tilde{C}_{t-h,t}^{Exp}$ and $\tilde{TC}_{t-h,t}^{Exp}$. Then we construct the regressors matrix as

$$\begin{aligned} X_{RV}^{Exp} &= \left(\widetilde{RV}_t^{Exp}, \widetilde{RV}_{t-7,t}^{Exp}, \widetilde{RV}_{t-30,t}^{Exp} \right), \\ X_C^{Exp} &= \left(\tilde{C}_t^{Exp}, \tilde{C}_{t-7,t}^{Exp}, \tilde{C}_{t-30,t}^{Exp} \right), \\ X_{TC}^{Exp} &= \left(\widetilde{TC}_t^{Exp}, \widetilde{TC}_{t-7,t}^{Exp}, \widetilde{TC}_{t-30,t}^{Exp} \right). \end{aligned}$$

Hence, the *HAR-Exp-RV-(T)(C)J* models are constructed by replacing *RV* and *C* in *HAR-RV(T)J* and *HAR-(T)CJ* models with their exponential smoothing versions. The *HAR-Exp-RV-(T)J* model is defined as follows:

$$\widetilde{RV}_{t,t+h} = \alpha + X_{RV}^{Exp} \beta_1 + X_{JS} \beta_2 + \varepsilon_{t,t+h}, \quad (29)$$

and *HAR-Exp-(T)CJ* is defined as

$$\widetilde{RV}_{t,t+h} = \alpha + X_{CS}^{Exp} \beta_1 + X_{JS} \beta_2 + \varepsilon_{t,t+h}, \quad (30)$$

where $X_{CS}^{Exp} = \{X_C^{Exp}, X_{TC}^{Exp}\}$.

3.3.5 Performance Evaluation

Apart from the t test for the significance of regressors and the adjust R^2 , we also use three other metrics to evaluate performance. The evaluation is based on the square form of RV , so for those results from the square-root and logarithmic models, we transform them back to square form for comparison.

The first metrics is mean square error (MSE) which is widely used in many cases, and it accounts the average Euclidean distance between the ex post realized variance RV and forecast result $\widehat{RV}$

$$MSE = \frac{1}{T} \sum_{t=1}^T (RV_t - \widehat{RV}_t)^2. \quad (31)$$

However, MSE is not robust to forecast errors as the realized variance scale changes. So we implement heteroskedasticity adjusted root mean square error (HRMSE) from [5]

$$HRMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T \left(\frac{RV_t - \widehat{RV}_t}{RV_t} \right)^2}. \quad (32)$$

Another metrics we use is QLIKE which gives consistent evaluation from different imperfect volatility proxies even though we only compare realized variance here.

$$QLIKE = \frac{1}{T} \sum_{t=1}^T \left(\log RV_t + \frac{\widehat{RV}_t}{RV_t} \right). \quad (33)$$

To summarize, we totally have nine regression equations due to matching estimators between $RV/C/RV^{Exp}/C^{Exp}$ and J/TJ . Also, we will test three forms of estimators, namely square form, square-root form, and logarithmic form.

3.3.6 Analysis of Regression Results

The empirical results reveal several insights in Table 5 based on BTC-G 5-min frequency $\alpha = 0.9999$, $c_\theta = 3$. Firstly, logarithmic models demonstrate superior performance in terms of R^2 , HRMSE, and QLIKE metrics when compared to other models. However, the analysis indicates that when considering MSE, logarithmic models do not exhibit a notable advantage over alternative models. This discrepancy suggests that the transformation from logarithmic to square form may lead to a larger scale for the forecast realized variance ($\widehat{RV}$) metric. Secondly, it is evident from our analysis that the coefficients associated with the daily, weekly, and monthly lagged continuous components (β_D , β_W , and β_M) are predominantly significant, with their

Table 5 Daily regression results on BTC-G, $\alpha = 0.9999$, $c_\theta = 3$, 5-min frequency

HAR : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{RV}_{t-1,t} + \beta_W \widetilde{RV}_{t-7,t} + \beta_M \widetilde{RV}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-RVJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{RV}_{t-1,t} + \beta_W \widetilde{RV}_{t-7,t} + \beta_M \widetilde{RV}_{t-30,t} + \beta_{JD} \widetilde{J}_{t-1,t} + \beta_{JW} \widetilde{J}_{t-7,t} + \beta_{JM} \widetilde{J}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-RVTJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{RV}_{t-1,t} + \beta_W \widetilde{RV}_{t-7,t} + \beta_M \widetilde{RV}_{t-30,t} + \beta_{JD} \widetilde{TJ}_{t-1,t} + \beta_{JW} \widetilde{TJ}_{t-7,t} + \beta_{JM} \widetilde{TJ}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-CJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{C}_{t-1,t} + \beta_W \widetilde{C}_{t-7,t} + \beta_M \widetilde{C}_{t-30,t} + \beta_{JD} \widetilde{J}_{t-1,t} + \beta_{JW} \widetilde{J}_{t-7,t} + \beta_{JM} \widetilde{J}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-TCJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{TC}_{t-1,t} + \beta_W \widetilde{TC}_{t-7,t} + \beta_M \widetilde{TC}_{t-30,t} + \beta_{JD} \widetilde{TJ}_{t-1,t} + \beta_{JW} \widetilde{TJ}_{t-7,t} + \beta_{JM} \widetilde{TJ}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-Exp-RVJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{RV}_{t-1,t}^{Exp} + \beta_W \widetilde{RV}_{t-7,t}^{Exp} + \beta_M \widetilde{RV}_{t-30,t}^{Exp} + \beta_{JD} \widetilde{J}_{t-1,t} + \beta_{JW} \widetilde{J}_{t-7,t} + \beta_{JM} \widetilde{J}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-Exp-RVTJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{RV}_{t-1,t}^{Exp} + \beta_W \widetilde{RV}_{t-7,t}^{Exp} + \beta_M \widetilde{RV}_{t-30,t}^{Exp} + \beta_{JD} \widetilde{TJ}_{t-1,t} + \beta_{JW} \widetilde{TJ}_{t-7,t} + \beta_{JM} \widetilde{TJ}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-Exp-CJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{C}_{t-1,t}^{Exp} + \beta_W \widetilde{C}_{t-7,t}^{Exp} + \beta_M \widetilde{C}_{t-30,t}^{Exp} + \beta_{JD} \widetilde{J}_{t-1,t} + \beta_{JW} \widetilde{J}_{t-7,t} + \beta_{JM} \widetilde{J}_{t-30,t} + \varepsilon_{t,t+1}$
HAR-Exp-TCJ : $\widetilde{RV}_{t,t+1} = \alpha + \beta_D \widetilde{TC}_{t-1,t}^{Exp} + \beta_W \widetilde{TC}_{t-7,t}^{Exp} + \beta_M \widetilde{TC}_{t-30,t}^{Exp} + \beta_{JD} \widetilde{TJ}_{t-1,t} + \beta_{JW} \widetilde{TJ}_{t-7,t} + \beta_{JM} \widetilde{TJ}_{t-30,t} + \varepsilon_{t,t+1}$

	Square form (i.e., $\widehat{RV} = RV$ and etc.)									
	HAR	HAR-RVJ	HAR-RVTJ	HAR-CJ	HAR-TCJ	HAR-Exp-RVJ	HAR-Exp-RVTJ	HAR-Exp-CJ	HAR-Exp-TCJ	
α	0.127 (2.056)	0.258 (3.977)	0.201 (2.044)	0.258 (3.977)	0.201 (2.044)	0.211 (2.865)	0.139 (1.335)	0.222 (3.133)	0.168 (1.670)	
β_D	0.399 (4.743)	0.504 (10.560)	0.390 (4.101)	0.504 (10.560)	0.390 (4.101)	0.787 (9.138)	0.634 (4.085)	0.802 (9.483)	0.701 (4.652)	
β_W	0.046 (0.604)	0.033 (0.448)	-0.009 (-0.063)	0.033 (0.448)	-0.009 (-0.063)	-0.530 (-2.667)	-0.515 (-1.459)	0.604 (2.903)	0.902 (2.858)	
β_M	0.393 (3.063)	0.307 (2.354)	0.620 (2.754)	0.307 (2.354)	0.620 (2.754)	0.616 (2.991)	0.846 (2.955)	-0.535 (-2.687)	-0.607 (-1.717)	
β_{JD}		-0.754 (-5.614)	0.019 (0.052)	-0.251 (-2.319)	0.410 (1.343)	-0.635 (-5.298)	0.067 (0.201)	-0.215 (-2.088)	0.397 (1.356)	
β_{JW}		-0.016 (-0.083)	0.142 (0.504)	0.017 (0.112)	0.133 (0.849)	-0.033 (-0.126)	0.049 (0.165)	0.041 (0.217)	0.121 (0.717)	
β_{JM}		-0.574 (-1.701)	-0.865 (-2.221)	-0.267 (-0.961)	-0.245 (-0.932)	-0.459 (-1.505)	-0.510 (-1.661)	-0.153 (-0.565)	-0.097 (-0.389)	
$adj-R^2$	0.296	0.326	0.300	0.326	0.300	0.321	0.301	0.324	0.303	
MSE	1.942	1.859	1.932	1.859	1.932	1.874	1.930	1.866	1.923	
HRMSE	117.823	156.201	110.672	156.201	110.672	137.424	91.297	139.459	102.919	
QLIKE	8.593	11.340	8.673	11.340	8.673	10.300	7.576	10.367	8.047	

	Square-root form (i.e., $\widehat{RV} = RV^{1/2}$ and etc.)									
	HAR	HAR-RVJ	HAR-RVTJ	HAR-CJ	HAR-TCJ	HAR-Exp-RV-J	HAR-Exp-RV-TJ	HAR-Exp-CJ	HAR-Exp-TCJ	
α	0.057 (2.203)	0.178 (4.496)	0.132 (2.963)	0.190 (5.146)	0.153 (3.452)	0.151 (3.201)	0.072 (1.410)	0.164 (3.940)	0.124 (2.638)	
β_D	0.513 (9.299)	0.513 (11.057)	0.445 (9.293)	0.516 (11.791)	0.432 (9.179)	0.705 (9.029)	0.615 (6.139)	0.738 (10.485)	0.661 (7.590)	
β_W	0.139 (2.820)	0.196 (3.340)	0.154 (1.270)	0.178 (3.223)	0.137 (1.449)	-0.149 (-1.083)	-0.226 (-0.966)	0.304 (2.837)	0.480 (3.306)	
β_M	0.218 (3.732)	0.153 (2.499)	0.476 (3.686)	0.139 (2.389)	0.296 (3.212)	0.314 (2.873)	0.642 (3.907)	-0.201 (-1.530)	-0.284 (-1.384)	
β_{JD}		-0.160 (-1.693)	0.115 (1.237)	0.033 (0.356)	0.329 (3.833)	-0.160 (-1.887)	0.151 (1.724)	0.002 (0.029)	0.323 (3.926)	
β_{JW}		-0.178 (-1.613)	-0.050 (-0.303)	-0.119 (-1.212)	0.025 (0.268)	-0.150 (-1.208)	-0.084 (-0.558)	-0.069 (-0.647)	0.036 (0.385)	
β_{JM}		-0.113 (-0.955)	-0.531 (-3.144)	-0.034 (-0.327)	-0.155 (-1.745)	-0.121 (-1.098)	-0.376 (-3.206)	-0.035 (-0.344)	-0.112 (-1.485)	
$adj-R^2$	0.532	0.546	0.541	0.549	0.544	0.540	0.537	0.547	0.547	
MSE	1.942	1.880	1.911	1.872	1.896	1.895	1.922	1.882	1.888	
HRMSE	33.568	45.473	26.863	47.813	20.640	39.860	20.861	40.099	19.794	
QLIKE	2.162	2.860	1.902	2.849	1.629	2.676	1.716	2.561	1.567	

	Logarithmic form (i.e., $\widehat{RV} = \log(RV)$ and etc.)									
	HAR	HAR-RVJ	HAR-RVTJ	HAR-CJ	HAR-TCJ	HAR-Exp-RV-J	HAR-Exp-RV-TJ	HAR-Exp-CJ	HAR-Exp-TCJ	
α	-0.256 (-6.740) 0.500	-0.066 (-1.342) 0.477	0.111 (0.886) 0.421	-0.090 (-1.533) 0.329	-0.181 (-1.392) 0.198	-0.113 (-2.306) 0.663	0.044 (0.346) 0.600	-0.429 (-3.869) 0.663	-0.163 (-1.287) 0.363	
β_D	(10.398) 0.323	(9.069) 0.377	(6.683) 0.461	(7.540) 0.343	(4.275) 0.393	(7.341) 0.159	(5.805) 0.229	(27.686) 0.333	(5.424) 0.043	
β_W	(4.716) 0.147	(4.724) 0.146	(4.667) 0.217	(5.024) 0.077	(5.189) 0.027	(0.964) 0.212	(1.248) 0.309	(3.936) -0.126	(0.477) 0.239	
β_M	(2.827)	(2.315)	(2.618)	(1.313)	(0.439)	(1.893)	(2.495)	(-2.556)	(1.946)	
β_{JD}		-0.184	0.394	0.363	1.021	-0.237	0.361	0.111	0.886	
β_{JW}		(-0.854) -0.820	(2.221) -0.833	(1.523) -0.761	(6.172) -0.537	(-1.201) -0.839	(2.442) -0.794	(0.528) -0.321	(5.841) -0.372	
β_{JM}		(-2.353) -0.861	(-2.971) -0.685	(-2.190) -0.219	(-2.157) 0.176	(-2.000) -0.865	(-2.944) -0.670	(-0.902) 0.402	(-1.750) 0.076	
$adj-R^2$	0.668	(-2.341) 0.676	(-1.901) 0.675	(-0.479) 0.663	(0.517) 0.669	(-2.636) 0.674	(-2.118) 0.672	(1.009) 0.666	(0.242) 0.674	
MSE	2.007	1.951	1.991	2.072	2.065	1.950	1.981	1.979	2.027	
HRMSE	15.837	16.262	15.563	25.412	6.448	12.854	11.808	19.593	7.967	
QLIKE	0.703	0.700	0.670	1.012	0.391	0.591	0.558	0.833	0.419	

significance levels diminishing as the lag increases. Notably, the daily coefficient β_D remains significant across all models. This observation is further supported by our autocorrelation function (ACF) plot on realized volatility (RV), indicating a pronounced positive autocorrelation.

Thirdly, in the context of forecasting models incorporating jump components (J), a consistent finding emerges: One-day lagged jumps exhibit a negative influence on future realized volatility ($\widetilde{RV}$). Conversely, there is a consistent positive correlation between one-day lagged transformed jumps (TJ) and future $\widetilde{RV}$. An intriguing observation arises when considering square and logarithmic models, revealing divergent outcomes in terms of significance. Specifically, in square form, the coefficient β_{JD} is consistently significant and negative across all models when using the jump component J , while it is consistently insignificant and positive across all models when employing the TJ . In logarithmic form, the scenario changes, where β_{JD} is significantly positive across all models when using TJ and remains consistently insignificant across all models when utilizing J . This intricate pattern suggests that a higher magnitude of jumps corresponds to lower RV the following day; however, in contrast, a higher logarithmic $\log(TJ + 1)$ results in an elevated logarithmic $\log(RV)$ the subsequent day. Fourthly, unlike the ambiguous findings observed with one-day lagged estimators, the coefficients β_{JW} and β_{JM} consistently demonstrate that $\widetilde{RV}$ has a significant reversal effect on weekly and monthly lagged jump components, respectively. This effect implies that when larger jumps occur during the previous week or month, the subsequent day is more likely to exhibit a relatively tranquil market environment.

Finally, comparing models utilizing separate jump components with the HAR model, we observe a 1–2% improvement in R^2 , diverging from [1]’s findings where $HAR - RVJ$ is superior. However, our results align with [9]. Despite this, models incorporating the J component perform notably worse in terms of HRMSE and QLIKE compared to models using TJ , indicating that the threshold method not only detects more jumps but also extracts more informative content. Furthermore, no compelling evidence is found to support any advantages conferred by the EWMA method.

4 Conclusion

In conclusion, our empirical analysis of Bitcoin and Ethereum markets at a 5-minute frequency reveals several key insights. Firstly, logarithmic models consistently outperform other models in terms of R^2 , HRMSE, and QLIKE metrics, although they do not exhibit a notable advantage in terms of mean-squared error (MSE). This discrepancy suggests that the transformation from logarithmic to square form may affect the forecast realized variance (RV) metric’s scale. Additionally, the significance of coefficients associated with daily, weekly, and monthly lagged continuous components underscores their importance in forecasting realized volatility, with daily coefficients maintaining significance across all models. The presence

of significant autocorrelation in realized volatility further validates these findings. Furthermore, the analysis of jump components reveals a consistent negative influence of one-day lagged jumps on future RV , while transformed jumps exhibit a positive correlation with future RV . Notably, the reversal effect of larger jumps on weekly and monthly lagged components implies a tranquil market environment following such occurrences. Comparisons with the HAR model indicate a modest improvement in R^2 , contrasting [1]'s findings, yet aligning with [9], while models utilizing jump components perform inferiorly in HRMSE and QLIKE metrics compared to those using transformed jumps, suggesting the latter's ability to extract more informative content. Overall, our analysis sheds light on the intricacies of cryptocurrency market dynamics and the effectiveness of different modeling approaches in capturing volatility dynamics.

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Part IV

Blockchain Analytics

Process Mining for Blockchain Analytics: Examining Blockchain Applications Through a Process Lens



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Abstract Process mining offers a rich tool set for analyzing event data which record information about events that occur during the execution of processes. Event attributes captured in those data include, at a minimum, the type of executed activities, the temporal ordering of events, and the process instance to which an event belongs, but can also cover additional aspects like the involved resource or the execution status of activities. Given the distributed, temporal, and concurrent nature of blockchain technology, process mining is a valuable addition to the blockchain analytics tool set, allowing applications to be analyzed through a process lens. In this chapter, we summarize the current state of the art regarding process mining for blockchain analytics. Starting from a process mining methodology for blockchain analytics, we provide an overview of available techniques and tools for the two most important analysis steps: *data extraction and preparation* as well as *data analysis and visualization*. We then discuss promising use cases and limitations of process mining for blockchain analytics and present an illustrative real-world case study involving a DApp deployed on the public Ethereum network.

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1 Introduction

Process mining [2] is increasingly used to monitor and improve operational processes. There are over 40 commercial process mining tools, next to open-source process mining frameworks like ProM, PM4Py, and bupaR, and more than half of the Fortune 500 companies are already using process mining.¹ A prerequisite for applying process mining is the availability of *event data* which provide information about the execution of operational processes and which can typically be extracted from enterprise information systems such as SAP, Salesforce, Oracle, and the like.

Similar to traditional enterprise information systems, many blockchain applications support operational processes. For example, there are many decentralized autonomous organizations for which blockchain applications define a set of transparent processes and rules that allow the organizations' members to control the organizations without requiring centralized leadership [20]. Blockchain-based prediction markets like Augur² enable users to obtain reliable information. That is, they define and coordinate a decentralized process for posing questions and collecting accurate answers. Similarly, more and more cross-organizational processes, e.g., in supply chains, are facilitated by blockchain technology by enforcing business rules and exchanging business information [18, 25]. While not the most significant application of blockchain technology, smart contracts of games like CryptoKitties³ implement processes that autonomously enforce the games' rulebooks.

Given the process nature of blockchain applications, it is logical to use the rich and mature set of analytical methods and techniques that process mining offers to, e.g., validate DApp implementations, identify security vulnerabilities, or understand user behavior. For instance, by inferring process models from event data, *process discovery algorithms* can visualize the structure of the process as it is implemented and utilized in the application. *Conformance checking* identifies process instances that deviate from a normative process model and thus can be used to detect unwanted deviations and security vulnerabilities. *Dotted charts* and *drift detection* allow us to examine the DApp behavior over time, e.g., to comprehend the influence of network traffic, transaction fees, or exchange rates on DApp usage. *Comparative process mining* can be used to compare different variants of a process or the same process over different periods. In addition to the control-flow perspective, process mining also supports the analysis of other process perspectives. For example, process mining can reveal patterns such as *handovers* and *delegation* to investigate the organizational perspective and user behavior. Alternatively, *social network analysis* can detect power users or clusters of users that often collaborate. Lastly, *predictive process monitoring* can be used to predict a running case's trajectory.

¹ cf. <http://processmining.org>, accessed: 14/10/2025.

² <https://augur.net>, accessed: 26/2/2022..

³ <https://www.cryptokitties.co>, accessed: 14/10/2025..

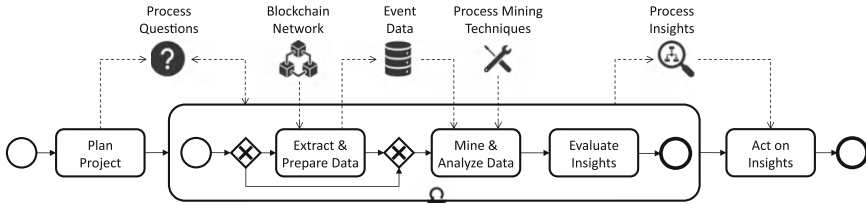


Fig. 1 A process mining methodology to analyze blockchain applications, based on [11]

While this list of techniques and use cases is not exhaustive, it demonstrates the potential of process mining. In this chapter, we hence provide an overview of methods, techniques, and use cases for analyzing blockchain applications through a process lens. To this end, we oriented the chapter toward the *process mining project methodology* (PM²) [11]. Figure 1 outlines a version of PM², instantiated for blockchain analytics, which shows the basic steps and artifacts. First, the project is planned by establishing an understanding of the examined DApp and the project goal. This step ultimately leads to a set of *process questions* which are then investigated by iteratively examining and refining the questions to generate *process insights* into the DApp’s behavior. In early iterations and in iterations where additional data are required but not yet available, event data is extracted from the blockchain network and prepared for analysis. The data are then mined and analyzed using process mining techniques, generating process insights that are evaluated against domain knowledge to validate their reliability. Finally, when all questions are answered, the project team acts on the insights, e.g., by communicating results or adapting the DApp.

In the following, we outline how event data can be extracted from blockchain networks and prepared for analysis in Sect. 2. Next, we provide an overview of process mining techniques in Sect. 3. Section 4 reviews potential use cases for process mining on blockchain data, clarifying the type of process questions that can be investigated with process mining and discussing its limitations. Section 5 then presents a real-world case study involving a DApp deployed on the public Ethereum network to illustrate the application of process mining for blockchain analytics and demonstrate the kind of insights that can be obtained. Finally, we close the chapter with a summary in Sect. 6.

2 Event Data

In this section, we outline how analysts can extract event data from blockchain networks. In Sect. 2.1, we introduce a generic logical data model for event data and show how blockchain data can be mapped to this model. Then, we give an overview of tools that support the actual extraction of data from blockchain networks in Sect. 2.2.

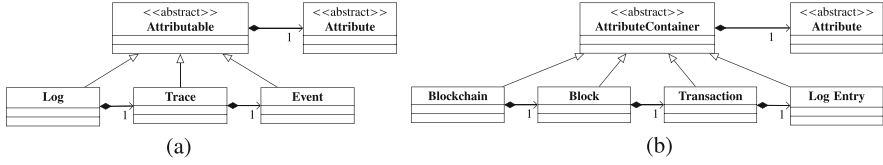


Fig. 2 A comparison of the data structures of event logs and blockchain data. **(a)** Event logs. **(b)** Blockchain data

2.1 Data Selection

To use process mining to analyze a blockchain application, event data must first be extracted from the application into a so-called *event log*. Independent of the specific analysis context, an event log only records event data for one particular process and structures these event data using the logical data model in Fig. 2a. That is, an event log contains a set of *traces*, where each trace represents a process instance, i.e., an execution of the process for a particular case. A trace itself comprises a list of *events* that capture information about the execution of particular activities within the process instance. For all three elements, *attributes* can be defined: Log attributes contain general information about the process or the event data itself; trace attributes record information about the entire process instance and/or the case; and event attributes characterize the respective activity execution. At a minimum, an event log must contain one trace for each process instance and must clearly assign the events to the respective trace. Moreover, there must be at least two attributes for each event: the activity type and a timestamp. While activity types capture the activity whose execution created the event, timestamps record the time at which events occurred and thus establish a temporal ordering of events. Commonly, event logs also include attributes of the involved resources, the execution status of activities, financial information such as costs, and other context-specific information.

Figure 2b shows that blockchain technology relies on a similar data structure. That is, a blockchain contains an ordered sequence of *blocks*. Each block contains an ordered set of *transactions* that were executed and included at some point in time.⁴ A transaction can also contain an ordered list of *log entries* that refer to transaction-specific events. In general, we can view the blockchain as one process instance, where blocks, transactions, and log entries are events that are temporally ordered according to their inclusion order. While this abstract view is helpful for blockchain-wide analysis, e.g., when examining the general network traffic, the examination of DApp or user behavior requires us to filter the blockchain data and map them to the conceptual data model for event logs. To this end, we must establish a *process notion* and determine (i) the process that we are interested in, (ii) the process instances for

⁴ Note that the notion of time in general, its precision, and time of transaction finality are actually difficult, technology-specific issues in blockchain systems. This topic is discussed further in Sect. 4.2.

which we want to collect data, (iii) the activities that occur within the process, and (iv) the attributes that are needed for analysis.

For example, consider the CryptoKitties DApp on the public Ethereum blockchain, which according to the game's developers is "a game centered around breedable, collectible, and oh-so-adorable creatures we call CryptoKitties"³. Viewing the lifecycle of CryptoKitties' as a process allows us to collect event data: (1) the addresses of the DApp's smart contract identify the process; (2) each individual CryptoKitty represents a separate process instance; (3) log entries provide information about events, i.e., important changes in a CryptoKitty's life such as its birth or a change in its ownership; (4) the block timestamps and the inclusion order of blocks, transactions, and log entries establish a temporal ordering of events; and (5) additional information about the CryptoKitties' DNA or owner can be captured as DApp-specific attributes.

2.2 Data Extraction and Transformation

Analysts need to implement *extract, transform, load* (ETL) pipelines to obtain the required event data, typically as scripts or programs in their preferred programming language and data processing platform. Those pipelines comprise three abstract steps. First, a connection to a data source must be set up and raw data must be extracted from it. In this regard, analysts must have access to a node in the blockchain network from which they can retrieve data via the native API (e.g., Ethereum's standardized JSON-RPC⁵ API), or via language-specific API wrappers such as the web3 wrappers for Ethereum (e.g., web3.js,⁶ web3.py,⁷ or web3J⁸). Alternatively, analysts can rely on databases that contain all network data. Here, extraction frameworks like Blockchain-ETL⁹ automate the network data transfer into databases. As this extraction requires significant compute power and space, analysts might choose to access potentially fee-based third-party databases like the Ethereum dataset hosted by Google BigQuery.¹⁰ Second, the raw data must be transformed, filtered, potentially decoded, and mapped to an event log data structure. Here, the analysts can rely on the transformation features provided by their preferred platform or programming language. Lastly, the event log must be exported in a format that process mining tools can load. In this regard, the IEEE *eXtensible Event*

⁵ <https://github.com/ethereum/execution-apis>, accessed: 14/10/2025.

⁶ <https://web3js.readthedocs.io/en/v1.7.0/>, accessed: 14/10/2025.

⁷ <https://web3py.readthedocs.io/en/stable>, accessed: 14/10/2025.

⁸ <https://docs.web3j.io/4.8.7/>, accessed: 14/10/2025.

⁹ <http://blockchainetl.io>, accessed: 14/10/2025.

¹⁰ <https://cloud.google.com/blog/products/data-analytics/ethereum-bigquery-public-dataset-smart-contract-analytics>, accessed: 14/10/2025.

Stream (XES) standard [1] for event logs and event streams is supported by many open-source and commercial process mining tools.¹¹

While there are many tools that analysts can rely on to extract event data from blockchain networks, the implementation of the ETL pipelines can be highly inconvenient and very slow [9] for multiple reasons. As outlined in Sect. 1, analysts need to establish a solid understanding of the DApp at the beginning of the process mining project. This can require significant efforts depending on their familiarity with the DApp. Moreover, the implementation and testing of the ETL pipeline is often exacerbated by the necessity to integrate a diverse set of tools, APIs and formats. Lastly, the efficiency with which the ETL pipeline can be executed ultimately depends on the available compute infrastructure and might consume a substantial amount of time, especially when additional data needs arise during the analysis. While the availability of DApp expertise and sufficient compute power are project-specific constraints, some frameworks simplify the definition and execution of ETL pipelines. The following are two such examples, created in part by the authors of this chapter. The *Ethereum Logging Framework*¹² (ELF) [15, 16] is an open-source framework for context-specific data extraction from Ethereum networks. It was later generalized into the *Blockchain Logging Framework*¹³ (BLF) [6], which can currently (as of 2022) connect to Ethereum and Hyperledger Fabric, and is extensible for other blockchains.

At heart, ELF and BLF are language processing systems that enable users to define *manifests* which are parsed, compiled, and executed by the frameworks. Both frameworks are specifically designed to ease the definition and execution of ETL pipelines for blockchain analytics by providing a range of core capabilities:

1. *Declarative specification*: The manifests hide low-level, implementation-specific functionality such as byte decoding, the composition of API calls, and connection details allowing analysts to focus on defining the operations needed to prepare the data for analysis.
2. *Technological independence*: ELF and especially BLF can extract data from different blockchain networks and technology.
3. *Selective data extraction*: The frameworks only extract relevant raw data, limiting the data volume that must be handled and stored, and thus the required compute power during extraction and downstream processing.
4. *Extensible transformation operators*: The frameworks offer basic data transformation operators and can be extended with custom operators.
5. *Multiple output formats*: Analysts can export data into textual logs, CSV, and XES, enabling the use of a wide range of analytical tools.

¹¹ <https://www.tf-pm.org/resources/xes-standard/for-vendors/tool-support>, accessed: 14/10/2025.

¹² <https://github.com/ChrisKlinkmueller/Ethereum-Logging-Framework>, accessed: 14/10/2025.

¹³ <https://github.com/TU-ADSP/Blockchain-Logging-Framework>, accessed: 14/10/2025.

Table 1 Excerpt of the CryptoKitties event log with the attributes *case ID*, *activity type*, *timestamp*, *variant*, and *block number*

Case ID	Activity type	Timestamp	Block number
...	...	...	...
640	Is born	19-11-2020 5:18:51	11286207
640	Is transferred	19-11-2020 5:18:51	11286207
641	Is transferred	19-11-2020 5:28:38	11286252
641	Is transferred	19-11-2020 5:52:08	11286361
642	Is transferred	19-11-2020 5:37:42	11286301
642	Cancel sale auction	19-11-2020 5:37:42	11286301
642	Is transferred	19-11-2020 5:54:21	11286371
642	Put up for sale auction	19-11-2020 5:54:21	11286371
...	...	...	...

6. *Batch and streaming export*: Analysts can choose to extract data in *batch* mode, i.e., to export all data at once, or in *streaming* mode, i.e., to continuously export data whenever a new block is appended to the blockchain.
7. *Contract dependency updates*: DApps often consist of multiple smart contracts that might be extended or exchanged at runtime to modify or update DApp functionality. Hence, data might need to be extracted from a varying set of smart contracts which might not be necessarily known when specifying the manifest. In these situations, the frameworks enable analysts to dynamically monitor and update the smart contracts from which data must be extracted.

Both frameworks have been applied to analyze various DApps, e.g., [6, 12, 13, 16]. Moreover, they were used to establish an open and publicly accessible collection of event logs¹⁴ [5] that aims at helping researchers and practitioners to understand the application domain and enables future research on process mining for blockchain analytics. This collection currently comprises event logs for four different DApps. For each dataset, it includes the ELF manifest, the XES event log, links to the DApp source code and website, a description of the XES log content, and preliminary analysis results. One of the DApps from this collection is CryptoKitties³, for which the event log was created by adopting the mapping logic from Sect. 2.1. An excerpt of the CryptoKitties event log is shown in Table 1.

3 Process Mining Techniques

Process mining combines ingredients from *process science* and *data science* and can be defined as follows: *Process mining aims to improve operational processes*

¹⁴ <https://ingo-weber.github.io/dapp-data/>, accessed: 14/10/2025.

through the systematic use of event data [2]. Using a combination of event data and process models, process mining techniques provide insights, identify bottlenecks and deviations, anticipate and diagnose performance and compliance problems, recommend countermeasures, and support the automation of repetitive work. Approaches to analyze and improve processes typically rely on *process models* that describe the activities in the process and their order. Traditionally, process models were created manually, often unable to capture the full complexity of the process as it was actually executed in real life. Process models can be created using process mining based on actual execution data to represent the as-is process execution, rather than an idealized, manually modeled version of the process.

According to Gartner, there are over 40 process mining vendors, e.g., Celonis, Signavio (now part of SAP), ProcessGold (now part of UiPath), Fluxicon, Minit, LanaLabs (now part of Appian), MyInvenio (now part of IBM), QPR, Apromore, and Everflow [14]. There are also several open-source tools such as ProM, PM4Py, and bupaR. Most of these tools can analyze the XES data described before. An up-to-date overview of tools can be found at www.processmining.org¹.

In this section, we provide an overview of typical process mining techniques. Note that while these techniques are generic, they can be applied to event logs extracted from blockchain networks using the approach presented in Sect. 2. Blockchain-specific applications of process mining are discussed in Sects. 4 and 5.

3.1 Process Discovery

Process discovery describes techniques used to generate process models from data stored in an event log. One perspective on a process is the control-flow perspective (i.e., the execution order of activities), on which this introduction focuses. In this chapter, we introduce two process discovery techniques: mining directly follows graphs (DFGs) and inductive mining (IM). Other well-known approaches are the α -algorithm (one of the first mining algorithms) and heuristic, fuzzy, and genetic mining.

DFGs are a popular, intuitive way for representing processes and captivate users, thanks to their ability to simplify complex control flows. DFGs can be generated with relative ease. In a first step, artificial start and end events are generated. To produce a DFG of the entire event log, a node is created for each activity in the event log. Individual nodes are then connected with a directed edge if in the same trace one activity is directly followed by another (or itself). Additionally, the artificial start event connects to each trace's first activity and each trace's last activity connects to the artificial end event. When repeated for all traces, that results in a visual representation of the control flow of the process. Optionally, filters can be applied to only show variants, edges, or activities that meet certain criteria (e.g., frequency counts) to reduce the complexity of the DFG. Due to their simplifying character, however, DFGs have received criticism and should be used with care [22]. A DFG for a section of the CryptoKitties event log is shown in Fig. 3. The nodes and edges

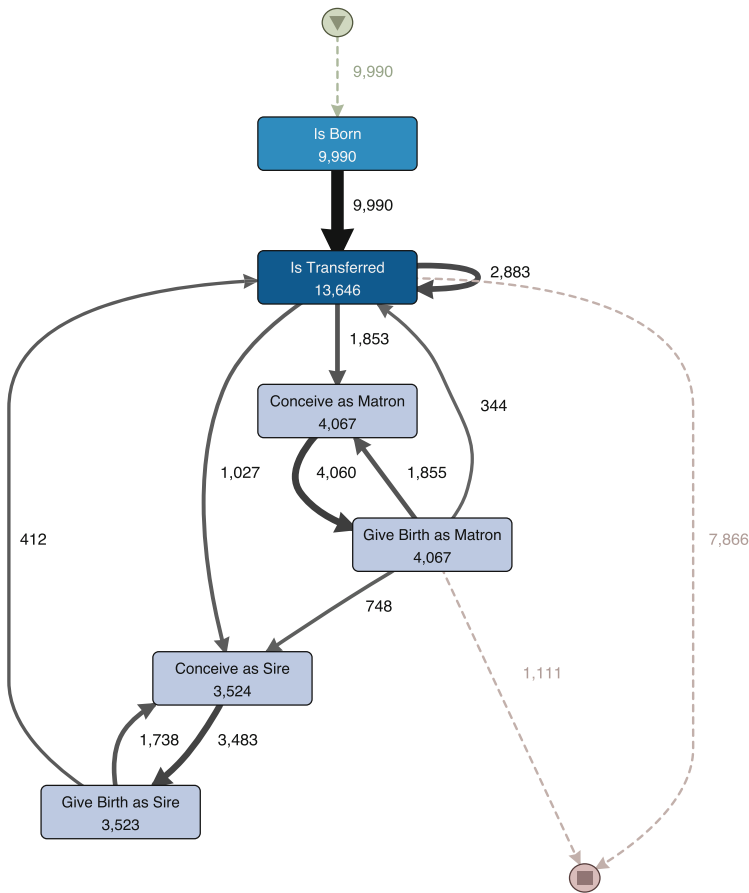


Fig. 3 Directly follows graph (DFG) for a section of the CryptoKitties event log (blocks 11280000 to 12243999) filtered for traces with “Born,” the six displayed events, and a 44.2% path filter

of the DFG in the figure are additionally annotated with the frequency of their occurrence. The artificial start event (green) and end event (red) are clearly visible.

The current state-of-the-art family of process discovery algorithms is called *inductive mining*. A vital feature of inductive mining is its guarantee to output a formally sound model, meaning guaranteed avoidance of deadlocks and other formal shortcomings in the process model. Inductive mining discovers a process model from an event log iteratively: Starting from the logic of a DFG, the miner splits the DFG (and the underlying event log into sublogs) by four criteria, *exclusive cut* (disjoining sections that appear mutually exclusively), *sequence cuts* (disjoining sections that follow each other), *parallel cuts* (disjoining sections that happen in exchangeable order), and *redo-loops cuts* (disjoining sections that contain repeated loops). These splitting criteria are repeatedly applied until the sublogs contain only one activity each. The relations of the activities are documented in accordance with

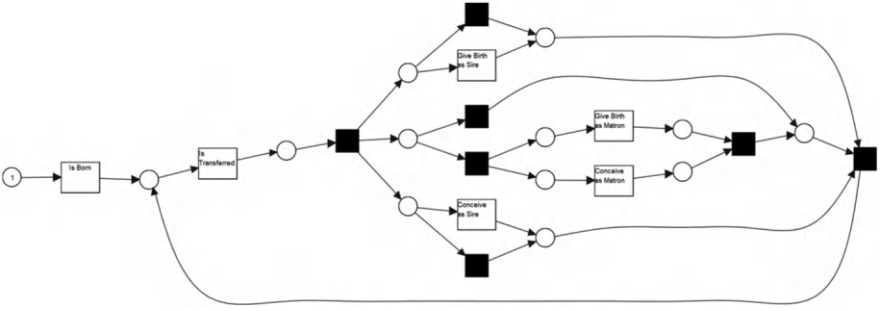


Fig. 4 Petri net, discovered with the inductive miner for a section of the CryptoKitties event log (blocks 11280000 to 12243999) filtered for traces with “Born” as the start activity and a 20% noise threshold

the cutting criteria in a structure called a *process tree* [17]. A process tree can easily be converted into other process notations, such as a Petri net, passing on the formal guarantees of the inductive mining algorithm. Figure 4 shows a Petri net that was discovered from the CryptoKitties log with an inductive miner. Compared to the DFG in Fig. 3, the Petri net uncovers actual parallel arrangements of the conceiving and birthing activities with no particular ordering.

3.2 Conformance Checking

Conformance checking techniques are used to compare process behavior represented in an existing process model and the behavior documented in an event log. The conformance checking aims to find similarities and deviations between the model and the log and quantify the process model’s *fitness* (a process model quality measure for the level of correspondence with the log) to the event log. The requisite process model might have been discovered from a section of the log itself, e.g., to compare the behavior of two user groups, or the process model might have been created manually, representing expected (normative) behavior. For conformance checking, activities in the event log are mapped to activities in the process model (usually by name). The most frequently applied conformance checking algorithms are *token-replay* and *alignment*, both described below.

Token-replay is a conformance checking technique that typically takes Petri nets as process model input. The Petri net notation defines the process behavior through rules around tokens that are transferred via transitions and places in the Petri net (see Fig. 4, circles are places, squares are transitions). In token-replay, quite literally, a token game is played, in which for each trace in the event log the token transfer through the Petri net is replayed or simulated. If a token transfer of a trace goes smoothly through the model, it has a fitness of 1 (i.e., fitting). If a token transfer of a trace does not go smoothly through the model, missing and

remaining tokens are documented and included in the fitness calculation that ends up below 1 (i.e., non-fitting) [4]. Thus, token-replay offers a case-specific conformance measure and helps identify problematic areas in the process model or problematic traces. However, this technique is Petri-net-specific and unable to compute optimal paths through the model for non-fitting cases, prohibiting complex performance diagnostics.

Alignment-based conformance checking techniques are the current state of the art. Fundamentally, alignments overlay “moves” in the traces of an event log and “moves” in the process model pair-wise. One “move” corresponds to an activity that is observed in a trace or depicted in the model. In an ideal scenario, all moves in one trace can be reciprocated with moves in the model (only synchronous moves). There may, however, be a deviation in moves. For example, if an activity is observed in a trace that is not part of the model (move in log only, no move in the model). Another example is an activity that cannot be observed in the trace that should be executed according to the model (move in model only, no move in log). These misalignments are documented for the trace and the process model. The goal of alignment algorithms is to find optimal alignments, i.e., with a minimal number of misalignments between traces and model. The fitness of the log and the model can be computed based on a cost function. In contrast to token-replay, alignments do not require a particular process model notation as input. Alignments also work for complex process models and can identify paths of non-fitting traces through the process model (with deviations), which allows for more complex analysis. However, the optimization in alignment algorithms makes them computationally intensive [2, 8].

3.3 *Enhancement*

Enhancement relays information about the actual process behavior back to the process model to *repair* or *extend* it. *Repair* of a process model is its adaptation to reflect actual process behavior better. For example, suppose the process model shows a consecutive order of activities that in the event log is documented in any order. In that case, the process model may be altered to reflect the actual process behavior. *Extension* of the process model uses data documented in the event log that exceeds the control-flow perspective. Possible extensions of a process model are bottleneck and performance analysis (if timestamps are available), social network analysis (if executing resources or user data are available), and decision mining (to attach data-based rules to choices in the model) [2].

3.4 *Additional Techniques*

An overview of the entirety of the events in the event log can be captured in a *dotted chart*. The dotted chart depicts time on one axis (usually horizontally on the

abscissa) and traces on the other (usually vertically on the ordinate). The resulting coordinate system is filled with dots, each representing an activity that can be assigned to a time and trace. The event dots can be filtered or colored by attributes (e.g., activity name). Additionally, events or traces can be sorted temporally (e.g., date time, trace duration, and timestamp of the start event). Dotted charts thus offer different perspectives on the event data with relative ease and help identify patterns in the event execution from a high-level perspective [21].

The execution of events over time is furthermore subject to fluctuation, meaning that at times, activities or entire branches of a process model might be executed more or less frequently, which will be documented accordingly in the event log. In process mining, as in related fields, this phenomenon is dubbed *drift*. Drift can be caused (among other reasons) by seasonal digressions, changes in user preferences, or policy adjustments, which can result in *sudden*, *gradual*, *recurring*, and *incremental* drifts or their superposition. Various techniques for detecting drift have been proposed, including drift point detection [7], visualizations of drift behavior and characterization of drift [26], and cause-effect analysis in drift [3].

There are obvious connections to neighboring disciplines such as machine learning and discrete-event simulation. Process discovery and conformance checking can be used to extract features from processes. For example, features such as remaining flow time and the number of deviations can be extracted for each case. Using mainstream machine learning techniques, a target feature can be related to descriptive features. This allows for predictions and explanations, e.g., cases get delayed when an activity is skipped. Such insights can be used to transform process models into simulation models that mimic the stochastic behavior seen in the event log. Using such models, it is possible to ask “What if?” questions.

4 Usage Scenarios

Analysts are equipped to conduct data-driven process analysis in the blockchain domain by combining the approach for event log data extraction from blockchains and the process mining techniques introduced in Sects. 2 and 3. In this section, we present usage scenarios in terms of the benefits and value arising from process analysis. Among others, we describe how shedding light on the actual execution of a process can help to understand technical design implications.

4.1 Current Application Areas

Understanding User Behavior An application, may it be implemented on a blockchain or other technology, typically exists to fulfill objectives for users and to serve certain business needs. Learning about processes that users follow to reach their objectives and learning about user interactions with an application

open opportunities for informed actions by end users (e.g., taking advantage of certain states of a smart contract) and application providers or owners (e.g., process improvements to increase business success). Analysts can discover a process model that represents a cumulative picture of the user journey using *process discovery techniques*. Figures 3 and 4 show process models as a DFG and a Petri net that resulted from applying process discovery techniques to the CryptoKitties event log. For a deeper understanding of the process execution, filters can be applied on the diagrams to find clues explaining decision points. In Fig. 3, it would be possible to drill down on cases with frequent “Is Transferred” activities to find out about users’ proclivities to transfer some CryptoKitties more frequently than others and learn about how the application’s incentives (e.g., CryptoKitty pricing) influence user behavior. Conformance checking can help determine behavioral differences between successful and unsuccessful users by comparing their respective event logs and process models. Using enhancement techniques, throughput times can be calculated for different subprocesses or user groups, also depending on external factors such as Ether price in the case of CryptoKitties.

DApp Implementation Validation Based on observing user behavior, the design of the application can be scrutinized. Analysts can evaluate if users follow intended control flows, e.g., by creating a normative process model from a DApp’s white paper specification and comparing it with the DApp’s execution data in the event log using conformance checking techniques. With those results, developers can improve the implementation to match intended user paths, and external stakeholders can validate the correctness of the smart contract implementation before deciding to use the DApp. In CryptoKitties, an event log holds information about whether CryptoKitties can indeed be transferred for Ether after breeding them, as the marketing material stated, or not. That may be critical information for users to join the DApp. Additionally, the impact of the incentive system on DApp usage can be assessed. Here, process discovery can help to reveal incentive deadlocks (i.e., no process participant has incentives to continue the process) or unintended incentives (i.e., incentives to behavior that is not in line with the original design idea).

Security Analysis Users of DApps rely on the blockchain as a neutral execution platform and on smart contracts as the implementation of the execution logic of processes baked in code. Particularly when smart contracts are used to administer blockchain cryptocurrency assets, vulnerabilities in the source code of a smart contract may result in immediate or indirect financial losses for users. Analyzing smart contract execution data with process mining techniques can help to uncover loopholes in the code. Take, for instance, process conformance checking: Based on a control-flow representation of the smart contract’s execution logic as a normative process model (e.g., flow chart), and the event log, conformance checking can uncover exploits once they occur. In code reviews, conformance-based information on the exploit can then be translated into prohibitive actions and patches. Candidates for security-sensitive deviations from the intended design that can be revealed with conformance checking include control-flow anomalies (i.e., activities occur or repeat unexpectedly) and time-based discrepancies (i.e., activities occur after

unexpected time intervals) [24]. Like DApp implementation validation, security analysis can be done as part of a development team's initiative for higher software security, but also by external parties that check if the process behavior documented on the blockchain reflects the behavior communicated by a development team, e.g., in white papers or in marketing material.

DApp Monitoring Monitoring is a central concept in business process management and is generally concerned with the analysis of event log data offline (process monitoring with historical data) or online (process monitoring with streamed data from active process instances) [10, p.414]. The blockchain data extraction tools ELF and BLF (described in Sect. 2) offer both, an option to extract data in a fixed block range for offline monitoring and a streaming mode for feeding event data of blocks that have recently been added to the chain to online monitoring solutions. On an operational level, monitoring dashboards help track the performance of active process instances. The number of observed active cases and throughput time can indicate critical states of an application for which alarms can be set. Monitoring of event sequences enables real-time process discovery and conformance checking. For example, to detect exploited vulnerabilities and to counteract them swiftly or to find out about starting process drifts and take preparatory steps to face them (e.g., increasing demand and price for a certain cross-breed of two CryptoKitties in your wallet). Also, instantaneous hand-over-of-work analysis is possible, e.g., to detect resources with critical behavior that should be isolated. Similarly, it is possible to identify a resource that is currently performing an activity, to influence process execution.

4.2 *Limitations and Challenges*

Using blockchain technology as a process execution platform is subject to limitations and poses challenges to process mining that need to be addressed, at times on a case-by-case basis. Limitations and challenges include the following:

1. An activity in the event log is by default assigned the timestamp of the block in which the corresponding event was written. That comes with three issues:
 - a. Information about the actual time of the execution of an activity remains unclear—there will be a gap between transaction creation/announcement and transaction inclusion in a block [23]. Depending on the application context, the time of the actual activity execution might be before transaction creation, transaction inclusion, or when the transaction is regarded as final (e.g., for an on-chain asset transfer, like the sale of a kitty). Although the time differences are typically measured in minutes, there might be deviations, e.g., transactions with a very low gas fee may not be included in blocks for longer time stretches without this time gap being represented in the event log by default.

- b. The block timestamp is set at different times, depending on the specific blockchain technology and the concrete configuration. For instance, Ethereum block timestamps mark when the block was assembled; in proof-of-work (PoW) configuration, the PoW calculation takes place afterward, taking 14 seconds as a median value (with substantial variation) on the public Ethereum blockchain—in other words, the timestamp is roughly 14 seconds before anyone, but the miner sees the block. A separate, private Ethereum blockchain might use proof of authority (PoA), where no complex computation is performed and the timestamp corresponds approximately to the time when the block was announced to the network.
 - c. The finality of a created block might, depending on the consensus mechanism chosen, be uncertain right after its creation. Temporary forks in the chain may result in discarded blocks. Thus, to safely assume the long-term integration of a transaction (i.e., an activity of a process) in the chain, a delay between block generation and data extraction from the block may be reasonable, e.g., for monitoring solutions.
2. Users of public blockchains are typically identified through account addresses. Accounts are pseudonyms hiding the personal identity of the users, and users can have multiple (walled) accounts so that user IDs documented in log entries are hard to link to individual users or persons. For process mining techniques such as social networks and hand-over-of-work analysis, transparency of the resources involved in the process helps with getting meaningful results. The usage of one-to-many pseudonyms on a blockchain reduces the reliability of such analysis because individuals cannot be uniquely identified with certainty.
3. Although the data extraction techniques introduced in Sect. 2 are extensive, they are not yet comprehensive. Events can only be queried, if they are emitted as specified in the smart contract code. Hence, smart contracts have to be designed to produce data for later analysis. However, this makes them expensive due to blockchain incentive mechanisms like gas cost. Additionally, higher transparency could be achieved with analyzing transaction execution in depth. Albeit resulting in higher complexity, doing so would allow extracting rich information from smart contract invocations, in addition to explicit events emitted from smart contracts.

5 A Case Study of Augur

Translating the usage scenarios described in Sect. 4 into an analysis contributes to increasing transparency of blockchain applications, as demonstrated in this section. This section presents a case study of the popular Ethereum DApp *Augur*²—a prediction and betting marketplace—to illustrate the application of process mining for blockchain analytics. The detailed analysis can be found in [12, 13].

5.1 Overview of Augur

Augur is a prediction and betting marketplace, where users can create bets (e.g., “Will Roger Federer win the 2021 Australian Open?”), and others can bet on the outcomes. Augur is implemented as a set of smart contracts on the public Ethereum blockchain; hence, all data are timestamped and transparent. Augur currently runs two versions: Augur v1.0 (launched on 2018-07-09) and Augur v2.0 (announced in April 2020 and launched on 2020-07-283). Once Augur v2.0 was deployed, most users migrated to it. However, some prediction markets are long-running and still active on Augur v1.0.

The Augur market process comprises four stages: market creation, trading, reporting, and settlement. *Market creation*: A market creator sets up a market for a future event, called “market event.” *Trading*: Traders place bets on the outcome of the market by buying shares for that outcome. *Reporting*: A designated reporter reports the outcome of the market event. If the designated reporter fails to report the outcome within three days of the market event, reporting is open to the public. Once an initial report is submitted, disputants can challenge the reported outcome by crowd-sourcing a dispute bond. If the dispute bond crosses a threshold, the crowd-funded outcome becomes the new tentative outcome. Otherwise, the tentative outcome becomes the final outcome. The market finalizes if a tentative market outcome is not successfully disputed within seven days. Augur also offers a fork event as the ultimate dispute resolution mechanism. It creates parallel instantiations of Augur based on each possible outcome of the market to which users can migrate. Forking is considered very disruptive and has not been triggered yet. *Settlement*: Once finalized, traders resolve their positions. Market creators also receive a market creation fee, and designated reporters receive a fee if their report becomes the final outcome. If the dispute is successful, disputants get rewarded once the market is finalized. Otherwise, they are rewarded after the unsuccessful dispute round. Augur smart contracts may emit 35 different types of events within these four stages.

We chose Augur for the analysis for several reasons, including the data availability, application design, and subsidiary information. Augur v1.0 was among the most popular Ethereum DApps at times, providing a substantial volume of data to analyze. Augur is designed such that a central logging contract emits detailed events. This allowed insights into user behavior and simplified data extraction. Subsidiary information on Augur is widely available, e.g., the Augur white paper [19] served as the basis for conformance checking.

5.2 Data Extraction and Transformation

Given that most markets run for months, and some even for years, a long observation time frame is crucial for a meaningful analysis. Thus we focused on Augur v1.0 and gathered data from its launch until its use was no longer recommended in July 2020.

Because Augur’s log entries are comprehensive, we focused solely on extracting information from log entries for the analysis. We wrote an ELF manifest to extract all events from the first execution of Augur v1.0 on 2018-07-09. We started the data extraction using ELF on 2020-11-12; hence, the last extracted event was on 2020-11-10. However, there was a noticeable decline in user interest once Augur v2.0 was announced in April 2020. To account for this, we removed 162 cases that were either created after Augur v2.0 was announced on 2020-04-02 or that were not finalized before its launch. This resulted in a total of 2,735 cases and 22,772 events. Data were saved as an event log in the XES format, where each log entry was represented as an XES event. Both the ELF manifest and the event log are publicly available.¹⁵

5.3 Analysis and Insights

Using the ProM process mining platform, we filtered the data and applied various process mining techniques to analyze Augur from control flow, conformance, and performance perspectives. Events are grouped using the case identifier (each case refers to a prediction market) and sorted using the timestamps. These traces can be grouped into variants by focusing only on the sequence of activity types. The Augur event log contains 414 variants, of which 35 variants have at least 10 corresponding cases and combined describe 2,203 cases. This follows a common pattern, in which event logs have a Pareto distribution, i.e., 80% of the cases are described by a minority of the variants (in Augur’s case less than 8.5% of the variants).

The dotted chart in Fig. 5 shows that many markets were created in the first months after the deployment of the Augur smart contract (July/August 2018). That period was followed by a steady flow of new cases until May 2019, when the arrival



Fig. 5 Dotted chart showing all 22,772 events. The vertical patterns indicate batching

¹⁵ <https://ingo-weber.github.io/dapp-data/augur.html>, accessed: 14/10/2025.

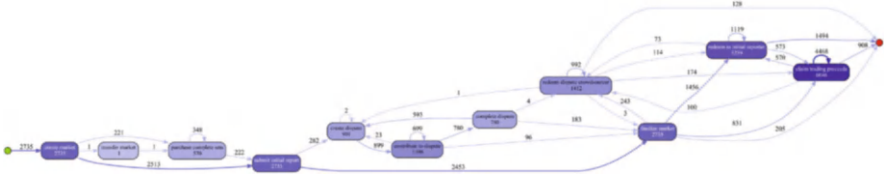


Fig. 6 Directly follows graph (DFG) for the whole event log

rate started to decrease. The vertical patterns indicate batching, i.e., shorter periods where the same activity occurs for many cases. Some of these batching patterns are highlighted in the figure. For example, on 2020-02-11, activity *claim trading proceeds* was executed 63 times for 53 cases within less than 60 minutes. Another example is the burst of the activities *claim trading proceeds* and *redeem as initial reporter* on 2019-07-07. These occur 148 and 98 times within a three-hour window, respectively. There are also horizontal patterns indicating a sequence of events for the same case in a short period. For example, for the “Ethereum Price at end of March 2019” market, we witnessed *redeem dispute crowdsourcer* and *claim trading proceeds* activities 4 and 132 times, respectively.

Figure 6 shows the discovered directly follows graph (DFG) for the Augur event log without filtering. The visualization as a process model shows the most common trajectories of markets (cases) through the process and represents the structure of the process as described in Sect. 5.1. Also, the DFG sheds light onto less frequent (and arguably more interesting) behavior such as the order of activities in the dispute phase.

Next, we focused on the conformance checking, for which we created a normative process model based on Augur’s white paper. We enriched it with information gained from discovery and conformance checking, as the Augur white paper did not cover all events included in the smart contracts (e.g., *DisputeCrowdsourcer-Completed* or *ratingProceedsClaimed*). Also, *completeDispute* only happens if a sufficient amount of stakes is contributed to a dispute; this information is not contained in the white paper. Therefore, multiple iterations were needed to create the normative process model. We then compared the reference model derived as a process tree and Petri net with the discovered process models. While a visual comparison showed that the reference model is close to the discovered models, we noticed differences. For example, a couple of activities were missing, while several others were reordered in the discovered process model [12, 13].

We then conducted a more detailed conformance analysis using alignment-based conformance checking. We noted that 224 cases have at least one deviation. 21,647 events in the log fit the model, and 1,125 events in the log did not fit the model. However, there are no moves on the model (i.e., missing events). 223 of the 224 deviating cases and 1,119 of the 1,125 log-only moves have multiple *redeem as initial reporter* events. We inspected the underlying blockchain transactions for a random sample of 20 of these occurrences and observed the following pattern in all 20 instances: (i) the first *redeem as initial reporter* event resulted in a payout, the

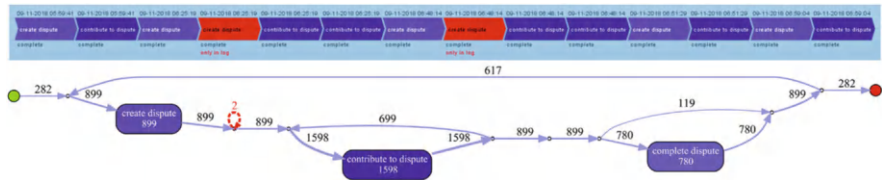


Fig. 7 Conformance checking results for the dispute phase including the activities *create dispute*, *contribute to dispute*, and *complete dispute*. Only one case is non-fitting

second did not; (ii) the first and second transactions came from the same account in all 20 pairs; and (iii) the pairs were close together (between 0 and 47 blocks, with a majority under 10 blocks). While the first event resulted in a payout, the second did not. Therefore, the logging on Augur could be made more precise here to differentiate successful, legitimate transactions from others. Given that these attempts were unsuccessful and the initial reporter had to pay fees, and the same reporter accounts repeatedly showed the same behavior, it is plausible that the reporter used an automated tool with a retry time to submit transactions. However, the tool may not have correctly implemented retry [12, 13].

Figure 7 shows conformance checking results for the dispute subprocess. In the upper part of the figure, we can see a single deviating case, where there are two instances of two subsequent occurrences of *create dispute* event without any other activities in between. This is not possible according to the reference model. This was a bug in Augur v1.0—albeit a noncritical one—as we established in discussion with Augur’s chief architect (see below). Note that the Augur smart contracts cannot be updated; in other words, there is no way to fix a bug for the running system.

Next, we focus on the performance perspective. Events have timestamps; therefore, enriching process models with timing information (e.g., waiting and service times) is trivial. However, for marketplaces like Augur, standard measurements like waiting times are less relevant because the duration is related to the nature of a given bet. In particular, users create markets for future events, and markets vary in terms of how far into the future the event is expected to take place. We inspected the top 100 completed traces concerning the waiting time before submitting the initial report to analyze this behavior. All these markets were created in 2018 but referred to events in 2019 and 2020. In addition to these analyses, we zoomed into the dispute phase and only considered the activities *create dispute*, *contribute to dispute*, and *complete dispute*. The mean duration of this phase is 15.4 days, the median duration is 7.1 days, and the longest duration is 111 days. This illustrates that time also plays a role in the analysis of betting markets.

The usefulness of the insights was confirmed by anecdotal evidence of Augur’s chief architect, particularly in terms of understanding user behavior and code validation. From his perspective, using process mining for blockchain analytics, and Augur in particular, provides value in three ways. First, it helps verify the design mechanisms and check for unintended behavior and bugs in the immutable code; immutability means that the code cannot be changed once deployed, and poses a

challenge from both a BPM and a software engineering perspective. Second, process mining provides a clear view of how an application is used, which is also helpful for designing updated versions of an application. Third, it has great potential for technical and economic security analysis; e.g., an auditor could create a model and conformance-check it against actual user behavior.

More details on methods and findings are available in [12, 13].

6 Summary

Many blockchain applications support cross-organizational processes, whether the process perspective is implicit or explicit. In this chapter, we outlined how process mining can be applied to analyze blockchain applications through a process lens. We discussed how analysts can extract event data from blockchain networks based on a process mining methodology. We then presented an overview of typical process mining techniques. Finally, we discussed specific use cases for process mining in the context of blockchain analytics and illustrated the application of process mining based on a real-world case study.

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Blockchain Meets Network Analytics: A Tale of Heuristics, Location, and Fraud Detection



Simon Trimborn and Le Yu

Abstract The information provided by blockchains of cryptocurrencies is immense and diverse. Blockchain information is represented by networks, the transaction network and the user network, approximated by the entity network. We review heuristics for constructing the entity network out of the transaction network and discuss how they have been used, improved, and developed over time. We introduce network analytics applied to blockchain data, which supports finance and economics research to look into location identification, fraud detection, and price investigations of cryptocurrencies, among other topics. We inspect how they make use of network analytics, often tailored to the specific properties of blockchains. By this comprehensive overview, we intend to aid research on the use of blockchain information to understand user behaviours and the corresponding price behaviours of cryptocurrencies.

JEL Classification: G00, C00

1 Introduction

Fourteen years ago, there was only one cryptocurrency. Now there are 13000. Nine years ago, it was soaked with suspicious reputation of illegality. Now, cryptocurrencies have huge market capitalisation, active trading communities, and indubitable interest from the mainstream.

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It has all started from the famed Bitcoin. Proposed by Satoshi Nakamoto in 2008 to get rid of trusted third parties in payments, this immensely popular and influential electronic decentralised currency depends on a peer-to-peer network with a proof-of-work system to verify the transactions [25]. Altcoins, cryptocurrencies other than Bitcoin, soon followed with Bitcoin as their technological prototype.

With the increase in the price for Bitcoin and other altcoins, people began to add them into their portfolios. Studying the financial properties of cryptocurrencies became important. How would cryptocurrencies perform when they are added to a portfolio consisting of other assets [28]? What are their diversification properties and financial factors [20]? How to identify a cryptocurrency's idiosyncratic risk [17]? Studies answering these questions tend to focus on the *time-series properties* of coins.

But there is another way of looking at things. Each cryptocurrency is powered by a blockchain. As the underlying platform and engine of a cryptocurrency, blockchain characteristics cannot be irrelevant to the financial characteristics of the cryptocurrency and to its overall success. Are user behaviours on the blockchain relevant to the price of a coin [11]? Will network measures and computing power of a blockchain impact the coin's expected returns [5]? Can we construct pricing models out of blockchain network characteristics [9, 20]? To answer these questions, we need to turn to the *network analytics* of blockchains and cryptocurrencies.

Network analytics of the blockchain are powerful tools to understand the behaviour of users in the ecosystem defined by any particular blockchain. Understanding user behaviour goes beyond its financial implication, since there is a lot of talk about its application outside the realm of cryptocurrencies, such as tourism [26], healthcare [22], and supply chain [31]. If such applications materialise, blockchain technology would have profound social and economic impact. But why would one want to use blockchain technology instead of established technologies? Catalini and Gans [7] provided a detailed analysis of the economics of the blockchain. In short, the magic of blockchains can be summarised by two points: They reduce the *cost of verification*, and they reduce the *cost of networking*.

Cost of verification arises inevitably in an economy of any respectable scale. Intermediaries increase the risk of "moral hazard", namely cheating, counterfeiting, and other behaviours that strain trust, in the words of [7], and information asymmetry in general in any real-world economy, in which most people do not have preestablished relationships. Blockchain technology helps in reducing the cost of verification by striking at its roots: Because they involve distributed ledgers, there is no intermediary involved. If an economic activity can be done completely online, the verification cost may be almost costless.

Cost of networking arises in bootstrapping, running, and scaling an economic network. The relationship between the cost of verification and the cost of networking is close but nuanced. Over time two types of blockchains arose: permissioned and permissionless. Permissioned blockchains can have a significantly reduced verification cost, but they do not necessarily significantly reduce the cost of networking: Unlike Bitcoin, they are not entirely public. Rather like traditional databases, they need an administrator role to govern the approved users of the

blockchain. Trust needs to be put on the administrator. By contrast, in permissionless networks, or public blockchains, trust is an unnecessary luxury. Without trusted intermediaries or privileged nodes of any kind, cost of launching and running a network is significantly reduced.

These cost reduction advantages of blockchains have the potential to boost applications in many areas. This further drives the need to understand how users interact on the blockchain [23], how illegal activity can be identified [15], how users obfuscate their behaviour [30], or to identify geographical user clustering [21, 32]. These topics and the resulting research questions can be tackled with *network analytics*.

This chapter introduces network analytics applied to blockchain data, which supports finance and economics research to look into location identification, fraud detection, and price investigations of cryptocurrencies, among other topics. We inspect how they make use of network analytics, often tailored to the specific properties of blockchains.

In Sect. 2, we distinguish between different kinds of networks relevant in the discussion of a blockchain of a coin. Section 3 then discusses the relationships between different networks and the heuristics to transform one into another. After knowing what the networks are, in Sect. 4, we are going to look at how they are used. Section 4.1 tracks down the location of cryptocurrency users: useful for economic insights but so much for anonymity claims. In Sect. 4.2, we filter out irrelevant transactions in cryptocurrency networks and keep only the “economically meaningful ones”, in a sense to be explained [21]. Section 4.3 is about fraudulent behaviours associated with cryptocurrencies: This is how cryptocurrencies got their bad names. Section 4.4 is about financial characteristics of coins, especially their pricing models. Section 5 concludes.

2 Network Types in the Blockchain

First, we discuss and introduce the different kinds of networks which appear or can be constructed from the blockchain. We will not introduce and explain Bitcoin and the blockchain before delving into the details of this study, since there is already a vast literature covering these topics. We refer you here to the comprehensive introductions into the features of Bitcoin and the blockchain by Böhme et al. [6] and Tschorsch and Scheuermann [33]. Given the Bitcoin scheme, there are three closely related networks we can look at:

- The transaction network
- The user network
- The entity network

A *network*, or a *graph*, is a set of objects in which some pairs are related in some relevant sense. The objects are called *points*, or *vertices* (singular: vertex) if you are a mathematician, or *nodes* if you are a computer scientist, or *sites* if you are a

physicist, or *actors* if you are a sociologist. Related pair of points is connected by *lines*, or *edges and arcs* (maths), or *links* (computer science), *bonds* (physics), or *ties and relations* (sociology).

Graphs are useful abstractions, and they are especially useful here. There are natural correspondences between elements of a graph (vertices and edges) and elements of the networks that we are going to look at, such as transactions and transacting and users and their relationships.

The *transaction network* captures the flow of bitcoins between transactions. This is Bitcoin's original, entirely public transaction graph. Each vertex is a transaction. A directed edge from vertex *a* to vertex *b* represents that an output of *a* is used as an input of *b*. Transaction graphs are also called address graphs.

The *user network* captures the flow of bitcoins between users. Each vertex is a user of Bitcoin identified by a public key. A directed edge from vertex *a* to vertex *b* represents an input-output pair of a transaction. The input's public key belongs to the user of the source, and the output's public key belongs to the user of the target. A user of Bitcoin is also called an *owner* or a *controller*, though we will stick to *user*. Analysis of the user graph elucidates the interactions between different economic players. It can tell us who the users are, where they are situated geographically, and which types of businesses they are involved in.

However, for all intents and purposes, the true user graph is unknowable. The Bitcoin scheme has the reputation of being anonymous. While the degree of anonymity it offers is a topic of hot debate, there is some truth to it. First, intrinsic to the Bitcoin protocol, the true identities of users are not linked with public keys. A single user can control many public keys, and a single public key can be controlled by many users. Second, user habits further obfuscate their identities. It is declared good practice for users to generate a new address for every transaction. People also mix and launder and gamble and steal.

Researchers managed to approximate the user graph well enough to derive insights that we want from it. The *entity network* is the working graph that researchers get as an approximation to the unknowable user network. User network is the real thing: Entity network is an unavoidable compromise. The networks amenable to study, therefore, are the transaction network and the entity network.

3 From Transactions to Entities

With the degree of anonymity offered by the Bitcoin protocol, it is doubtful whether we can ever achieve a full trace of users from their transactions. Yet some trace is better than none. We can achieve a good approximation of the user graph, dubbed the entity graph. There are three major heuristics to contract the transaction graph into the entity graph:

1. The heuristic of multi-inputs
2. The heuristic of change addresses

3. The heuristic of off-network information

Other minor ones abound.

The work “heuristic” is proper here. In contrast to a proof or an algorithm, a heuristic is a special approach to problem solving that is experimental and relies heavily on trial and errors. None of the heuristics that we shall see is foolproof. They are empirical, context dependent, ad hoc even. However, it can be argued that they work reasonably well, and the entity graph they generate offers us important insight.

3.1 Multiple Inputs

Heuristic 1 *For multi-input transactions, treat the addresses of the inputs as the same user.*

Heuristic 1 was first identified by Nakamoto himself in [25]. He realised this when discussing anonymity:

Some linking is still unavoidable with multi-input transactions, which necessarily reveal that their inputs were owned by the same owner. The risk is that if the owner of a key is revealed, linking could reveal other transaction that belonged to the same owner.

Why is it reasonable to link multiple inputs? Because each sender needs to know the private key belonging to each public key used as an input to the transaction. The public key is known publicly to everyone on the blockchain. Users require a private key to prove that a transaction, signed with a public key, belongs to them. Hence, users have an incentive to keep their private keys as safe as possible. Whoever possesses the private key has access to the funds signed with the public key. If a collection of public keys were controlled by multiple entities, then they would need to reveal their private keys to each other. The idea is that this is unlikely, though not impossible.

With hindsight, we now know that this assumption does not hold infallibly. Benignly, web wallets, for example, pool many private keys. They would be mistakenly identified as a single entity. Maliciously, by taking care to make sure that his multiple addresses are never used in a single transaction, a user can deliberately conceal the true connection between his addresses. As a result, as pointed out by Makarov and Schoar [21], the multi-input heuristic provides a lower bound for the actual number of users.

Heuristic 1 of multi-input is implemented by Reid and Harrigan [29]. True to Nakamoto exegesis, their overriding concern was with the breakdown of anonymity. Taking the above cited passage as a hint, they merged any vertices with undirected edges where each edge joins a pair of public keys that are inputs to the same transaction. From the transaction graph, they contracted an entity graph.

They immediately identified crucial differences in features between the transaction and the entity graphs. The transaction graph has neither multi-edges nor loops.

It is also a directed acyclic graph, which means that the output of a transaction can never be an input to the same transaction, either directly or indirectly. On the other hand, the entity graph is much messier. It does contain multi-edges (a user sends multiple transactions to himself), loops (a user sends to an account of his own and then sends back) and directed cycles.

Heuristic 1 using multi-inputs is popular and became the de facto default method for contraction from the transaction graph into the entity graph [19], in their data preprocessing, first contracted the transaction network using the multi-input heuristic before further processing the data with off-network information. In [21], to get their “most complete information about crypto entities that have been used in academic research up to this point”, they scraped cryptocurrency blogs, websites, and the database Bitfury Crystal Blockchain. They then contracted the transactions graph based on the multi-input Heuristic 1 and got 1,032 entities, including 63 mining pools, which further illustrates the fact that this heuristic gives us a lower bound. Other notable studies that used this heuristic include [23, 30] and [4].

3.2 Change Address

Heuristic 2 *The change address is controlled by the same user as the input address.*

The *change address* is an offspring of a peculiar idiom of use of the Bitcoin scheme. I want to buy a bottle of water worth \$1, but I only have \$5 notes with me. I give you a \$5 note and you give me back \$4 of change. Similarly, change addresses are used to give money back to the input entity in a transaction. But unlike cash transactions, in which case you give the money back into my hands, Bitcoin change addresses are generated by the Bitcoin client. These change addresses are not chosen by the input entity and should not be reused by the user. As a result, it is likely that the change address and the input address are controlled by the same entity. If we can identify these change addresses reliably, then by linking the change address to the input address, we can further contract the transaction graph.

This line of thought is pursued by Androuraki et al. [3]. They called the change address the *shadow address*. The main idea is that if a transaction has two outputs, with one public address and one newly generated address (the shadow), then the shadow address belongs to the entity who initiated the transaction. The thought is that it is likely that the input entity is receiving change. However, the assumption they relied on in their study is not maintainable and is in fact routinely violated. In real life, entities issue transactions to different users, much more than two, all the time, for example, in mining pools or bets on gaming sites.

As a result, [23] took the matter more stringently. They imposed four conditions before identifying an address as a change address: (1) this address cannot have appeared before, (2) the transaction cannot be a coin generation, (3) there is no self-change address, and (4) for all outputs in the same transaction, only this address is making its first appearance.

Recall that Heuristic 1 of multi-inputs relies on assumptions that may not be true. If anything, Heuristic 2 of change address is even less robust. The existence and features of the change address are heavily dependent on the present idiom of use and are not inherent in the Bitcoin protocol itself.

Heuristic 2 is also tricky to implement. After [23] ran the analysis, they ended up with a giant super-cluster containing 1.6 million public keys, including those of the popular services Mt. Gox, Instawallet, BitPay, and Silk Road.

As a result, they imposed further conditions. Sometimes the same change address is used twice, and if the second use is with a new address, the new address would be falsely labelled as the change address. They got rid of those problematic transactions to make the heuristic more robust. Heuristic 2 based on change address, as a result, depends on the unique situations within the blockchain.

And Heuristic 2 has several variants. Another notable one was introduced by Athey et al. [4]. They based their heuristic on an observation of human psychology. If I give you change, for example, it is unlikely for me to give you tortuous amounts, or in [4] words, amounts that are “cognitively-difficult”, such as \$4.94275. More likely, I would just generously say: “forget about it, let me give you \$5”. Athey et al. [4] therefore conjectured that if a transaction has two outputs, and if one of them has three more decimal places than the other (3 is a heuristic and is not based on intrinsic reasons), then the output with more decimal places is the change address. They supplemented [23] one-time change address heuristic with their own decimal-place heuristic to contract the Bitcoin transaction graph.

Apart from developing the Heuristic 2 of change address, [23] also actively engaged in Bitcoin transactions with various services. Because they, of course, knew what public keys they themselves have used, the public key on the other end can be explicitly tagged as belonging to the service they are interacting. They also scouted in forums and other places online to look for addresses claimed publicly. These activities cannot be found in the network themselves: They are squarely within the realm of the third heuristic, to which we now turn.

3.3 Off-Network Information

Heuristic 3 *Anonymity can be further broken down by off-network information.*

It is impossible to live without leaving some trace. Although users are not linked to their Bitcoin addresses explicitly, there are always some giveaways. Sometimes this off-network information is enough to link addresses to users. A good contraction does not only rely on features of the transaction network, but on extra information that is circumstantial and not intrinsic to the network itself.

This line of thought is first pursued by Kaminsky [18] in concurrence with [29]. Kaminsky exploited the fact that Bitcoin is an electronic currency requiring Internet connections. His assumption is that the source of a transaction is the first node that informs you of that transaction. Kaminsky proposed to map IP addresses to Bitcoin

public keys by making connection to all public peers in the network simultaneously. Again, this assumption is not infallible, though Kaminsky is reasonably confident about it: “[this is] more or less true, and absolutely over time”.

Reid and Harrigan [29] integrated some other network information to de-anonymise users. They considered a wide variety of sources to glean their off-network data. For example, they scraped the Bitcoin Faucet, where IP addresses are published together with the history of recent giveaways. They also searched in Bitcoin Forums where users attach public keys to their signatures, Twitter streams, and user-generated public directories. They acknowledged the ad hoc-ness of their endeavour and that a larger, more centralised Bitcoin service provider can perform the same analysis with their user information more reliably and on larger scale. But their attempt is an invaluable proof of concept. By integrating such a rich variety of off-network data to the entity network, it is possible to construct the entity network as a very good approximation to the real, unknowable user network.

Apart from discussing the multi-input Heuristic 1 and the change address Heuristic 2, [23] also considered off-network data to aid the contraction. Other than passively spending time in Bitcoin Forums and other online places to find user disclosures, which they served “manual due diligence” and regarded as “less reliable”, they actively transacted with a wide variety of Bitcoin services. Because they know which user they are transacting with, they can tag users by observing the addresses they used. They mined with major mining pools, kept money in wallet services, engaged in bank exchanges and non-bank exchanges, purchased physical and digital goods with vendors, gambled with poker sites (but not dices because they already advertise their public keys), and interacted with mix and laundry services, where some services sent their own coin back and others simply stole their money.

Lischke and Fabian [19] is another example of using both Heuristic 1 of multi-inputs and Heuristic 3 of off-network information. After contracting the transaction network using Heuristic 1, they considered IP addresses and business tags from the initiator of the transactions. To deal with Tor and proxy nodes, they downloaded all current Tor server and the addresses of Tor server exit nodes, which resulted in the finding that around 1% of nodes might be used a Tor network. Their study is therefore also a refinement of Kaminsky’s pioneering talk [18], which did not deal with proxies.

4 Blockchain Analytics in Finance and Economics

Networks on the blockchain in hand, we are ready for the economic and financial implications of the blockchain.

4.1 Location Analysis

Identifying the geolocation of users is one of the most important goals of Bitcoin network contraction. The mainstream method of tracking down users' geolocation is by analysing their IP addresses. This line of analysis has a venerable history. Reid and Harrigan [29] already realised that the Bitcoin Faucet can map users to geolocated IP addresses, and it would be interesting to know where people are using Bitcoin. They went as far as producing a map plot of the users receiving bitcoins from the Faucet. This is done in larger scale by Donet Donet and Perez-Sola [12], which we will discuss later in more detail in the connection with Bitcoin address propagation.

Knowing the whereabouts of users is interesting in itself. There are, however, deeper reasons for performing geolocation analysis. Geographical concentration can bear risks for the functioning of the blockchain system. As [21] pointed out: "geographical concentration increases the risk that a private or a state actor in one part of the world [...] could gain control over the network and inflict large losses on the general public and financial institutions if they are holding bitcoins". Such a situation could arise due to several situations. For example, if the combined computing power of all users of a blockchain network in one town is large enough, they could take together control of the network. Also if a significant number of miners are located in the same area and a power outage appears, the blockchain network would be seriously disrupted due to the inability of these miners to contribute to its operations for the duration of the power outage. Consequently, geographical analysis of blockchain users provides important insights into the risks the network is exposed to.

There are four prominent ways to track the geolocation of a blockchain user:

- Relaying node IP tracking (Sect. 4.1.1)
- Bitcoin address propagation (Sect. 4.1.2)
- Off-network information collection (Sect. 4.1.3)
- Miner geotracking via cashout behaviour (Sect. 4.1.4)

Let us turn to them one by one.

4.1.1 Relaying Node IP Tracking

Nodes in the Bitcoin transaction network are not all created equal. Some of them act as coordination nodes, or "relaying nodes", which transmit the information of newly recorded transactions to all users. The addresses of these relaying nodes are publicly known: One can track their IP addresses by connecting to them. And the relaying node which relays a transaction first should be the one closest to where the transaction originated. This line of thinking gives us a viable way to estimate the origin of a transaction, which is a variant of Heuristic 3 above.

Heuristic 3.1 *Transaction takes place where the node first informs about it.*

Though this Heuristic 3.1 of IP addresses is reasonable, it holds true only without the widespread use of proxies.

Lischke and Fabian [19] performed a comprehensive mapping between IP addresses and geolocations. They extracted IP addresses of Bitcoin users from Blockchain.info and identified 40,329 distinct geolocations. Of course, as all heuristics, the link between transactions and IP addresses is not an infallible one. They reported that around 70% of the transactions can be linked to an IP address. As of 2016, at the time of their writing, based on the geolocations of the IP addresses, they identified the active markets in terms of the number of transactions and the associated value of bitcoins. The US takes an undeniable lead, followed by Germany. With a sharp decrease, France, Russia, and Canada are the next contenders.

Can we go further than throwing darts onto the map? Trimborn et al. [32] looked at the time-dependent impact of Bitcoin transactions in different geolocations. They again followed the above Heuristic 3.1. After grouping their data continent by continent, they further broke each continental group into 10 according to transaction sizes. Not surprisingly, Antarctica is not worth a mention. They therefore ended up with 60 groups, 10 for each of the remaining continents, Africa, Asia, Europe, North America, Oceania (Australia), and South America.

By focusing on transaction data of these 60 groups from 25 Feb 2012 to 17 Jul 2017, a time series of daily log accumulated transactions, they already made interesting observations using just descriptive statistics. In short, European and North American transactions were large and steady. In terms of transaction sizes, Europe and North America took the lead, followed by Asia and Oceania, and then by Africa and South America. In terms of volatility, Europe and North America have steady transaction sizes, whereas the other continents were very volatile, with days of zero transactions on end.

With a time series of Bitcoin transaction records to hand, an obvious question to ask is: “Do Bitcoin transactions in the past influence Bitcoin transactions in the future?” “Of course” is the answer, but how can we study it?

Trimborn et al. [32] constructed a time-dependent model to analyse the 60 groups across time and space. The model considers each group as potentially influential for all the other groups. Via regularising the group informative value for the transaction behaviour on other continents, the group with a steady impact on all the other groups remains. Across time, they analysed the occurrence and vanishing of structural influence in the Bitcoin blockchain over the years. 2012 is particularly active. In 2012, Bitcoin first captured the imagination of the mass. Together with its activity, its price shoot up from \$5.27 to \$770 by 2014. However, as other cryptocurrencies, or altcoins, began to emerge, Bitcoin had its hibernation and did not show network effects from 2013 to 2015. Something out of the ordinary must have happened in 2016. Not only did Bitcoin again turned active, the price again went up.

Across space, their analysis is a pronounced reminder of the true international character of Bitcoin. Asia, as the place where most mining farms located, often has the greatest media attention. Europe and North America, as developed regions and financial centres, are also assumed to have a lot of Bitcoin flows. However,

they showed that South America and Africa are showing network effects of significant magnitude. In fact, apart from 2014, South America is outperforming North America and Europe. Through the mining farms in Asia, people in every corner of the world are contributing to the Bitcoin blockchain.

4.1.2 Bitcoin Address Propagation

Each Bitcoin node aims to maintain at least eight connections. The maximum connection is usually set at 125 (*maxconnections*). A caveat is that, as reported in [24] in a study of Bitcoin's public topology by studying Bitcoin's peer-to-peer link using their technique of AddressProbe, many nodes in fact exceed the number of the supposed maximum connections persistently over 80 times. These high-degree nodes are primarily mining pools and wallets.

Bitcoin nodes, like employees with roll-call attendance responsibilities, announce their own IP address every 24 hours (*addr* message) throughout the network. Nodes can also actively ask for IP addresses using the *getaddr* message. Those nodes receiving the *getaddr* message would duly response. Instead of disclosing everything they know, however, nodes only include 23% addresses they know randomly, and in any case not above 1000 of them. By doing this, the Bitcoin protocol deliberately tries to mitigate the risk of information eclipsing. Information eclipsing can happen spontaneously or maliciously. If a villain dominates the environment of a node, what information gets to the node will be solely at his mercy. He may decide to report the correct information, but he may also decide to double-spend for fun. By disclosing only a small part of the IP addresses a node knows, that node is protecting itself from the threat of the villain which comes with the intention of filling up a node's neighbour with compromised IPs in mind.

The *getaddr()* command gives us another way to find out where Bitcoin users are located in the world. Donet Donet and Perez-Sola [12] extracted more than 800,000 IP addresses by exploiting precisely the *getaddr* messages. Everyday at 9am from 30 November 2013 to 5 January 2014, they issued *getaddr()* command to a set of seeds and then recursively to the nodes connected to those seeds. By thus discovering the neighbours of the initial nodes, and their neighbours, and their neighbours' neighbours, they collected a giant list of IP addresses running a Bitcoin node.¹

From this list, [12] were able to identify the geographical location of Bitcoin nodes. Not surprisingly, the United States and China are where most of the nodes located, followed closely by Germany, Russia, and the United Kingdom. They also calculated the Bitcoin adoption rate, normalising the above with the

¹ It is noteworthy that [18] in his talk already proposed the method of "recursively ask every node about every other node it knows about" using the *getaddr* message and proposed to start from hardcoded seeds. This seems to be a case of independent rediscovery on the part of [12] because they seem to be unaware of this.

number of internet users in each country, and the list of countries appeared quite differently. Now the leading countries are the Netherlands, Norway, Finland, and the Czech Republic. And their summary is a true testimony to the viral popularity of Bitcoin: “there are Bitcoin nodes all over the world, with very low populated areas and underdeveloped countries being almost the only exceptions”. Reminiscent of NASA’s breathtaking photo of the Earth’s city lights, in their geolocation plot, discovered nodes thickly dotted all the continents, with the sole exception of Antarctica.

4.1.3 Off-Network Information Collection

Athey et al. [4] conducted a geolocation analysis on Bitcoin users without relying on their IP addresses. Rather, they used the random forest algorithm, which is a generalisation of the simple decision tree, to identify the regions of transactions.

They got their training data sample from Bitcointalk.org, which is a website filled with user gossip about their geolocations. They scraped it and got 2995 addresses labelled with geolocations. They then grouped all countries into four groups, not following geographical continent demarcation specifically: the Americas, Europe, Asia, and Eastern Latin America. After training with random forest, they used their model to predict the origin and shares of the full data.

About the performance of the model, they reported that: “it may be surprising that the error rate is as low as it is, given the small size of the training dataset and the fact that a large share of users engage in only a few transactions per user”. With a larger or more representative training dataset, organisations and agencies that are interested, by virtue of conducting a similar analysis, have the prospect of making very accurate predictions of the geolocations of users. This is good news for some (e.g. law enforcement) and bad news for others (e.g. criminals or those who otherwise wish to remain anonymous).

Finally, it should be noticed that [4] random forest classification does not seem to be as fine-grained as the approach based on IP addresses. Random forest is only possible to place entities into large geographical bins, but not pinpoint them on the map.

4.1.4 Miner Geotracking via Cashout Behaviour

Makarov and Schoar [21] proposed a heuristic to track down the geolocation of Bitcoin miners in particular. They focus on the exchanges in which miners cash out their rewards. Their heuristic is based on the thought that miners are likely to send their rewards to an exchange in the region in which they are themselves physically located. Their heuristic is therefore:

Heuristic 3.2 *Miners are physically located in the area where they send their rewards to.*

By following this Heuristic 3.2, they distinguished between four categories of exchanges: Chinese, US-Europe, International, and Other. Chinese and US-Europe are just what their names advertised. Others include those transactions that operate in areas other than China, the USA, and Europe. The International category involves trans-jurisdiction exchanges, so tracking down these flows may not give us valuable information about the whereabouts of the miners. As of 2021, they found that Chinese miners carry the most weight in the landscape, responsible for around 70% of all the transactions.

Apart from geolocational analysis, Bitcoin network contraction also has several other applications. Other popular analyses include:

- Industry type: What uses are bitcoins being put by individuals and companies? Athey et al. [4]’s analysis identified 13 major uses, with exchanges being the top use by far. Interestingly, in second place is a cluster of industries labelled by them as “unknown”. Lischke and Fabian [19] did further analysis in this respect. Going above a cross-section characterisation, they identified a changing application pattern of bitcoins in different businesses. Under their analysis, donations, wallets, and gambling are important.
- User type: Who are using bitcoins? Are people short-term users or long-term users? Are people transactors, miners, or merely investors? Athey et al. [4] looked into this question and discovered that long-term, frequent users are in fact only a minority. Rather, most people tend to hoard their bitcoins. Investors are therefore most important in contributing to Bitcoin transaction volume. Their analysis confirmed the earlier findings of [23]: Although people began to spend their bitcoins since 2011, hoarders still exist in legions. Ron and Shamir [30] gave a very early analysis of Bitcoin’s user type. At the early time of their writing, they found that 78% of all bitcoins were left in addresses that only received but never sent out bitcoins.

4.2 *Blockchain Analytics for Price Investigations*

Why is it important to investigate cryptocurrency pricing? Apart from the usual reasons for conducting studies on pricing, cryptocurrencies are special in this respect. Many of them are unbacked by any underlying assets or supporting foundation/company, and its exchange rate is therefore governed by different fundamentals than other assets. As a result, the pricing series of cryptocurrencies attracted much attention.

In the absence of regulation, it is plausible to postulate that Bitcoin prices are governed by “economic fundamentals”, i.e. supply and demand. This is precisely what [4] thought. They built a model which demonstrates that the Bitcoin exchange rate is grounded in supply and demand, and exchange rate rises with usage. Their central theoretical conclusions are:

- There is a unique equilibrium exchange rate in each period determined by supply and demand provided that: (a) no investor is present and (b) all agents adopt Bitcoin eventually.
- If investors buy bitcoins, the effective supply for users decreases and the market equilibrium price increases.
- Bitcoin adoption and exchange rates are influenced by beliefs about Bitcoin.

In short, more intense activity on the blockchain means an increased demand for the use of it.

As they themselves noted, in reality, things are perhaps more complicated than in models. In particular, their model does not take into the account the competition provided by altcoins, or traditional banking itself. Another thing to note is that they did not take into account of state variables of users other than beliefs. Heterogeneous beliefs and speculative bubbles, something that Bitcoin may be especially prone to, are also not considered.

In addition, for each cryptocurrency, its blockchain and the behaviour of users on it impact its price. Following this thought, [27] constructed a model to study the price equilibrium of Bitcoin as a function of its blockchain characteristics, such as the hashrate. A higher hashrate indicates that miners are dedicating more computing power to the network. It is sensible to assume that hashrate and price are related. More computing power dedicated to a cryptocurrency is associated with higher costs, and so the reward needs to be higher to make up for this. However, the reward is fixed and, in the case of Bitcoin and many other cryptocurrencies, halves regularly. As a result, if hashrate increases, the price has to go up. If the price does not go up, miners would just lower their dedicated computing power, namely lower the hashrate. Price and hashrate are in equilibrium.

Financial factors to explain movements of cryptocurrencies on the exchanges are also interesting. A financial factor is a constructed variable which is proven to have an impact on asset prices. The most famous financial factors were discovered by Fama and French [13] which comprise, among others, the book-to-market ratio. This factor describes the difference between the market value of things owned by a company and their value in the accounting books. If the assets of a company worth more than they are in the accounting books, this will influence the price of the company's shares.

What are the financial factors for cryptocurrencies? Bhambhwani et al. [5] constructed such factors for the network size of the blockchain using unique addresses and the hashrate (proxy for computing power). They showed that these two factors are positively associated with the return structure of the cryptocurrencies in their sample.

This empirical analysis of the positive relationship between the hashrate and the price supports the theoretical model of [27] and [4]: A higher number of unique addresses imply a larger number of active users. And a larger number of active users raise demand for the cryptocurrency in question and therefore increases its price.

Following the same path, [9] built an asset-pricing model and constructed financial factors based on the growth in addresses of the underlying blockchain.

Their findings also point towards that network activity and size are important for the price of a cryptocurrency, which further support the theoretical model of [4]. They constructed five weekly re-sorted portfolios, sorted by quintiles of the growth in transactions with balance, and growth in total transactions. In a long-short analysis, the portfolios sorted by their financial factors produced a statistically significant difference in returns.

They also constructed financial factors for the growth in the total on-chain transaction volume, measured once in terms of tokens and also in terms of the total value of tokens in USD. However, for these factors, the difference in returns was statistically insignificant. They cannot explain the cryptocurrency price dynamics.

4.3 Filtering for Relevant Transactions

Not every transaction on the blockchain is relevant for analysis. Based upon the target of one's study, different types of transactions are important. To be able to extract the relevant transactions from the blockchain, we have to consider first the transaction patterns which commonly appear. For example, if one is interested in identifying who holds currently a coin, the initial and final transactions of the coins history are relevant. The rest can be omitted hence simplifying the analysis. Bitcoin transaction patterns can be grouped into the following categories, as comprehensively traversed by Ferrin [14]:

For single transactions, Bitcoin follows:

- Peel transactions. This is the most common type of transaction on the Bitcoin blockchain. They can have any number of inputs, but only two outputs. One output is the receiver's, and the other is the change address.
- Sweep transactions. They combine multiple inputs into one output: Many inputs are swept into one.
- Distribution transactions. They can have any number of inputs, but they have more than three outputs. They are mostly used when a single organisation pays many parties.
- Relay transactions. They have one input and one output. This is also a popular pattern. Bitcoins can be moved from one address to another without leaving traces such as change addresses, which are seized by analysers using, e.g. Heuristic 2 to break down anonymity.
- Self-spending transactions. They can have any number of inputs and outputs. Their distinguishing feature is that one or more of the input addresses also appears as output addresses in the same transaction.
- Meta-transactions. In those transactions, external data is inserted into the transaction to enhance the stability and trustworthiness of the Bitcoin blockchain.
- Joint transactions. In contrast with sweep transactions, where addresses are swept into one, in joint transactions, different transactions are combined into one larger transactions using some protocol.

Note that some of the transaction patterns can be layered. For example, a transaction can both be a peel and be a self-spending one: Someone careless may just use his old address as the change address. One can also sweep into his own address, making it both sweep and self-spending. There can also be meta-peels and meta-relays.

For multiple transactions, Bitcoin follows:

- **Peeling chains.** They are a series of peel transactions. The start of a peeling chain is usually an address that contains a significant amount of bitcoins. Then, a small amount is peeled from it, and a small amount is peeled from the remainder, and so on. Peeling chains are the focus of study of several important works, including [21, 23, 30].
- **Green addresses.** In these transactions, bitcoins pass through a single publicly known address that is trusted by the receiver of the coins.
- **Mixing clouds.** Also known as tumblers, mixing pools, and washers. They are multiple interconnected joint transactions. As the names suggest, mixing cloud transactions aim to be very difficult to trace.
- **Tunnelling.** These are the transactions that have connections outside of the blockchain in question. This is a technique advertised by some Bitcoin services for anonymity: Bitcoins are shifted from account to account and finally to another address of the original owner.

Some of the outlined transaction patterns can be utilised to cluster different transaction/addresses into a single entity. In particular, for example, Heuristic 2 depends on change addresses. Other transaction patterns, such as meta-transactions, joint transactions, green addresses, mixing clouds, and tunnelling are designed to obfuscate the true users behind the scene.

Chang and Svetinovic [8] went beyond Heuristics 1 and 2 and performed clustering using different Bitcoin transaction patterns. In peels, they identified the owner of the sending address and the owner of the change address (in effect, Heuristic 2). They identified all addresses that participate in self-spending transactions. They grouped all the relay addresses. For sweeps, they first identified the transactions that have more inputs than average and grouped the input and output clusters together if they are different in number.

Chang and Svetinovic [8]'s work is a vivid reminder that as Bitcoin transaction patterns evolve, our heuristics to contract the transaction graph also need to change. Although apart from Heuristic 1 and Heuristic 2 and their extensions as outlined in Sect. 3, the other clustering methods did not seem to have caught on, it is worth reminding ourselves that they are certainly not the only heuristics. Other heuristics based on other Bitcoin transaction patterns may give better clustering results that better approximate the user graph. It is most valuable when we use multiple heuristics together to make the most of the advantages of all.

The peeling chain is the most interesting among all transaction patterns. As briefly described above, peeling chains typically begin with addresses that have significant amounts of bitcoins in them, and then in subsequent transactions tiny amounts gradually peel off into a myriad of addresses after hundreds, even

thousands of hops. It is also noteworthy that peeling chains sometimes can be chained themselves: Tiny amounts can be aggregated again into a new account, forming the beginning of another peeling chain.

Ron and Shamir [30] already noticed the peeling chain pattern, although they did not use this term. They call peeling chains “fork-merge patterns.” They noticed that a “frequent scenario” in Bitcoin involves the transferring of bitcoins to many intermediate addresses (the forks), and those addresses are merged into another address (the merge). Ron and Shamir [30] investigated a particular case, where an entity, owning 90,000 bitcoins, transferred this amount of tokens using 90 different addresses in 90 transactions, and all back to itself: an example of a peeling chain and a self-loop.

Peeling chains received a greater scrutiny in [23], where Heuristic 2 is first being introduced. By utilising Heuristic 2, [23] were able to follow a peeling chain in detail. At each hop, they followed the change address to the next hop, and at each step, they were able to recognise the “meaningful recipient”, i.e. the peel, of the transaction, namely, the address that is not the change address.

Makarov and Schoar [21], like [8], consider further heuristics based on transaction patterns of bitcoins. They consider two based on peeling chains.

Heuristic 4.1 *Addresses on a peeling chain belong to the same entity.*

Heuristic 4.2 *Backtrack volume on a peeling chain to the original address. Discard intermediate addresses.*

Implicitly, [23] and most of the others that addressed this issue are following Heuristic 4.1 [1, 30]. Makarov and Schoar [21], however, focused on Heuristic 4.2, using a recursive algorithm for the backtracking. They used peeling chains to get rid of the great number of intermediate addresses, which allows one to get rid of much spurious volume on the Bitcoin blockchain.

What is “spurious volume” on the Bitcoin blockchain? Makarov and Schoar [21] take it to mean the transactions that are not economically meaningful. Economically meaningful transactions are those that involve true financial transfers between two parties. Alice buys a bottle of water from Bob and pays him \$1: This is economically meaningful. On the other hand, Alice may transfer her money between her own different bank accounts out of boredom: These transfers are then spurious in [21]’s eyes.

Why is it important to distinguish between spurious and economically meaningful transactions? One consideration is very practical. If we can get rid of the spurious transactions on the Blockchain, the remaining network would be much pared down, reducing computational burden. Further, and more importantly, by focusing on the economically meaningful volume, we can get a fairer picture of how Bitcoin is used.

Makarov and Schoar [21] branded all the intermediate addresses on peeling chains as spurious. By discarding all the intermediate addresses, they pared down the database from 869 million addresses to 640 million addresses. After clustering, they obtained 189 million clusters, within which 116 million contained only transactions with addresses used once. Addresses appearing only once on the

blockchain challenge any analysis of the transaction graph since they cannot be linked with other addresses without off-chain information.

4.4 Fraud Identification

Scamming cryptocurrency users, stealing tokens, manipulating exchange rates: These fraudulent behaviours are commonplace in the cryptocurrency market. We observe liquidity, but this liquidity can be a fake. This is achieved by wash trading, looked at by Cong et al. [10] and Amiram et al. [2].

Even the actual price of a cryptocurrency can be manipulated. Griffin and Shams [16] detailed a famous case. During the market frenzy of 2017, the price of Bitcoin was lifted by the trading activities of a trader with Tether, another cryptocurrency. Using the blockchains of Bitcoin and Tether, they found that Tether was sold on various exchanges in exchange for Bitcoin at market downturns. Bitfinex was a favourite place. This resulted in an increase in Bitcoin price. Sinisterly, this behaviour was so systematic that there are ample reasons to believe that the price was manipulated.

Especially during the early days of cryptocurrencies, it was commonly assumed that the vast majority of transactions are linked to illegal activities. Silkroad was a case in point: This platform using bitcoins sold all kinds of illegal substances and services. Foley et al. [15] investigated the extent of illegal activities financed by bitcoins. Their estimate was that 46% of the transactions in their data sample are linked to illegal activities. They linked blockchain data with known Bitcoin seizures by law enforcement agencies, and users from darknets and users identified in darknet forums involved in the use of bitcoins for financing illegal activities. Fortunately, they further found that with the emergence of altcoins that are more privacy preserving, and with an increased mainstream interest in cryptocurrencies, the share of bitcoin-financed illegal activities decreases.

Bitcoins themselves can be stolen. Turning back to our previous discussion of the peeling chain, see Sect. 4.3: It is natural for criminals to peel. A thief may steal bitcoins, but since the Bitcoin blockchain is public, he has nowhere to hide if the original owner tracks down his sinful address. Things are very different if he peels away: He can create multiple addresses and hop bitcoins along them. Unless the owner is truly determined, this practice obfuscates the whereabouts of the money and is difficult to track.

Meiklejohn et al. [23] were truly determined. Armed with Heuristic 2, they looked into three case studies, all criminal in nature, of the peeling chain:

- (1) The Bitcoin theft on 11 Apr 2012 is one of the earliest examples of Bitcoin crimes. 3,171 BTC were stolen from the gambling site Bitcoin. The thieves were patient. They left the money in their address silently and waited until Bitcoin rose in price. Finally, from the main address, they started a peeling chain on 15 Mar 2013, in which one peel went to Bitcoin-24, and another peel went to Mt. Gox.

- (2) The Bitcoinica theft in May 2012 also involved a peeling chain, with peels sent to known exchanges, BTC-e, CampBX, and Bitstamp.
- (3) The Bitfloor theft is the most sophisticated amongst the three. The thieves may have read [30] and followed the fork-merge pattern. After an initial peeling chain, the small amounts of peels were aggregated, and another peeling chain ensued. Meiklejohn et al. [23], however, followed the forks and merges closely and followed all the later chains through. Their effort allowed them to observe that some peels were, again, sent to popular addresses, including Mt. Gox, BTC-e, and Bitstamp.

Peeling chains, and the close study of them, therefore offer law enforcement another chance to de-anonymise the “most motivated Bitcoin users”, in the words of [23], namely thieves. Even if only a small portion of the peels flow to exchange sites, by following those peels closely, we may have a chance to track down the true identity of the criminals.

5 Conclusion

The information provided by blockchains of cryptocurrencies is immense and diverse. This information has been used to analyse transaction flows, user behaviours, and the prices of cryptocurrencies. Blockchain information is represented by two types of networks, the transaction network and the user network, approximated by the entity network. We reviewed heuristics for constructing the entity network out of the transaction network and discussed how they have been used, improved, and developed over time.

Finance and economics research based on blockchain information focused on location identification, fraud detection, and price investigations. These topics made crucial use of network analytics tailored to the specific blockchain properties. By this comprehensive overview, we intend to aid research on the use of blockchain information to understand user behaviours and the corresponding price behaviours of cryptocurrencies.

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Part V
Blockchains and Finance

Blockchain Transaction Decision: Through the Lens of Mempool



Ankush Agarwal, Cathy Yi-Hsuan Chen, and Elsa Hernandez

Abstract In the Bitcoin, blockchain miners use computational resources to validate transactions and are rewarded with newly created Bitcoin and transaction fees appended by users for their transactions. The priority of a transaction is determined, among others, by the fee appended to it. Profit-maximising miners choose transactions that pay higher fees. As trading volume in Bitcoin increases, network congestion has become a key issue in the viability of Bitcoin as a medium of exchange. As congestion leads to large waiting times and soaring transaction fees, users with small-value transactions may no longer find it profitable to record transactions. We aim to examine the viability of Bitcoin blockchain when users pay no transaction fees so that there is no priority given to transactions. Users' decisions to record a transaction through the Bitcoin blockchain are modelled through a queueing game in a processor-sharing system. The symmetric equilibrium threshold of this game sheds light on the limit of the queue length at which self-optimising users who seek to minimise transaction costs find it economically inviable to record a transaction through the Bitcoin blockchain. This analysis highlights the key role that transaction fees play in sustaining an increasing number of transactions in the Bitcoin blockchain.

1 Introduction

Since its inception in 2009 by an anonymous group of developers, Bitcoin has emerged as the most successful cryptocurrency with a market capitalisation of 117.81 billion USD in April 2020. It is an open-source, peer-to-peer (P2P) digital

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currency with no central authority or issuer. It has served over 500 million transactions, and it is the first and largest decentralised cryptocurrency [1, 2]. Its invention is revolutionary in the sense that the double-spending problem can be solved without the need for a third party [3]. Through technological solutions such as peer-to-peer (P2P) connectivity and cryptographic algorithms, Bitcoin and cryptocurrencies, in general, allow for a decentralised organisation that provides operational security and transparency [4].

As a peer-to-peer electronic cash system, Bitcoin allows online payments to be sent directly from one involved party to another without going through an institution. The transaction process in the Bitcoin system can be briefly described as follows. Transactions that are recorded by users flow into the memory pool (mempool) where they sit until they are confirmed by miners and then added to a block. Per the Bitcoin protocol, blocks can contain a limited amount of transactions. In order to decrease the time a particular transaction spends in the mempool (that is, unconfirmed), a user may choose to attach a certain amount as a fee to their transaction that acts as an additional form of compensation to miners, who in turn may select the transactions with the highest paying fees to verify and include in a block. It is possible for transactions to be processed without fees; however, they may sit in the mempool for long periods of time before they are confirmed and added to a block [5]. In order for miners to add a block to the blockchain, they must work on solving a computationally mathematical puzzle, the difficulty of which is adjusted to keep the rate at which blocks are added fixed at a rate of 10 minutes.

Given the rate at which blocks are added to the chain, increases in the volume of Bitcoin trading have given rise to a scalability problem [6]. That is, with a capacity of processing around seven transactions per second, increases in the number of Bitcoin transactions into the mempool have generated a congested traffic that results in longer average times [6]. While the emergence of transaction fees paid to prioritise transactions has been an important development in the evolution of Bitcoin, the manner in which transactions fees work is still being developed and is likely to change over time [7]. Nonetheless, as Bitcoin continues to sustain a large trading volume, it is clear that the risk of a large backlog waiting in the transaction queue is significant. Without the payment of transaction fees, users may face long waiting periods before their transactions are confirmed. At a given point, waiting times may become elevated enough to the point that users no longer find it viable to use the Bitcoin blockchain to record a transaction. While other cryptocurrencies may provide for an alternative, increased trade volume in those alternatives would likely lead to the same considerations. Moreover, as the network grows busier, competition by users to confirm transactions leads to an upward trend in the amount of transaction fees paid. This may also result in potential users finding the recording of a transaction through the Bitcoin blockchain economically infeasible.

In order to better understand the role that waiting times and transaction fees play in the participation of users, it is important to first understand how the Bitcoin blockchain could operate in terms of users' incentives to continue to use this service without resorting to the payment of transaction fees. Here, we consider the choice faced by a user who wishes to record a transaction and may choose between doing so in the Bitcoin blockchain or through a bank. The model presented here

follows the work of [8], who analyse the decision problem faced by users who observe the number of participating customers in a processor-sharing system and decide to join this queue or opt for a fixed-service rate alternative. This decision problem is modelled as a non-cooperative game, and [8] show how to compute the unique threshold equilibrium strategy. The Bitcoin blockchain environment in this scenario is taken as a kind of processor-sharing system, with the fixed-cost option representing the usually fees that are charged by banks. The analysis sheds light on the limit of the queue length at which self-optimising users who seek to minimise transaction costs will it more to resort to recording their transactions through a bank rather than through the Bitcoin blockchain. This places a theoretical upper bound on the mempool transaction count after which no new transactions would be recorded. As such, the results also highlight the importance of transaction fees to sustain an increasing number of new transactions in the Bitcoin blockchain.

The rest of this chapter is organised in the following way. Section 2 provides an overview of the blockchain technology. Next, Sect. 3 presents the theoretical model, analysis framework, computation of the symmetric equilibrium, which is then followed by a numerical example. Finally, in Sect. 4, concluding remarks and a discussion on the limitations and future extensions of this study are presented.

2 Transactions on the Blockchain: a Step-by-Step Illustration

The underlying technology behind the working of blockchain is illustrated here through an example of the flow of a sample transaction using the example of the Bitcoin blockchain. While Bitcoin is just one of many applications of blockchain technology, blockchain design was the main innovation underlying Bitcoin, and it is covered most extensively in the literature. The following is a simplified description of the Bitcoin system. Throughout it is assumed that user *A* makes a transaction to transfer x Bitcoins (BTC) to user *B*. The following steps illustrate how this transaction is processed (i.e., added to the Bitcoin blockchain):

Step 1 User *A* joins the Bitcoin peer-to-peer (P2P) network and makes a public transaction in which it is detailed that she wishes to reassign a number of BTC from one or more of identities that she controls to user *B*. Identities can be verified using public-key cryptography, which ensures the anonymity of users by storing the mapping of a user to his or her public keys on that user's node only¹ [9]. Each transaction is uniquely identified by a hash, and therefore the hashes of the previous transactions that show proof of ownership must also be included. In addition to this, it is possible for *A* to specify the payment of a transaction fee that will be assigned to the node that succeeds in adding the block containing the transaction of *A* to the chain. The private key of *A* is used to digitally sign this information, which is then broadcasted to the nodes in the system.

¹ For an analysis on the limitations of anonymity for Bitcoin users, see [9].

Step 2 Transactions that are broadcasted to the system, where they sit in the memory pool waiting to be verified by the nodes in the network, collected into a block and then hashed in a Merkle tree to a root hash. A Merkle tree is a tree in which every leaf node is marked with a data block and every non-leaf node is marked by the cryptographic hash of the labels of the concatenation of its child nodes [10]. This method allows to verify transactions without downloading the entire block since “a user needs to keep a copy of the block headers of the longest proof-of-work chain [...] and obtain the Merkle branch linking the transaction to the block it’s timestamped in”. In Bitcoin system, priority mechanism is implemented, in which the priority value of a transaction is determined according to transaction attributes such as remittance amount, data size, coin age, and fee [5]. Moreover, when selecting transactions to bundle into a block, miners may prioritise certain transactions over others if, for example, they provide a better mining reward in the form of transaction fees. The Bitcoin Core protocol limits blocks to 1 MB in size, and each block contains at most some 4,000 transactions [11].

Step 3 Each node simultaneously and independently tries to add the block—which in addition to the transactions consists of a timestamp and a nonce—to the blockchain by finding a PoW for the block. This process is commonly known as *mining*, and users who participate in this process are referred to as *miners*. PoW can be thought of as a solution to a difficult mathematical puzzle, and it requires that miners perform computationally expensive tasks. As part of the Bitcoin consensus rules, there is a difficulty retargeting algorithm that adjusts the difficulty of finding PoW so that the expected time between two block solutions is set at 10 minutes [12, 13]. To encourage nodes to pay for the cost related to the considerable computational effort required to solving this cryptographical puzzle, the miner that succeeds in finding the PoW is awarded with block rewards (newly created BTC) and/or transaction fees.²

Step 4 When a node finds a PoW, it broadcasts the block to all nodes for validation. A block is said to be valid if it satisfies the following properties: (i) all transactions in the block are valid, (ii) it contains a valid timestamp (i.e., the previous block is valid), and (iii) the PoW is satisfied [13].

Step 5 Once the block is accepted by the nodes in the network, it is added to the blockchain. Miners express acceptance of the block by using its hash when working on the next block [15]. A transaction is then said to have been confirmed.³ Since various miners may produce multiple blocks based on the same timestamp, nodes

² The supply of Bitcoins is such that they are generated at a predictable rate, whenever a new block is added to the Bitcoin blockchain, and it is capped at approximately 21 million [14]. The block reward in Bitcoin is halved after the discovery of every 210,000 blocks, which takes around four years to complete. As of writing, the most recent halving occurred on 11 May 2020, causing the block reward to fall from 12.5 to 6.25 bitcoins.

³ Sources typically recommend considering a Bitcoin transaction final only after six confirmations, to assure that the transaction is truly recorded in a permanent part of the blockchain [1].

may broadcast different versions of the next block simultaneously. In this case, the consensus rule dictates that the longest valid blockchain (i.e., the one with most accumulated PoW) is the canonical one and miners work on extending it [13]. Miners are incentivised to follow the longest chain rule so that their rewards are accepted by all other users.⁴

3 Transaction Decisions and the Mempool

In this section, we consider a setting in which no transactions offer a fee to miners and homogeneous users who wish to record a transaction can choose between a banking facility with fixed service times and the bitcoin blockchain.

Transactions recorded via the bitcoin blockchain arrive to the mempool, where miners work on adding blocks of B transactions to the blockchain at a constant service rate of μ . We view the mining capacity as a kind of processor-sharing system, where the service intensity for each block of transactions is equal to $v(x) = \mu/x$, where x is the number of blocks in the mempool. The mempool is assumed to have a maximum capacity of M blocks of transactions, which may be finite or infinite. Transactions recorded via the bank alternative entail the payment of a fixed transaction cost of θ and are validated immediately. Individual transactions arrive sequentially, with i.i.d. interarrival times with finite mean B/Λ .

Upon arrival, customers observe the number of blocks of transactions sitting in the mempool. For each user $i = 0, 1, \dots$, a *strategy* (or *decision rule*) is given by a function $u_i(x) : \mathbb{N} \rightarrow [0, 1]$, where $u_i(x)$ denotes the probability of recording a transaction through the bitcoin blockchain when there are $x \in [0, M]$ blocks sitting in the mempool (and choosing the bank alternative with complementary probability $1 - u_i(x)$). We denote by U the collection of decision rules and by $\pi = (u_0, u_1, \dots) \in \Pi \triangleq U^\infty$ the vector of decision rules of all users (referred to as a *policy*). Self-optimising users wish to minimise their respective expected transactions costs, given by

$$\bar{C}_i(x, \pi) = u_i(x)\bar{w}_i(x, \pi)\delta + (1 - u_i(x))\theta, \quad (1)$$

where $\bar{C}_i(x, \pi)$ denotes the expected waiting time for user i when there are x blocks in the mempool upon his arrival and δ denotes the cost of waiting per unit of time. We assume that the parameters δ, μ, θ, B , and M are common knowledge.⁵ Note that for all transactions grouped into a given block, individual waiting times coincide with the waiting time the block spends in the mempool. Since a user's waiting time depends on the unknown actions of all subsequent transactions arriving (i.e., on the

⁴ Bitcoin imposes a 100-block delay before newly mined block rewards can actually be spent. This rule does not apply to transaction fees.

⁵ Note that under the assumption that the parameter B is common knowledge we may refer to users observing the number of transactions or blocks of transactions interchangeably.

decision rules of all subsequent users), the decision problem can be studied as a non-cooperative game.

We focus on finding the symmetric equilibrium of the game, and following the work of [8], we compute the unique threshold equilibrium strategy. We are interested in the limit of the queue length at which self-optimising users who seek to minimise transaction costs will find it more profitable to resort to recording their transactions through a bank rather than through the bitcoin blockchain. The analysis allows to pinpoint a theoretical upper bound on the mempool transaction count after which no new transactions would be recorded. As such, the results highlight the importance of transaction fees to sustain an increasing number of new transactions in the bitcoin blockchain. The following section characterises the main equilibrium points.

3.1 Nash Equilibrium Solution

Considered here are strategies in which users choose to record a transaction through the bitcoin blockchain if and only if the number of blocks in the mempool is less than some given constant. These decision rules are referred to as threshold strategies and are formally defined as follows:

Definition 1 A decision rule u with threshold $g = L + q$, $L \in \mathbb{N}$, $q \in [0, 1]$, has

$$u(x) = \begin{cases} 1 & \text{if } x < L, \\ q & \text{if } x = L, \\ 0 & \text{if } x > L. \end{cases} \quad (2)$$

A customer who employs this rule joins the mempool if the number of blocks in the mempool, x , is smaller than L , while if $x = L$, she does so with probability q . Otherwise, she records the transaction with the bank. Whenever $g \in \mathbb{N}$ ($q = 0$), the strategy is said to be pure. Otherwise it is mixed. A Nash equilibrium policy of the game is one in which the threshold strategy adopted by all users is an optimal response for each of them, and no one can gain by unilaterally deviating from this strategy. Formally, for each policy π , let π^{-k} denote the collection of all decision rules in π , excluding the u_k , and let $[\pi_k | u'_k]$ be the policy which replaces u_k by u'_k . Then:

Definition 2 A policy $\pi = (u_0, u_1, \dots)$ is a Nash equilibrium if

$$\bar{C}_i(x, [\pi^{-i} | u_i]) \leq \bar{C}_i(x, [\pi^{-i} | u'_i]) \forall i, x \geq 0, u'_i \in U.$$

We now present the main results on the existence and uniqueness of the symmetric equilibrium.

Theorem 1 (Theorem 1 [8])

1. For any equilibrium policy $\pi^* = (u_0, u_1, \dots)$, each decision rule u_i is a threshold rule.
2. A symmetric equilibrium policy $\pi^* = (u^*, u^*, \dots)$ exists, is unique, and u^* is a threshold rule.

Lemma 1 (Lemma 3 [8]) Let π be an arbitrary policy, and let U_i^* be the set of decision rules for user i that are optimal against π :

1. Any $u_i^* \in U_i^*$ is a threshold rule, with finite threshold, and is given by

$$u_k^*(x) = \begin{cases} 1 & \text{if } \delta \bar{w}_i(x, \pi) < \theta \\ q_x & \text{if } \delta \bar{w}_i(x, \pi) = \theta \\ 0 & \text{if } \delta \bar{w}_i(x, \pi) > \theta \end{cases}, \quad (3)$$

where $0 \leq q_x \leq 1$ is arbitrary.

2. Consequently, the set U_i^* is given as follows. Let L^* be the smallest nonnegative integer such that $\delta \bar{w}_i(L^*, \pi) \geq \theta$. If $\delta \bar{w}_i(L^*, \pi) = \theta$, then $U_i^* = \{[L^*, q] : 0 \leq q \leq 1\}$. Otherwise, U_i^* consists of the single threshold rule $[L^*, 0]$.

It follows from Theorem 1 that in this setting there exists a unique equilibrium strategy threshold g^* that will dictate users' entry decisions into the bitcoin mempool. Moreover, as Lemma 1 shows, the mixed or pure strategy equilibrium will be dictated by users comparison of the expected transaction costs resulting from the waiting time faced in the mempool before a transaction is validated and the transaction fee charged by a bank.

3.2 Computation of the Symmetric Equilibrium Threshold

Given the results presented previously regarding the existence and uniqueness of a symmetric equilibrium, this section considers the computation of the equilibrium threshold. This is characterised in terms of the expected waiting times in the bitcoin mempool under threshold policies. Explicit calculations are presented for the case of Poisson arrivals, with closed-form solutions illustrated through a numerical example.

3.2.1 Characterisation

Lemma 2 presents the procedure to determine the equilibrium threshold $[L^*, q^*]$. It can be seen that whenever the transaction cost from validating through the bitcoin blockchain is lower than the fee charged in the bank alternative for any number

of transactions sitting in the mempool, users will always choose to opt for the blockchain alternative. In line with this, whenever waiting times in the mempool and/or the cost of waiting per unit of time are large enough so that the bank alternative always represents a lower transaction cost, users will never opt to record their transaction through blockchain. Both cases represent pure strategies of the game. Next to this, an equilibrium mixed strategy with threshold $[L^*, q^*]$ dictates that, given that all users adopt such strategy, any individual user is indifferent between validating their transaction through the blockchain or through the bank.

Lemma 2 (Lemma 4 [8]) *The equilibrium threshold $[L^*, q^*]$ may be determined by the following procedure:*

1. If $\delta\bar{w}(M-1, [M]^\infty) < \theta$, then $L^* = M, q^* = 0$. Otherwise,
2. $L^* = \min\{L \geq 0 : \delta\bar{w}(L, [L, 1]^\infty) > \theta\}$:
 - a. If $\delta\bar{w}(L^*, [L^*, 0]^\infty) \geq \theta$, then the equilibrium threshold is $[L^*, 0]$.
 - b. If $\delta\bar{w}(L^*, [L^*, 0]^\infty) < \theta$, then the equilibrium threshold is $[L^*, q^*]$ where $0 < q^* < 1$ is the unique solution of

$$\delta\bar{w}(L^*, [L^*, q^*]^\infty) = \theta. \quad (4)$$

3.2.2 Calculations

Following [8], with Poisson arrivals and given the constant mining capacity μ , fixing $[L, q]$, and letting $\bar{w}(x) := \bar{w}(x, [L, q]^\infty)$, the service times $\bar{w}(L, [L, q]^\infty)$ can be determined to obtain the solution q^* of Eq.(4). Then $\bar{w}(x)$, $0 \leq x \leq L$, is the solution of the following set of $L+1$ linear equations:

$$\bar{w}(x) = \frac{1}{\alpha} + \frac{\mu x}{\alpha(x+1)}\bar{w}(x-1) + \frac{\Lambda}{\alpha}\bar{w}(x+1), \quad 0 \leq x \leq L-2, \quad (5)$$

$$\bar{w}(L-1) = \frac{1}{\alpha} + \frac{\mu(L-1)}{\alpha L}\bar{w}(L-2) + \frac{q\Lambda}{\alpha}\bar{w}(L) + \frac{(1-q)\Lambda}{\alpha}\bar{w}(L-1), \quad (6)$$

$$\bar{w}(L) = \frac{1}{\mu} + \frac{L}{L+1}\bar{w}(L-1), \quad (7)$$

where $\alpha = \mu + \Lambda$. That is, the $w(x)$ equals the expected remaining waiting time of any customer present when there are $x+1$ blocks in the mempool.

By (8), $\bar{w}(x)$ can be expressed as

$$\bar{w}(x) = a(x)\bar{w}(0) + b(x), \quad 0 \leq x < L, \quad (8)$$

where recursively substituting (8) into (5) the coefficients $a(x)$ and $b(x)$ yield the coefficients

$$a(x+1) = \frac{1}{\Lambda} \left[(\mu + \Lambda)a(x) - \mu \frac{x}{x+1} a(x-1) \right], \quad x \geq 1, \quad (9)$$

$$b(x+1) = \frac{1}{\Lambda} \left[(\mu + \Lambda)b(x) - \mu \frac{x}{x+1} b(x-1) - 1 \right], \quad x \geq 1, \quad (10)$$

with initial values $a(0) = 1, a(1) = (\mu + \Lambda)/\Lambda, b(0) = 0, b(1) = \Lambda^{-1}$. Then using (6)–(8) yields

$$\bar{w}(0) = \frac{\mu^{-1} - \frac{1}{L+1}b(L-1) + q^{-1}(b(L) + b(L-1))}{\frac{1}{L+1}a(L-1) + q^{-1}(a(L) - a(L-1))}. \quad (11)$$

$w(x)$ can be obtained from (8)–(10). The equilibrium threshold can be calculated by computing L^* by Lemma 2, where $q^* = 0$ in case (1) or (2a), and $0 < q^* < 1$ is the unique solution of (4) in case (2b). Obtaining this solution and using (7), (8), and (11) yield the following solution for q^*

$$q^* = \Psi^{-1} \left[\left(\frac{a(L) - a(L-1)}{a(L-1)} \right) \left(\left(\frac{\theta}{\delta} - \mu^{-1} \right) - \frac{L}{L+1}b(L-1) \right) + \frac{L}{L+1}(b(L) - b(L-1)) \right], \quad (12)$$

where $\Psi \triangleq \mu^{-1} - \frac{\theta}{\delta} \cdot \frac{1}{L+1}$. It is easy to see that the equilibrium probability of any user validating a transaction through the bitcoin blockchain is increasing in the mining capacity, μ , as well as in the fixed transaction cost set by the bank alternative. On the other hand, as the cost per unit of waiting increases, the equilibrium probability decreases. This illustrates the fact that, taking into account that μ is fixed, as users value their time more highly, they will be less likely to be willing to have their transaction in the mempool. Similar to this, the higher the transaction costs of validating a transaction through a bank, the more attractive it is to resort to the bitcoin blockchain alternative.

3.2.3 Numerical Example

We consider a numerical example to illustrate the previous results with parameters $\mu = 0.1, \Lambda = 0.09, \theta = 3$, and $M = \infty$. Using data available from blockchain.info, we obtain a range of estimates for the cost of waiting per unit of time, δ , as

$$\delta = \frac{Fees}{1/\mu},$$

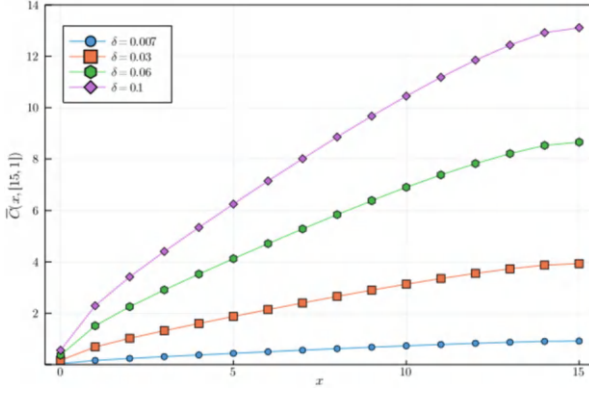
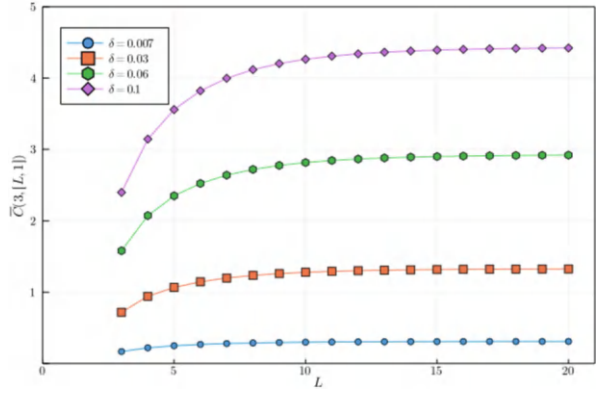


Fig. 1 $\bar{C}(x, [15, 1])$ as a function of x for different values of δ

Fig. 2 $\bar{C}(3, [L, 1])$ as a function of L for different values of δ



where $1/\mu = 10$ is the average time taken to add one block to the blockchain, and *Fees* refers to average fees per transaction.

Figure 1 illustrates the transaction cost $\bar{C}(x, [15, 1])$ as a function of x for different values of δ . As $[L, q] = [15, 1]$ by Eq. (1), changes in the transaction cost hold for arbitrary values of θ and are determined by the cost of waiting through the service duration, which in turn is monotone increasing in L . For increasing values of δ , it can be seen that the transaction cost associated with the waiting time in the mempool is higher than the transaction fee paid at the bank even when the number of customers sitting in the mempool is relatively low. This illustrates the fact that customers who place a higher value on their time will more quickly turn to the bank alternative when there is no option to prioritise their transaction in bitcoin. Intuitively, these would be the types of customers who would be willing to pay higher fees to reduce their waiting time, whilst still reducing their transaction costs. Similarly, Fig. 2 shows the transaction cost when $x = 3$ as a function of the threshold L .

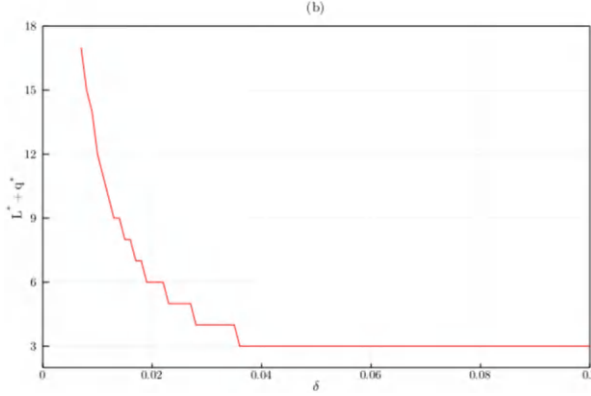


Fig. 3 $g^* = L^* + q^*$ as a function of δ

Finally, Fig. 3 presents the equilibrium threshold $[g^*]$ as a function of δ , where a real number of the form 10.34 means $L^* = 10$ and $q^* = 0.34$. L^* is obtained from Lemma 3 as the first integer from which $\bar{V}(L, [L, 1])$ exceeds θ/δ . L^* is a decreasing function of δ and approaches 3. Intuitively, as δ increases, the equilibrium threshold for the length of the queue that a customer must observe to decide to record their transaction in the bank will decrease since the cost of waiting in the bitcoin queue is increasing. It can be seen that for $\delta \rightarrow \infty$ the queue length approaches a constant. In our setting, the individually optimal threshold decreases to $L^* = 3$.

Throughout it is clear that the parameter for the cost of waiting, δ , plays a key role in the decision to record a transaction through bitcoin given waiting times. For large enough values of δ , customers will be less willing to record a transaction through the bitcoin blockchain even when the number of transactions sitting in the mempool is not significant. Through this preliminary analysis, we have illustrated the fact that some customers will be willing to pay high transaction fees in order to prioritise their transactions and avoid large waiting times.

4 Conclusion and Discussion

This chapter focused on understanding the viability of the Bitcoin blockchain without the payment of transaction fees for users to prioritise transactions. We investigate the decision problem faced by users when choosing to record a transaction through the Bitcoin blockchain or through a traditional banking alternative in a processor-sharing queueing game. The analysis focuses on the trade-offs between these two alternatives and is guided by the expected transaction costs incurred in by users. Following the work of [8], the Nash equilibrium solution was characterised with

the corresponding calculations presented for the case with Poisson arrivals and a constant mining capacity. In equilibrium, it is shown that a user's probability of recording a transaction through the Bitcoin blockchain is increasing in the mining capacity and in the fixed transaction fee set by the traditional banking alternative and decreasing in the cost of waiting per unit of time. In other words, given the fixed expected time between two block solutions set by the Bitcoin protocol, as users grow more impatient or as the network becomes more congested, one could expect users to be less likely to resort to using Bitcoin. Moreover, through a numerical example, the equilibrium threshold of users' decision rule was explicitly calculated and presented as a function of the cost of waiting. This example illustrates the existence of a theoretical upper bound on the mempool transaction count after which arriving users would be discouraged from resorting to Bitcoin to process their transactions.

These results are in line with the behaviour observed in real life. For example, a report by Barysevich and Solad [16] found that dark web users were switching from Bitcoin to alternative cryptocurrencies due to slow processing and high fees. The analysis found that since 2017, Bitcoin payments were economically infeasible due to a tenfold increase in transaction costs, an amount that can account for as much as 30 percent of smaller payment amounts. Here it was assumed that users did not have a possibility to pay a transaction fee in order to prioritise their transaction, thereby having their expected waiting time until confirmation necessarily increase as new transactions arrive. This scenario, although far from the workings of Bitcoin in practice, is particularly relevant for small payments transactions since transaction fees paid for such transactions are likely to be very small. This will hold specially in periods in which the transaction volume increases and transaction fees are pushed through the roof. Next to this, an important issue in the wider implementation of permissionless distributed ledger technologies has to do with the incentives schemes that have to be in place for record-keepers to continue to maintain and secure the blockchain. This issue is of particular importance in Bitcoin, since coinbase rewards are expected to disappear by the year 2140. After this, the entire existence and security of the chain will have to be based on transaction fees, or on an alternative reward scheme. However, prohibitively high transaction fees are likely to discourage the use of Bitcoin for small-value transactions, thereby impeding its further (widespread) adoption as a medium of exchange.

Overall, the results presented here highlight the importance of transaction fees for maintaining the Bitcoin blockchain, as well as the important role transaction fees play on the attractiveness of using Bitcoin as a speculative investment or as an alternative currency and medium of exchange. Nonetheless, several limitations to this chapter deserve comment. First, although the use of queueing theory to study the mechanism of the Bitcoin system is common within the blockchain economics literature, here the mining capacity is modelled as a PS system, which results in excluding some of the most relevant characteristics to Bitcoin. Mainly, this assumes that as new transactions arrive to the mempool, the mining capacity is equally divided among new and existing transactions waiting to be confirmed. In reality, assuming heterogeneous users, the Bitcoin system may be better portrayed by a

queueing game with random order service (ROS). Second, the model presented here does not incorporate the Bitcoin price into user's decision-making process. Given the past high volatility of the price of Bitcoin, it is natural to assume that this is a key factor that users would take into account when using Bitcoin as a medium of exchange. Third, while here it is assumed that the arrival of transactions into the mempool is fixed at some given rate, in reality network congestion in Bitcoin tends to be highly volatile.

Altogether, these limitations present an avenue for further research within the topic of transaction fees in the Bitcoin blockchain. These of course vary within their degree of complexity. For example, the sojourn time distribution of M/M/1 PS queue and waiting-time distribution in M/M/1 random order service (ROS) queues have been shown to be identical, up to a multiplicative constant, and the equilibrium threshold can be computed in a similar way, so that the model presented here could be modified to one with ROS quite straightforwardly while obtaining similar results [17, 18]. Next to this, the effect of the volatility in the Bitcoin price level could be incorporated by considering a stochastic game-theoretical model. A more complex issue would involve relaxing the assumption of a constant arrival rate of transaction, as systems in which the distributions of arrival rates are derived from observed data or time dependent are difficult to analyse via queueing theory [19]. Finally, in addition to the above-mentioned limitations and the scope for further research that they present, a natural next step in the analysis presented in this chapter is to extend the study to allow for the payment of transaction fees by users in order to better understand the behaviour of users, as well as the role of transaction fees in the Bitcoin blockchain.

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Part VI

Modern Applications

Blockchain Business Development



Katarina Krüger

Abstract This chapter shows how blockchain technology can create real value for companies, especially small- and medium-sized enterprises (SMEs). It first introduces key ideas, including public and permissioned blockchains, consensus methods, and smart contracts in clear, non-technical language. Next, it gives a step-by-step plan to analyse business processes, map stakeholders, and run a simple utility analysis to see which process is the best place to start. The study then explains technical points such as system architecture, data storage (on-chain versus off-chain), and links to current IT systems. A special section presents Decentralised Autonomous Organisations (DAOs) as a new way to manage joint work without middlemen. Throughout, the author stresses that blockchain is not a goal itself, and that it helps only when it cuts costs, saves time, or opens new business options. The framework offered here helps managers judge when and when not to use blockchain in practice.

1 Introduction

There is more and more awareness that blockchain technology can be of interest to companies and thus the economy in general in many ways.¹ Large companies find it much easier to turn to and test new technologies. However, the situation is different for the so-called SMEs (Small and Mid-Size Enterprise). Companies that belong to this category focus on day-to-day business—there is little or no time for testing new technologies as long as the testing/research of new technology is not a logical incremental product or process improvement in their own core competence.

¹ <https://www.hardwareinside.de/wie-die-bitcoin-technologie-das-geschaef-der-zukunft-veraendern-wird-64165/> Retrieved on 01.03.22.

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Nevertheless, it is also necessary for SMEs to deal with this technology in order to understand the mechanisms of action on the one hand and to be able to make decisions on the other hand as to whether and, if so, in which context this technology should be used in the company. To do this, awareness must first be raised and developed in order to gain access to this technology, which is much more than just transferring a bitcoin (or a fragment of a bitcoin) from one wallet to another. On top of that, the development within this technology is very dynamic, and it can almost be called negligent to only watch this development from the side-lines.

It can be deduced from this that the use of Blockchain-Technology should not be an end in itself. It only makes sense if the Blockchain-Technology can also create actual added value. It is therefore very important to analyse the company's own processes. It should be clearly stated at this point that not every process is improved through the use of a Blockchain-Technology!

At the same time, digital transformation is generating more and more data. This data—used wisely—can represent added value in the company. For example, in the course of cooperations, this data can play an important role—and with it the securing of this data and its authenticity as well as the question of who is allowed to access which data. Here in particular Blockchain-Technology can be highly relevant for small- and medium-sized enterprises. However, this requires a basic knowledge of this technology.

This article is structured as follows. First, “Blockchain-Technology” explains briefly the characteristic of the technology, including the distinction between public and permissioned blockchains, consensus mechanism and smart contracts. “Process analysis” explains methods that should be used to identify the processes that help create value through the use of Blockchain-Technology. Third, “The Concept of DAOs” helps to understand the DAO and why companies should take this concept into consideration for creating better and at the same time more reliable cooperation-pattern with partners in industry. The final part concludes and provides an outlook.

2 Blockchain-Technology

The first association with the term blockchain is often the topic of cryptocurrency.² However, blockchain technology is about much more than cryptocurrencies, even though there are now thousands of different currencies.³

A blockchain is a form of database system in which the transactions made in a peer-to-peer network are stored chronologically in blocks. This peer-to-peer network enables the shared and synchronised storage of data and uses a continuous chaining of data/transactions secured by cryptographic methods.⁴

² Xu, et al. [1], p. 3.

³ According to Statista.com and investing.com, there are approximately 10.516 different currencies available by March 2022.

⁴ Prinz, Wolfgang et al. (2017), p. 8.

This results in the same information being available at each participant (node) of the network when these participants act as part of the infrastructure.⁵ Therefore, transaction management is not centralised, but takes place in a distributed network.⁶

The data is stored as the so-called “cryptographically chained blocklists”, and it is not possible to manipulate or subsequently forge transactions within this network. The consensus mechanism used by the network decides how the transactions/data are verified and validated to become valid and stored on the chain.⁷

Businesses can use this. Companies could use the blockchain not only to store data, but also to define and automate processes with the help of “smart contracts” in such a way that they cannot be manipulated, but are traceable.

Blockchains have certain characteristics and enable different forms of application, depending on the intention of the corresponding community or network. First of all, blockchains can be divided into public/private or permissioned and permissionless systems according to the degree of publicity and the degree of access restriction.⁸ The Bitcoin blockchain is a public network in which everyone can participate and see which transactions are added. This blockchain system has no central authority and the decisions on whether a transaction is valid (or not) are made by the participants of the network via the consensus mechanism Proof-of-Work (PoW). There are no access restrictions in this type of system—rather, anyone can join and participate in the network as a system.

Private systems, on the other hand, have access restrictions and you have to register to be accepted into the system and participate. In addition, role profiles determine which actions which participant is allowed to perform. For example, it can be determined that one type of role only has read rights, while another is involved in the process of reaching consensus and thus validating transactions, etc. The mixed form of a private blockchain with restricted access allows for efficient consensus mechanisms since it can be assumed, among other things, that a certain level of trust exists between the participants of the private (possibly consortium) blockchain.⁹

In addition to the definition of access and access restrictions, there are other features that characterise the blockchain. Blockchain applications are described as secure and resistant to forgery. This security is achieved through various mechanisms. As already explained, it is a so-called peer-to-peer network. This means that all participants enabling the infrastructure of the network always have the

⁵ Nodes are independent computers of the network that communicate and synchronise with each other, verify the transactions of the network and confirm the formed blocks via the consensus mechanism determined in the network. Users of the network do not necessarily have to be nodes—they can simply be users of the infrastructure provided. To use an analogy: Today, anyone can buy a ticket for a train connection from A to B without having to build the stations, lay the tracks and bring the waggons onto the corresponding route. The rail network exists and can be used by everyone. It is similar with a blockchain network. You don't have to participate as a node to be able to use the network.

⁶ Bheemaiah [2] p. 54.

⁷ Casino et al. [3], p. 56.

⁸ Vukolic (2017): Rethinking Permissioned Blockchains, <http://www.vukolic.com/rethinking-permissioned-blockchains-BCC2017.pdf>, retrieved 18.03.22.

⁹ Shrimali/Patel [4]: p. 3.

same level of knowledge about the transactions—the participants have synchronised ledgers. For attackers, the incentive to attack such a system is very low because it is simply not technically possible (at present) for the attacker to identify all participants (nodes) in the system and attack them simultaneously. However, the attack on only one (or even several) node does not shake the system as a network. In this context, one also speaks of “no-single-point-of-failure”.¹⁰

The transactions are resistant to forgery because the transactions stored in the blocks are encrypted via hash value functions. This means that deviations from the original are immediately recognisable because the hash values of a changed file no longer match the original. Therefore, a blockchain solution can be used to build up a trustworthy history since a subsequent change of data is prevented or at least documented.

Another important component is formed by the so-called smart contracts which means that Blockchains can “automate messages by the addition of code snippets”.¹¹ The Bitcoin blockchain is initially the basis on which the entire ecosystem has developed. With Ethereum as the infrastructure provider, it has become possible to transfer not only financial transactions but also contractual content via the so-called smart contracts within a network. These computer programmes enable data and transactions to be both stored and executed on the blockchain. This means that smart contracts can hold assets managed by the blockchain, transfer them or call up other smart contracts managed on the blockchain and execute them according to an “if-then logic”. It is important not to be misled by the term—smart contracts are “just” computer programmes that usually regulate and coordinate control logics. However, it is not a contract in the legal sense.¹²

A frequently related criticism of blockchain technology in general and in particular of the consensus mechanism “proof of work” used by the Bitcoin network is the low scalability as well as the high energy consumption in the so-called mining.^{13, 14} Triggered by the high demand for secure and non-manipulable data storage systems in industry, among others, research is being conducted into innovative, scalable and energy-efficient systems in order to dispel the existing criticism. Here, too, the first successes can be seen, which conversely means that this technology will leave its niche existence.¹⁵

¹⁰ Crosby et al. [5]: p. 9.

¹¹ Morabito [6]: p. 4, 15.

¹² De Filippi/Wright [7]: p. 78.

¹³ Mining in this context means that a computer-generated crypto puzzle must be solved by the nodes (computers) in the network. The node that solves the mathematical puzzle first receives the reward for the calculated solution and publishes its result. The other participants in the network check the result and confirm it if it is correct. The block is now added to the existing chain of blocks. At the same time, a new puzzle is created that must be solved. The special feature of the Bitcoin blockchain is that the difficulty of the puzzle depends on the number of transactions and participants and that only the person who solves the puzzle first receives the reward.

¹⁴ Morabito [6]: p. 65.

¹⁵ Author’s note: The tense situation of the labour markets, which is prevalent at least in the industrialised nations, will accelerate automation and digitalisation—there are simply not

In order to successfully integrate blockchain systems into existing IT systems, interaction with the existing systems and the integration of external data is a mandatory prerequisite. Two approaches are considered state of the art:

1. External data (e.g. text documents, images, videos or even excerpts from multimedia databases) can be integrated by using hash values¹⁶ as a “fingerprint”. To do this, the hash value of the external data must be compared with the one stored on the blockchain. If the two hash values are identical, then it is the original file. However, if the external data has been manipulated, then it no longer matches the hash value of the value stored on the blockchain. In this way, discrepancies and the smallest changes can be detected immediately.
2. The use of the so-called oracles enables the integration of external data by first having smart contracts log and verifies the external data before integrating it into the blockchain based on plausibility checks or through certification.

Which approach is the “better” one for a company cannot be generally deduced. The question of purpose is in the foreground and can be filtered out via the process analysis described below.

3 Process Analysis

Whether and for which business process the integration of a blockchain solution is suitable cannot be answered in general terms. Rather, a variety of analyses are required in order to generate added value for the company with the integration.

In addition, the integration of a blockchain-based solution should not be an end in itself. It is becoming increasingly clear that blockchain technology can be understood as a linking technology that unfolds its full potential in combination with other digital technologies.¹⁷ The combination of blockchain technology with sensors/Internet of Things (IoT), artificial intelligence (AI), etc. appear very promising in this regard.

enough qualified available employees on the labour market. The digitalisation of processes promotes the amount of data, which is to be classified as an important capital of a company. Blockchain technology makes it possible to secure this and still be able to work cooperatively and collaboratively with other companies.

¹⁶ A hash value is a unique bit string that can convert any digital information into a hash value using hash functions. Complementary properties such as the “one-way property” extend the security mechanisms used in the context of blockchain technology.

¹⁷ Sandner/Gross/Richter (2020): <https://www.frontiersin.org/articles/10.3389/fbloc.2020.522600/full>, retrieved 03.03.22.

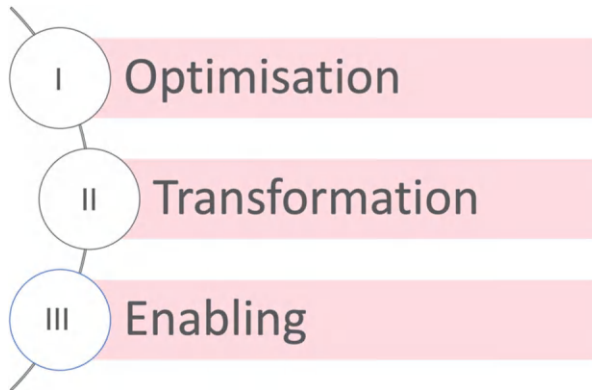


Fig. 1 Sub-aspects within the framework of the process analysis

International market research institutions estimate the potential to range from USD 39 billion (e.g. Statista¹⁸) to USD 69 billion (MarketsandMarkets¹⁹). It is thus no longer just a niche technology, but actually one that must be seriously considered by all companies.

Even if blockchain technology is initially a (special) type of database, the abundance of variations becomes clear at the latest when considering how and with what characteristics this database should be used in the company. In addition, critics may object that many existing blockchain infrastructures on the market are only poorly scalable, or if so, then quite expensive. But, waiting for the perfect solution causes a loss of knowledge building. In order to understand the potential of this technology in different scenarios, it is necessary to engage with the technology.

A structured approach helps in this challenge!

At the beginning of a discussion, it is important to visualise the trigger.

Who came up with the idea that implementing blockchain-based applications is a smart solution for the company and why?

What should be improved and for what reason?

By means of a process analysis, these questions can be answered at least on a higher level. It is also important to understand the internal “trigger”. Thus, the obvious purpose of a process analysis can be broken down into the following sub-aspects (Fig. 1):

All three sub-aspects can be understood as triggers for process analysis, whereby process optimization is to be understood as the improvement of existing processes. These can be already digitalized processes that are handled via bi- or multilateral

¹⁸ <https://www.statista.com/statistics/647231/worldwide-blockchain-technology-market-size/> retrieved 22.02.22.

¹⁹ https://www.marketsandmarkets.com/practices/blockchain.asp?gclid=EAIaIQobChMIgKeUks-T9gIVi813Ch3WxgOHEAAyAAEgJj1fD_BwE, retrieved 22.02.22.

interfaces and show potential for optimization, or they can be analogue processes that are digitalized and thus optimise the entire process.

The transformation of processes, e.g. from analogue to digital, is also a streamlining of process steps. This makes processes more efficient and, for example, requires no or only a reduced involvement of intermediaries.

As an enabler, the use of technology enables systems to carry out processes, transactions, etc. that were not technically feasible before. This may be due to a lack of hardware or insufficient infrastructure.²⁰

The reason for process analysis and thus the consideration of which service or process within a production should be looked at and improved is often economically and to a certain extent also organisationally motivated. In the following, some methods are outlined that are based on the author's reference book.²¹ The methods used form a combination of classic and agile project management and can thus be understood as a hybrid approach.

From this, the question can be derived as to how the procedural principles are to be determined. Whether from the general to the specific or the other way round must be clarified in the team. What do your processes look like in general, or do you pick a specific process and then analyse it further. In addition, the team should be clear whether they want to follow the top-down approach or actually the bottom-up approach to process analysis. Top-down in this context would be to act "from the rough to the detail"—this helps to get an overview first and not to immediately burden oneself with the detailed questions. The principle of bottom-up means that the individual tasks and work packages are integrated and summed up to produce an overall result that is characterised by a high degree of accuracy and planning reliability. By means of this method, there is a connection to the operative core processes and the possibility to develop models that can be implemented in the short term.

According to the division into the three sub-aspects listed above, a business performance must first be clear about what is to be achieved with a potential implementation of a blockchain solution:

Is it about improving existing (digital) processes, or about transforming analogue processes into digital processes, or are new business ideas and fields to be explored, thanks to new technological possibilities?

Perhaps it is also about the combination of the partial aspects. Only then must it also be determined by the management which aspect should contribute to the overall context and with which priority.

²⁰ Concepts for cryptocurrencies already emerged in the early 1990s of the last century, the technically secure implementation only took place in 2008 with the protocol of Satoshi Nakamoto, who succeeded in improving the previous approaches to the effect that no "double spending" of a digital unit is possible. Otherwise, it is easy to copy a digital date over and over again. Avoiding double spending allows bitcoin (and all other cryptocurrencies that follow this approach) to have intrinsic value.

²¹ Adam [8]: Blockchain-Technologie für Unternehmensprozesse.

Processes within companies or cross-organisational processes with other companies need special security mechanisms. Usually, security measures are characterised by system and media discontinuities. The result is a high degree of intransparency, fragmentation, data silos and lengthy process times. In order to reduce or even eliminate the effects of these consequences, the use of blockchain technology lends itself.

However, it is not “the” Blockchain-Technology, but a system with many variations. The following steps can help to achieve greater overall clarity from a business perspective.

This clarity will then enable management to formulate precise requirements for the technical design and, accordingly, the requirements for a potential set of specifications and requirements.²²

To approach this complex issue, it is advisable to assemble interdisciplinary teams in order to be able to integrate different aspects sufficiently well and to prevent “silo thinking”.²³

4 Stakeholder Analysis

Let’s assume that the interdisciplinary team has been found and now the actual work phase is to start.

We start with the stakeholder analysis. In the management literature, there are many approaches to defining stakeholders, as the term can be interpreted very broadly. In this context, a distinction should be made between internal and external persons or stakeholder groups.²⁴ The following definition should apply in the course of appropriate stakeholder management:

*Stakeholders are all those groups and individuals who can influence the achievement of the goals of an organisation or a project or who are influenced by the organisation or the project.*²⁵

²² In classic project management, a requirement specification is the description of the client’s requirements for the services to be provided within the scope of the project. These requirements are listed by a contractor in a performance specification. These documents serve as part of the contract between the contractor and the client.

²³ In an interdisciplinary team, the boss does not have to or should not necessarily take on the role of team leader. It could be to the detriment of creativity within the team. To find out who can best take on which role in the team in order to make the best use of the entire spectrum of knowledge available in the team. Finding out the appropriate role could be done according to a test by Belbin (<https://www.belbin.com/about/belbin-team-roles>). Belbin has divided the structure he created into three main categories, each of which covers three team roles (nine team roles in total). The test is based on a questionnaire and allows team members to be assigned to one of the nine roles. The wise combination of different team roles strengthens the team and the output.

²⁴ Jakoby [9]: p. 87.

²⁵ Freeman [10]: p. 27.

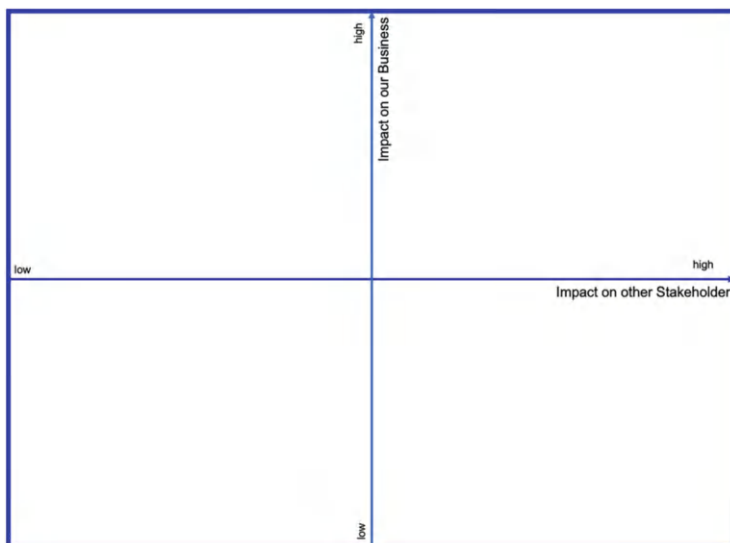


Fig. 2 Stakeholder map

Stakeholders influence the organisation as well as each other.

In order to be able to carry out this analysis in an open-ended way, it is advisable to first identify the stakeholders using the brainstorming method.

Note: Please keep in mind what is meant by brainstorming in general: ideas are to be collected that are not yet evaluated while writing them down! Hand out small pieces of paper in the team. Each team member should now write down all the stakeholders that come to his/her mind within a given time (e.g. 5 min). One stakeholder per note. After the time has elapsed, all the notes are compared and the duplicate mentions are eliminated.

The stakeholders found in the course of the brainstorming method can now be categorised via a stakeholder map. This stakeholder map is divided into four quadrants.

The top right quadrant shows the highest relevance, and the bottom left quadrant shows the lowest relevance for the project (Fig. 2).

The positioning of the stakeholders should first be done along an axis. For example, the vertical Y axis could be used first to determine an order from “high” to “low”. This order should be discussed in the team. As this order meets with agreement within the team, the second axis level can be developed. To do this, the stakeholders on the Y axis are now moved horizontally according to their impact.

The result will be a stakeholder map showing the most important stakeholders in the upper right quadrant. Once these stakeholders have been identified, it becomes easier to define the needs of these identified stakeholders. And already now one can see that a company may not be able to cover all needs. This raises the question of whether there is a need for a new alignment of business models, or whether enough

other important stakeholders can be satisfied with the generally existing business model. With the stakeholders listed in the upper right quadrant, one can discuss within the team which services and products are important for these stakeholders.

After the stakeholder analysis has been done, a service and product analysis could follow. Recording the current process is the starting point for further steps. What these might look like in detail can be seen in the authors' main work.²⁶

5 Utility Analysis

Utility analysis belongs to the family of analytical decision-making methods and therefore has a systematic approach to obtaining and evaluating information for a decision-making process. With the utility analysis, comprehensible and reproducible results are generated. These results are prioritised and thus also indicate alternative courses of action.²⁷

With the help of the utility analysis, we can therefore compare processes with each other and make a statement about which activity is the most suitable for the implementation of a blockchain-based solution. For this, the current process steps must be known as well as the desirable ones. Moreover, not every core process will deliver the same amount of added value through the use of a blockchain solution. However, with the help of the utility analysis, these processes can be weighted and thus compared with each other accordingly. The weighting must therefore be determined in the same way as the associated criteria. Here, too, it is advisable to determine the criteria in a team.

According to Table 1, the processes identified in the team are listed first. The drivers leading to these processes are derived from the partial aspects (see above): for example, the transformation from analogue to digital processes could be listed here. This goes hand in hand with the so-called routine processes, many of which are still carried out manually and therefore have a lot of potential for optimisation. For example, process steps along the supply chain can still be tracked manually and the goal could be to create a digital secure and automated solution here. Interface management is of high importance at the latest when software from different providers is used in the company. Here, the orchestration of interfaces is

Table 1 Definition of potential processes

Process No. 1	Tracking
Process No. 2	API
Process No. 3	Governance
Process No. 4	XXX
Process No. 5	XXX

²⁶ Adam [8]: p. 79.

²⁷ Kühnapfel [11]: p. 6.

Table 2 Criteria-determination

Criteria	Determination	3	2	1	0
Transparency	Review	All processes	Partly all processes	Internally	none
Tamper-Proof	Review	Very secure	Secure	Medium	not secure
Cost	Estimation	Low	Low-medium	Medium-high	high
Access Authorisation	Query	Read-only	Read & write	Read & write	Admin-rights
Velocity	Measurement	High	High - medium	Medium-low	Low

of elementary importance. Under the “governance” process, the use of blockchain technology, for example, makes it easier and faster to check the accuracy of the constantly increasing amounts of data. These are all just small examples to make the use of utility analysis comprehensible.

After the processes have been named, the next step is to determine the criteria. Here, too, it is advisable to first brainstorm the criteria in the team. Then the criteria found can be categorised: which of the criteria are an absolute must, which can be defined as should criteria and which are desirable? After you have done this categorisation, think about an evaluation scheme: not all defined criteria are equally important to achieve the overall benefit. In this case, it is a kind of “grading system” with the expression “zero” to “three”. And yes, even the weighting you now determine will always have a subjective character, but that is why you discuss it within the interdisciplinary team how to grade which criteria. Here, “zero” means the least favourable and “three” the best. From a pragmatic point of view, it is recommended not to complicate the evaluation of the criteria. The following table shows an example:

Subsequently, with the help of a preference matrix, the criteria can be weighed against each other, so that the criteria are ranked in percentages according to their importance. Using the above example (see Table 2), the preference matrix is shown in the figure below (Table 3).

According to the preference matrix, a first partial result is obtained. This is then underpinned with a weighting. Again, the request to discuss this weighting per criterion in the team in order to reach a consensus. (Author’s note: It is advisable to make a fairly robust division using increments of 5. Here in the example, one point = 5% and thus six points = 30%). Overall, this serves as orientation and the inherent subjectification can be broken up through discussion in the team).

Following this, both tables can now be combined and findings derived (Table 4):

Now the processes are interlocked with the corresponding criteria. For the evaluation of the processes, the already introduced “grading scheme” from “zero” to “three” is also introduced.

Table 3 Preference-matrix

Criteria	Transparency	Tamper-Proof	Cost	Access Authorisation	Velocity
Transparency		1	1	1	0
Tamper-Proof	1		0	1	0
Cost	2	2		1	1
Access Authorisation	1	1	1		0
Velocity	2	2	1	2	
Total	6	6	3	5	1
Measurement	30 %	30 %	15 %	25 %	5 %

Table 4 Utility analysis

Criteria	Tracking	API	Compliance	Measurement
Transparency	1	2	2	30%
Tamper-Proof	1	2	1	30%
Cost	2	2	2	15%
Access Authorisation	1	2	2	25%
Velocity	1	1	1	5%
Total	6	9	8	
Weighted Results	1,2	2,05	1,75	
Ranking	3	1	2	

If one now multiplies the weighting with the grade and adds up the values per process, then one receives a clear order and thus the indication of which process one should concentrate on first in order to improve (or expand) processes in the company using blockchain technology.

According to the example given here, interface management is the process with the highest vulnerability in the company. Accordingly, improving this process will fundamentally reform the value creation in the company.

6 Prerequisites/Requirements (the Next Level)

Let's assume you have carried out all the analyses necessary for your business and have arrived at usable findings in your interdisciplinary team. You can make a solid assessment of the potential that your chosen solution approach offers.

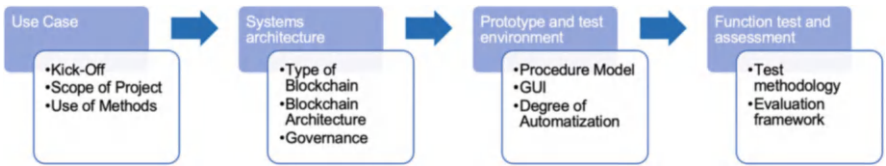


Fig. 3 Processual development of a blockchain solution

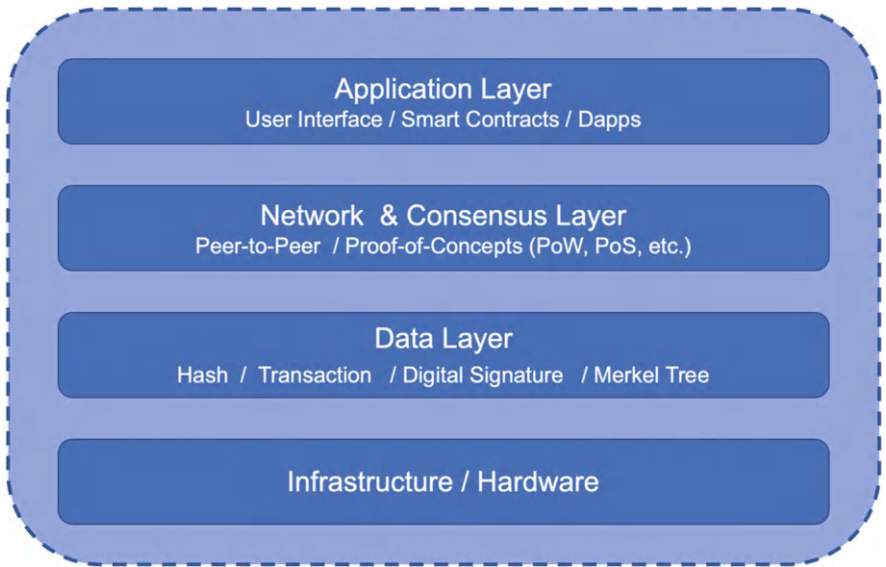


Fig. 4 Blockchain architecture—example

The next step is to design the so-called system architecture to define the best blockchain model for your use case. This includes the technical requirements analysis, the detailed analysis of the existing system landscape and the modelling of the system architecture. Based on this concept, the so-called mock-ups (partially functional imitations of the user systems) can then be simulated. In addition, the design of the graphical user interface can be determined in a further team discussion. All of this leads to the development of a prototype in the next step by comparing it with the previously determined technical requirements. These requirements can be described very well with the help of a so-called requirements specification. Here, it is necessary to divide into functional and non-functional requirements that are placed on the technological solution. In this way, decisions are also made about the automation of certain workflows and decisions regarding the use of smart contracts. Prototypes can be developed and tested on the basis of these decisions (Fig. 3).

In addition, a distinction must be made between the different levels and the blockchain architecture (Fig. 4).

Depending on the infrastructure provider with which a solution is to be implemented, different characteristics will be found on the infrastructure layer.²⁸

For example, the Ethereum blockchain, which is a widely used open-source blockchain solution that is continuously being developed by a large and active community, can be used to show how the different layers are interwoven.

The data layer reflects how the transactions generated in a blockchain system are arranged and stored in the peer-to-peer network. Transactions made within the network are bundled into blocks and then, after validation, represented as a linked list of blocks. Each block contains a hash with the information of the previous block as well as a timestamp, a nonce, the version number of the block and the current difficulty level. Transactions are digitally signed on the blockchain with a private key. Anyone in possession of the public key can therefore verify the signer. Digital signatures can be used to prevent the manipulation of transactions.

In order to enable communication between the nodes, the P2P network is required on the network layer. Within the architecture, this layer ensures that the nodes in the network discover each other, communicate with each other and synchronise the content in order to keep the blockchain up to date. The way in which the content is synchronised depends on the consensus mechanism selected by a corresponding network.

Finally, there is the so-called application layer, which comprises all the applications that users have to use to interact with the blockchain solution. This layer consists of scripts, APIs, user interfaces and frameworks. Thus, the Blockchain-Network belongs to the so-called backend system and this system is connected to the application layer via APIs.

To dovetail these different layers in a meaningful way and at the same time to define which requirements are to be fulfilled and how, can be presented via a so-called requirements specification. (For example, in order to meet the requirements for public and decentralised storage, it may not always be the best idea to store all content on the blockchain itself. The disadvantage of this is that the scalability of the blockchain is affected by this storage of all content and the size of the blockchain grows faster. In addition, it is also more expensive.²⁹ (According to Schönhals: “However, on-chain storage is de facto not applicable due to the high costs and

²⁸ Blockchain infrastructures can be roughly differentiated according to the use or non-use of cryptocurrencies. Both Bitcoin and the Ethereum blockchain use cryptocurrencies to pay for storing transactions and executing smart contracts. Hyperledger Fabric, on the other hand, is an example of a blockchain platform that does not require cryptocurrency for its basic functionality. However, through suitable smart contracts, the concept of a cryptocurrency can also be fulfilled on Hyperledger.

²⁹ Without having to forego the advantages of decentralised data storage, it is possible with the so-called InterPlanetary File System (IPFS) to operate a node that only stores and makes available selected content. If the contents are stored off-chain on IPFS, they can always be retrieved directly from IPFS nodes with the hash. However, IPFS nodes do not store all content forever. When using IPFS, one depends on there always being at least one IPFS node that makes the text available on a voluntary basis. In the meantime, there are projects that make it possible to pay IPFS nodes to store a file in order to ensure that it is available for a certain period of time or permanently.

the limited block size. Data can also be managed off-chain, which is significantly cheaper and more efficient”³⁰) Well-formulated specifications create the necessary clarity to ultimately obtain a technical implementation that can actually be used.

7 The Concept of DAOs (Decentralised Autonomous Organisation)

Another important concept that can be realised with blockchain is the creation of a decentralised organisation that autonomously executes decisions based on smart contracts. Hierarchical levels for approving decisions are obsolete. This approach makes it possible to establish a new approach to how work and, above all, collaboration is regulated. Interestingly, a DAO cannot be legally registered as a company because a DAO has no managers or directors.³¹ The Ethereum Foundation already had its first, albeit only partially successful, experiences with this concept in 2016.³² Within only 28 days, the DAO collected around 150 million USD from 11,000 different participants who wanted to vote according to their established rules on which project should receive how much of the collected money as an investment.³³

But why could this approach be interesting for companies? The way business models will be built in the future is currently changing not only because of the ongoing transformation into digitalisation.³⁴ We see that, for example, geopolitical events require new cooperation. This makes the coordination of joint service provision, the joint financing of assets and the use of joint resources even more strategically important than before. Here, the blockchain-based approach via a decentralised autonomous form of organisation promises a new opportunity for cooperation because this type of organisation can make tamper-proof decisions.

Blockchain technology can thus also be understood as an institutional technology that enables value-creating cooperation.³⁵

Typically, the cooperation of individuals and or companies is determined by governance: rules and (corporate) nomenclature are established, taking into account the general legal situation, in order to bind formal and even informal networks to these rules. However, asymmetrical power relations cannot be ruled out. The

³⁰ Schoenhals et al. (2019): p. 4679.

³¹ Sims [12], p. 1.

³² DAO/Ethereum: in this context, people talk about “The DAO” because it was the first and at the same time the one that spectacularly illustrated that errors and gaps in software programming cause great damage. Regardless of the fact that “The DAO” was never successfully realised, however, the concept shows the potential possibilities.

³³ Metha et al. [13], p. 117.

³⁴ Tapscott & Tapscott [14], p. 91.

³⁵ Argus/Samson [15], p. 16.

literature describes this in the course of the Principal-Agent-Theory.³⁶ With the help of blockchain technology, systems (companies) and institutions can now network decentrally as organisations worldwide and design work processes more efficiently, as an intermediary is not necessary for orchestration. The “orchestration” of this newly created organisation is based on the rules that the members have given themselves. Thus, voting and agreements can take place via the programmed set of rules. For this purpose, smart contracts are programmed that are operated via a blockchain infrastructure to enable immutability as well as autonomous operation of the smart contracts.

It is indisputable that every organisation requires a set of rules according to which the organisation is organised. A good analogy is provided by classic board games such as “Monopoly” or “Ludo”. The rules must be defined before the game begins and then followed (e.g. how many times one may roll the dice to start the game with a “six”, etc.).

In a board game, participants monitor each other to ensure that the rules are followed. In the case of the DAO, smart contracts take over this set of rules in the form of an immutable software code or “if-then” sequence³⁷ based on a consensus reached between the members.³⁸ Members of this DAO are issued tokens that they can use to submit proposals within this DAO, participate in voting, and share in the tasks and success of the DAO. These tokens are sold by the community of DAO founders as a reward in exchange for capital or resources, whereby the DAO community determines the purpose for which a token can be acquired (i.e. purchased) or received. Just like the purpose, the number of tokens to be issued can be determined at the beginning. In the case of the Bitcoin blockchain, the coins (as tokens) are limited to 21 million.

The purpose of the tokens regulates what rights such a token contains. For example, the token can be a placeholder for the right to a share of the profits. Or, the token makes it possible to participate in certain decision-making processes within the DAO. The design is diverse and should be based on what this DAO pursues as its “raison d’être”.³⁹

According to Voshmgir, the Bitcoin network is the first true decentralised, autonomous organisation coordinated through a consensus mechanism by members of the Bitcoin blockchain.⁴⁰ Systems of this kind, operated through blockchain infrastructure and code-based smart contracts, can be understood as open source

³⁶ Dillerup/Stoi [16], S. 23.

³⁷ Smart contracts are computer programmes that use an “if-then logic” to ensure that recorded agreements between people and organisations are secured, enforced, and settled. For this purpose, smart contracts require a blockchain for execution.

³⁸ DAOs are online platforms that not only provide convenient access to content for their members, but at the same time manage the collaboration of members by having a programmed (or a whole set of smart contracts) governance contracts.

³⁹ Sims [12], p. 21.

⁴⁰ Voshmgir [17], p. 105.

organisations that enable collaboration in what is known as a peer-to-peer approach. The wisdom and knowledge of the individual can be put to much greater use and thus strengthen the community, i.e. the organisation.

8 Conclusion

Dealing with the topic of Blockchain-Technology cannot and must not be reserved “only” for IT specialists or a few technology-savvy people in the company. However, in order to use this technology strategically and operationally in a company, one needs diverse and interdisciplinary knowledge. Knowledge about the technical necessities and technical requirements as well as knowledge about the business and process relations in the company must be made accessible.

With all the enthusiasm about new technologies and here about Blockchain-Technology, however, it must not be forgotten that the use of this technology should ultimately create added value for the company. This could be, for example, the digitalisation of non-automated processes as well as the perception of new cooperation opportunities with other market participants. The so-called stakeholder analysis can be helpful in making an initial assessment and discovering potential.

Use cases can be filtered out within the framework of a process analysis. For this purpose, it is important to define both a goal (i.e. what is to be achieved) and with which measures the goal can be achieved.

The use of blockchain technology creates the possibility for companies to transfer values, data and information in a secure and traceable way. There is no longer a need for a large platform provider. New business models will develop through the use of blockchain technology. For example, proof of origin of raw materials, documents and certificates of authenticity, proof of work and identity in inspection processes, or even in accounting, many processes can be improved using this technology. The examples listed here represent an incomplete list.

Once the impulse has been given in the company to deal with this technology, an interdisciplinary team is needed to filter out both the use case and the added value. Blockchain technology is not an end in itself, but with the constantly increasing amount of data due to the digitalisation transformation, necessities arise that justify the use of this technology despite critical voices.⁴¹ However, the goal must be when using this blockchain technology:

- Better value creation
- Focused use of resources and concentration on what is relevant
- Flexibility in the choice of infrastructure
- Choice between hybrid storage of data

⁴¹ Criticisms include the high energy consumption, the lack of scalability, the lack of interoperability, the long waiting times for confirmation of a transaction and the sometimes very high costs for carrying out a transaction.

- High security standards in real operations
- Fast results

Even if what is technically feasible may not correspond 100% to the ideas, it is absolutely necessary to deal with this technology in order not to lose touch with modern corporate management with all the diverse prerequisites and to preserve the company's own opportunities in the market.

In addition, blockchain technology can be used to set up a decentralised, autonomous organisation that can organise processes and workflows in the company in a completely new way as an online platform without intermediaries. It makes it possible for participants to have a say because they have a say in consensus-building, i.e. how decisions are made within this DAO. This may seem scary at first when looking at these concepts from the world of hierarchically organised companies. But the world of work is changing! Tomorrow's companies are organised differently than today's are. In order to work efficiently in the company, to be able to retain talent and to carry out meaningful work, routine work will have to be delegated more and more to bots. Control will be redefined and assigned. Market-dominating platform monopolies are already under strong pressure today (e.g. latest European digital law ⁴²)—DAOs can set up a counterweight, as it is not controlled and dominated by an operator of a digital marketplace, but by the participants participating in it.

Additionally, anything beyond routine work will generate added value for the company—the question remains how to organise this. From the authors' point of view, the use of blockchain technology makes it possible to answer this question.

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The Implication of DLT and Blockchain: Legal and Financial Aspects



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Abstract As a representative of distributed ledger technology (DLT), blockchain holds enormous potential to reform cyberspace architecture by revolutionizing information storage, circulation, and exchange through decentralization, transparency, and de-identification. This blockchain-driven information revolution has significant implications for legal and financial frameworks, providing tools for brokerage, finance, legal enforcement, and regulation. Ordinary participants can become traders, miners, retailers, and customers simultaneously, breaking the market barriers and reducing the information gap within the community, as all records are visible to all participants.

This study explores the implications of the blockchain-driven information revolution from legal and financial perspectives. It first investigates the legal implications of blockchain, examining the notion of trust and a new understanding of electronic evidence and the free flow of data. The study then uses blockchain in finance as a case study to illustrate its practical applications, discussing emerging financial models, corporate governance, sharing mechanisms, and the influence of blockchain on central banks and digital currencies. The research concludes that blockchain is

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much more than a technology; it represents a community, acting as a source of trust and a new architecture for cyberspace.

1 Introduction

The past few years have ushered in an enormous shift of digitization in our daily life, transforming the ways in which individuals communicate, transact, and establish trust through information, while fundamentally reshaping economic and legal infrastructures [1]. With the emerging AI Agent and Large Language Model, trust among individuals and society is more fragile than ever [2]. However, with the advent of Distributed Ledger Technologies (DLTs) [3], represented by blockchain, it provides us with a cryptography-based trust mechanism. Due to its unique framework and mechanism, blockchain reforms cyberspace and transforms the storage, circulation, and exchange of information through decentralization, transparency, and pseudonymization [4]. These features deeply influence society from a centralized social world to a brave decentralized world. Such a decentralization mechanism through technology shakes the foundation of society, i.e., trust, where all the trust is secured by every one of us rather than any authority we know from reality. Therefore, the information transformation empowered by blockchain also has a profound impact on our legal and financial development in cyberspace [5].

The first shift enabled by blockchain is about the regulatory paradigm of legislations in cyberspace [6–10]. Because of the decentralization of the blockchain framework, trust among individuals and the architecture of cyberspace are all reestablished through cryptography, which alters the foundation of legislation in cyberspace [6]. Accordingly, there has been a corresponding shift in the subject of legal regulation (from the single regulatory body to multiple regulatory bodies) and changes in the approaches and methods to legal research.¹ Blockchain refreshes and integrates many legal notions (e.g., trust, transparency, and evidence), which brings the revolution to the legal practice (e.g., electronic evidence and as a data repository to boost data protection and data free flow) [11].

Besides that, the blockchain-enabled information revolution also enables many new applications in the business and financial area [3, 4]. As the market for Bitcoin—the forerunner of blockchain applications—has continued to grow at a formidable pace since its genesis block was mined in 2008 [12], it has transformed the financial market and posed a significant challenge to traditional centralized systems. However, Bitcoin does not unleash the full power of blockchain, as the blockchain means so much more than just cryptocurrency [4], which was already an earth-breaking application from the decentralization idea. Meanwhile, in the area

¹ Legal regulations have changed from simply relying on legislation mechanism to multiple regulatory approaches. For example, legislators have to consider the function and implications of code in cyberspace and how code can be used as a tool to supplement legislation. It is also called “code as law.” This part has been further discussed in Sect. 3.1.

of business and finance, blockchain brings more revolutionary changes [13]. For example, smart contract powered by blockchain could improve the efficiency of the financial market and optimize corporate governance. Moreover, the fact that various central banks are establishing digital currencies further demonstrates the influence of blockchain on the financial sector [14]. However, it is also worth noting that the application of cryptocurrency could bring new risks from cybercrime and money laundering perspectives. Therefore, more stringent financial regulation is required.

This chapter focuses on the main research question: **what are the implications of blockchain-driven information revolution from legal and financial perspectives?** The first contribution of this chapter is to define the blockchain and conclude its feature from a social sciences perspective. Secondly, through the analysis of the impact of blockchain on law and finance, this chapter contributes providing a more comprehensive analysis of the blockchain-driven impact on society and social sciences to current scholarships.

The chapter is structured as follows: A brief introduction to the definition of blockchain and its features that could have a great impact on society and social sciences is presented. Then the chapter investigates the legal implications from blockchain-driven information revolution from methodological impact (e.g., code as law) to theoretical (e.g., impact to trust notion) and practical impact (electronic evidence and data repository to boost data free flow). In addition to law, the financial and economic envisioning of blockchain is discussed from corporate governance, central bank, and digital currencies (CBDC) and anti-money laundering (AML).

2 The Definition of Blockchain and Its Features from the Social Sciences Perspective

As for the terms “blockchain,” it is an old concept of chaining data together proposed from the mid-1980s [13]. Now it appears to be the pillar of booming cryptocurrency technology, as it enables a trustless, decentralized, and tamper-proof ledger [12]. As its name suggested, the data is stored in the form factor of blocks [12] and then chained by the previous block’s hash to ensure data integrity and security. Hence the later block can locate itself from the current block’s hash value, which is a unique identifier based on the content in the block. In most cases, such distributed manner and consistency of blockchain is decided by consensus protocols, a.k.a., consensus algorithms [15], under which the formation of data processing and storage is embedded when each node runs.

Unlike the traditional centralized system, trust in the blockchain system is established through cryptography [12], and the blockchain can be roughly divided into public chains and private chains, based on their access settings. In the case of public chain, anyone is free to join and participate in the core activities of the blockchain network. Therefore, in the context of the public chain, blockchain allows any user to store a local copy of the blockchain and propose new blocks, which lay down the foundation for a trustless environment.

Another most appealing aspect of blockchain technology is the degree of transparency that it can provide [13]. From the design philosophy, blockchain is designed as a transparent framework in which anyone can join the network (especially in public chain) and, therefore, is able to see all information on the chain. In the mechanism of DLT, data is grouped into each block and bonded with the existing blocks through hash check and consensus algorithm. The data on the chain includes different metadata, for instance, the hash of all transactions contained in the block, a timestamp, and a hash of the previous block. Such a hash-based cryptographic chain makes the records traceable and appended-only characteristics and increases transparency and accountability. Through the traceability of information, blockchain could safeguard the transparency that nothing is unduly modified without recording the changes, and, as a result, everything is transparent.

Additionally, blockchain could enable self-tamper-proof and self-authentication applications [12, 13]. Due to every block being linked and verified through the hash algorithm, data integrity and security can be guaranteed. If it is necessary, the blockchain network could trace back to the history transaction without any third parties. Therefore, blockchain-based applications can achieve self-tamper-proof and self-authentication.

With these features, blockchain has shaped the recording of humanity, with a new world that no records can be easily altered, erased, or disguised in every single transaction of blockchain participants. In the traditional thinking of power, information asymmetry can be regarded as one of the most important sources of power [16], which people are classified based on their disclosure of information. The blockchain-driven information revolution has transformed it permanently, as the transparency opens up all possibilities for ordinary participants to outlook the society with no gaps or asymmetry, stepping forward from the paper-based medium, and beyond the scope of the Internet, where the information is still held by a minority of users [17]. The origin of the panoptic tower, which was first proposed as a prison model [18, 19], provides analytical thinking of the utopia that everyone is watched by a surreal tower that knows all secrets of the member, which, in fact, is not far away from daily political practice in a surveillance state. Individuals can also be regulated by the blockchain, just like any centralized government. The panoptic tower is represented by the governors of blockchain, which see through the identities of blockchain network users and enforce the order based on their data. The blockchain has been given the instrumental way of power in terms of philosophy. The panoptic model can be resurrected to the reality, essentially helping develop the sophisticated cryptographically safe panoptical instrument for everyone without tampering the much concerned privacy, human rights, and freedom [20].

A panoptical instrument for personal behaviors is the tool to reach a society with better order, discipline, honesty, and reality. During the progress of recording more behaviors on the blockchain, for instance, the daily transactions and interactions with civil services, commercial transactions, juries' records, records of lawsuits, healthcare records, etc., the value carried by blockchain increases, and the enforcement of social order can be greatly eased. The transparency of personal data in certain fields related to the governor provides an opportunity to enrich the concept

of surveillance [16], discipline, and punishment [18]. Meanwhile, being gazed by authority, individuals are motivated to self-discipline and behave responsively to society and community, embracing the new reality of transparent society. These traits empower blockchain with a significant function for data recording, altering the structure of information storage and circulation and having a profound effect on the whole society. The next section analyzes this effect in depth from a legal standpoint.

3 The Impact from Blockchain to Law

The advent of blockchain alters the cyber framework and transforms information storage, circulation, and exchange through decentralization, transparency, and pseudonymization. This information transformation has accordingly posed new opportunities and challenges to the way that law normally regulates online information-related areas. This section will discuss the key question: what impact does the blockchain-enabled information revolution have on the legal sector? It will analyze the theoretical and practical impact of blockchain on law, including the cyber law theory, the foundation of trust, blockchain-based electronic evidence, and blockchain-based data repository.

3.1 From Code Is Law to Code as Law: Blockchain as an Alternative Architecture

As Lawrence Lessig put forward in his book “Code: And Other Laws of Cyberspace,” the regulation of the Internet is not only about the legislation but also includes social norms, market forces, and particularly the architecture, which is so-called pathetic dot theory [21]. In the pre-Internet era, the world was normally regulated and affected by four main factors: Legislations are enacted by the government to regulate behaviors directly, social norms are shaped by community gradually, and market regulates rational individuals and architecture subtly influences behavior [21]. This phenomenon has been deepened in the Internet age. Online giants, as an intermediary, has directly influenced users’ behavior. They can set rules, code of conduct, and even enforce penalties. This is because the architecture—as a coder that shapes both physical and digital worlds—can easily influence users’ behavior and choices by default.

The emergence of blockchain technology presents new opportunities and challenges [22, 23]. On the one hand, the blockchain-based mechanism can increase the legal regulation by default, which means that code can be a kind of manner to achieve legal objectives. On the other hand, the blockchain-based legal system causes certain problems as traditionally, and the law is woven by natural language

rather than machine-readable code. It leads to the issues of how to translate the legal language to machine-readable code, which contains the spirit of the law in a less flexible context [6].

From Lessig's theory, the answer to the machine is the machine. In other words, "the best way to regulate a code-based system is through code itself" [6]. In the context of the blockchain-based legal system, it can indeed set up a new framework of code-based regulation to regulate people, companies, and machines. Through blockchain technology and relevant smart contract, legal rules can be interpreted simply as machine-readable rules that could be automatically executed by architecture. To this extent, code-based technical rules could achieve the legal function and play the same role as legislation [10]. By the same token, computer code not only can be a complement to law but also can be another form of law in cyberspace. Filippi and Hassan describe this tendency as from "code is law"² to "code as law"³ [7].

3.2 Blockchain as a Complement to Law in Building Trust

In the traditional world, trust among individuals, entities, and government is normally established and maintained by law [24]. Through legal protection and obligation of trustor's rights and assets, trust can be established [8, 24, 25]. If we delve into why the law is respected, one of the important reasons is that law is endorsed and represented by the trust and trustworthiness of the state. Citizens can reach a consensus in democratic mechanisms to form legislations, and the violation will be punished. However, after the emergence of the Generative AI, it has brought the complicated trust dilemma again, especially in the online environment between citizens, government, large private entities, and emerging technologies [26]. People may always ask similar questions: can we trust the online news regarding the pandemic? Can we trust the man who said he is vaccinated and does not wear facial masks? Can we trust the COVID contact tracing app that does not leak users' sensitive data to third parties, although the app is developed by the government? Can we trust telecommunication technology when we are working at home? Those are all emerging but long-standing issues. Generally, if people want to trust something or someone, there should be multiple trust sources to prove one fact [8]. People, for example, can listen to the public service broadcaster, scientists, friends, or online gossip networks for information on vaccines and masks. However, the Internet and digital technologies have broken this balance.

² "Code is Law" is developed from Lessig's theory, which refers to using code to implement specific rules into technology.

³ "Code as Law" refers to relying on the technology itself to both define and implement mandatory laws.

On the Internet, multiple news sources are emerging every day, but users cannot distinguish the truth. Many transactions are conducted without authenticity. Sztompka and Balázs argue that, prior to the Internet era, people primarily relied on communal trust, public trust, and private trust to establish social cohesion and reliability [27]. For example, the familiarity that consists of ethnic or religious relations, the community with shared location or shared past, is becoming unreliable to establish the communal trust [8]. By the same token, public trust infrastructures such as public education, public service media, and public press are challenged and undermined by a flow of online key opinion leaders (KOLs) and online news platforms.

Regarding the private trust infrastructures, it often establishes the trust through their services or commodities on the market. The independent entities, for example, banks, insurance companies, and other independent companies, establish trust for a service fee. However, with the emergence of Internet giants, few oligarchies occurred. Because of the prominence and monopoly in the industry, if the trust could be established remains doubtful [28].⁴ However, blockchain and similar systems create a restrictive and prescriptive technical environment that establishes trust through embedded traceability.

Without the endorsement of a third party or state, blockchain can also establish trust among individuals. Such trust is not built from legislation but from the cryptography and architecture of blockchain. With such cryptography-based trust, blockchain can act as a complement to law to establish trust that is embedded in cyberspace by design. Blockchain allows users to store a local copy of the blockchain and propose new blocks, which lay down the foundation for a trustless environment. In contrast, nowadays, trust is built in the trusted third party. In the physical world, trust is realized through personal reputation, social norms, or government endorsement, for instance, the legislations and contract [25]. For example, the reason why clients trust their bank is that their bank as a legitimate organization is not worthwhile to harm clients' interests [8]. More importantly, such trust is also built partially on the previous experience and mostly based on the recourses when something bad happens. Such recourses include immediate response and guidance, regulatory intervention, and private legal action. However, in the circumstance where both parties are strangers, legal compliance would generate expensive transaction costs from a law and financial perspective. Similarly, users need an online intermediary to establish trust to trade in cyberspace. However, such centralized architecture causes various issues regarding the online gatekeepers, including data privacy, anti-competition, etc. [7]. By contrast, blockchain can overturn the traditional architecture and redefine online trust by design [29]. This trustless environment can be established through three overlapping dimensions in that blockchain: users, nodes, and miners.

In the permissionless blockchain, for example, the Bitcoin system can allow anyone to generate a public private key pair and a Bitcoin address through open-

⁴ For example, Facebook users receive losses due to massive personal data breaches.

source software [12]. The software broadcasts the users' transactions over the network, and miners will incorporate the transactions into blocks. Meanwhile, anyone, including users themselves, can be a node via relevant software and store the blockchain archive. Nodes store a local copy of the whole chain. Full nodes store a full copy of the chain, and the light nodes will only store partial records in order to verify transactions. Nodes will be connected and maintained with each other through a P2P network. It also checks the previous transactions to ensure the current transaction is valid. After the new block is valid, the node will add it into the copy of that chain and broadcast it to the whole network. Through this mechanism, the whole procedure of the transaction will be recorded in a decentralized manner. Miner works to distribute the transactions into a block and broadcast blocks across the whole chain while rewarding with the new cryptocurrency, for instance, Bitcoin. Of course, such distributed manner and consistency of blockchain is decided by consensus protocols under which the rules are embedded when each node runs.

Because the records on the blockchain cannot be duplicated, manipulated, or faked, it increased visibility in every user. Such immutable and timestamped records of transactions endow blockchain with the ability to promote trust through each other because of the "endorsement of code." Therefore, blockchain can establish trust without third parties in cyberspace since all transactions are verified and recorded across the whole network.

3.3 Blockchain-Based Electronic Evidence: New Understanding of Evidence in Cyberspace

Blockchain-based electronic evidence is the concrete manifestation of the trust system built by blockchain in cyberspace [30]. With the burgeoning development of Internet industries, traditional forms of evidence and forensic methods have been challenged, and electronic evidence (e-evidence) is gradually being used and admitted [31]. In order to be legally recognized, the e-evidence normally needs to satisfy three conditions: objectivity, relevance, and legality [31]. The objectivity indicates the independent and objective existence of the evidence without subjective change. The relevance means the existence of an objective link between the evidence and the facts to be proved. The legality means that the collection and use of the evidence must be in accordance with legal procedures. However, because e-evidence is composed of digital data, it is more vulnerable to tampering, deletion, and instability compared to traditional forms of evidence [30]. Moreover, most e-evidences are time-sensitive, and without a quick and easy way to lawfully obtain and preserve them, it will be significantly more difficult to recover the evidence.

Currently, there are three ways to obtain and store e-evidence lawfully. Firstly, the e-evidence is obtained and stored by the individual itself or representative by taking photos, downloading, or taking screenshots [32]. It is easy and inexpensive to operate, but the evidentiary capacity and probative force are weak and less likely

to be admissible in court. Therefore, e-evidence obtained by taking screenshots on individuals is insecure and may lead to inadmissibility and even the risk of losing the case.

Secondly, the e-evidence can be obtained through notarization [32]. It means that the notary will certify the authenticity and legality of civil actions, facts, and documents in accordance with the statutory procedures upon application by a natural person, legal person, or other organization. Due to the high credibility of the notary, it largely solves the problem of the authenticity and legitimacy of e-evidence and has become one of the most common ways to store e-evidence in legal practice. However, there are also some drawbacks. For instance, it requires a high time cost. In the process of notarization, it is necessary to make an appointment for notarization, commission a notary, go to the notary office, communicate with the notary about the details of the operation, etc., which all cost time, and due to the easy tampering of e-evidence, if there is a long wait, it is likely to lead to the loss of evidence [32]. Apart from that, the cost of notarization might be high. For example, in a copyright infringement dispute, the amount of compensation for an image may only be a few hundred dollars, but the cost of e-evidence forensics will require from a thousand dollars, and the high cost of forensics certainly makes it more difficult to defend individuals' rights.

Thirdly, the e-evidence can be verified through timestamp technology to ensure authenticity. For example, in Article 11 of Chinese Provisions of the Supreme People's Court on Several Issues Concerning Trial of Cases by the Internet Courts 2018 [33], it recognizes that the Internet Court shall confirm the authenticity of electronic data submitted by the parties, if it can be proved by technical means of evidence collection, such as electronic signature, trusted timestamp, hash value verification, blockchain, etc. Currently, the common timestamp is that the service provider issues a certificate to prove the authenticity and integrity of e-evidence without access to its content [30].

Although the timestamp is quite cheaper than the notarization, there are many uncertainties and drawbacks. Firstly, the validity and admissibility are inferior to that of the notarization. This largely depends on the discretion of the judge to decide whether such e-evidence can be accepted. Additionally, if there is any inconsistency between the notarized evidence and the timestamped evidence, the judge is more likely to accept the notarized evidence. Secondly, the scope of the use of the timestamp is limited. For example, web evidence forensics requires the following requirements need to be met: (1) The website must be opened by PC, (2) the web content must be static, and (3) there is no requirement to log in.

Overall, there is currently no unified approach that can be applied across all scenarios and circumstances. In the face of the dilemma of e-evidence's authenticity and integrity, because of the characteristics of blockchain, it can effectively eliminate the vulnerability of e-evidence to tamper and greatly reduce the necessity for judges to have relevant technological knowledge. Because blockchains are append-only digital databases where information can be added to ensure the integrity of data [29]. One good example of the Non-fungible Token (NFT) has come into the practice of copyright and has been accepted as a major reference and copyright

certificate for the ownership of the data and content, which effectively prevents hacking and other activities that may destroy evidence [34]. Unlike the traditional complex and costly copyright legal regime [35], NFT can enable digital asset trading accompanied by the deed of copyright license. Therefore, blockchain inherently makes e-evidence self-tamper-proof and self-authentication endorsement to ensure its authenticity and integrity [30].

Moreover, blockchain not only can serve as a repository to store e-evidence but also can act as a platform to generate e-evidence, store e-evidence, and notarization of e-evidence [32]. Such a combination of generating, preserving, and notarization platform is a better reflection of the trust system established by blockchain. For example, the Chinese Hangzhou Internet Court has already employed such a system where a consortium blockchain-based system connects courts, notarial agencies, forensic institutions, and all users [30]. The application can also be used for daily transaction services for enterprises or individual users. When a dispute occurs, the enterprises or individual users can directly use the application to show the e-evidence with a notarial certification. Therefore, it can avoid the issue of authenticity and integrity essentially.

Nonetheless, blockchain-based e-evidence systems present challenges to the legal framework [31], as not all jurisdictions currently recognize e-evidence generated and stored via blockchain technology. It may lead to the new legal amendment of current evidence forensics laws in the EU or China. Thus, the law must keep pace with technological developments to ensure that, on the one hand, the law can still regulate technology and, on the other hand, technology could more easily achieve legal objectives. Meanwhile, during the design stage of blockchain-based evidence systems, it also requires the combined efforts of technical staff and legal practitioners to ensure that blockchain e-evidence is also fully compliant with the requirements of legal practice.

3.4 Blockchain as a Tool to Boost Data Free Flow

One of the legal objectives of GDPR is that it shall boost the personal data free flow. Just like oil, the value of data lies in its free flow and use. Therefore, many jurisdictions are dedicated to establishing such digital data marketplace where personal and non-personal data can be shared, traded, and free flow. For example, European Commission considers that the data marketplace can be used to unlock the value of data for the Digital Single Market and also render the EU more competitive in the context of artificial intelligence [36]. Similarly, in China, the data trade marketplaces are booming. Many data marketplaces, for example, Shanghai Data Exchange Centre [37] and Guiyang Global Big Data Exchange Centre [38], are taking shape. Data marketplace powered by blockchain is able to offer users confidentiality and convenience for data sharing by listing their data on the marketplace [39]. Potential buyers are eligible for using the data with authentication records from data provider when the trade has been secured by the marketplace [40].

Blockchain as a digital database that is shared and synchronized, maintained through a consensus algorithm and stored on multiple nodes, can offer considerable advantages to build up such data trade marketplace while providing extra protection to data privacy. Firstly, blockchain is a decentralized framework, which means that the data exchange can be operated without a central trusted intermediary. Because blockchain could enable the zero-trust mechanism [41], it can be used for the transaction of data and make data free flow [42]. Such a mechanism can allow participants to provide proof of a statement without disclosing identity information [43]. For example, the blockchain-based system can reveal whether a transaction has occurred without revealing the public key and the value of the transaction.

Furthermore, the blockchain-based data governance tool can establish a smart contract, which can automate the data sharing, and also reduce transaction costs [40]. With the support of blockchain technology, it is apparent that the data marketplace is easier and safer to build up and stimulate data sharing and the digital economy. Meanwhile, such a system can also be used to achieve legal objectives of users' data rights.

However, it is worth noting that blockchain and DLT must never accomplish these goals automatically [9]. Rather, they must be intentionally designed through interdisciplinary collaborations between engineers and lawyers. Otherwise, there are many potential conflicts between blockchain and law.

4 Implications of Blockchain from a Financial Prospect

As one of the largest industries for blockchain applications, the financial sector is particularly noteworthy. Blockchain brings a new revolution in information processing, storage, and exchange, enabling many new applications in the economic and financial industry. For example, blockchain in the finance regime has been widely accepted in the form of cryptocurrencies. However, blockchain has more potential that goes beyond the scope the cryptocurrencies. It has reshaped the world with its encrypted nature and opens up the opportunity for every individual with the power to mine or mint tokens that can be used in every possible way. Tokens are the essential carrier of the value of blockchain. It makes sense when people pass on tokens to perform transactions on the blockchain. It brings the breakthrough of the distribution system, where the individual has the total control of its earnings and earns extra fees when serving the network. Since the distribution of tokens is direct and transparent, it equalizes the roles of participants and creates the unique jurisdiction of the legal person in the blockchain aspect, where the actual man is mapped with an encrypted identity used for the legal person. In this case, blockchain has more potential in the economic and financial industry. For instance, the blockchain-based application could contribute to corporate and community governance and collaborative production without revealing the roles of participants. Such a framework breaks the market barriers and reduces the

information gap, bringing new opportunities and challenges. Cryptocurrency and anti-money laundering, as a typical representative, is worth noting. Such blockchain applications can pose potential risks and be detrimental to society.

4.1 Blockchain and Corporate Governance

Smart contract is the critical technology to improve the efficiency of financial markets.⁵ Simple extensions enable blockchain to keep self-executing transactions. Regarding the automaticity and time-saving feature, using blockchain could resolve a number of long-standing issues originating from corporate governance area [44]. Even more important, blockchain has the potential capability to yield unparalleled transparency, identifying the ownership proportion of debt and equity by investors or senior management and minimizing the risk of malfeasance on the part of listed companies, exchanges, and regulators. For market participants, the execution of blockchain allows for faster, economic acquisitions of shares, but with far less confidentiality than under the centralized system [45]. Financial records would become far more transparent, reducing the chances for profits manipulation and other accounting tricks, and related party transactions such as shadow banking would be far more accessible.

The execution of blockchain also promotes the speed and cost of trading in the financial market. In the United States, stock trades typically take three business days to settle and transfer ownership from seller to buyer. Market attendees such as brokers and individual and institutional investors are engaged, involving direct and indirect fees (via commissions and the bid-ask spread separately). However, a blockchain-based stock transaction would be instantaneous and erase the existence of these middlemen. While stock exchanges will likely continue to connect buyers and sellers in some other form, liquidity would skyrocket, owing to cheaper costs and fast trading time. Based on the existed literature [46–49], liquidity is a critical threshold with the funding flow and caters to the interest demand of fund managers and investors. It has a high likelihood to generate high frequency trading generally through the low cost of trading under the application of blockchain.

Besides transparency and low trading cost, blockchain technology has been presented as a voting platform for all sorts of the election. Numerous studies, including [50–52], have shown the existing difficulties with corporate elections, including inaccurate voter lists, limited ballot distribution, and even chaotic vote counting. In a blockchain system, the token can be regarded as a valid ticket to record the choices by voters. As claimed by Kshetri and Voas [53] and Wright and De Filippi [54], voting under blockchain could encourage shareholders to engage actively in corporate governance and vote on a broader range of issues more

⁵ Smart contracts are programs recorded on a blockchain that executes when certain circumstances are satisfied.

frequently. Under the similar spirit, election fraud, especially the rumor on the 2020 US president election, can effectively be avoided under the blockchain-based voting system.

4.2 Blockchain and Central Bank Digital Currencies (CBDCs)

Central banks and global financial regulators are debating how blockchain will affect their own technology underpinnings, as well as the underlying technology of the commercial banks and financial institutions. Considering the fact that representative cryptocurrency Bitcoin is undertaking a recognized part of the financial landscape, worldwide central banks are becoming more interested and taking seriously the potential application of digital money. Feared by the possible risk of losing sovereign currency issuance, central banks in the UK, the USA, the EU, New Zealand, and elsewhere are considering whether it is possible to establish their own central bank digital currencies (CBDCs). And the extensive discussion and conception are published in a flood of white papers [55–58].

Although CBDC is commonly connected with DLT, the majority central bank attitude toward such linkage is disapproval, considering the centralized payment system is the foundation of absolute control of currency. However, a blockchain-based ledger has a number of potential breakthroughs that can be utilized independently of one another, allowing the author to discuss the particular features of the blockchain system that are most relevant and useful in a CBDC setting. Smart contract can be independently implemented over a number of different kinds of ledgers, including a centralized system. Further, it could achieve security and efficiency targets through restricting the broad function accessible inside a smart contract programming language. Besides that, cryptography could also be employed to strengthen the CBDC system. Based on the white paper from [55], it is necessary to upgrade the cryptographic approaches since particular cryptographic functions can be weak over time and vulnerable to high-turnover cyberattacks. In a nutshell, an indisputable fact is that the innovation of blockchain technology triggers the revolution of the financial system and boosts the invention of CBDC. Although it is still in the early development phase in a majority of countries, the People's Bank of China completed salary payouts in the digital yuan utilizing blockchain technology in 2020 and plan to popularize it in the Olympic Winter Games [59].

4.3 Bitcoin, Cybercrime, and Anti-money Laundering

Although blockchain and cryptocurrency have shown diverse advantages, there are still some risks associated with the application of such new technologies. Regarding the decentralized feature, network users with pseudonyms make them more easily subject to cybercrime and illegal activities [60]. Based on [61]'s finding, over the

period of January 2011 to February 2020, there were 76 attacks on cryptocurrency exchanges with an average stolen value of US\$ 29.19 million per event. The likelihood of attacks relates to cryptocurrency market prices, evident by the crypto bubble period in 2018. The authors [62] find that approximately one-quarter of Bitcoin users are involved in illegal activity.⁶ A representative example is the closure of the silk road.⁷ They estimate that around \$76 billion of illegal activities per year involve Bitcoin (46% of Bitcoin transactions).

More broadly, investigations of Internet search engine data have found a correlation between interest in cryptocurrency use and anti-money laundering (AML) and know-your-customer (KYC) rules [63]. While Bitcoin theoretically enables money launderers to move illicit monies more quickly, cheaply, and covertly than ever before [64], there is less proof that such unlawful operations are really being carried out through the cryptocurrency market, but still arise the public attention.

Therefore, the challenges that CCs theoretically pose to global AML efforts have been routinely touted as actual threats by financial regulators and law enforcement. The Bank of International Settlements (BIS) [65], the International Monetary Fund (IMF) [66], and the SWIFT Institute [67, 68] expressed the concern of using cryptocurrency to facilitate the money laundering. Not just cryptocurrency, the implementation of ICO for firms issuing a new digital coin to investors may be monitored by criminal securities fraud laws. There is also a possible link between digital currency transactions and money laundering, terrorist funding, economic sanctions, and anti-corruption, which in the high likelihood can be monitored by the US Department of Justice (DOJ) [60].

Despite several limitations and unintended consequences for the economy and finance, blockchain or decentralized governance networks are viewed as inventive and ultimately more successful than old centralized means of coercion in an era of rapid and unexpected technology development [69]. By implementing a risk-based strategy, the Financial Action Task Force (FATF)⁸ achieves an effective balance between the existing hazards and potential that crypto-coins now bring fate [70, 71].

⁶ It normally refers to drugs, weapons, malware, and illegal pornography.

⁷ In 2013, the US Federal Bureau of Investigation (FBI) closed down the original Silk Road. Silk Road 2.0 was forced to shut down in 2014 as a result of a multinational operation involving law authorities from 17 different nations. Silk Road 3.0 was detected in mid-2016, highlighting the longer term limits of multinational law enforcement crackdowns.

⁸ The Financial Action Task Force (FATF), headquartered in Paris, is an international organization comprising of 35 member states and 2 regional organizations. Around 170 nations are affiliated with the FATF, which is widely recognized as the center of the global anti-money laundering legislation. This includes associate and observer members, as well as members of FATF-style regional groups.

4.4 Blockchain and Big Data in Financial Sector

Big data is one of the key drivers of the financial and economic industry. Through big data, large, diverse, and complex data sets can be used to provide financial services to banking companies around the world with solutions to long-term business challenges. However, the current storage mechanism is difficult to guarantee data integrity and security in its whole life cycle. If blockchain is regarded as an innovative methodological cognition rather than just a technology, it also demonstrates great potential in combination with other technologies. For example, the integration of blockchain technology and big data has the potential to provide a slew of fascinating opportunities [72]. Although blockchain has not been widely investigated in terms of big data management, it should be seen as two sides of the coin. As [73] states, not only does blockchain increase the size of big data, but it also improves its security and value by structuring and preparing it for big data analytics. While blockchain is focused on storage data, big data is concentrated on data analyses and forecasts from a large amount of data. Alternatively, it can be regarded as quantity for big data and quality for blockchain. Therefore, the data gathered and authenticated on the blockchain by big companies will add value to enterprises [74]. There are multiple blockchain-based big data analytics enhancements, stemming from benefiting the corporate governance and eliminating exposure.

More specifically, the exposure contains fraud prevention, agency cost, information asymmetry, and reverse causality. As blockchain technology enables to implement the real-time accounting, the business' entire ledger would then be available to interest parties like shareholders, lenders, creditors, or others [45]. The instant accounting statements replace the existence of quarterly financial statements and accounting audits. Furthermore, the real-time accounting feature can relieve the dilemma for small- and mid-size enterprises (SMEs) to access external financing, especially for shadow banking. Equipped with instantaneous accounting records through blockchain-based big data platforms, bank groups and relevant financial institutions can easily grant loans to SMEs without complicated risk assessment procedures.

5 Conclusion

The principle and design philosophy of blockchain have high relevance in the legal and financial sectors and could have a great impact on society as a whole. The blockchain-driven information revolution refreshes the cognition of trust and challenges the traditional centralized source of governance. Meanwhile, blockchain provides the community with a new architecture where individuals are more self-discipline, and the power of governance is bottom-up.

In the context of legal regulation in cyberspace, blockchain, as an alternative cyber architecture, refreshes the legal cognition of “code as law” and enriches the

legal enforcement in cyberspace. Meanwhile, blockchain refreshes the traditional concept of legal foundation, for example, trust. It reestablishes trust notions through cryptography. Therefore, it could embed legal values and notions into the cyber architecture. Because blockchain embeds these concepts in the cyber architecture, it enables a more convenient e-evidence system in practice. Furthermore, blockchain shares similar objectives as data protection law, in which both of them are intended to protect users' data protection rights and boost the data free flow. To a certain extent, blockchain could help data protection law achieve its legal objective through privacy by design.

As to the implication of blockchain to the economy and finance, this chapter found that blockchain not only has a great impact on the cryptocurrency field but also could enable new applications in corporate governance. Blockchain could break the market barriers and reduce the information gap between participants. It also found that although CBDC is commonly connected with DLT, the majority central bank attitude toward such linkage is disapproval, considering the centralized payment system is the foundation of absolute control of currency. It is also worth noting that the application of such new technology and cryptocurrency brings new risks from cybercrime and money laundering perspectives.

It is apparent that blockchain will lead the legal and financial fields into a more connected and flowing manner, where the information is transparent and consistent, and the trust is established by cryptography. However, we are keen to see a more connected scene of blockchain-enabled interdisciplinary cooperation, to make the data flow effortlessly and easily accessible. It still needs cross-disciplinary collaboration from a legal, financial, and social scientific perspective.

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Applications of Blockchain in Healthcare



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Abstract Despite great advances in storage technology and the exchange of health information and data, the lack of easy access to this information by patients and clinicians has long been considered a barrier to transparency and efficiency in healthcare delivery. Among the emerging information communication technologies, blockchain is a type of distributed ledger technology that creates a shared, immutable, and chronological record of transactions. Blockchain technology has several characteristics, including decentralized storage and data management infrastructure, high security, privacy protection, accessibility, transparency, and interoperability.

There are several potential applications of blockchain in healthcare. The first is health information exchange and interoperability. Blockchain's decentralized data storage technology can allow immutable patient records to be maintained across different medical organizations. A second application of blockchain involves saving personal health information from wearable devices and the Internet of Things (IoT). Third, blockchain's distributed identity management model can ensure data integrity and information sharing across medical systems and provide high security and privacy. Fourth, the immutable and interoperable nature of blockchain presents innovative opportunities for medical supply chain management. Fifth, the introduction of the blockchain information system will simplify the medical insurance claim process. Sixth, blockchain can facilitate the upload and integration of experimental data records from different sources due to its characteristics of immutability and transparency.

There have been new attempts to apply blockchain technology to control the SARS-CoV-2 pandemic throughout the world. The decentralized architecture, high transparency, high interoperability, and high confidentiality and security of the blockchain system meet the requirements for national epidemic prevention, surveillance, and management in Taiwan. If the blockchain surveillance system

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operated by the Center for Disease Control (CDC) detects a clear growing trend of infections after analyzing data from primary care clinics, hospitals, pharmacies, and long-term care facilities, the healthcare authority could take immediate actions to control the outbreak.

Blockchain technology can improve healthcare delivery in a decentralized, immutable, transparent, and secure manner, and it has the potential to provide long-term benefits as it develops and integrates with other emerging technologies. It can not only protect existing electronic medical records but also provide a powerful tool for patients to take control of their health data.

1 The Unmet Need for Healthcare

With the rapid development of medical information, the amount of medical and health data is increasing rapidly. This digital information is generated not only by medical providers but also by other related institutions such as personal mobile devices, pharmacies, medical insurance companies, research institutions, nursing homes, long-term care facilities, and even fitness centers. The development, storage, exchange, and application of this information play important roles in different levels of healthcare systems such as strategy making, resource allocation, financial management, quality control, and most important, clinical care [1].

Despite great advances in storage technology and the exchange of health information and data, the lack of easy access to this information by patients and clinicians has long been considered a barrier to transparency and efficiency in healthcare delivery. Electronic medical records (EMRs) or electronic health records (EHRs) are the most commonly used data tools in healthcare organizations around the world today. Paper-based medical records have been gradually replaced by EMRs; however, for security and confidentiality, most EMRs are owned by medical providers, and medical information is stored separately among healthcare providers, including primary care clinics and hospitals. In other words, each healthcare provider owns only a part of a person's medical information, and, therefore, no provider can see the entire picture. Unless approved by medical institutions, patients and other third parties, it is difficult to obtain relevant medical information. A lack of adequate medical data may encourage repeat testing, consume additional medical resources, and delay important treatment. Limitations in terms of medical data sharing also erode patients' autonomy, as many patients are left without timely access to their data, and, therefore, they are unable to make informed decisions promptly.

In addition to the aforementioned limitations, the current exchange and sharing of medical information also face the following challenges: First, medical information is easily tampered with, leading to fraud or medical disputes. Second, the interoperability between disparate repositories is quite limited. Third, manual processing

increases operational costs and expenses for patients and healthcare providers. Fourth, cyber-attacks or system breaches increase uncertainty.

Among the emerging information communication technologies, blockchain-based data systems have been widely discussed in terms of their ability to improve the accessibility, security, and interoperability of EMR [2].

2 Characteristics of Blockchain

Blockchain is a type of distributed ledger technology that creates a shared, immutable, and chronological record of transactions [1], and it can ensure data fidelity and security. In blockchain technology, information is collected into groups, called blocks, which contain sets of information. Blocks have a certain storage capacity, and when filled, they are closed and linked to previously filled blocks, forming a chain of data called a blockchain. All new data added after a filled block is connected to the blockchain are compiled into a new block, which will also be added to the chain once filled.

There are several characteristics of blockchain technology [1, 2].

The first characteristic is decentralized storage and data management. The entire blockchain network lacks centralized management but has a distributed end-to-end network structure. As a result, data preservation and authorization are not centralized by one organization or healthcare provider as in traditional EMR, allowing for increased aggregation of medical data from disconnected sources.

The second characteristic is improved autonomy and efficiency. Blockchain uses consensus-based specifications and protocols that enable all nodes to freely exchange data. Therefore, healthcare stakeholders and participants can use medical data in a free but safe way. Approved participants are granted access to the blockchain, which allows patients and doctors to access appropriate medical data more efficiently compared to traditional centralized data structures. Due to the immutable and distributed nature of blockchain, a public shared ledger can also facilitate health information exchange [2].

The third characteristic is high security and privacy. Data on the blockchain is encrypted using asymmetric cryptography and controlled by a dual-key system, that is, a public key and a private key, and the proof-of-work mechanism ensures that the data is not easily accessible.

The fourth characteristic is blockchain's accessibility, transparency, and interoperability [2]. All of the transaction records on the blockchain are open and transparent on the entire network. The visibility and data auditing are high, and the data flow and related performance between medical institutions in the same blockchain network are enhanced. As a result, the transaction cost is expected to be lower than that of the traditional data management model [3–7].

3 Potential Benefits and Applications of Blockchain in Healthcare

1. Health information exchange and interoperability

The decentralized data storage technology of blockchain allows immutable patient records to be maintained among different medical organizations [8, 9]. Each organization provides medical data to the blockchain “block” and has the right to share data from other medical institutions on a secure network. Blockchain-based data networks also allow patients to access their medical information while allowing them to grant this access to third parties, usually primary care clinics or health insurance companies. The distributed nature of blockchain facilitates the sharing of medical information among different data systems across organizations and enables inter-institutional data exchange, which reduces friction between isolated databases and the cost of intermediaries [10, 11].

In Taiwan, Lo et al. [12] developed a blockchain-based data network for integrating patient electronic medical records that were originally stored in hospitals and primary care clinics to integrate medical data for patient referral. This system promotes medical referral services to enhance trust among patients, primary care clinics, and hospitals. As telemedicine has become mainstream during the COVID-19 pandemic, blockchain-based health data storage systems will play an important role in protecting patient privacy and ensuring integration and interoperability.

2. Saving personal health information from wearable devices and the Internet of Things

In addition to the information provided by medical organizations, there is considerable personal health data regularly collected from wearable devices, smartphones, and the Internet of Things (IoT). A blockchain-based network can use distributed ledgers to receive authenticated activity information from these wearable devices. This health information is different from the traditional information included in medical records and is often ignored when visiting a doctor. There are a growing number of studies showing that information from wearable devices may reduce patients’ need for more complex treatments, thereby reducing the number of emergency room visits and hospitalizations as well as healthcare expenditures [8]. These devices are especially important for people with chronic diseases such as diabetes, hypertension, chronic obstructive pulmonary disease, and cardiovascular diseases. The systematic collection and integration of health information from wearable devices and the IoT can be used to track patient health and make timely decisions regarding treatment.

3. Digital identity management

Because medical information is sensitive and confidential, data security is critical. Traditional identity management adopts a centralized mechanism and is vulnerable to malicious attacks [13]. The distributed identity management model

of blockchain ensures data integrity and information sharing across medical systems and can also provide better security and privacy [8].

4. Medical supply chain management

The immutable and interoperable nature of blockchain provides innovative opportunities for medical supply chain management. The shared ledger system also allows for greater transparency and traceability, and when applied to health-care systems, goods such as pharmaceuticals can be recorded on the supply chain and service flow records, providing better visibility and timeliness [8]. Another application is drug supply chain management. Due to the poor verification and traceability during drug manufacturing, transportation, and delivery, counterfeit drugs are prevalent. If the circulation and transportation of drugs are recorded on the blockchain and authenticated at every level, it will provide better in-time monitoring and prevent the supply and circulation of counterfeit drugs. The combination of blockchain and medical IoT may be an important development direction for blockchain technology in the field of supply chain management. The considerable health data generated by IoT devices can also be stored in different systems. With the help of blockchain technology, patient data can be stored off-chain, but these data can be read through the pre-set permissions of the blockchain. This strategy will be useful in healthcare because the exchange of patient health information becomes more efficient without intermediaries [2].

5. Automation of financial transactions and insurance claim processes

The introduction of the blockchain information system will simplify the claims process for medical insurance claims [2]. At present, patients spend considerable time applying for claims from medical insurance companies after seeing a doctor, and they also need to apply for many medical documents and medical records at the hospital, which is time-consuming and inefficient. There are many intermediaries during the entire process that increase the risk of data leakage. Process automation driven by blockchain systems provides insurance companies and hospitals with a platform that can streamline the process of medical claims and insurance write-offs from a shared data network, improving transaction efficiency, procedural transparency, and reducing transaction costs. A decentralized immutable ledger system on a blockchain can make payment and claims records more transparent, facilitating future audits and reviews. In this sense, the most important manipulations of data exchanges and payment transfers between insurers and their stakeholders may become easier and cheaper than the status quo.

6. Clinical research

Medical research is very important in terms of providing a better quality of healthcare. Because clinical and experimental data are inseparable from high-quality medical research, the source and reliability of data are crucial to researchers [2]. Due to the immutability and transparency of blockchain, the upload of experimental data records and data sharing can be accelerated. Currently, medical records are typically managed in a centralized manner, and the data is not coherent between different medical organizations. It may be difficult

for researchers to obtain comprehensive patient data. Researchers can legally and efficiently access and collect data for research purposes from a shared medical ledger system with participants' authorization [11]. Participants in research can receive rewards or credits from research contributions or payments by granting different levels of permissions, encoded and stored by smart contracts on the blockchain. The blockchain can assist in accessing the subjects' data, and the researchers can also access and pay through the system. Research design and data reconciliation during clinical trials will become easier, helping to resolve medical and health-related issues.

4 Potential Applications of Blockchain in Pandemic Control in Taiwan

Taiwan has a robust medical care system; it is a government-run single-payer health insurance system that uses world-renowned information technology for a population of approximately 23 million. It is the ideal place to develop a blockchain-based health information system on a national basis.

The rapid spread of COVID-19 around the world since 2020 has drawn widespread attention to the prevention and control of the pandemic, and there have been new attempts across the world to apply blockchain technology to control epidemics in general [2, 11]. Mashamba et al. [14] proposed a self-testing and tracking system for infectious diseases based on the coupling of blockchain and artificial intelligence. Based on the blockchain, ICD-9 or ICD-10 diagnosis codes were used to map the cause of death as well as observe disease trends and patterns in different geographical locations over time. Blockchain can also collect personal data through the use of a public and private key system, and blockchain's DeepTrace system can provide anonymous personal identification while allowing regulators and healthcare providers to contact those at potential risk of infection [15]. Blockchain can also be used to streamline the clinical trial process for vaccines and drugs, increase public awareness, transparently track donations and fundraising efforts, and act as a reliable data tracker.

In response to the worldwide pandemic caused by the SARS-CoV-2 virus, Taiwan has successfully used the advantages of information technology and big data in pandemic control since 2020. Comprehensive data collection and application have provided invisible but very powerful support for frontline immigration officers, medical personnel, household officers, and infection control policymakers during daily tasks such as rapid quarantine, large-scale testing, early diagnosis, and timely treatment. Blockchain technology has also assisted hospitals and clinicians in identifying whether a patient has records that indicate they traveled abroad at the point of care. This powerful information system has also helped all Taiwanese to purchase face masks efficiently. The COVID-19 pandemic has not come to an end. Instead, it may resurface at any time through a different variant virus. The

current infection control information system must be upgraded to provide better service for all people and healthcare workers. The decentralized architecture, high transparency, high interoperability, high confidentiality, and security of blockchain meet the requirements of the national epidemic prevention, surveillance, and management system in Taiwan.

The blockchain-based information system is independent of the existing national health insurance and household registration system. All citizens and legal residents from other countries will be enrolled in this system. Immigration records, flight information, the results of repeated PCR tests, quarantine or isolation records, health status in quarantine centers, address of residence after being released from isolation, contact tracing records, medical records, and vaccination records are examples of data that can be uploaded and saved on the blockchain-based data chain. Each organization can contribute necessary information and access that information within the system. As a result, immigration officers at the harbor or airport can identify whether incoming immigrants have received a complete vaccination as well as the results of the latest PCR or rapid COVID-19 antigen screening.

The blockchain-based data system is different from current large-scale and centralized databases. Most importantly, the blockchain system has better confidentiality and security in terms of data protection. Otherwise, the impacts could be very significant if pandemic information were leaked or hacked. Therefore, although there is a similar data system in Taiwan, the public is wary of breaches of personal information.

In addition to the aforementioned applications, the development of a blockchain-based database is important in terms of the collection and analysis of data as well as the system's ability to provide more useful information so that senior managers can provide timely decisions and take appropriate action. Using pandemic control as an example, if the surveillance system run by the CDC noticed a clear growing trend of infection outbreaks in Taiwan after analyzing data from primary care clinics, hospitals, pharmacy stores, and long-term care facilities, it would need to take immediate action to control the outbreak.

5 Challenges and Limitations

While blockchain technology can enhance the exchange of health data, transparency, security, healthcare efficiency, and medical research, several practical limitations must be addressed before widespread implementation. The primary challenges are data throughput speed and file size limitation. Due to the current technical limitations of designing blockchain applications, the data block size is still limited, causing doubts regarding the scalability of such platforms; therefore, until now, only key information will be exchanged using blockchain technology. Second, patients have the right to access their medical records without the approval of the medical organizations that provide and input patients' original data. Using blockchain, patients can further authorize third parties to access and download

their medical information, which is different from the current regulations of most medical organizations. This challenge will be difficult in terms of medical record management in many hospitals. Medical laws also need to be amended to stipulate legal liability.

6 Expectations and Conclusion

Blockchain technology can improve healthcare delivery in a decentralized, immutable, transparent, and secure manner. Blockchain has the potential to provide long-term benefits as it develops and integrates with other emerging technologies. It can protect existing electronic medical records and provide a powerful tool that allows patients to take control of their health data. Blockchain can also expand to various value-added medical services to further promote health and timely resolution of ailments.

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A Survey on Blockchain-Based Federated Learning (BCFL) Frameworks



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Abstract Federated learning (FL), proposed by Google in 2016, has attracted increasing interest from academia and industry to address the data shortage issue of machine learning due to privacy concerns. The emerging FL paradigm *outsources* model training tasks to data owners instead of collecting data and saving them to a central repository. In an FL setting, a central server coordinates the learning rounds by dispatching the model under training to each client, which trains the model using her data. Thus, FL preserves user data privacy while satisfying machine learning needs. However, adopting a centralized server requires participants' trust in the server. Moreover, federated learning does not address how to incentivize data owners to participate. Without proper reward/penalty systems, it is hard to ensure all clients behave and provide quality data for training. With features such as immutable distributed ledgers, smart contracts, and integral incentive mechanisms, blockchain has shown promises to be used with FL to solve the trust and incentive issues. This chapter reviews recently proposed Blockchain-based Federated Learning (BCFL) framework architectures and categorized them into three design patterns according to blockchain scaling mechanisms. We characterized each pattern's applicability, advantages, and disadvantages from the network architecture, privacy and security,

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and incentive designs. These analyses provide insights for future framework design and three open research areas for further study.

1 Introduction

Rapid advances in data analytics, machine learning, cloud computing, and communication technologies drive the need to collect more and more data to fuel various intelligent engines. These engines are primarily data analytics or machine learning models that are trained using the collected data to make better decisions for given services. The more data collected, the smarter the services individuals or organizations could benefit from. However, after several data breaches [1, 2], people are aware of potential vulnerabilities in data security and privacy while trading their data for convenience or *free* services. They began to express concerns about the ability of data collectors to safeguard the data collected and protect their privacy. Governments have mended their data security legislative procedures for privacy and national security protection, such as the General Data Protection Regulation (GDPR) in Europe, the California Consumer Privacy Act (CCPA) in the USA, and the Personal Information Protection Law (PIPL) in China. While these regulations help to protect user data, they make data collection, access, and sharing even more difficult. This lack of constant access to diverse data affects the accuracy and impartiality of these intelligence engines. Worse yet, building applicable models without enough data can be challenging!

McMahan et al. [3] proposed a new Federated Learning (FL) paradigm to facilitate knowledge sharing while protecting data privacy and reducing communication and storage costs. The proposed FL scheme shares the model under training with participants or *clients* to extract knowledge from their local data without collecting it. A training model is sent to each client to train the model with her data iteratively. A central coordinator, or *server*, aggregates the training results in each round and redistributes the updated model to each participant for subsequent training rounds. This iterative training process continues until the server determines that the model has converged and stops the training process.

Keeping data local reduces data privacy concerns. In addition, not all raw data need to be sent to the cloud for processing. Even traditional machine learning relies on data filtering at the edge before collecting data in the cloud to alleviate communication overheads and reduce storage costs. Furthermore, the raw data is usually orders of magnitude larger than intermediate training updates. Therefore, another advantage of FL is that it generally takes fewer communication overheads and storage costs to train a model.

Despite all the potential advantages, FL comes with new challenges:

1. The dependency on a central FL server has raised system fault tolerance and trust issues. Moreover, the central FL server could pose a system throughput bottleneck and hinder system scalability.

2. FL models are trained over a network iteratively, and thus, they are subject to a broader attack surface. Some malicious clients have attempted to poison the model under training with false data or model updates. Moreover, research [4] has shown that an *honest but curious* FL server could reconstruct client data using only client model updates.
3. There is a lack of reward and penalty mechanisms to encourage or discourage active and proper participation in providing quality data and model training.

Blockchain [5], with its features such as smart contracts, decentralized and immutable ledgers, and built-in reward systems, makes integrating blockchain with FL a promising solution to those FL challenges. Adopting blockchain as its underlying network infrastructure, FL does not need to rely on a central server, thus alleviating the single point of failure and model aggregation throughput bottleneck. Moreover, any updates exchanged and user actions are traceable and resort to participants' accountability. Furthermore, blockchain can incorporate a reward or penalty mechanism to compensate and influence the behaviors of FL participants. As a result, a new area of internetwork research called *BlockChain-based Federated Learning* (BCFL) has emerged to facilitate the contextual needs of FL systems, especially when dealing with the system, privacy, and security vulnerabilities.

This chapter first reviews FL and highlights its challenges in the system and statistical heterogeneity, and vulnerabilities in privacy and security. Then, we review recently proposed BCFL framework architectures and categorize them into three design patterns according to common blockchain scaling mechanisms. We characterized each pattern's applicability, advantages, and disadvantages from the network architecture, privacy and security, and incentive designs. We then provide a general discussion in the hope that these analyses provide insights for future framework design and three open research areas before we conclude the chapter.

2 Federated Learning

A typical FL setup consists of an *FL server* or *coordinator* and some *clients* who are data owners willing to participate in training a model with their private data collaboratively. The FL server delivers and maintains the global model under training to selected clients and orchestrates multiple rounds of training process between clients until the model converges. Clients train a global model with their local data in each round of training and send the resulting updates to the server. The FL server validates client updates, aggregates all client updates, checks for model convergence, and redistributes the updated global model for the next round of training if the model has not yet converged. This iterative training process continues until the training process converges.

The following steps summarize an FL training process and its actors' behaviors:

1. **Task publishing.** A training task, including its model and hyperparameters, is published by a *task owner* to an FL server.
2. **Client selection.** In each round, the server follows specific guidelines to select a set of qualified clients, i.e., dataset owners, to help train the model with their local datasets.
3. **Global model distribution.** Once the set of clients is selected and ready to perform the task, the server distributes the model, and hyperparameters to the selected clients.
4. **Local model training.** Each client trains the received model with its local dataset and sends the model updates to the server.
5. **Global model aggregation.** Upon receiving a preset number of client updates, the server performs an aggregation algorithm to update the global model. If the global model has not yet converged, go to step 2 for the next round of training.

Like most machine learning problem formulations, the FL problem can be formulated as follows. First, we define a loss function $f(w) = l(x, y; w)$ for all samples (x, y) and model parameters w . Let us assume that there are K clients and the client i has its dataset D_i and $n_i = |D_i|$, where n_i is the number of samples in D_i . Thus, the objective of an FL is to find the best w^* that minimizes the loss function, $f(w)$, which is a weighted sum of $F_i(w)$ for all i in K , i.e.,

$$\operatorname{argmin}_{w^*} f(w) = \sum_{i=1}^K \frac{n_i}{n} F_i(w) \quad \text{where} \quad F_i(w) = \frac{1}{n_i} \sum_{j \in D_i} f_j(w)$$

By exchanging model updates instead of raw data between each client and the server, FL systems decouple model training from the need to access training data directly to preserve privacy. Moreover, it achieves better communication efficiency as the model updates are orders of magnitude less than the raw data.

Notice that the above formulation assumes that the partition D_i is a subset of the overall training data with the same distribution, so $E_{D_i}[F_i(w)] = f(w)$. The objective of FL is to learn the w^* that minimizes $f(w)$, which is a weighted average of the local loss function, $F_i(w)$, for all i in K . The above approach is called *Federated Averaging*, or *FedAvg*, in [3]. Although identical and independent distribution (IID) data distribution is commonly assumed by distributed learning algorithms, it does not hold well in real-world FL application scenarios. That is, F_i may not approximate f due to the non-IID nature of the data distribution. The non-IID data distribution in FL settings, or *statistical heterogeneity*, has caused issues such as *objective inconsistency* [6] that affect the accuracy of the global model under training.

2.1 Types of Federated Learning

For applicability clarification, FL is usually categorized into different types to reflect various statistical heterogeneity of the datasets and the heterogeneities of participants' systems.

Yang et al. [7] classify FL into the following types according to how the training data is distributed between the *feature space* and *sampled clients*: *horizontal (or sample-partitioned) FL*, *vertical (or feature-partitioned) FL*, and *federated transfer learning*. The horizontal FL type is suitable for data collection from similar feature spaces but different in samples. The vertical FL type fits those situations where data are collected from similar samples but with different features. Moreover, the federated transfer learning type applies to situations where the datasets differ in samples and feature space. Inspired by transfer learning, the FL server's knowledge of local models is transferred across domains.

Moreover, FL can be classified based on the availability of data and the number of participants into *cross-silo* and *cross-device* FL [8]. The cross-silo FL refers to the scenarios where a small group of clients actively participates in a collaborative training task, *e.g.*, a group of hospitals. All clients make their resources available throughout the training session. Furthermore, the cross-device FL refers to the cases involving a larger group of participants. The FL server randomly selects a group of clients from the participants for each round of training. Some clients may drop out of a training session without completing a training round.

In addition, from the model aggregation timing perspective, FL can operate in *synchronous* or *asynchronous* modes [9]. The former refers to the FL server starting the next round of training after receiving all or most client updates. In contrast, the latter operation starts a model aggregation as soon as a local model update becomes available. Although the synchronous FL operation could lead to better model accuracy, it faces challenges from heterogeneous system environments FL often operates over. For example, its performance is subject to dropout clients and *straggling clients*, slower devices that take longer to complete a training round due to limited system resources [10]. To cope with these challenges from systems heterogeneity and client availability, asynchronous FL performs the model aggregation with one client update at a time to achieve better scalability but at the expense of model stability and accuracy [9].

2.2 Vulnerabilities of Federated Learning

Unlike centralized machine learning systems, an FL system involves many parties over a network and, hence, faces a broader attack surface. Any compromised central server or participating clients could launch attacks during the learning process. This subsection reviews intrinsic FL system vulnerabilities and known vulnerabilities in privacy and security.

2.2.1 System Vulnerabilities

- **Fault tolerance and scalability.** The design of relying on a centralized server poses a few concerns. First, it presents a single point of failure. Second,

it may become a computational bottleneck when performing aggregation for model updates as the number of clients increases. Moreover, the *hub-n-spoke* communication pattern between the server and clients creates a communication bottleneck over the multi-round training process.

- **Systems and statistical heterogeneity.** With its distributed learning settings, FL faces systems and statistical heterogeneity challenges [11] since not all participants have the same computing resources, and each client's data may have intrinsic differences. As discussed at the beginning of this section, the IID assumption by FL does not fit well with real-world application scenarios [10]. Naïvely taking a weighted average over model updates as suggested in FedAvg [3] could cause *objective inconsistency* [6], which may lead the global model and local models to converge to different optima.
- **Communication overheads.** Although FL only exchanges the model updates, the transmission of bursty model updates could still take a heavy toll on the communication bandwidth depending on the model's size, which could be in 10s or even 100s of gigabytes. Moreover, the larger the number of participating devices and the limited network resources, the more likely the communication overhead becomes a system bottleneck in an FL system.
- **Lack of incentive mechanism.** A critical success factor for FL is to provide enough incentive for participants, so they are willing to participate continuously in training with their quality datasets. Models trained with high-quality data give fast convergence and better performance. Unfortunately, the original proposed FL does not incentivize clients with any built-in reward or penalty mechanisms. This lack of incentive design can cause devices to drop out or attract attackers to participate in training to steal additional client information.

2.2.2 Privacy and Security Vulnerabilities

Recent studies [8, 12] have proved that the model updates can reveal private information about clients. Moreover, FL settings can give rise to various threats from many actors as participants can freely join and leave a training round [13]. Thus, the usual objective of attacks on an FL system is to extract private information from victims or force the resulting model to behave differently than intended. Attacks can introduce false data or *backdoors* [14], inducing misclassification or rendering the model unusable.

A malicious FL server could exploit the FL system because it could select and regulate the participating clients, access the model updates from clients before aggregation, and aggregate the model updates in each round of training. Likewise, malicious clients could launch their attacks by exploiting the aggregated model received from the server, the local raw data on which the client trains the model, and the training process and the resulting gradient updates.

FL attackers' targets are either to infer private data, *i.e.*, privacy attacks, or to manipulate the training model, *i.e.*, security threats. We list a few common privacy attacks and security threats below.

Privacy Attacks

Inference attacks aim to extract and learn additional FL participants' private information during training from the model updates. The inference attacks could infer class representatives, membership, training data properties, and training samples and their corresponding labels. Inference attackers usually inspect and derive additional information from the received gradients without interfering with the training process.

Zhu et al. have shown in their DLG work [15] that it is not safe to share gradients and constructed experiments to successfully reconstruct client samples and corresponding labels in contrast to most beliefs. Zhao et al. [16] further improved the DLG work to extract the one-hot ground-truth labels from the gradients for any differentiable model trained with a cross-entropy loss function.

Hitaj et al. [17] and Wang et al. [18] proposed Generated Adversarial Networks (GAN)-based attacks that aim to accurately recreate the training samples used to train the model and their associated labels from the client-side and the server-side, respectively. Client-side GAN attackers masquerade as participants to influence the learning process and trick victims into publishing more sensitive information [17]. Client-side GAN attacks are powerful due to their passive nature. Constantly injecting mislabeled training samples leads the victims to provide more training data to learn to distinguish the correct labels, significantly benefiting the iterative improvement of the attackers' discriminators.

The server-side GAN attacks have some limitations. Unlike the client-side GAN attacks, attackers can hide among the clients. The malicious server aims to reconstruct training samples of a specific target while performing FL services as a regular server does. Wang et al. [18] proposed a framework, mGAN-AI, that explores the situation where a malicious server obtains the targeted victim's data samples based on its submitted local model updates.

GAN attacks are more invisible and pose a more significant privacy threat to FL systems since they cannot be easily detected.

Security Attacks

Security attacks refer to the cases in which the attackers try to manipulate the model under training to suit the attackers' purposes. It can introduce exploitable backdoors [14] in the model or even corrupt the entire model.

Data or model poisoning attacks are used to achieve the model corruption attacks. Data poisoning attacks are achieved by the adversary damaging participants' data under its control. In contrast, model poisoning attacks the model under training by sending false gradients to the FL server for model updates. Attackers can deceive the model under training by implicitly embedding designed local samples in the FL process. It can also launch model poisoning attacks by uploading specific gradient updates to compromise the model under training.

Based on the targets of model poisoning attacks, poisoning attacks can be classified into two categories according to the target of attack: (1) *untargeted attacks*; (2) *targeted attacks*. The former affects the model's performance under training, while the latter attacks the targeted subtasks without affecting the others. Both attacks attempt to tamper with the model under training maliciously. Label-flipping [19] is a typical data poisoning attack, which changes the label of honest training samples to another class while retaining the data feature unchanged. The authors also show that existing defenses in ML do not apply to FL settings, especially with Sybil attacks.

Model poisoning attacks are realized by transferring false updates to the aggregator or inserting a backdoor into the global model [14], disrupting the security aggregation process and compromising the availability and performance of the global model. Moreover, Cao *et al.* [20] studied a distributed poisoning attacks problem, in which more than one attacker could collude and inject malicious training samples into their respective local models.

Generally, untargeted model poisoning attacks are more likely to be detected than targeted ones. Moreover, data poisoning attacks are viewed as subsets of model poisoning attacks since data poisoning ultimately changes model parameters that are incorrectly sent in any given iteration.

2.3 Mitigations of Federated Learning Vulnerabilities

Vulnerabilities due to system design issues must be addressed through design changes, while cryptographic primitives can address the privacy and security issues.

2.3.1 Mitigation of System Vulnerabilities

Decentralizing the centralized FL server could eliminate the single point of failure and increase system scalability and reliability. Thus, blockchain technology has been proposed to integrate with FL [21–28] to address server decentralization and provide rewards for participants. We defer the details of blockchain-based FL until a later section.

In addition, techniques of gradient compression [29] or allowing only the relevant client outputs to be sent to the FL server [30] to alleviate communication overhead have been proposed to reduce the communication overhead. Konecny *et al.* in [29] demonstrated how a lossy compression method reduces communication costs. Li *et al.* [31] proposed another compression method using an autoencoder compressor to encode data before transferring gradients. The first server trains an autoencoder and releases the encoding part to each client while keeping the decoding part secretive so that clients encode their gradients with the encoder before transferring them to the server.

Shokri et al. [32] proposed only exchanging a subset of gradients to improve communication efficiency. They proposed a gradient selection method to submit the most critical gradients and add a randomized weight selection variant. A randomized variant is also proposed in an FL setting by [29] to increase communication efficiency. Yoon et al. [33] proposed a different gradient subset approach and subsequently demonstrated that sending sparse and selective gradients can improve performance when dealing with non-IID datasets.

The gradient compression techniques do not only reduce communication overheads; it also reduces the available information the server can extract from clients and thus mitigate inferential attacks.

Furthermore, recent *asynchronous FL* [9] approaches are proposed to cope with systems heterogeneity and increase system scalability.

2.3.2 Mitigation of Privacy and Security Vulnerabilities

With open training rounds and a potentially large number of participants in multi-device FL settings, it is naïve to assume that all clients are honest and will train the assigned model according to the predefined FL parameters. Therefore, FL systems must cope with situations where dishonest clients submit false training results that cause negative impacts on the global model under training. Meanwhile, they should try to prevent the so-called *honest-but-curious*, or *semi-trusted*, server from extracting additional client information from client updates.

A set of privacy and security vulnerabilities mitigation cryptographic primitives such as differential privacy (DP) [34], secure multi-party computation (MPC) [35], homomorphic encryption (HE) [36], and the newly developed Trusted Execution Environment (TEE) [37] and Secure Aggregation (SecAgg) [38] are adopted to mitigate the privacy and security vulnerabilities in FL. Effective mitigation strategies involve adopting different combinations of these primitives to cope with vulnerabilities.

Differential Privacy (DP) [34] is a lightweight privacy protection approach that adopts obfuscation mechanisms to add noise to sensitive data. Moreover, DP provides rigorous provability of privacy. Although the loss of data quality caused by the noise added for each user is relatively low, there is an inevitable trade-off between privacy and model performance. In FL setups, clients can adopt DP to efficiently add noise to their updated parameters before sending them to the server to avoid inverse data retrieval. Likewise, the server can add noise to the aggregated model before redistributing it to the clients for the next round of training.

Secure Multi-Party Computing (MPC) [35] protocols allow two or more participants to jointly compute a function without disclosing any data to the other parties. Most of the existing secure MPC protocols are based on the garbled circuit, secret sharing, oblivious transfer, and homomorphic encryption in semi-honest, malicious, and publicly verifiable security models. Bonawitz et al. [38] proposed using secure MPC to protect client updates and cope with client dropouts in an FL setting.

Homomorphic Encryption (HE) [36] relies on cryptographic operations, where mathematical operations can be applied to encrypted messages and obtain the same results as if those mathematical operations were applied to the same messages in plaintext. HE is particularly useful for ensuring privacy against untrusted servers. Although applying HE to the FL model updates between clients and the server ensures no performance penalty in model convergence, HE has significant scalability drawbacks due to its considerable computation overheads and implementation challenges. Furthermore, the sizes of encrypted models increase linearly with each supported homomorphic operation, thus, increasing communication costs.

A Trusted Execution Environment (TEE) [37] is a trusted environment for executing code on an untrusted remote compute server. The remote server performs the specified computation in an isolated *enclave*, a tamper-resistant process environment that provides integrity and confidentiality for code execution. A TEE provides three fundamental properties: *confidentiality*—the state of the code’s execution remains secret unless the code explicitly outputs a message; *integrity*—the code’s execution cannot be affected except by its input; *measurement/attestation*—the TEEs can prove to a remote party what code (binary) is executing and what its starting state was, defining the initial conditions for confidentiality and integrity. Thus, TEEs can be used as FL’s trustable computing outsourcing platforms for model validation or aggregation.

Secure aggregation (SecAgg) [38] can be used to ensure that the FL server obtains the aggregated result while guaranteeing that clients’ updates are not revealed to the server. Moreover, they are robust to client dropouts during the training, and they could scale to accommodate many participants and update vector lengths. Thus, SecAgg enables FL clients to submit their model updates securely to the server for aggregation. To further ensure the aggregated result does not reveal additional information to the server, local DP can be used to protect clients’ privacy.

3 Blockchain-Based Federated Learning

As discussed in the previous section, the original FL design has three fundamental system limitations. First, it relies on a central server to function correctly, which could be a single point of failure and limit the system’s scalability. Second, it also requires all participants to trust their data security and privacy to the server. Third, it does not reward or penalize participants according to their contributions. This lack of incentive design directly impacts and reduces the willingness to participate and the opportunity to access quality training data.

With its decentralized, cryptographic primitives-based, trusted environment and built-in reward system, blockchain has been identified as one of the promising secure infrastructure technologies to address the FL system and privacy and security vulnerabilities.

This section first explains how we categorize surveyed BCFL framework architectural patterns into three types: *Basic BCFL Framework*, *Layer-1 BCFL Frame-*

work, and *Layer-2 BCFL Framework*. We then describe and analyze each framework pattern and its pros and cons.

3.1 Taxonomy of BCFL Framework Architectural Patterns

Blockchain technology has shown great potential to tackle critical security and trust challenges in various applications, e.g., cryptocurrency, Internet-of-Things, and edge computing. However, the scalability limitation in the initially proposed blockchain severely affects its wide adoption due to the so-called *trilemma of decentralization-security-scalability* [39], which states that trade-offs are inevitable between three critical properties: decentralization, security, and scalability. In other words, the trilemma states that we can only have two out of decentralization, scalability, or security, but not all three. However, decentralization is the core and the nature of blockchain, and security is an essential property expected by the users. Hence, achieving higher scalability becomes the main challenge.

The objective of existing blockchain scalability solutions is to process a high number of transactions per second (i.e., throughput) without sacrificing security and decentralization. These solutions can be classified into two categories: *Layer-1 scaling* (e.g., sharding [39], bigger blocks, and DAG [40]) and *Layer-2 scaling* (e.g., payment channels [41] and side chains [42]). The Layer-1 scaling is designed to improve the throughput of blockchains systematically. The approach is to optimize a blockchain system to handle the growing amount of work by reducing the communication and computation overhead, vertical scaling by adding resources to a single node, or horizontal scaling by adding more nodes to the Blockchain. The Layer-2 scaling, *a.k.a., off-chain scaling*, proposes mechanisms to process certain transactions outside of the blockchain, e.g., micropayment transactions, and only records important transactions, e.g., final balances, *on-chain*.

Hence, we categorize the surveyed BCFL framework patterns into the following three types of frameworks, as shown in Fig. 1:

- **Basic BCFL Framework.** The single-chain design pattern is the most intuitive integration of blockchain and FL, e.g., BlockFL [21], Biscotti [22], and Swarm Learning [23]. It leverages consensus algorithms and smart contracts to achieve FL operations.
- **Layer-1 BCFL Framework.** The Layer-1, or Multi-chain, BCFL framework is first proposed by Majeed et al. [24] in their FLchain framework for smart city applications. Later, this pattern is expanded with *sharding* and *hierarchical* chains like SRB-FL [25] and ChainsFL [26] for system scalability consideration.
- **Layer-2 BCFL Framework.** The Layer-2 or Off-chain Compute BCFL framework is either based on the payment channel, e.g., FLchain [25], or the side chains to perform model aggregation in an *off-chain trust computing with an on-chain recording* fashion. The side chains approach can further select an architectural choice from two runtime environments: to run over any outsourcing computers, e.g., Arbitrum [27], or to run in TEEs, e.g., Ekiden [28].

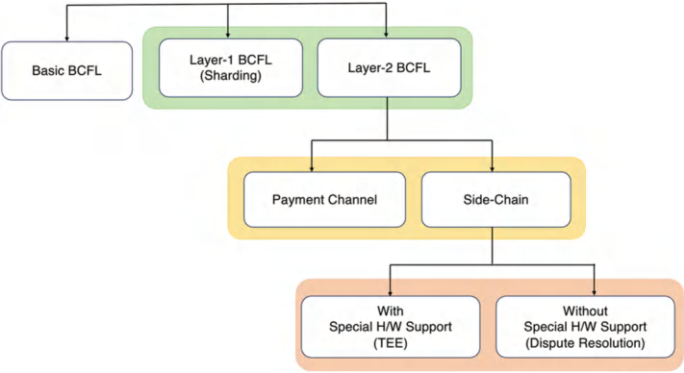


Fig. 1 Taxonomy of BCFL (Blockchain Federated Learning) framework architectures. The hierarchy depicts how BCFL evolves from a basic form toward Layer-1 (sharding-based) and Layer-2 frameworks. Layer-2 solutions branch into payment-channel and side-chain designs, further categorized as with TEE-based hardware support or without TEE support (dispute-resolution-based)

The following subsections will elaborate on each pattern with proposed examples and provide the pros and cons analysis.

3.2 Basic BCFL Framework

Kim et al. first proposed a BCFL framework called *BlockFL* [21] to enable participants to exchange device local model updates and win rewards. Its decentralization approach mitigates the FL single point of failure problem. It allows any participant though potentially untrustworthy devices in a public network, to contribute and join the FL training rounds. Moreover, the more network participants it gets, the better the model quality.

BlockFL is a very intuitive approach to integrating BC with FL. Later, Martinez et al. [43] added validation error threshold metrics and reward mechanism adjustment and started researching BCFL frameworks we called *Basic, or Single-Tiered, BCFL Framework*.

The conceptual framework of the Basic BCFL Framework is shown in Fig. 2, and it has the following characteristics:

- All FL participants are members of a blockchain. They are either *lightweight nodes* (or *wallets*) or *full nodes* in the underlying blockchain.
- All nodes or randomly selected groups of nodes play the FL validator or the aggregator roles in each round to validate and mitigate malicious clients. Moreover, respective consensus are formed within the validators and aggregators to avoid single server resource constraints and malicious behaviors.

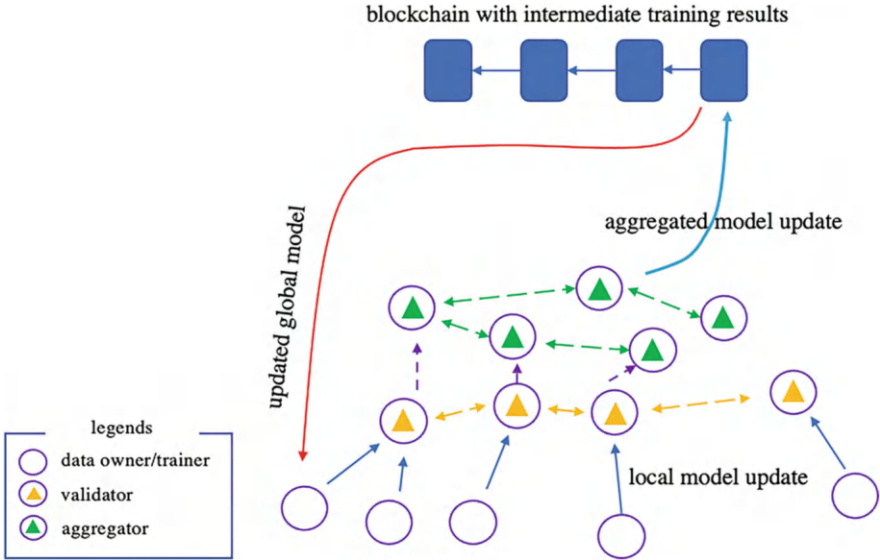


Fig. 2 Blockchain-based model update process in BCFL. Clients perform local training and submit model-update transactions to the chain; validator nodes verify updates (e.g., via agreed checks) and, through consensus, approve them. An aggregator committee then combines the validated updates (e.g., FedAvg) to produce the global model. Intermediate updates and roundwise aggregated models are immutably recorded on-chain for auditability, and the new global model is redistributed to clients for the next round

- The immutable ledger of the blockchain *stores* all the intermediate training updates and the final global model for auditing. Notice that *storing* updates does not mean storing model updates *per se*. Most frameworks store off-chain filestore handles, e.g., IPFS [44], Amazon S3, *on-chain* for system efficiency and scalability.
- Rewards are distributed among FL participants based on each one's contribution.
- Depending on the applicability of cross-device FL or cross-silo FL, a permission-less or permissioned chain could be used.

Biscotti [22] is an example of a cross-silo Single-Tiered BCFL Framework. This work incorporates cryptographic primitives such as differentiate privacy, digital signature, Shamir's secret sharing [45], and verifiable random function (VRF) [46] schemes *on-chain* to protect federated learning processes and achieve privacy preservation.

Biscotti takes the following measures to ensure that the federated learning process converges into a global model in a privacy-preserving fashion.

- Each block in the blockchain represents a single iteration of training and the resulting state of the global model at each iteration.
- Each client *masks* their update with DP noises. The noises are collected from a set of noising peers by each client using VRF.

- An FL *verification committee* selected from blockchain nodes validates the masked updates to defend against poisoning. Each member in the verification committee signs the commitment to the peer's *unmasked* update if it passes the Multi-Krum test [47].
- Once the majority of the committee signs an update, the signed update is divided into Shamir secret shares [45] and given to an FL *aggregation committee*.
- The FL aggregation committee from blockchain nodes aggregates the updates. Those who contribute a share to the final update receive an additional reward from the system.

Participants in Biscotti verify each other's contributions to prevent poisoning input. The cryptographic primitives used in the system prevent information leakage attacks. Moreover, the randomly selected verifier and aggregator roles are mutually exclusive to prevent potential collusion among peers. The reward system in Biscotti adopts a *stake-to-play* rule to penalize malicious participants.

An example of a cross-device Single-Tiered BCFL Framework is *Swarm Learning* (SL) [23]. SL uses a permissioned Ethereum to perform BCFL among participating medical organizations. Those medical organizations collaboratively classify CORVID-19, tuberculosis, leukemia, and lung pathologies without violating privacy laws. Each participant must register to the SL framework, and only pre-authorized participants can execute transactions.

3.2.1 Pros

- Using decentralized mechanisms for validation and aggregation mitigates the potential single point of failure and computation bottlenecks.
- Using consensus algorithms and reward systems to establish a trustable environment reduces and mitigates malicious server-side or client-side poisoning attacks.
- Recording signed intermediate training results on-chain provides an auditable log and holds participants accountable for cross-device cases.
- Biscotti integrated common FL privacy and security countermeasures into its cross-silo framework design. In contrast, Swarm Learning showed a semi-trusted, *e.g.*, bound by legal agreement, cross-device BCFL design.

3.2.2 Cons

- The surveyed frameworks in this type mainly deal with synchronous FL and do not deal with dropouts and stragglers' issues due to system heterogeneity.
- In cross-silo cases, BCFL frameworks that use public chains introduced third-party nodes, *i.e.*, miners, in the FL training loop, which could lead to model leakage.
- Although using a secure reputation for devices participating in learning could improve the system security and privacy, effective model update validation, and malicious updates detection are open issues to FL research. Without a fair

reward system, BCFL frameworks, especially the cross-silo FL cases, may not be sustainable.

- The use of distributed consensus over the entire network increases communication overheads and the communication delay (block time) for an FL training.

3.3 Layer-1 BCFL Framework

Sharding [48] is one of the most practical Layer-1 scaling techniques for achieving horizontal scalability as blockchains grow in size. Blockchain nodes are divided into small groups called *shards*. Each shard can run its consensus protocol independently and maintain a blockchain instance. On the top level, the main chain, or *beacon chain*, is maintained by a group of *shard managers* or *leaders* of shards. Each shard manager periodically commits a digest of the shard chain to the main chain. The *parallelism* of sharding theoretically *scales out* both transaction throughput and network size.

Recent research has applied blockchain with sharding to BCFL frameworks. We call this type of BCFL framework *Layer-1* or *Multi-Tiered BCFL Framework*. The Multi-Tiered BCFL Framework could be used to train a single task or multi-task in parallel. The framework works in a hierarchical BCFL fashion when training a single FL task. In contrast, in the multi-task case, each FL task is trained in a shard, and the *main chain* acts as a verifiable repository for the intermediate training results of the model under training.

The general framework of the Multi-Tiered BCFL Framework is as shown in Fig. 3, and it possesses the following characteristics:

- All participating nodes are subdivided into shards.
- In the single-task training case, members in a shard jointly train the same global task but keep their intermediate results in the blockchain within the shard. All the shards then perform a global model aggregation at the main chain level before the next round of training starts.
- In the case of multi-task training, members of each shard jointly train a unique task and maintain the intermediate results in their local blockchain within the shard. Once a task model is trained, the final model is recorded to the main chain.

ChainsFL [26] is a single-task example of the Multi-Tiered BCFL Framework. To avoid high communication complexity and improve scalability caused by a large-scale blockchain, ChainsFL proposed a two-level BCFL architecture. For a given training task, FL clients are grouped into shards. The leader aggregates the locally trained model updates, based on consensus, in a synchronous FL fashion. They then submit it to the main chain, where an asynchronous FL aggregation is performed to derive a global model for the shard in the following training round.

A task publisher of ChainsFL invokes a smart contract to release the FL task and its hyperparameters. Then the smart contract creates the genesis transaction (denoted as g_0) of the main chain, which includes this information and a test

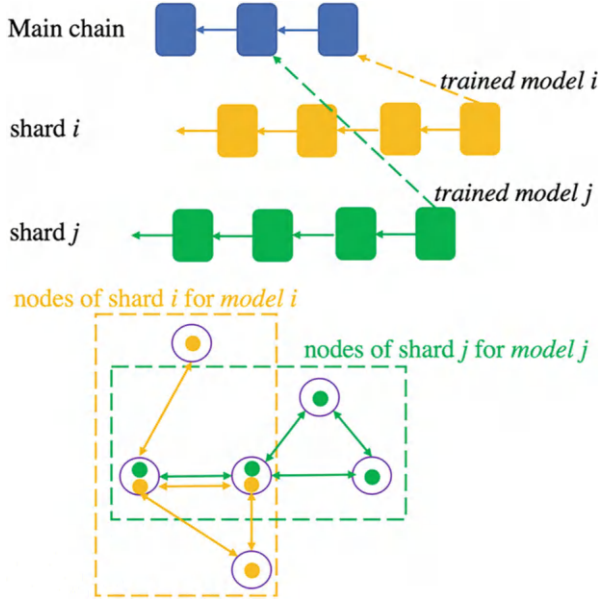


Fig. 3 Multi-tiered BCFL (Blockchain Federated Learning) framework pattern. The figure illustrates how multiple shards operate under a main blockchain, each shard maintaining its own federated model training process. Local updates within shard i and shard j are aggregated independently, and their trained models are periodically committed to the main chain. This hierarchical sharding structure enables parallel training and scalable coordination across distributed BCFL networks

dataset. Meanwhile, this smart contract will trigger related shard networks to run the training task. The leader of the shard will pull the genesis transaction. Then the configurations and requirements of this task will be stored in the shard chain to initiate the *shard training process* as follows:

1. *Device Selection*: The leader of a shard chain selects participants for the shard training based on availability, such as power status and network status, which are reported by devices periodically. Then, these devices will be authorized to access the shard chain to download the basic model for local training and upload the transactions.
2. *Local Update*: The device runs the training process using its data based on the basic model downloaded from the shard chain. After a preset number of local epochs, the locally trained model will be sent to the shard chain as a transaction.
3. *Model Aggregation*: Each node in the shard receives the local model update transaction and validates its accuracy against the published test data in g_0 . When enough valid updates are received, the leader aggregates local updates using FedAvg [3] to form the updated shard model of this round. The updated shard model is published in a block, which all nodes can download in the shard for the next training round. Due to its synchronous FL design, a training round may

be abandoned if the preset timer expires, or the number of updates received is insufficient.

4. *Commit a Shard Model to Main Chain:* When a shard has completed rounds needed for an iteration, the latest shard model is packed as a transaction of the main chain by the leader and commits it to the main chain. The main chain is a DAG with a tangle graph [49]. The leader commits the shard model to the main chain by selecting and validating tips of the tangle graph to form a new global model. This shard-main chain interaction continues until the model converges.

Next, we take SRB-FL as an example of a multi-task Multi-Tiered BCFL framework. SRB-FL [25] uses multiple shard chains to train different tasks and the main chain to keep all the final trained models. Each model is assigned to a group of devices belonging to the same shard to train. Once these devices have completed their training with local data, they record their final model updates to the main chain for future auditing purposes. The training tasks in shards conduct synchronous FL training iteratively. Consequently, several models can be trained in parallel.

Furthermore, to improve the reliability of devices, SRB-FL designed intra-reputation and inter-reputation processes to represent the degree of FL devices' trustworthiness based on multi-weighted subjective logic. Subjective logic is a probabilistic logic that evaluates the trustworthiness of different entities. It uses *opinions*, in probability, to determine a subjective belief through statements such as *belief*, *disbelief*, and *uncertainty*. Every verification is weighted after being collected from a shard to the main chain for an accurate reputation update. The weight of verification is set based on multiple factors. Devices with higher reputation values are selected to train an FL model; hence, SRB-FL avoids being misled by malicious devices.

3.3.1 Pros

- This type of BCFL framework leveraging layer-1 scaling is more scalable than the Basic BCFL Framework type.
- The approach that ChainsFL trains a task using asynchronous FL by combining a DAG tangle graph main chain with multiple shard chains addresses system and statistical heterogeneity issues and further improves the efficiency of the model training.
- The multi-task SRB-FL approach is suitable to combine with edge computing facilities for many smart city applications, *e.g.*, intelligent traffic systems and internet of vehicles, to train multiple regional specific models in parallel.
- The intermediate training results and final models are recorded on-chain for auditing purposes.

3.3.2 Cons

- Sharding was proposed to resolve the blockchain trilemma. However, sharding in the surveyed BCFL frameworks does not follow the original proposal to

address cross-shard transactions atomically. Hence, the security may need to be examined, especially in those multi-task cases.

- Although sharding allows a small number of nodes to form a shard for faster consensus, it could be more vulnerable to security threats since the number of attackers could dominate a shard easier than the blockchain. The validation mechanism in the main chain should be designed to mitigate this kind of vulnerability.

3.4 *Layer-2 BCFL Framework*

Another option for increasing the number of transactions handled by a blockchain is layer-2 scaling. Blockchains are used in this approach to process transactions off-chain but settle them on-chain. For instance, the Lightning Network [50] and Raiden Network [41] create payment channels for off-chain micropayment transactions but settle them on-chain. Another layer-2 scaling mechanism is based on a sidechain, such as Plasma [42], which develops off-chain/on-chain applications.

The goal of layer-2 scaling is to reduce latency and minimize interactions between users and the blockchain during the transaction process. Thus, it does not directly increase the throughput of blockchains. We refer to this type of BCFL framework as *Layer-2 or Off-Chain Compute BCFL Framework*.

Some BCFL frameworks use the *payment channel* approach, such as FLchain [24], while others use a *sidechain*, such as Arbitrum [27] and Ekiden [28]. The Off-Chain Compute BCFL Framework has the following characteristics:

- All transactions or computations are performed off-chain, and the results are recorded on-chain.
- Any dispute of a transaction or computation result is resolved within the off-chain channel or the sidechain.
- The off-chain trusted environment could be built by a blockchain as in the case of payment channels, e.g., FLchain, or by sidechain with or without Trusted Execution Environment (TEE), e.g., Ekiden, Arbitrum.
- All intermediate and final training results are recorded to the main chain, which is immutable and auditable.

3.5 *Payment Channel-Based Layer-2 BCFL Framework*

The FLchain [24] is a multi-task Layer-2 BCFL Framework leveraging the Hyperledger Fabric channels. Channels are private subnets used for peer off-chain communication. Only channel-associated peers are entitled to read, submit, validate transactions, and reach consensus in the channel. A separate ledger is maintained for each channel. The consensus is also applied on a per-channel basis. When a new

task is published, a channel and its channel-specific ledger, including its genesis block, which contains all the operating parameters and model hyperparameters, is created for each task's learning model.

FLchain provides an independent channel for each learning task. The consensus and ledger for storing the local model updates are channel-specific. The global model state trie is also maintained on a per-channel basis. The global model state trie can securely and unblemished store the global model weights in a Merkle Patricia Tree. The global model state trie can be regenerated and verified at any iteration from the genesis block up to the top block in the blockchain of a particular channel. The global model updates are computed, validated, verified, and stored on-chain.

3.6 Sidechain-Based Layer-2 BCFL Framework

The *sidechain scaling* approach is another commonly adopted Layer-2 scaling. Its objective is to scale smart contracts by taking the load off the blockchain while maintaining the promises of decentralization.

Arbitrum allows parties to create off-chain smart contracts using code to specify the behavior of a virtual machine (VM) that implements the contract's functionality. Arbitrum incentivizes parties to build consensus off-chain on what a VM would do so that Arbitrum miners need only verify digital signatures to confirm that parties have agreed on the behavior of a VM and *rollup* the results to the blockchain.

Arbitrum adopts an approach called *optimistic rollup* to increase the throughput of the underlying Ethereum network. Optimistic rollup refers to the situation that when an assertion is posted on-chain, it does not contain proof to guarantee its validity. Instead, the validator making that assertion posts a bond for some time to see if anyone wants to challenge the assertion on-chain. The challengers need to post a bond to challenge the assertion. If the assertor is wrong, they lose their bond. The assertion is accepted and becomes final if the challenge window expires with no successful challenges. Thus, Arbitrum allows honest parties to advance the VM state on-chain even if the parties cannot reach a consensus off-chain. Figure 4 shows the operation diagram of the Arbitrum protocols.

The verifier (or miners) will identify and penalize the dishonest party through the challenge-based protocol if a party lies about the behavior of a VM. Moving the verification of VMs' behavior off-chain dramatically improves smart contracts' scalability and confidentiality.

Similarly, Cheng et al. [28] propose an *off-chain compute, on-chain recording* framework called Ekiden. However, it requires the off-chain smart contract compute nodes to support TEE. The mechanism to secure off-chain smart contracts with blockchain operations is shown in Fig. 5.

A client sends the contract code to a selected compute node when creating a contract. The compute node loads the contract code into a TEE (*contract TEE*) and starts the initialization. The contract TEE obtains new cryptographic keys from a

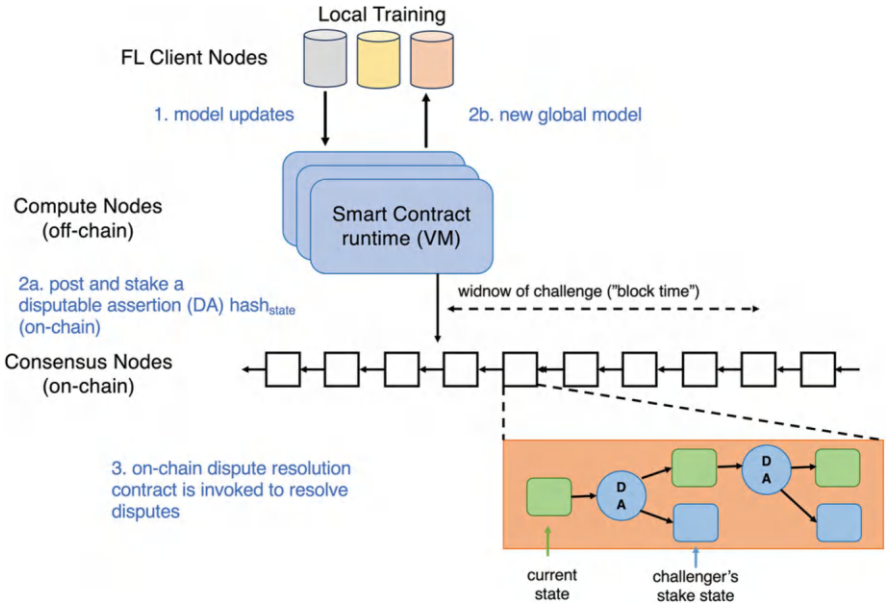


Fig. 4 Sidechain-based Layer-2 BCFL framework (Arbitrum). FL client nodes perform local training and send model updates to the smart-contract runtime on compute nodes, which aggregate them into a new global model. The compute nodes post and stake a disputable assertion (DA) hash on-chain, opening a challenge window during which consensus nodes can verify the result. If challenged, an on-chain dispute-resolution contract is invoked to determine the valid state. This mechanism ensures verifiable aggregation and trustless coordination across federated participants

key management committee (KMC), which publishes the public key along with an attestation that certifies the association between the newly created contract and the public key with which users can submit encrypted inputs. The compute node then sends the contract code, public key, initial state, and TEE-generated proofs, *i.e.*, attestation, to consensus nodes, verifying the attestation and incorporating the contract as part of the blockchain.

Once a contract is initialized, users can obtain the public key associated with the contract from the blockchain. They can then create a request with their input encrypted under the contract's public key. The request is sent to a compute node of the client's choice. Then, the contract TEE obtains the corresponding cryptographic keys from the KMC to decrypt the previous state and the client's input before it starts the execution. When the execution is completed, the contract TEE generates an output, a new encrypted state, and a cryptographic proof for correct computation. Finally, the output is delivered to the client only if the consensus nodes approve the new state. Consensus nodes verify the correctness of a state update before accepting the new state as valid and placing it on the blockchain.

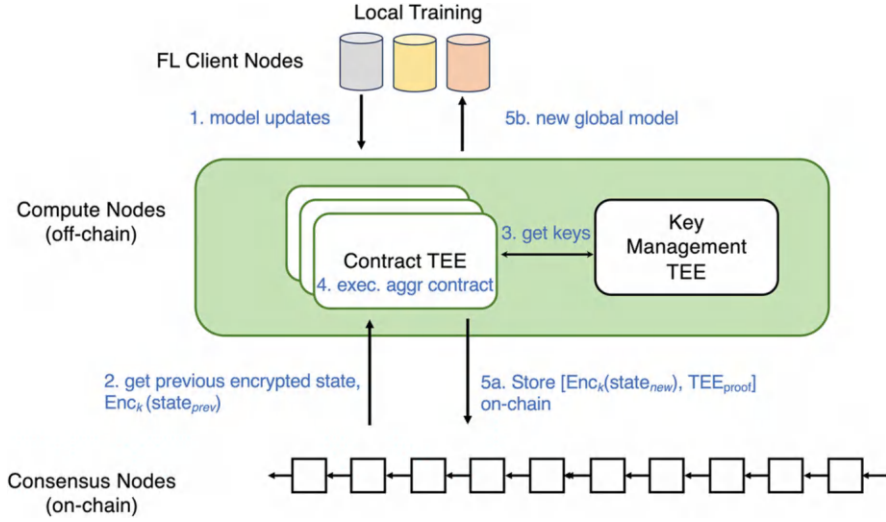


Fig. 5 Sidechain-based Layer-2 BCFL framework (Ekiden). FL client nodes perform local training and send model updates to off-chain compute nodes operating within Trusted Execution Environments (TEEs). The Contract TEE executes the aggregation contract using keys supplied by the Key Management TEE. Encrypted states and proofs are periodically stored on-chain for consensus verification, ensuring confidentiality, integrity, and verifiable aggregation in federated model updates

3.6.1 Pros

- The separation and collaboration of off-chain and on-chain smart contracts allow for offloading complex computation from all blockchain nodes. Moreover, the underlying blockchain can maintain auditable training results and handle occasional disputes on-chain.
- The off-chain smart contract compute nodes can be nodes with specific capabilities, e.g., TEE and GPU, to perform required tasks.
- The *stake-to-play* challenging protocols deter malicious actors.
- When deployed in FL settings, the model aggregations performed by smart contracts within the contract TEE in each FL round can mitigate the semi-trust or malicious server attack issues.
- The secure aggregation computation in TEE requires fewer overheads than Fully Homomorphic Encryption or Multi-Party Computing. Moreover, it does not need to make a trade-off between added noises and model performance as DP does.
- This type of framework can support not just FL but emerging confidential computing [51] as well.

3.6.2 Cons

- When applied to FL settings, the expiry of the dispute resolution timer for the on-chain challenge protocol may be impacted by the system heterogeneity. Setting it too short may pose a security threat or suffer from extended training unnecessarily if too long.
- Currently, TEE is under development. There are known attacks [52], *e.g.*, cache-based side-channel attacks and page-fault attacks, to the enclave. Moreover, enclaves may be killed at the host OS level [52], resulting in service disruption at the compute nodes.
- Enclaves run with library OS, which has limited support for multitasking [53].

4 Discussion

The primary objective of FL is to extract knowledge from silo datasets without violating participants' privacy or being subject to security threats. The main challenges are to provide a trustable environment and enough incentives for participants with good data to take in the knowledge extraction process. Meanwhile, protecting the system from malicious actors injecting false data or extracting additional information from other participants is a crucial success factor for FL.

Although BCFL frameworks alone cannot resolve all the FL intrinsic system, privacy, and security challenges, they provide means to address trust computation and reward for quality data and behavior issues. Regardless of the limitations of the three types of frameworks, each could be applied to solve FL tasks in the proper contexts. For example, the single-tiered Basic BCFL Framework may be suffered from inherited blockchain scalability issues. However, it is suitable for the swarm learning application to build a BCFL network using a permissioned chain among a limited number of medical organizations. Moreover, both Layer-1 and Layer-2 BCFL Frameworks not only scale the underlying blockchain operations but add flexibilities to extend needed secure and auditable FL operations for cross-silo and cross-device FL contexts.

Besides those inherent FL challenges and some of their mitigation listed in Sect. 2, designing fair reward systems that work with the client selection, consensus, and dispute challenge protocols is another open research topic to BCFL. A sound reward system is not just for providing fair compensation to encourage participation but is also directly related to system security and can be used to deter malicious participants.

Currently, there are only centralized FL open-source frameworks such as flower [54] and FedML [55] available for research. A decentralized BCFL open-source framework that considers emerging environments and contexts, *e.g.*, 5/6G edge computing facilities and IoV, could benefit future BCFL research. Moreover, a flexible BCFL research framework that modularizes algorithm and protocol

implementations could provide more insights while selecting suitable algorithm and protocol combinations for application contexts.

Furthermore, the sidechain-based layer-2 BCFL frameworks open more flexible applications beyond FL. With its off-chain trust compute and on-chain immutable recording approach, the layer-2 BCFL frameworks and proper reward systems could serve as a foundation for confidential computing outsourcing platforms. Both TEE technologies and on-chain dispute challenge protocol are emerging and subject to limitations and vulnerabilities that remain to be addressed.

5 Conclusion

Although the FL and BCFL are still in their infancy, with the rising privacy concerns, they have become mainstream for machine learning to access private data. In this study, we adopt the blockchain scaling taxonomy to classify surveyed BCFL frameworks into three categories, characterize each type with examples, and analyze its pros and cons. We identify three open areas that may further advance this research field: a fair reward system, an open framework, and a blockchain-based confidential computing platform that goes beyond FL applications.

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Blockchains and Cybersecurity



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Abstract In today's era of information explosion, data is being constantly created and accumulated, making it a valuable resource. Storing these large and sensitive resources securely and efficiently introduces a number of challenges. Blockchain, a decentralized digital ledger known for its tamper-proof and transparent characteristics, has great potential to improve data collection and storage. This chapter will discuss how blockchain technology can be used in data sharing and verification services. For example, it can be combined with software-defined networks (SDNs) to enable secure and intelligent data sharing in Intelligent Transport Systems (ITSs). Additionally, researchers can use a blockchain-based platform with smart contracts and batch verification to maintain scalability and the integrity of firmware updating files. Lastly, a decentralized DDoS data exchange platform over a consortium blockchain, called SOChain, can be used to address issues of trust and fairness in DDoS blacklist sharing. Overall, we believe that blockchain will play a crucial role in the current data landscape due to its secure and powerful features.

1 Introduction

With the advancement of technology, data sharing in various systems has made information analysis more convenient and diversified. For example, drivers in vehicular ad hoc networks (VANETs) can obtain many kinds of information such as driving record pictures or real-time road conditions through the transmission or sharing of data [13]. Medical data collected by hospitals can be shared with privacy protection policy so that a doctor can better understand a patient's condition and possible solutions [10]. The system log generated by website servers or the underly-

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ing programs of the critical system can be shared so that the system administrators can grasp the operating status of the entire system [5]. Moreover, the popularity of cloud technology has made data sharing unrestricted, and related applications have emerged one after another. From the reasons above, the importance of data sharing is increasing day by day.

The above research only focuses on how data analysis technologies can be used prudently for further prediction. The challenge of how to securely and privately store and share data still remains. According to a statistical report published by the software company Intact [14], a total of 256 personal data breaches occurred during the decade from 2011 to 2020, which is 382% more than the 67 cases from 2004 to 2010. Ineffective protection of information security (IS) led to hacker intrusion, causing a risk of personal data leakage. For example, user data has been stolen or lost from the platform.

In 2008, Satoshi Nakamoto proposed the concept of Bitcoin [9]. Bitcoin employs the blockchain technology which is a decentralized digital ledger with anonymity, tamper-proof, decentralized records, and transparent sharing. Those merits are very attractive in terms of security and privacy enhancement. In 2014, Ethereum has been proposed to improve the program ability. Smart contract (or chaincode) is a set of program codes stored and run in blockchain. Thanks to the merit of tamper-proof, the logic of a program cannot be modified, which can enhance the credibility. Many related studies have been proposed in the past few years [7, 12]. In [12], the authors stored the encrypted electronic medical data in a decentralized interplanetary file system (IPFS) to ensure the security of the storage platform and to solve the problem of single node failure. The authors of [7] have implemented a system model for efficient data collection and secure sharing. This model combines blockchain and deep reinforcement learning (DRL) to enhance the maximum data collection for ensuring the security and reliability of the shared data. As a result, the blockchain technology can be applied into several fields to strengthen the security aspect.

In this chapter, we introduce some research on blockchain platforms for data analysis and data sharing in various fields. First, we present a combination of blockchain and SDN for secure and intelligent data query service in Intelligent Transport Systems (ITSs) [15]. In terms of implementation on Internet of Things (IoT) devices, we present a novel firmware update platform with smart contract and batch verification to keep its scalability and security [2]. The last example is SOChain, a decentralized DDoS data exchange platform built on the consortium blockchain, which is proposed to overcome the challenges of trust and fairness [16]. We believe that these example applications are able to make readers quickly and effectively use blockchains to collect data for subsequent analysis.

2 Blockchain-Based Privacy-Preserving and Sustainable Data Query Service over 5G-VANETs

With the advancement of various sensors and communication technologies, the popularity of Intelligent Transport Systems (ITSs) has become a future trend [11].

Traditional ITSs are mostly managed by centralized systems. However, a big volume of data is stored in a centralized system, which may cause data loss and system errors due to a single point of failure. Moreover, as the rise of people's awareness of information security, the privacy and security of data sharing are also important issues. On the other hand, the lack of reward mechanisms may discourage data owners from sharing data. Those are why blockchain technology comes in. The features of blockchain technology including decentralization, incentive token, and anti-tampering can effectively improve the reliability of the ITS.

Although the blockchain can achieve specific security effects, there are still some problems that need to be considered [6]. For example, the content and calculation process of smart contracts are public to each node. Also, due to the blockchain's characteristics that need to reach consensus, the efficiency issue may be a bottleneck. What is worst is blockchain can only store small amounts of data and perform simple calculations. Therefore, taking advantage of blockchain's benefits and getting rid of the above problems is a challenging task.

In this study [15], the authors proposed an ITS architecture that combines multiple decentralization technologies and software-defined networks (SDNs). In addition to providing a reward mechanism, this platform employs 5G SDN technology to achieve efficient data transmission speed and low latency. It also proposed a series of encryption mechanisms offering the searching ability to accomplish the purpose of privacy protection.

The authors briefly introduced several important units in this system.

- **Trusted Authority (TA)**

TA is in charge of system initialization, including generating key pairs to all units in the system and announcing system parameters.

- **Data Owners**

Data owners can upload the message which is encrypted and stored into IPFS. Also, the data owners will send the encrypted keyword to the smart contract.

- **Data Users**

Data users can get the messages in need by searching with the encrypted keywords.

- **Blockchain Oracle**

The primary purpose of Blockchain oracle is to serve as the data owner, while the data owner may be offline. To achieve this effect, the oracle should get the necessary information from the data owner and offer security parameters to smart contracts.

- **IPFS**

IPFS provided a decentralized platform to store data with large file sizes.

As shown in Fig. 1, the authors introduced the workflow of the system. It can be roughly sorted out into seven steps.

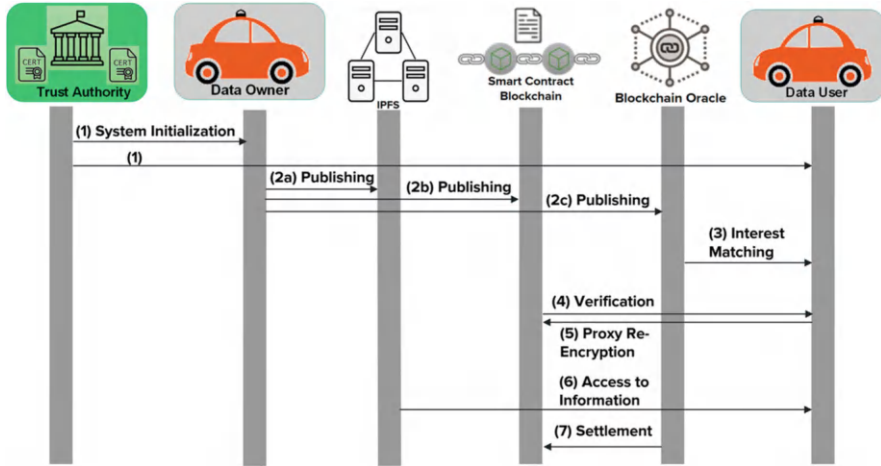


Fig. 1 System architecture

1. System Initialization

TA will publish the system parameters in this step. All users generate key pairs for the main system and oracle. To generate the incentive coin, TA will deploy the smart contract and initially give some coins to each user.

2. Traffic Information Publishing

The data owner encrypts the video record of the road by a temporary session key, uploads it to the IPFS, and gets the following URL. Then, the data owner sends a transaction to the smart contract with the encrypted keyword $CT_i(KW)$ and corresponding URL. The blockchain oracle can get the shared key for performing the proxy re-encryption algorithm.

3. Interest Matching

The data user computes his own encrypted keyword $CT_j(KW)$ and sends it to the blockchain oracle. After receiving it, the oracle will perform a trapdoor algorithm for the equality test. Once getting those matching messages $CT_{1..k}(KW)$, the oracle will return them to the data user. The data user chooses the corresponding index of $CT_i(KW)$ and sends it to the blockchain oracle.

4. Verification

After receiving the index, the blockchain oracle will generate and send trapdoor i to j ($T_{oi \rightarrow j}$) to the smart contract. The data user will send $CT_j(KW_j)$, $T_{j \rightarrow oi}$, and enough coins to the same smart contract. The smart contract will check whether the parameters for verification of KW_i and KW_j are matched. This step is to leave a record in blockchain for future audit.

5. Proxy Re-encryption

If the payment is confirmed and the verification has passed, the blockchain oracle will produce the rekey $rk_{oi \rightarrow j}$ for the data user and send it to the smart contract.

6. *Information Access*

The data user generates the re-encrypted ciphertext and gets the temporary key $T K$ by running the proposed ReEncrypt algorithm with the rekey. The data user can decrypt the encrypted video records from IPFS with $T K$.

7. *Settlement*

After the above steps are finished, the data owner and blockchain oracle will get their payment from the smart contract, which has stored some coins by the data user.

After defining the system architecture, the authors evaluate whether the entire framework meets the original design goals by analyzing the security properties:

- The system's confidentiality is proved based on the Diffie-Hellman assumption, and the employed hash functions are collision-resistant.
- The searchability and privacy are shown by cryptography with bilinear pairing operations.
- Sustainable data accessibility is ensured by blockchain oracle, which will take the role of data provider when the data provider is offline.
- The feature of the blockchain component guarantees the decentralization and transparency merits.

The experiments of this study illustrated the feasibility of this system. The key content of experiments includes the following:

• *Simulation Configurations*

The authors used SUMO and ns-3 to simulate the network with and without using the SDN technique. After the simulation results are obtained, the relevant delay values suggested that the performance is significantly improved after applying SDN.

• *Computation Cost Analysis*

The authors estimated the time complexity of each phase in the encrypting process and made a comprehensive evaluation of the delay results using the above experiments.

• *ON-OFF Model*

The authors adopted a two-state Markov chain to simulate the online behavior of the data owner. The experimental results show that no matter whether the data owner is online or not, their model will reply with a shorter delay.

• *Gas Consumption*

Their smart contracts are deployed in the Ethereum testnet, and the consumption of gas is less than in other implementations.

From the above experimental results, it can be inferred that this study provides a secure and privacy-preserving solution for ITS. Also, with the aid of 5G and SDN, the proposed scheme can achieve shorter response time than the current implementation.

3 Autonomous and Malware-Proof Blockchain-Based Firmware Update Platform with Efficient Batch Verification for Internet of Things Devices

With the increased use of Internet of Things (IoT) devices, the facing security challenges are also raising. For example, distributed denial of service (DDoS) attacks often take advantage of IoT devices to launch attack. In 2016, a notorious DDoS attack broke out across North America for a while [3]. The source of DDoS network flow came from several IoT devices including printers, IP cameras, and baby monitors. Therefore, how to enhance the security of IoT devices is an imperative issue. Blockchain is an emerging technology with unique features, such as tamper-proof, consensus, and traceable. In Ethereum, smart contracts can execute simple logic checks and program operations, which lays a good foundation for the practical use in different aspects. The outcomes of a smart contract are executed by each node with high credibility. In [2], the authors proposed a blockchain-based firmware updating platform. The platform is equipped with efficient batch verification function to keep scalability and maintain security. The key contributions are listed as follows:

- A smart contract provides the download URL and keeps checksum of the latest firmware file to achieve the tamper-proof feature.
- An independent malware scanning is performed by VirusTotal to ensure the uploaded firmware security.
- The batch verification mechanism can verify multiple signatures at the same time to achieve high scalability.

In Fig. 2, the proposed architecture and message flow are presented. Basically, the proposed system consists of the following roles:

- **Blockchain Full Node**

This node installed the blockchain platform to execute the blockchain operations including transaction sending, transaction verification, consensus execution, and smart contract deployment.

- **Client Node**

This is a thin device such as IoT device. This lightweight device is equipped with blockchain wallet to protect its own private key. It is assumed that the client node is with tamper-proof storage memory for the better protection of private key. The client node can perform the update/query firmware transaction in blockchain network.

- **Storing Node**

The peer-to-peer storage protocol is implemented in the storing nodes for saving the firmware. For example, Swarm, IPFS, and BitTorrent are the famous peer-to-peer protocols.

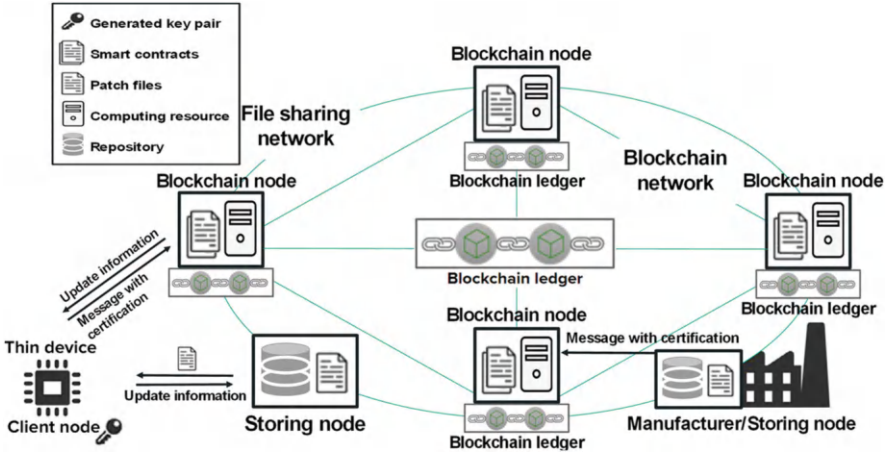


Fig. 2 System architecture

• *Manufacture Node*

A manufacture node is an update server maintained by an IoT manufacturing company, which is responsible to release the latest firmware to blockchain. Basically, the manufacture node will also serve as the roles of blockchain and storing node.

There are four phases: (1) system setup and registration, (2) manufacture update firmware procedure, (3) device query and update procedure, and (4) transaction and batch verification. In the setup and registration phase, the certificate authority (CA) is responsible for generating system parameters, the master secret key and the corresponding public/private key pair. The hardware security module (HSM) is used in client and manufacture nodes to save the private keys. The HSM will generate the pseudo-identity and corresponding public/private key for its own manufacture/client nodes.

In the manufacture firmware updating procedure, a smart contract is responsible for examining the integrity of the uploaded firmware. Moreover, the smart contract also integrates with VirusTotal API for malicious code detection, which can ensure that the uploaded firmware is secure. In the device query and updating procedure, the authors design another smart contract to execute the batch verification and to check whether the updating procedure is necessary.

In the transaction and batch verification phase, how to verify the single transaction and batch ones is presented. When a popular firmware for mitigating the famous attack is released, thousands of transactions may pour in at a short period. By the proposed batch verification, the verification burden of the smart contract can be alleviated. Note that a peer-to-peer storage protocol such as Swarm or IPFS is adopted to save the uploaded firmware for keeping the spirit of decentralization.

	Traditional	Lee and Lee (2017)	Ours
Firmware integrity	Half	Yes	Yes
Malicious code detection	No	No	Yes
Scalability	Poor	Yes	Yes
Verification efficiency	Poor	Poor	Good
Autonomous	Half	No	Yes

Fig. 3 Functionality comparisons

In terms of functionality comparisons, Fig. 3 shows the unique functionality of the proposed scheme. As you can see, a traditional firmware update procedure cannot ensure the integrity because of the centralized server architecture. For malicious code detection, the proposed scheme can integrate the VirusTotal scanning to guarantee the code security. In scalability, the traditional scheme is a central server with poor scalability. Thanks to batch verification, the proposed scheme can have better verification efficiency. To achieve autonomy, the proposed scheme leverages smart contracts to execute the update procedure, which is not feasible in Lee and Lee [4].

4 SOChain: A Privacy-Preserving DDoS Data Exchange Service over SOC Consortium Blockchain

Distributed denial of service (DDoS) is a malicious attempted attack. Attackers usually take advantage of multiple compromised, stolen, or compromised computer systems as the source of their attack. This attack will overwhelm the target server or its surrounding infrastructure. An effective solution is blocking abnormal traffic to the target server, service, or network. These compromised machines may include computers and other network-related resources, such as Internet of things (IoT) devices.

In September 2017, one of the largest DDoS attacks has occurred. The attackers targeted Google services, sending fake requests to 180,000 web servers, which causes the size of 2.54 Tbps for responses to Google in a short period. Google Cloud then revealed the incident in October 2020. However, the attack was not a one-time occurrence. In fact, the attackers had launched several DDoS attacks against Google’s infrastructure in the past 6 months to this incident. Another large DDoS attack occurred in September 2016, targeting OVH, a French company that provides web hosting services worldwide. The attack was launched with a huge botnet of 145,000 IoT devices, including routers, closed-circuit television (CCTV), and network-connected IP cameras. The peak data flow was close to 1Tbps, which was the largest DDoS attack at that time.

The security mechanisms on IoT devices are relatively poor. Once these IoT devices are exploited, the hackers may bring devastating attacks. So, this problem should not be trifled with. In view of this, many methods have been proposed to reduce the influence of DDoS attacks. However, it is important to note that most antivirus software still treats blocking IP addresses on DDoS blacklists as the most effective solution. For this reason, many fields are deploying information sharing and analysis centers (ISACs) to host security operations centers (SOCs) incident lists, which then correlate, build intelligence, and share across different domains. Unfortunately, the exchange of information between SOCs is not immediate, and it is not usually mandatory for SOCs to proactively update information about their detected DDoS attacks [8]. Therefore, the establishment of an information-sharing platform for sharing information about DDoS attacks is not only urgent but also conforms to the trend.

As the blockchain technology has got matured over the last decade, it has the potential for a wide range of applications, along with decentralized networks and tamper-proof features. In addition, the permissioned blockchain only allows licensed nodes to join, so unauthorized nodes cannot access the blockchain system, thus significantly improving the efficiency and auditability of transactions. Since the blockchain technology eliminates intermediaries and is also a federation that does not require third-party intervention, some security schemes have incorporated blockchain in their proposed approaches.

In [16], the authors proposed a novel consortium blockchain platform that puts external information into the blockchain to enhance the criticality of data validation. Moreover, the network monitoring function in the SOC is used to verify the uploaded DDoS information. In addition, since the data stored in the smart contract is almost transparent, which leads to a series of privacy problems, the proposed framework implements a two-level Bloom filter mechanism, which not only solves the privacy problem of each SOC in the smart contract but also improves the performance and efficiency.

Figure 4 presents the design architecture of the proposed system. It consists of four elements, which include SOC, smart contract, Swarm, and decentralized oracle service.

1. *SOC*

Here, SOC is responsible for integrating and analyzing security events to ensure information security. In the proposed framework, each SOC is treated as a node in the consortium blockchain. Each SOC can act as an uploader, buyer, or verifier. The uploader is to upload the collected DDoS information to earn tokens as a reward. The buyer queries the list of DDoS attack IPs stored in the platform to see if there are any suspicious IPs worth ordering. The verifier checks the validity of the uploaded DDoS information by anomalous network flow.

2. *Smart Contract*

A smart contract is a special agreement, in which the terms of the agreement are stored in the public database in the form of code, and automatically executed

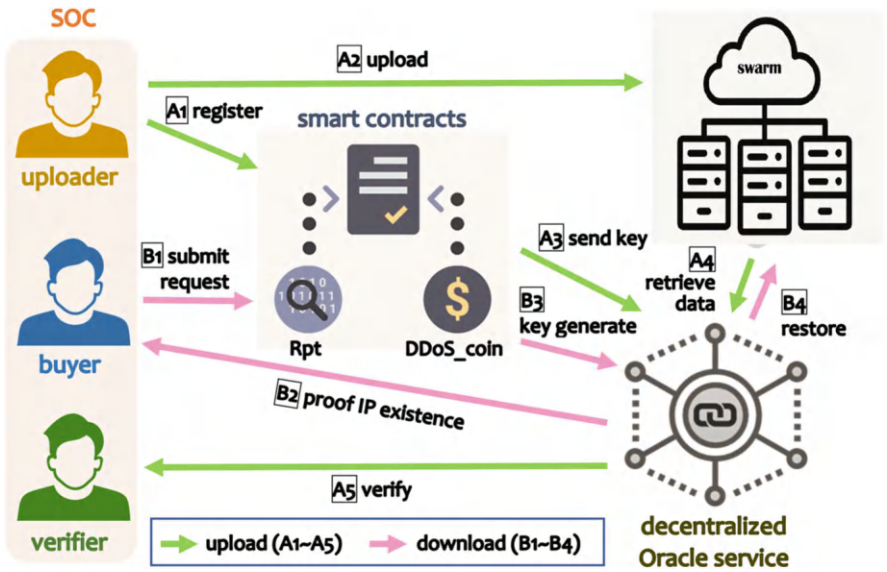


Fig. 4 System architecture

on the blockchain without being changed. In this framework, the smart contract plays the key role of controlling the entire process on the platform.

3. *Swarm*

This is an Ethereum-based system of peer-to-peer (P2P) network nodes that creates decentralized storage and communication services. Here, the Swarm is used to store the files of the encrypted DDoS IP list.

4. *Decentralized Oracle Service*

An oracle is a bridge that can push off-chain data to the blockchain. The oracle service can be categorized into centralized and decentralized, with the major difference in how data is obtained and audited. To avoid single node failure or malicious oracle problems, decentralized oracle services are introduced in this approach [1]. There is a mechanism to ensure that the data pushed to the smart contracts is trustworthy.

We briefly describe the operation process of the proposed system. First, suppose that many SOC's intend to join the consortium blockchain platform and share the security information they collect. As mentioned above, a SOC can play the role of an uploader, a buyer, or a verifier. The more information a SOC contributes, the more tokens it can receive. In order to advocate fairness, two types of reputation points are created: one representing positive reputation, while the other representing negative reputation. Suppose there is a SOC registered to the platform, and it has a small number of tokens. The initial value of both types of reputation points is set to 0. When an uploader wants to upload some DDoS information, it must first deposit some tokens into a specific smart contract to avoid abuse of the platform. A portion

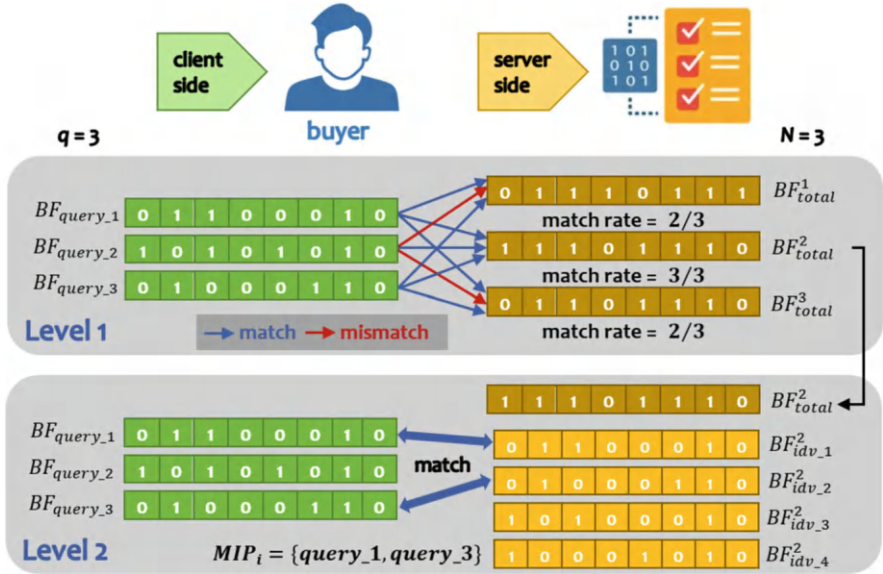


Fig. 5 Example of a dual-level Bloom filter

of this deposit will be paid to the verifier who endorses the data. If the verifier does not work honestly, the negative reputation of the verifier will increase. This abnormal behavior will eventually be recorded on the blockchain and removed from the endorsement list. In a sense, effective incentive mechanisms are the key to the success of a data exchange platform.

The uploaded information is encrypted to ensure better privacy. However, it makes the retrieval of the specific data more difficult. Therefore, as shown in Fig. 5, an additional two-level Bloom filter is designed to improve the efficiency of identifying the most relevant IP lists while maintaining privacy. Suppose the buyer detects some abnormal IP addresses and wants to query which IP list files stored in the platform are useful for their abnormal IP addresses.

The function of the first level is mainly responsible for identifying the matching rate between all the abnormal IP addresses detected by the buyer and all the unexpired IP lists stored in the platform. In the server side, the platform will generate a Bloom filter denoted as BF_{total}^i for each uploaded IP lists. In the client side, the buyer will produce BF_{query_i} for his/her detected abnormal IP addresses. After performing the first-level Bloom filter, the platform can rule out the specific uploaded IP lists, which do not contain enough useful information. However, due to the problem of error rate of Bloom filter, this scheme further devised a second-level Bloom filter.

In the second level, the buyer can further confirm whether the detected abnormal IP addresses exist exactly in the specific IP list. Each IP address stored in the uploaded IP list is also equipped with an individual Bloom filter $BF_{div_x}^i$. Therefore,

the platform further compares BF_{query_i} with each $BF_{idv_x}^i$. Finally, the buyer can make a decision to place an order based on the accurate match rate. In addition, to avoid the centralization of a single oracle, the approach based on [1] is adapted to implement a decentralized oracle service [17] in the framework to avoid the malicious behavior of a single oracle.

Generally speaking, with the development of the Internet, DDoS attacks are becoming more and more diversified and difficult to prevent, especially for large official websites and online trading platforms, which are the preferred targets of attackers. However, DDoS attacks have caused huge losses to many vendors and enterprises. As the nodes of blockchain are decentralized, each node has complete blockchain information to verify the validity of the uploaded data in other nodes. Even if an attacker breaches a node, the rest of the nodes can maintain the entire blockchain system normally and can effectively resist DDoS attacks.

This study [16] proposed a DDoS information exchange platform based on the consortium blockchain, attempting to protect privacy effectively and keep it fairly. In this study, the smart contract is used to control the operational process of the platform to effectively overcome the trust problem. Furthermore, an incentive system is proposed, i.e., tokens are issued to reward the data endorsers and to pay validation fees to the verifiers. Notably, to protect the privacy of the uploaded and purchased DDoS information, this framework also develops a novel dual-level Bloom filter to effectively improve the identification efficiency. The extensive performance and security analyses show that their proposed solution is a decent option for future DDoS information exchange platforms.

5 Conclusion

With the evolution of the wave of digitalization, the demand for data applications is increasing. How to obtain/utilize big data obtained through different sources has become one of the most important issues today. The blockchain with the unique characteristics of decentralization, transparency, traceability, and anonymity can keep privacy and security, making it have a wide range of application potential. In this chapter, we introduce the research on data sharing of blockchain in different fields, so that readers can quickly grasp the know-how of using blockchain for data collection and subsequent analysis. We found that some challenges associated with big data can be overcome with blockchain technology and vice versa. With the help of blockchain technology, it not only reduces the risk of being leaked but also makes the data more valuable and auditable.

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Part VII

The Blockchain Ecosystem

The Use of Blockchain Technology in the Compliance Function



Mark L. Shope

Abstract This chapter explores the opportunities and challenges of utilising blockchain technology to enhance compliance monitoring, regulatory compliance, and due diligence. Blockchain technology may offer the tools needed to usher sophisticated businesses into a new technological era, but ideas about how blockchain technologies fit into current compliance regimes continue to evolve. When new and fundamentally transformative technology meets traditional ways of thinking, there tend to be contradictions to our schema, and the revelation of possibilities is not entirely apparent. Thus, this chapter will attempt to discuss new possibilities of blockchain in the compliance monitoring, regulatory compliance, and due diligence functions. Attention will be given to six possible advantages of using blockchain technology in the compliance function of an organisation: (1) transparency, (2) immutability, (3) real-time availability of information, (4) security, (5) uniqueness, and (6) proof of compliance. This chapter will also discuss challenges to using blockchain technologies and conclude with suggestions.

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1 Introduction

This chapter explores blockchain technology solutions¹ as they relate to compliance monitoring, regulatory compliance, and due diligence.² For purposes of this chapter, “compliance monitoring” includes self (internal) or third-person (external) monitoring (e.g. companies utilising blockchain technology to monitor internal revenue and expenditures or franchisors utilising blockchain technology to periodically inspect franchisee revenue reporting); “regulatory compliance” relates to government monitoring and interaction (e.g. compliance reviews by government agencies that utilise blockchain technology to determine whether covered entities comply with relevant regulations); and “due diligence” relates to new client, employee, or business partner monitoring (e.g. utilising blockchain technology when conducting due diligence of a new banking client or JV/M&A partner or during customer onboarding or user registration). These concepts can move among and engage with each other in ways unique to the particular business.³ The overall purpose, however, of organising these concepts in this way is to sort out the potential stakeholders involved (internal, third-party, government), the specific activity relating to those stakeholders, and any reporting that is necessary among those stakeholders.⁴ This chapter will attempt to

¹ Works discussing distributed ledger technologies and their specific use cases generally benefit from disclaimers and qualifications. The author urges readers to consult the Standard Blockchain Disclaimer available at Mark L. Shope, *Distributed Ledger Technology in International Trade: Rethinking the Role and Necessity of the Customs Declaration*, 5 STAN. J. BLOCKCHAIN L. & POL’Y 114 (2022) and Mark L. Shope, *The Bill of Lading on the Blockchain: An Analysis of Its Compatibility with International Rules on Commercial Transactions*, 22 MINN. J.L. SCI. & TECH. 163 (2021).

² This chapter will not discuss blockchain technology in detail or explain its technological components. For an explanation of blockchain technology, see ARVIND NARAYANAN ET AL., *BITCOIN AND CRYPTOCURRENCY TECHNOLOGIES: A COMPREHENSIVE INTRODUCTION* (2016), Jean Bacon et al., *Blockchain Demystified: An Introduction to Blockchain Technology and Its Legal Implications* (Queen Mary Univ. of London, Sch. of Law, Legal Studies Research Paper No. 268/2017, 2017), <https://ssrn.com/abstract=3091218>.

³ For a discussion surrounding conceptualising and measuring compliance, see Christine Parker & Vibeke Nielsen, *The Challenge of Empirical Research on Business Compliance in Regulatory Capitalism*, 5 ANN. REV. L. & SOC. SCI. 45 (2009).

⁴ For an example of the scope of how technologies can be utilised with bank and financial institutions compliance, G.A. Walker notes that regulatory technologies can be used to improve “identity management, risk management, regulatory compliance, and financial crime control, as well as regulatory, supervisory, and macro-prudential analysis[, and] may improve risk data aggregation, modelling, scenario analysis, and forecasting; payment transaction monitoring; client and legal person identification; internal culture and behaviour monitoring; financial market trading; and new regulatory identification”. G. A. Walker, *Regulatory Technology (Regtech)—Construction of A New Regulatory Policy and Model*, 54 INT’L LAW. 1, 37–38 (2021) (internal citations omitted).

discuss issues as they relate to all three concepts and will generally refer to them as the “compliance function”.⁵

2 Potential Advantages of Using Blockchain Technology for Compliance Functions

Blockchain technologies can be used to record data entries that are immutable,⁶ verifiable,⁷ and traceable.⁸ The characteristics of immutability, verifiability, and traceability help provide a reliable tool for compliance monitoring, offering relevant stakeholders data to make faster and more efficient assessments of compliance.⁹ These data housed on the blockchain can enhance the processes and procedures necessary to act upon compliance incidents.¹⁰ Essentially, an “audit-ready” status could be maintained to anticipate (and act upon) any compliance issues.¹¹ Among the universe of blockchain technology uses, there are several key advantages of using blockchain technology in the compliance function of an organisation, among these are: (1) transparency, (2) immutability, (3) real-time availability of information, (4)

⁵ The “compliance function” and similar concepts have been defined in a number of ways by a number of authorities. See Sean J. Griffith, *Corporate Governance in an Era of Compliance*, 57 WM. & MARY L. REV. 2075, 2085–86 (2016). Some examples include: “the means by which firms adapt behavior to legal, regulatory, and social norms” *id.* at 2075; “creating and managing policies and procedures around ethics and compliance to uncover and prevent misconduct” Michele DeStefano, *Creating A Culture of Compliance: Why Departmentalization May Not Be the Answer*, 10 HASTINGS BUS. L.J. 71, 87 (2014); and “prospectively ensuring adherence to laws and regulations through the use of monitoring, policies, and other internal controls” Eric C. Chaffee, *Creating Compliance: Exploring A Maturing Industry*, 48 U. TOL. L. REV. 429, 429 (2017).

⁶ Zachary L. Catanzaro & Robert Kain, *Patients As Peers: Blockchain Based EHR and Medical Information Commons Models for HITECH Act Compliance*, 44 NOVA L. REV. 289, 315 (2020) (“The immutable nature of a blockchain ledger provides a consistent and redundant audit trail for reporting compliance”).

⁷ Depending on the blockchain, some information could be verified on the blockchain, other information requires external verification. See Shlomit Azgad-Tromer, *Crypto Securities: On the Risks of Investments in Blockchain-Based Assets and the Dilemmas of Securities Regulation*, 68 AM. U. L. REV. 69, 96 (2018) (“In some blockchains, the underlying sources of value can be verified automatically on the blockchain, while other blockchains require off-chain verification”).

⁸ C. Alden Pelker et al., *Using Blockchain Analysis from Investigation to Trial*, 69 DOJ J. FED. L. & PRAC. 59, 67 (2021) (“At its most basic level, blockchain analysis is not that much different than tracing funds from one bank account to another”).

⁹ In the securities context, see Richard Pan, *Blockchains, Securities, and Sections 11 and 12 of the Securities Act*, 15 N.Y.U. J.L. & BUS. 453 (2019).

¹⁰ Pedro Magalhães Batista Wolf-Georg Ringe, *Dynamism in Financial Market Regulation: Harnessing Regulatory and Supervisory Technologies*, 4 STAN. J. BLOCKCHAIN L. & POL’Y 203 (2021) (discussing regulatory and supervisory technologies).

¹¹ Regarding stock ownership, see Kevin V. Tu, *Blockchain Stock Ledgers*, 96 IND. L.J. 223 (2020).

security, (5) uniqueness, and (6) proof of compliance. The following will discuss each of these advantages, offering examples of potential practical use cases of each.

2.1 Transparency

Blockchain technology enables the recordation of transactions in a transparent environment.¹² This includes blockchains open to anyone (permissionless) or blockchains limited to certain participants (permissioned).¹³ Many of the blockchains contemplated in this chapter require a permissioned type of blockchain, calling for access to be kept by a trusted third party or group of parties.¹⁴ However, even permissioned blockchains (ones which only authorised members can participate) can in theory achieve a certain degree of meaningful transparency utilising zero-knowledge proofs and/or fully homomorphic encryption.¹⁵ For example, certain reviews by third parties could be conducted on encrypted data, eliminating or diminishing privacy, trade secret, or other various concerns relating to sharing sensitive data.¹⁶ There are, however, persistent issues relating to management override of internal control that pose challenges for blockchain record management.¹⁷ In other words, management or other actors within a company can still manipulate records by entering bad information onto blockchains

¹² Steven Young, *Changing Governance Models by Applying Blockchain Computing*, 26 CATH. U.J.L. & TECH. 53, 63 (2018) (“When code expresses an organization’s governing principles and rules on the blockchain, that organization’s foundation becomes verifiable, transparent, and nearly impossible to change without permission of the voting parties (i.e. no extra powers for government actors)”).

¹³ Lawrence J. Trautman & Mason J. Molesky, *A Primer for Blockchain*, 88 UMKC L. REV. 239, 254–55 (2019).

¹⁴ See Jean Bacon et al., *supra* note 2, at 26 (“If a permissioned system is stored by a [Trusted Third Party], it can withhold access to the blockchain archive and grant permission to view blocks or entries only in specific cases”).

¹⁵ See Tatsuo Mitani & Akira Otsuka, *Traceability in Permissioned Blockchain*, IEEE (2020). For a discussion on permissionless and permissioned blockchains in the corporate governance context, including advantages and disadvantages of each, see Alexandra Andhov, *Corporations on Blockchain: Opportunities & Challenges*, 53 CORNELL INT’L L.J. 1 (2020).

¹⁶ The future of crypto: IBM makes a new leap with Fully Homomorphic Encryption, IBM Research Blog, <https://research.ibm.com/blog/fhe-progress-milestones>. Providing personal data may be a major challenge to implementing blockchain solutions, especially in the new environment of GDPR. This important subject, however, is beyond the scope of this chapter.

¹⁷ Nadine Rückeshäuser, *Do We Really Want Blockchain-Based Accounting? Decentralized Consensus as Enabler of Management Override of Internal Controls*, in Leimeister, J.M.; Brenner, W. (eds.): *Proceedings der 13. Internationalen Tagung Wirtschaftsinformatik (WI 2017)*, St. Gallen, S. 16-30 (2017).

(intentionally or unintentionally).¹⁸ In this case, however, smart contracts could be used to automate cross-validation of the data on the blockchain.¹⁹ An efficiently functioning blockchain environment could automatically aggregate, analyse, and cross-validate information from multiple sources, reducing (if not eliminating) the problem of management override.²⁰ In addition, there are other consensus mechanisms available for permissioned blockchains,²¹ including the outsourcing of the validation mechanism.²² In any case, meaningful transparency does exist and is a key aspect of blockchains, boosting levels of compliance and access to crucial information when necessary.²³

2.2 Immutability

Blockchain technology, in general, creates an immutable ledger of entries,²⁴ and there are technological devices utilised in blockchain technology to increase its

¹⁸ Adam Sulkowski, *Blockchain, Business Supply Chains, Sustainability, and Law: The Future of Governance, Legal Frameworks, and Lawyers?*, 43 DEL. J. CORP. L. 303, 322 (2019) (noting that “data verification is only useful if the original record is reliable. Or, put another way: ‘garbage in, garbage out’”).

¹⁹ Ryan M. Mardini, *Point of Intersection Where Blockchain Meets Bankruptcy: Can the Ingenuity of Blockchain Restructure and Streamline the Bankruptcy Process?*, 3 WAYNE ST. U.J. BUS. L. 8, 34 (2020) (discussing tokenising real estate investments and eliminating intermediaries through smart contract automation).

²⁰ See generally Lennard Segers et al. The use of a blockchain-based smart import declaration to reduce the need for manual cross-validation by customs authorities (2019), Proceedings of dg.o 2019: 20th Annual International Conference on Digital Government Research (dg.o 2019), June 18, 2019, Dubai, United Arab Emirates (2019), <https://doi.org/10.1145/3325112.3325264> (discussing aggregating information from different sources in the import context).

²¹ Heather Hughes, *Blockchain and the Future of Secured Transactions Law*, 3 STAN. J. BLOCKCHAIN L. & POL’Y 21, 33 (2020) (“Permissioned blockchains may use a consensus mechanism other than proof-of-work. Hyperledger Fabric, for example, provides a number of consensus algorithms available to participants who use a Hyperledger platform”).

²² Parma Bains, *Blockchain Consensus Mechanisms: A Primer for Supervisors* (IMF, 2022), at 12.

²³ Iris H-Y Chiu & Ernest WK Lim, *Technology vs Ideology: How Far Will Artificial Intelligence and Distributed Ledger Technology Transform Corporate Governance and Business?*, 18 BERKELEY BUS. L.J. 1, 17 (2021) (“These efficiencies, driven by the operation of computational logic on a single ledger and the transparency of the ledger to all participants, can augur greater changes at the executive and corporate governance levels”).

²⁴ Marc Pilkington, *Blockchain Technology: Principles & Applications*, in RESEARCH HANDBOOK ON DIGITAL TRANSFORMATIONS 225, 233 (F. Xavier Ollerros & Majlinda Zhegu eds., 2016. But see Angela Walch, *Blockchain’s Treacherous Vocabulary: One More Challenge for Regulators*, 21 J. INTERNET L. 1, 1 (2017) and A.J. Bosco, *Blockchain and the Uniform Electronic Transactions Act*, 74 BUS. LAW. 243, 249 (2019) for alternate views on the concept of immutability of the blockchain.

integrity (creating a tamper-resistant environment).²⁵ Increased integrity and an essentially immutable ledger make changing the ledger nearly (but not absolutely) impossible.²⁶ Indeed, information entered onto the blockchain could be manipulated, organisations could keep parallel accounting books off-chain, and management override, discussed above, is a persistent issue.²⁷ All of this is true (and most systems, no matter how secure, are susceptible to bad actors, manipulation, and override); however, an audit process could, for example, aggregate transaction information across multiple blockchains and information sources (both on-chain and off-chain) to verify entries on one particular blockchain and utilise triple-entry accounting with blockchain.²⁸ Another way to accomplish this would be to utilise smart contracts programmed with the logic of relevant compliance regulations.²⁹ In this sense, Lawrence Lessig's observations in 1996 are informative:

If the regulator wants to induce a certain behavior, she need not threaten, or cajole, to inspire the change. She need only change the code—the software that defines the terms upon which the individual gains access to the system, or uses assets on the system. If she wants to limit trespass on a system, she need not rely simply on a law against trespass; she can implement a system of passwords. If she wants to limit the illegal use of copyrighted material, she need not rely on the threat of copyright law; she can encrypt the copyrighted material so only those intended to have access will have access. Always in principle, and increasingly in practice, there is a code (as in software) to assure what the code (as in law) demands, which means always in principle and increasingly in practice, law is inscribed in the code.³⁰

Lessig's observations highlight the possibility of technological authority over the regulatory process by use of code, passwords, or encryption and, by extension, the compliance process.³¹ In short, smart contracts could be used as a monitoring

²⁵ Phillip Shaverdian, Comment, *Start with Trust: Utilizing Blockchain to Resolve the Third-Party Data Breach Problem*, 66 UCLA L. REV. 1242, 1277 (2019). But see Rebecca M. Bratspies, *Cryptocurrency and the Myth of the Trustless Transaction*, 25 MICH. TECH. L. REV. 1, 25 (2018) (discussing the level of tamper-resistance using the blockchain).

²⁶ Jesse Marks, *Distributed Ledger Technologies and Corruption the Killer App?*, 20 COLUM. SCI. & TECH. L. REV. 42, 44 (2018) (“[The blockchain] is a peer-to-peer system, with no central authority managing the transaction flow. Because records are stored in many places, because no single user has the ability to change a given record, and because all nodes in the network are responsible for validating attempted updates or transactions, the end result is an immutable ledger that is virtually impossible to falsify”).

²⁷ Fiammetta S. Piazza, *Bitcoin and the Blockchain As Possible Corporate Governance Tools: Strengths and Weaknesses*, 5 PENN ST. J.L. & INT'L AFF. 262, 296 (2017) (“Indeed, parallel accounting, be it on a side ledger or in paper form, would still offer companies an ‘under the table’ opportunity to circumvent reporting duties. Accordingly, this system does not accomplish the result it purports to achieve: spotting and impeding self-interested and fraudulent transactions”).

²⁸ Cynthia Cai, *Triple-entry accounting with blockchain: How far have we come?* (2019).

²⁹ Morgan N. Temte, *Blockchain Challenges Traditional Contract Law: Just How Smart Are Smart Contracts?*, 19 WYO. L. REV. 87, 99-100 (2019), citing Pierluigi Cuccuru, *Beyond Bitcoin: An Early Overview on Smart Contracts*, 25 INT'L J. L. & INFO. TECH. 179, 184-86 (2017).

³⁰ Lawrence Lessig, *The Zones of Cyberspace*, 48 STAN. L. REV. 1403, 1408 (1996).

³¹ For a discussion of technological solutions imbedding rule of law values (and the extent to which the rule of law is actually promoted), see Monika Zalnieriute et. al., *The Rule of Law “By Design”?*, 95 TUL. L. REV. 1063 (2021).

tool³² because smart contracts can be programmed with actions necessary for the compliance function, including reporting and dealing with potential anomalies.³³

2.3 Real-Time Availability of Information

Real-time availability of information (and corresponding speed of the digital environment)³⁴ can increase the time that transactions are (and could be) cleared by reducing frictions.³⁵ For example, the US Securities and Exchange Commission standard settlement cycle for most broker-dealer securities transactions is T+1, and in theory this could be shorted to T+0 on the blockchain.³⁶ Regulators could utilise real-time information to make real-time decisions potentially using smart contracts to assist in determining and implementing proper responses.³⁷ Banks could make real-time decisions relating to customer onboarding.³⁸ Banks could also use this real-time information in determining whether a transaction should be completed or blocked due to economic sanctions or other considerations.³⁹ Stakeholders in the healthcare industry could improve the efficiency of health information exchange.⁴⁰ Energy markets could utilise blockchain technology to enhance real-time settlement

³² Chiara Picciau, *The (Un)predictable Impact of Technology on Corporate Governance*, 17 HASTINGS BUS. L.J. 67, 89 (2021) (“Business rules and accounting standards could likewise be encoded as smart contracts in a blockchain, facilitating internal monitoring and external auditing services, thus preventing possible violations from the outset”).

³³ Marcelo Corrales, Paulius Jurcys & George Kousiouris, *Smart Contracts and Smart Disclosure: Coding a GDPR Compliance Framework* (2021).

³⁴ Anthony J. Biller, David M. Chambers, *Distributed Ledger Trademark Registry? A Proposal for One Ledger to Rule Them All!*, 22 J. INTERNET L. 12, 12-13 (2018) (noting examples of blockchain applications that benefit from time elements).

³⁵ David Mills et al., *Distributed Ledger Technology in Payments, Clearing, and Settlement* 34 (Fed. Reserve Bd. Fin. & Econ. Discussion Series, Working Paper No. 95, 2016).

³⁶ New “T+1” Settlement Cycle – What Investors Need To Know: Investor Bulletin (March 27, 2024), <https://www.sec.gov/oiea/investor-alerts-and-bulletins/new-t1-settlement-cycle-what-investors-need-know-investor>.

³⁷ Yueh-Ping (Alex) Yang & Cheng-Yun Tsang, *Regtech and the New Era of Financial Regulators: Envisaging More Public-Private-Partnership Models of Financial Regulators*, 21 U. PA. J. BUS. L. 354, 402 (2018).

³⁸ Yvonne Lootsma, *Blockchain As the Newest Regtech Application-the Opportunity to Reduce the Burden of KYC for Financial Institutions*, Banking & Fin. Services Pol’y Rep., August 2017, at 16.

³⁹ *Id.*

⁴⁰ Zlate Dodevski, Sonja Filiposka, Anastas Mishev, and Vladimir Trajkovik, *Real Time Availability And Consistency Of Health-Related Information Across Multiple Stakeholders: A Blockchain Based Approach* (2021).

of trading certificate of origin.⁴¹ In this sense, robust third-party blockchain oracles would need to be utilised and connected to smart contracts on a blockchain.⁴² The combination of “on-chain code and off-chain infrastructure” can provide robust systems that can “react to real-world events and interoperate with traditional systems”.⁴³ In the compliance function, real-time monitoring is essential for taking appropriate action, including implementing preventative measures and, if necessary, issuing disclosures.⁴⁴

2.4 Security

Blockchain technology’s decentralised and immutable nature make it a highly secure system.⁴⁵ The security of the blockchain system makes it reliable for government regulators and other stakeholders who need to access the information on the blockchain.⁴⁶ The secure way that blocks of information are added to the blockchain makes it an ideal technology to ensure that only data that should be on the blockchain are added.⁴⁷ This is premised on the idea that measures are in place to prevent management override and actors adding false information onto the blockchain.⁴⁸ Mentioned above, this could be accomplished by aggregating

⁴¹ Claire Henly, Sam Hartnett, Sam Mardell, Buck Endemann, Ben Tejblum & Daniel S. Cohen, *Energizing the Future with Blockchain*, 39 ENERGY L.J. 197, 210 (2018) (“A peer-to-peer system for [certificates of origin] trading would enable real-time settlement and reduce internal administration and auditing costs”).

⁴² Morgen E. Peck, Blockchains: How They Work and Why They’ll Change the World, IEEE Spectrum 28 (Oct. 2017), at 34.

⁴³ Chain.Link, What Is a Blockchain Oracle?, <https://chain.link/education/blockchain-oracles> (September 14, 2021).

⁴⁴ Onnig H. Dombalagian, *Preserving Human Agency in Automated Compliance*, 11 BROOK. J. CORP. FIN. & COM. L. 71, 80 (2016) (“As regulators heighten real-time monitoring, individual firms must follow through with real-time operational and compliance controls as well”).

⁴⁵ Marcia Narine Weldon, Rachel Epstein, *Beyond Bitcoin: Leveraging Blockchain to Benefit Business and Society*, 20 TRANSACTIONS: TENN. J. BUS. L. 837, 883-84 (2019).

⁴⁶ Yuta Takanashi, Shin’ichiro Matsuo, Eric Burger, Clare Sullivan, James Miller & Hirotoishi Sato, *Call for Multi-Stakeholder Communication to Establish a Governance Mechanism for the Emerging Blockchain-Based Financial Ecosystem*, Part 2 of 2, 3 STAN. J. BLOCKCHAIN L. & POL’Y 117 (2020) (discussing the regulability of cyberspace).

⁴⁷ Jean Bacon et al., *Blockchain Demystified: A Technical and Legal Introduction to Distributed and Centralised Ledgers*, 25 RICH. J.L. & TECH. 2, 50 (2018) (“[Blockchain platforms] feature distributed storage and adding of new blocks managed by complicated consensus protocols and resource-intensive mining to ensure that no single party controls the addition of new blocks”).

⁴⁸ Alexander Savelyev, Copyright in the Blockchain Era: Promises and Challenges 15 (Nat’l Res. Univ. Higher Sch. Econ., Working Paper No. 77, 2017), <https://ssrn.com/abstract=3075246> (noting that “it is necessary to manage expectations of copyright owners and users accordingly: they have to admit that blockchain does not guarantee trustworthiness of information originating from the off-chain reality, only immutability of information generated within blockchain itself”).

information (both on-chain, among-chain, and off-chain) to cross-verify entries on the blockchain.⁴⁹

2.5 Uniqueness

The idea with uniqueness and the blockchain covers a few areas. It includes authentication by unique digital signatures and hash values that cannot be easily duplicated.⁵⁰ In situations where a “guarantee of uniqueness” or “guarantee of singularity” is required (e.g. in negotiable electronic shipping documents), blockchain technology inherently includes the technology that guarantees these ideas.⁵¹ This idea of uniqueness also covers unique custody and control.⁵² Possession of a private key is, in theory, strong (although not conclusive) evidence that the holder is the only entity with control of the particular electronic record.⁵³ Regulators and stakeholders can have a high level of comfort that the owner or controller of a particular record is, in fact, *the* owner or controller of that record.⁵⁴ Therefore, regulators and stakeholders can have a high level of comfort that transactions related to a particular record are in fact transactions that emanate from a particular record.⁵⁵

⁴⁹ For an illustration of how a given transaction *x* could be verified on a given block *b*, see Marten Sigwart, Philipp Frauenthaler, Taneli Hukkinen & Stefan Schulte, Towards Cross-Blockchain Transaction Verifications (White Paper, 2019) (“First, the destination chain checks that the block header of block *b* exists. . . . Second, it is verified that block *b* is currently part of the longest PoW chain. Then, it is verified that block *b* is followed by at least *n* succeeding blocks. . . . Further, the destination chain checks that the header of block *b* and the *n* succeeding block headers are not locked anymore. . . . If all of the above verifications are successful, the Merkle proof of membership certifying the inclusion of transaction *x* in block *b* is verified”).

⁵⁰ Scott J. Shackelford & Steve Myers, *Block-by-Block: Leveraging the Power of Blockchain Technology to Build Trust and Promote Cyber Peace*, 19 YALE J. L. & TECH. 334, 345 (2017) (“Since digital signatures are unforgeable, when a transaction looking to spend the money in a given account comes in, it is first checked for a valid digital signature; if the signature verification fails, then the transaction is discarded as illegitimate. If, on the other hand, the signature is valid, we know that it was sent by someone who has the corresponding signing key, and thus technically has spending authority”).

⁵¹ Christian Albrecht, *Blockchain Bills of Lading: The End of History? Overcoming Paper-Based Transport Documents in Sea Carriage Through New Technologies*, 43 TUL. MAR. L.J. 251, 264 (2019) (“While the issuance of a single record solely by a decentralized system was not possible at the dawn of electronic bills of lading, the blockchain’s “guarantee of uniqueness” offers this possibility”).

⁵² Nina Kilbride, *Custody, Control, and Assurances: Improving Legal Operations with Blockchain*, Bus. L. Today, December 2018, at 1, 4.

⁵³ See, e.g. Christian Albrecht, *supra* note 51, at 272 (“Exclusive control of a token on the blockchain is evidenced by a private key”).

⁵⁴ For an elaboration on challenges with blockchain technology and evidence law, see J. Collin Spring, *The Blockchain Paradox: Almost Always Reliable, Almost Never Admissible*, 72 SMU L. REV. 925, 945 (2019).

⁵⁵ Evidence could also be accompanied by a written declaration of a qualified person. See, e.g. Vermont Blockchain Enabling Act, VT. STAT. ANN. tit. 12, § 1913 (2018).

2.6 Proof of Compliance

Because of the nature of blockchain technology, organisations should be able to rely on the technology as proof of compliance (if, in fact, the evidence on the blockchain confirms compliance); not only could this be proof for regulators, but it could also be proof for a potential business partner, employer, or internal department.⁵⁶ For example, in the JV or M&A context, business partners who already use virtual data rooms could create a blockchain data room and provide access to third parties needing access to this room.⁵⁷ This blockchain data room could conceivably be programmed with code (in the form of a smart contract, and with the assistance of an oracle or cooperative artificial intelligence) that would test for and report back on certain compliance issues.⁵⁸ The code could be continuously monitoring in real time, without the need to actually initiate an audit because of a suspect transaction or event.⁵⁹ Rather, a well-programmed smart contract would be capable of monitoring events and transactions in real time.⁶⁰ Organisations could obtain immediate and secure access to customer KYC data and other compliance data necessary and unique to company operations.⁶¹ For example, “companies using permissioned

⁵⁶ Marcia Narine Weldon, Rachel Epstein, *supra* note 45, at 879 (“Because blocks are validated and recorded, companies could conceivably use the technology to provide proof that they have complied with various steps required by regulation”).

⁵⁷ Craig Rosenthal et al., *Using Legal Technology and Creative Structuring to Get Deals Done*, ACC Docket, September 2020, at 60, 62–63 (“With an increase in the number of documents exchanged between parties in an M&A transaction and the number of parties needing access to the relevant documents, VDRs have largely replaced physical data rooms in modern M&A transactions. A VDR is a secure online platform for document storage and distribution that allows multiple parties to simultaneously access relevant documents”).

⁵⁸ MUHAREM KIANIEFF, *BLOCKCHAIN TECHNOLOGY AND THE LAW: OPPORTUNITIES AND RISKS* 42 (2019) (discussing smart contracts and the role of oracles).

⁵⁹ Ajinkya M. Tulpule, *Enforcement and Compliance in a Blockchain(ed) World*, 1 CPI Antitrust Chron., Jan. 2017, at 3, <https://ssrn.com/abstract=2906465> (“Transaction data held on blockchains is created contemporaneously with the associated commercial transaction in real time and, therefore, provides a much clearer picture of the conditions of the market that exists at a specific point in time”).

⁶⁰ *But see* Danielle D’Onfro, *Smart Contracts and the Illusion of Automated Enforcement*, 61 WASH. U. J.L. & POL’Y 173 (2020).

⁶¹ Eamonn Maguire et al., *Could Blockchain be the Foundation of a Viable KYC Utility?*, KPMG 3 (2018), <https://assets.kpmg.com/content/dam/kpmg/xx/pdf/2018/03/kpmg-blockchain-kyc-utility.pdf>.

blockchain and smart contracts could automatically update records when shares are issued, sold, or transferred, which will provide shareholders, regulators, and other investors with real-time access to information”.⁶² This could include automatically filing a Form 8-K (a report public companies must file with the US Securities and Exchange Commission to announce major events) with the SEC.⁶³ Logically, transparent and easily accessible information on the blockchain could also reduce the need to even file a Form 8-K because the information necessary to include in the Form 8-K would already be open to the public and government regulators on the company’s blockchain.⁶⁴ A smart contract could highlight for the public or regulators material events in real time.⁶⁵

3 Challenges of Using Blockchain Technology for Compliance Functions

3.1 Understanding, Organisation, and Culture

There are often frictions to existing norms that are created when adopting new and transformative technology.⁶⁶ First, new technologies require individuals with adequate expertise on these new technologies.⁶⁷ Second, new technologies require individuals with subject matter expertise (i.e. expertise in the compliance function

⁶² Marcia Narine Weldon, Rachel Epstein, *supra* note 45, at 886; *see also* Federico Panisi, Ross P. Buckley & Douglas Arner, *Blockchain and Public Companies: A Revolution in Share Ownership Transparency, Proxy Voting and Corporate Governance?*, 2 STAN. J. BLOCKCHAIN L. & POL’Y 189, 191 (2019) (“Indeed, blockchain could enable the tracking of share ownership through the complete settlement cycle, enhancing the ‘shareholder democracy’ of listed companies and benefiting their corporate governance and the market in their shares”).

⁶³ Form 8K, <https://www.sec.gov/about/forms/form8-k.pdf>; U.S. Securities and Exchange Commission, Fast Answers, at <https://www.sec.gov/fast-answers>.

⁶⁴ Joan MacLeod Heminway & Adam J. Sulkowski, *Blockchains, Corporate Governance, and the Lawyer’s Role*, 65 WAYNE L. REV. 17, 42 (2019).

⁶⁵ David A. Wishnick, *Reengineering Financial Market Infrastructure*, 105 MINN. L. REV. 2379, 2425 (2021) (“Regulators and supervisors who have access to real-time, machine-readable trade data will be in much better positions when it comes to risk regulation and conduct policing than their peers were in prior days”).

⁶⁶ Frank-Jürgen Richter and Gunjan Sinha, *Why Do Your Employees Resist New Tech?*, HBR (Aug. 21, 2020), at <https://hbr.org/2020/08/why-do-your-employees-resist-new-tech>.

⁶⁷ Report of the Commission on the Geopolitical Impacts of New Technologies and Data, Atlantic Council, at <https://www.atlanticcouncil.org/wp-content/uploads/2021/05/GeoTech-Commission-Report-Full.pdf>.

of their specific line of business) to understand how the new technology can or cannot be incorporated into their compliance function.⁶⁸ Third, new technologies (sometimes) require new regulations or clarifications to existing regulations, which requires regulator expertise.⁶⁹ In short, there is often a disconnect between the domain of knowledge required for new technologies and the domain of knowledge required for the specific compliance function.⁷⁰ This is often an internal organisational issue, that is, new connections must be made within organisations between individuals with expertise in the new technology and individuals with expertise in existing compliance processes.⁷¹ New ways of thinking and connections are also required both internal to organisations (how individuals in the organisation are interacting internally with regard to new technologies) as well as external to the world (how individuals in the organisation are embracing new interactions with third parties, including government entities).⁷² All of these issues require individuals and organisations to actively find ways to incorporate new technologies into their thinking, which often requires a lifetime of learning and discovery.⁷³

3.2 Interoperability

One of the challenges to widespread and logical use of blockchain technology relates to interoperability challenges.⁷⁴ The blockchain ecosystem is largely siloed,

⁶⁸ For a discussion of the compliance function continuum, see Compliance modernization is no longer optional: How evolved is your approach? Deloitte, at <https://www2.deloitte.com/content/dam/Deloitte/us/Documents/regulatory/us-compliance-modernization.pdf>.

⁶⁹ Wulf A. Kaal & Erik P.M. Vermeulen, *How to Regulate Disruptive Innovation-from Facts to Data*, 57 JURIMETRICS J. 169 (2017).

⁷⁰ Dorothy Leonard-Barton and William A. Kraus, Implementing New Technology, HBR (Nov. 1985), at <https://hbr.org/1985/11/implementing-new-technology> (“As a rule, one organization develops the technology and then hands it off to users, who are less technically skilled but quite knowledgeable about their own areas of application”).

⁷¹ *Id.* (“The implementation manager has to integrate the perspectives and the needs of both developers and users”).

⁷² See generally, Tech Trends 2022: Peering through the lens of government, Deloitte, <https://www2.deloitte.com/content/dam/Deloitte/us/Documents/public-sector/us-gps-government-tech-trends-2022.pdf>; and Gov2020: A Journey into the Future of Government, Deloitte, <https://www2.deloitte.com/content/dam/Deloitte/au/Documents/public-sector/deloitte-au-ps-gov2020-journey-future-government2-130315.pdf>.

⁷³ For a discussion of the influence of new technologies on our idea of ourselves and the rule of law, see Jeanne Gaakeer, *Chapter 7 Control, Alt, and/or Delete? Some Observations on New Technologies and the Human*, 25 IUS GENTIUM 135 (2013).

⁷⁴ Rafael Belchior, Andre Vasconcelos, Sergio Guerreiro & Miguel Correia, *A Survey on Blockchain Interoperability: Past, Present, and Future Trends* (2021).

that is, there are compatibility issues when sharing and accessing information on a particular blockchain.⁷⁵ In this case, cross-chain oracles could be used to bridge the gap between blockchains and off-chain information.⁷⁶ According to the National Interoperability Framework Observatory (NIFO), there are a number of layers and components of interoperability, which includes four layers of interoperability: “legal, organisational, semantic and technical” (as well as “a cross-cutting component of the four layers”, integrated public service governance; and “a background layer”, interoperability governance).⁷⁷ From a technology point of view, there are interoperability issues “inter-chain”, that is, between different chains belonging to the same blockchain system (homogeneous blockchains) and different chains belonging to different blockchain systems (heterogeneous blockchains); and off-chain, that is, between blockchains and off-chain databases.⁷⁸ It is unlikely that any one blockchain platform will dominate an industry, which means interoperability issues will need to be integrated into any compliance systems.⁷⁹ This involves, among other things, creating standards, good governance, a legal framework, and trustworthy consensus and authentication mechanisms.⁸⁰

3.3 Reporting Challenges and Liability

A major legal issue that needs to be addressed by all stakeholders in the compliance function is the extent to which actors can rely on the blockchain/smart contract to avoid liability for non-compliance or non-disclosure if the actor relies on the blockchain to contain relevant information or a smart contract to properly identify an anomaly.⁸¹ For example, as previously mentioned, companies using a permissioned

⁷⁵ Blockchain and Interoperability: key to mass adoption, <https://www.finextra.com/blogposting/18972/blockchain-and-interoperability-key-to-mass-adoption>; Blockchain Interoperability: Why Is Cross Chain Technology Important?, <https://101blockchains.com/blockchain-interoperability/>.

⁷⁶ Les Wilkinson & Curtis Capeling, How to Understand Blockchain, ACC Docket, September 2018, at 66, 69 (“Very often, smart contracts use ‘oracles’ to provide off-chain information (such as proof of payment or performance, or data from devices in the Internet of Things) necessary to the execution of a smart contract”).

⁷⁷ National Interoperability Framework Observatory. European Interoperability Framework (2020), <https://joinup.ec.europa.eu/collection/nifo-national-interoperability-framework-observatory/3-interoperability-layers>; New European Interoperability Framework: Promoting seamless services and data flows for European public administrations (2017), at https://ec.europa.eu/isa2/sites/default/files/eif_brochure_final.pdf.

⁷⁸ Rafael Belchior, Andre Vasconcelos, Sergio Guerreiro & Miguel Correia, A Survey on Blockchain Interoperability: Past, Present, and Future Trends (2021).

⁷⁹ Interoperability, World Economic Forum, <https://widgets.weforum.org/blockchain-toolkit/interoperability/index.html>.

⁸⁰ *Id.*

⁸¹ Joan MacLeod Heminway & Adam J. Sulkowski, *supra* note 64, at 17-18 (discussing how blockchain technology may impact corporate governance considerations).

blockchain and smart contracts could automatically update corporate records and, if necessary, could automatically file a Form 8-K with the SEC.⁸² To what extent can the company or its board and senior management avoid liability if the smart contract fails to identify a material event that should have been disclosed to the SEC?⁸³ Or to what extent can the company or its board members and senior management avoid liability if the smart contract fails to determine that an event requires filing an 8-K?⁸⁴ Also, can the board substitute its knowledge for that of a smart contract or algorithm used on the blockchain?⁸⁵ For example, item 5.01 on the 8-K form (Changes in Control of Registrant) states, “If, to the knowledge of the registrant’s board of directors, a committee of the board of directors or authorized officer or officers of the registrant, a change in control of the registrant has occurred, furnish the following information . . .”⁸⁶ Should this be changed to “If, to the knowledge of the registrant’s board of directors, a committee of the board of directors, authorized officer or officers of the registrant, [as informed by an algorithm or other digital protocol], a change in control of the registrant has occurred, furnish the following information . . .?”⁸⁷ In the due diligence environment, one issue would be to what extent contracts can be rescinded or deals unwound if the blockchain or a smart contract did not uncover an anomaly that it should have. All of these questions would need to be addressed either by contract, regulation, case law, or other interpretive guidance.

4 Conclusion: Building Trust, Demolishing Bureaucracy

Building trust and demolishing bureaucracy is part of the title of an article written by John Palfreyman, who states:

Blockchain offers a new approach to enhancing transparency and collaboration between governments, business and citizens. By registering assets and recording ownership changes on a distributed shared ledger, the quest for transparency and right to privacy needn’t be at odds. Important information can be shared widely, seamlessly engendering true openness. Transactions can be verified close to real time and combined with the extensive use of privacy services make fraud and cybercrime very difficult. Smart contracts—once

⁸² Form 8K, *supra* note 63.

⁸³ Gregory Benson Jr., *Implications of Adopting Blockchain Technology on International Sales Transactions*, 39 N. ILL. U. L. REV. 486, 499 (2019) (noting that “questions linger on the legal implications of blockchain such as the lack of jurisdiction, the liability of blockchain technology, enforceability of smart contracts, and implications to due diligence”).

⁸⁴ *See generally*, Joan MacLeod Heminway & Adam J. Sulkowski, *supra* note 64, at 40.

⁸⁵ *Id.*

⁸⁶ Form 8K, *supra* note 63.

⁸⁷ Some considerations would be how the algorithm was programmed, trained, or designed (either intentionally or unintentionally), and whether the algorithm was “reasonable”. *See* Karni Chagal-Feferkorn, *The Reasonable Algorithm*, U. ILL. J.L. TECH. & POL’Y, Spring 2018, at 111, 129.

verified and deployed—will always execute. Reneging on a contract becomes a thing of the past. This all goes towards amplifying transparency, building trust and breaking down bureaucracy between government, business and citizens.⁸⁸

The idea of trust-building at its core asks us to work together to use the technologies and philosophies inherent in the blockchain to our advantage, the result of which is to demolish unnecessary bureaucracy. This chapter has briefly explored how blockchain technology could enhance compliance monitoring, regulatory compliance, and due diligence. Necessary among these solutions is the use of smart contracts and oracles to automate cross-validation of the data on a given blockchain. In order for the blockchain environment to function, information should be automatically aggregated, analysed, and cross-validated from multiple sources. This includes analysing information across multiple blockchains and information sources and programming smart contracts with the logic of relevant compliance regulations. For example, blockchain data rooms could be created and programmed to test for and report back on certain compliance issues.

Among the benefits of utilising blockchain technologies are the advantages of having real-time availability of information on the blockchain. This, including the relative speed of the digital environment, can increase the time to review transactions to make decisions and implement proper responses. This requires the use of robust third-party blockchain oracles and trust that the owner or controller of a particular record is the owner or controller of that record because of the use of public/private key technology.

Finally, serious attention should be given to the inherent disconnect between the domain of knowledge required for new technologies and the domain of knowledge required for the specific compliance function. In this sense, new connections must be made within organisations between individuals with expertise in the new technology and individuals with expertise in existing compliance processes. Targeted efforts should be made for more cross-domain communications. Attention should also be given to interoperability issues for both inter-chain and off-chain blockchains. Clear rules on liability should be further explored relating to reliance on blockchain tools, defining the extent to which actors can rely on the blockchain/smart contract to avoid liability for non-compliance or non-disclosure if the actor relies on the blockchain to contain relevant information or a smart contract to disclose this information.

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⁸⁸ John Palfreyman, Government Blockchain: Building Trust, Demolishing Bureaucracy (Feb. 7, 2017), <https://www.linkedin.com/pulse/government-blockchain-building-trust-demolishing-john-palfreyman>.

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Who Cares About Decentralized Finance? Evidence from DeFi Attacks



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Abstract This chapter provides an overview of historical decentralized finance (DeFi) protocol attacks. We empirically analyze the impact of DeFi attacks on the underlying blockchains' valuation. Using event study difference-in-differences regressions, we show that treated blockchains, relative to control group blockchains, experience an increase, rather than a decrease, in the market capitalization of the native cryptoasset in the days following the attack. Our results suggest that single DeFi security issues, once resolved, enhance the value of the underlying blockchain. We discuss our findings in the context of the vast blockchain and DeFi protocol ecosystem and its security.

JEL Codes: G23, O16, O33

1 Introduction

Decentralized finance (DeFi) protocols have emerged as one of the major categories of applications in the blockchain ecosystem, gaining economic relevance and attracting substantial capital deployment (e.g., total valued locked TVL in DeFi applications such as decentralized exchanges, borrowing and lending protocols, among others) [see, e.g., 7]. However, this success comes along with an economic interest in attacks on the security of blockchains and the applications they host. For example, the European Securities and Markets Authority (ESMA) highlights that

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DeFi comes with a major risk of exploitation.¹ Meanwhile, centralized exchanges and borrowing and lending institutions, i.e., CeFi institutions, are competing fiercely with their decentralized counterparts on liquidity and user convenience. Many major centralized custodians, for example, Mt. Gox and Binance, have been hacked, and significant user funds have been stolen directly from them. While DeFi's noncustodial access to complex financial intermediation protocols offers a clear advantage over centralized options such as the security against double-spending attacks [12], it may still suffer from economic protocol exploits, among other shortcomings.

Many DeFi attacks exploit vulnerabilities specific to a particular blockchain or its smart contract platform. If these vulnerabilities are not promptly addressed and patched, it can undermine trust in the blockchain's security, affecting the value of its native cryptoasset. Any attack on the blockchain's infrastructure layer will therefore inevitably test the demand for the underlying native cryptoasset, which then will be reflected in the asset's exchange rates on cryptoasset exchanges. Likewise, users adopt blockchains because of the decentralized applications (dApps) that they host [see, e.g., 9] as they derive economic utility from being able to use digital marketplaces, speculate using decentralized finance platforms, or for token issuance [7]. Thus, if dApps' viability is undermined by economic or other attacks that undermines the security and functioning of the platform [cf. 5, 13], the demand for the underlying cryptoasset may be relatively reduced as users can migrate to other blockchains' dApps with similar functionality to avoid recurrence of the security issues.

Therefore, one hypothesis of this study is that DeFi attacks result in negative valuation effects on the underlying blockchain's native cryptoasset needed to operate on a given blockchain.

However, the relative impact of DeFi attacks on the underlying blockchain's native asset valuation may be insignificant or even positive. DeFi ecosystems are interconnected, and an incident in one DeFi project can have a spillover effect on others so that an attack on one platform will inevitably (negatively) affect others as well, resulting in negligible relative change in market valuations [4, 8].

When an adverse event occurs, such an event attracts media attention, which may even lead to more people learning about the associated DeFi application or infrastructure and ultimately increase the demand for that blockchain. In addition, blockchain technology, including the DeFi application, heavily relies on the community for system development. In response to a DeFi attack or failure, the respective blockchain user communities and project developers can work together to solve the problem and enhance the overall resilience of the DeFi ecosystem through improved security measures. This will have a positive impact on the native cryptoasset, even if the attack appears to be a negative event in the very short term.

¹ https://www.esma.europa.eu/sites/default/files/2023-10/ESMA50-2085271018-3349_TRV_Article_Decentralised_Finance_in_the_EU_Developments_and_Risks.pdf.

The second hypothesis of this study is that DeFi attacks may result in positive valuation effects on the underlying blockchain's native cryptoasset.

In this chapter, we provide an overview of DeFi protocol attacks and test our hypotheses by exploring how investors in the underlying blockchain's native cryptoasset react to such exploits and attacks on DeFi protocols. Instead of analyzing individual DeFi tokens, we focus on the valuation of the native cryptoasset to understand the impact of vulnerabilities in decentralized financial applications (protocols) on the underlying smart contract blockchain. Such blockchains allow us to compare the effects of attacks on different blockchain ecosystems. We consider attacks on DeFi protocols as a potential source of systemic risk to the entire blockchain-specific ecosystem, especially if its native asset prices are affected by a single DeFi protocol attack.

We collect a comprehensive sample of DeFi attacks from the [REKT database](#) and estimate (event study) difference-in-differences regressions to assess the impact of DeFi attacks on the market valuation of affected blockchains. In doing so, we observe whether there are differential effects on affected (treated) platforms compared to various control groups of blockchains. In our models, we consider multiple control groups (e.g., largest blockchains) and restrict our analysis to certain types of DeFi exploits (e.g., significant losses, Ethereum only) to enhance the robustness of our models and main finding.

Our key finding is that when major DeFi protocols on a blockchain encounter attacks that lead to financial losses for their users, the valuation of the blockchain's native cryptoassets (e.g., ETH for Ethereum) increases instead of decreasing after the events. This suggests that the blockchain infrastructure views these incidents as positive developments, as they may attract more new users, lead to more resilient smart contract code, thorough audits, and assurance that the developer community is able to swiftly cope with individual economic or technical exploits of blockchain-based applications.

Section 2 provides an overview of DeFi exploit categories relevant to understanding the nature of the events we study. Section 3 introduces our data set and empirical methodologies we use to test hypotheses related to blockchain market valuation effects of DeFi attacks. Section 4 discusses our findings from the empirical analyses, and Sect. 5 concludes.

2 Categories of DeFi Exploit Events

While DeFi brings enormous technological advances to the financial markets, it is subject to many novel risks due to its intertwined nature and lack of oversight. Such risks may arise from, among others, external attackers, the founders or staff, or problems of governance, operation, or techniques. Given the relatively recent emergence of DeFi, the literature on the security and valuation effects of DeFi attacks is scarce. [17] provide a systematization of the knowledge, highlighting taxonomies and different types of DeFi attacks, including network, consensus, smart

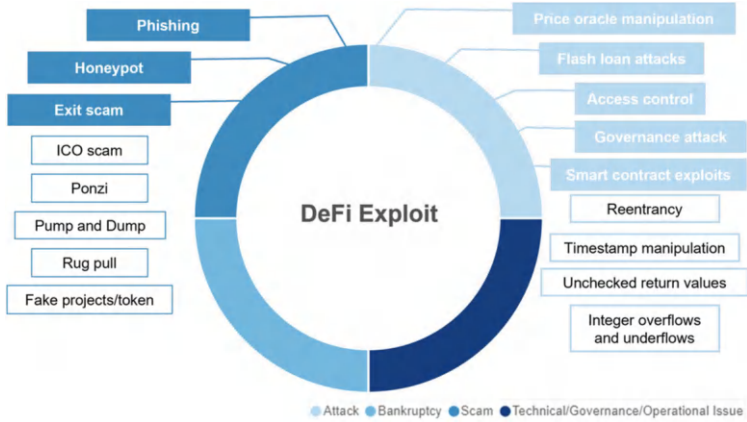


Fig. 1 Categories of DeFi exploits

contract, or protocol attacks as major categories with potential security issues. Hornuf et al. [6] provide an alternative taxonomy for Ethereum blockchain-related attacks. We consider the impact of attacks at the DeFi application level (i.e., not infrastructure or consensus issues) and evaluate whether attacks on protocols impair the valuation of the underlying blockchain’s native cryptoasset, across multiple blockchains. Figure 1 illustrates the categories of some of the most common DeFi exploits [10, 17]. Four types of exploits are listed, many of which are unique to the DeFi space. The categories are by no means exclusive and may overlap.

Attack DeFi applications are built on a smart contract-enabled blockchain to ensure the automated execution of transactions [16]. Smart contract is, therefore, the backbone of DeFi applications. Once the code is deployed, it cannot be altered. If the code is flawed, the flaws can be exploited by attackers. DeFi attacks, therefore, majorly target the vulnerabilities of the code or the design of the smart contracts to disrupt the proper functioning of the smart contract and drain the funds.

Reentrancy is one of the most catastrophic types of attacks in the DeFi space and has been studied mainly in computer science research from a technical perspective [3, 14]. A reentrancy attack occurs when a smart contract interacts with an external contract before completing its state updates. An attacker creates a malicious contract to act as the “external contract.” When the original contract interacts with it, the malicious contract callbacks or “reenters” a function (often a withdrawal) in the original contract, exploiting the delay in state updates. The importance of reentrancy attacks is illustrated by the 2016 DAO attack, in which the attackers exploited 3.6 million Ether (approximately \$50 million). The event had a significant impact and led to the hard fork of Ethereum [11].

Flash loan attack is another special attack that exists in the DeFi environment. Flash loan uses the unique structure of the blockchain to provide a way to borrow without collateral [5, 13]. It allows the borrower to take the loan and repay it within

one blockchain transaction. If the borrower fails to repay the debt by the end of the transaction, the transaction fails and is reversed. Such a design allows the borrower to gain massive liquidity without providing upfront collateral. However, a malicious borrower could leverage the fund to exploit the vulnerability of other smart contracts or DeFi protocols (e.g., market price manipulation) to gain an unfair advantage before the borrower pays back the loan by the end of the blockchain transaction.

Scam Unlike attacks, scams are usually not purely technical. They involve more comprehensive structures and plans with the intent to defraud. Within this category, in comparison to phishing and honeypot, exit scams have a more greater impact and can result in significantly more financial loss.

Rug pull is one of the most impactful exit scams on investors. It is usually a complete project with a native token and some eye-catching features designed to attack investors. The scammer then tries to hype the project by fraudulently promoting the project by creating websites and social media accounts with misleading information, using wash trading to inflate the price of the token, etc. Once the scammer has raised enough funds, they abruptly abandon the project and leave with the money from investors. The impact of sophisticated scams is not limited to the financial losses, but can also weaken investor confidence in similar projects and the blockchain platform.

Technical/Governance/Operational Issue Governance or operational issues can occur during the operation of a project, mainly due to internal negligence or mismanagement without the intention of deception. Technical issues can be rooted in the flawed design of the products, the reduced reliability of the infrastructure and the smart contract, or problems with external systems (e.g., Oracle issues). This can result in financial losses but, more importantly, trigger external attacks if not correctly addressed in a timely manner.

Bankruptcy Bankruptcy is the insolvency of a company, usually centralized, but is closely associated with the DeFi project. It is not a stand-alone risk, but a potential outcome of various risks and exploits in the DeFi ecosystem, including the exploits we discussed above. However, the unique nature of the DeFi environment makes the project particularly vulnerable to market volatility and the current ambiguous regulatory phase. For example, in November 2022, the collapse of one of the largest cryptocurrency exchanges, FTX, shook the market and led to the subsequent bankruptcies of related companies, including Genesis and BlockFi.

3 Data and Methodology

3.1 Data

Our event set is from the [REKT database](#), which covers relevant detected DeFi and CeFi attacks that have resulted in attackers extracting economic value from individual entities or DeFi protocols, causing protocol stakeholders to experience

financial losses. It lists over 3000 events, and the corresponding amount of US dollar funds lost as a result of an attack. Our sample includes all exploit events listed in the REKT database between September 1, 2012, and June 1, 2023, with more than \$10 million in financial losses. This way, we obtain 143 exploit events, including 56 CeFi events and 87 DeFi events affecting 11 chains (Avalanche C-Chain, Binance Smart Chain, Cronos, MultiverseX (Elrond), Ethereum, FANTOM, Terra Classic/Terra, Polygon, Ronin, Solana, TRON).²

For each DeFi event, we collect the event date, the business model of entities, the type of the exploit event, financial losses in US Dollar, and affected blockchains (only for DeFi events).³ In addition, we include bitcoin (non-smart contract blockchain) and 49 smart contract blockchains that are never targeted by attack events as our control group sample. For each blockchain, including the treated and control group blockchains, we identify its native asset and collect daily market capitalization values from coinmarketcap.com.

Figure 2 gives an overview of the events on a quarterly basis. The number of events remained stable until 2020 when the number of event started to increase substantially. The majority of events were attacks, mostly exploiting the technical vulnerabilities in the systems for financial gain. These were followed by scams, which were mostly perpetrated by the founders who created the projects with the intention of committing fraud.

The financial loss curve is largely influenced by single high-loss events. The most significant event was the Terra-LUNA crash which happened in May 2022 and caused approximately \$ 40 billion financial loss to investors. TerraUSD (UST), a US dollar-pegged algorithmic stablecoin launched by Terraform Labs, maintained its peg through an underlying mechanism that regulated the supply of LUNA, the native token of the Terra network. Such algorithmic stablecoins rely heavily on public confidence in the coin and can have high volatility, despite their sophisticated designs [18]. On May 7th, the UST began to lose its peg to the US dollar, triggered by a series of large withdrawals of UST, causing a bank run. The selling pressure led to a “death spiral” that eventually drove the LUNA price to zero.

The second largest event is the rugpull scam perpetrated by the founders of a South African centralized crypto platform, Africrypt. In April 2021, shortly after the platform collected the funds from investors, investors were notified that their accounts had been frozen due to a hack. Shortly after the claim, the founders disappeared with 69,000 BTC (\$3.6 billion) collected from the investors. In April 2023, the founders were arrested in Zurich, but denied the fraud and claimed the funds were missing due to a hack. And the third largest event is a major hack on the darknet marketplace Silk Road. In 2012, the hacker pulled off the theft by exploiting a flaw in Silk Road’s bitcoin withdrawal mechanism for sellers and was able to steal

² Note that we treat Layer 2 solutions as individual chains if they have their own consensus process.

³ The March 29, 2022, attack event on Ronin chain is excluded from our data set for further modeling due to the lack of market capitalization data for RONIN.

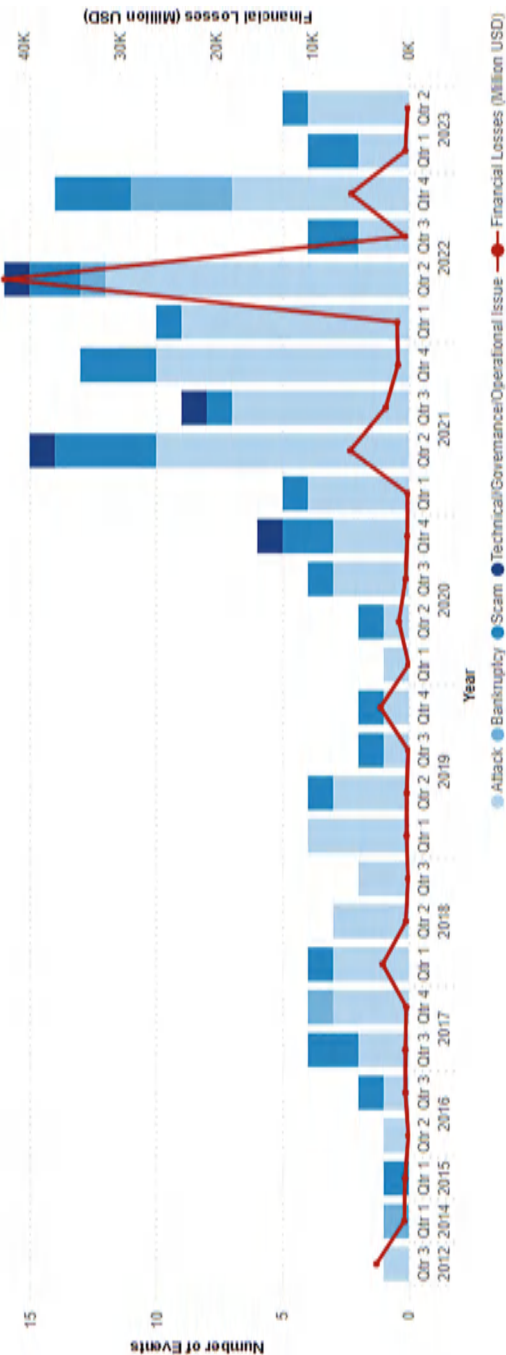


Fig. 2 Timeline of crypto-related exploit events between 2012 and 2023

50,000 BTC (\$3.36 billion) from the system [15]. Nine years after the event, the hacker was arrested, and most of the lost funds were recovered.

3.2 Methodology

To evaluate the impact of DeFi hacks on respective blockchains' market values, we perform two-way fixed-effect (TWFE) DID regressions in which we regress daily market capitalization of the blockchain's native cryptoasset on treatment dummies as well as chain- and time-fixed effects for each cohort (i.e., each event has its own fixed effects).

$$\text{MCAP}_{k(i),t(i)} = \alpha_{k(i)} + \lambda_{t(i)} + \beta (\text{POST}_{t(i)} \times \text{TREATED}_{k(i)}) + \rho_{k(i),t(i)}.$$

$\text{MCAP}_{k(i),t(i)}$ is either the total market capitalization of blockchain k in event i or the ratio of its market capitalization and Bitcoin's market capitalization, which captures the valuation effect on the treated chain relative to the non-smart contract blockchain Bitcoin. The coefficient β of the DID term, $\text{POST}_{t(i)} \times \text{TREATED}_{k(i)}$, captures the change in native asset market capitalization (or ratio) following the treatment of a blockchain by a major DeFi attack.

We also run (TWFE) DID event study regressions [cf. 1] in which we estimate treatment coefficients for each time period before and after the event in time $t = 0$. That is, for each event i , we estimate

$$\text{MCAP}_{k,t} = \alpha_{k(i)} + \lambda_{t(i)} + \sum_{l=-L, l \neq 0}^L \mu_l D_{k(i),t(i)}^l + \varepsilon_{i,t},$$

where $D_{k(i),t}^l$ is the treatment dummy for blockchain $k(i)$ in cohort i in DeFi hack (cohort) being $L = \{10, 15\}$ days from the start of treatment.

To build control groups of unaffected blockchains to compare to treated ones within each cohort (event), we follow three strategies. First, we sample all blockchains that are not affected by event i and do not experience overlapping events in the respective time window of length L (i.e., blockchains not treated in event i). Second, for each event, we construct two control groups. The first one includes all the not effected blockchains mentioned in first strategy. In the second control group, we restrict our sample of control chains to the largest four (as of market caps one day before the event window starts). This alleviates concerns that mainly smaller chains are used as control observations to compare against mainly larger chains that are affected by DeFi security issues. Third, in order to identify the factors of event scale and platform characteristics, we run all of the analyses above in three types of event settings: (1) full event sample, (2) the events with more than 100 million US Dollar in financial losses ("large events"), and (3) all events on the Ethereum blockchain.

4 Results and Discussion

4.1 TWFE DID Regressions

Our analysis begins with TWFE DID regressions. The results of these regressions are detailed in Table 1, where 24 models are presented, each varying in their estimation windows, event settings, and choice of control groups.

Table 1 TWFE DID regressions with estimation windows of 10 days and 15 days

	10 Days				15 Days		
	Mcap blockchain		Mcap ratio		Mcap blockchain		Mcap ratio
	A All chains				A All chains		
Post X treatment	5,244,491,842	*	0.0069	**	4,703,499,605		0.0072
Prob.	0.0613		0.0227		0.1472		0.0400
R ²	0.0076		0.0200		0.0044		0.0150
	B Four largest chains				B Four largest chains		
Post X treatment	4,723,051,490		0.0061	*	3,881,056,266		0.0056
Prob.	0.1440		0.0563		0.2922		0.1317
R ²	0.0138		0.0258		0.0028		0.0085
	Set 2—Large Events						
	A All chains				A All chains		
Post X treatment	2,547,524,131		0.0025		2,653,221,133		0.0029
Prob.	0.6241		0.6587		0.7131		0.6750
R ²	0.0029		0.0200		0.0200		0.0037
	B Four largest chains				B Four largest chains		
Post X treatment	(2,181,311,825)		0.0042		57,002,336		0.0050
Prob.	0.7672		0.4414		0.9951		0.4710
R ²	0.0020		0.0116		0.0000		0.0108
	Set 3—ETH events						
	A All chains				A All chains		
Post X treatment	11,398,000,000	**	0.0137	***	10,837,000,000	**	0.0140
Prob.	0.0118		0.0064		0.0426		0.0186
R ²	0.0507		0.0724		0.0278		0.0535
	B Four largest chains				B Four largest chains		
Post X treatment	11,398,000,000	**	0.0137	***	10,837,000,000	**	0.0140
Prob.	0.0137		0.0076		0.0470		0.0211
R ²	0.0511		0.0730		0.0280		0.0541

Significance levels are denoted *** p<1%, ** p<5%, * p<10%

For the all-event sample, we find a positive effect of DeFi exploits events within a 10-day window on the blockchain market capitalization when benchmarked against all control chains. In comparison, when estimating the market capitalization ratio of the blockchain against bitcoin, a significant positive effect is observed across both 10- and 15-day windows. Narrowing down the control group to the four largest blockchains, the positive effect only remains within the 10-day window for the market capitalization ratio.

Interestingly, when we solely consider large events that result in financial losses exceeding \$100 million (Set 2), regardless of the composition of the control group, the events exhibit neither a significant impact on the market capitalization of the blockchain platform nor on the market capitalization ratio of the blockchain to bitcoin. Notably, all eight models of ETH events present statistically significant positive impacts on native asset market values.

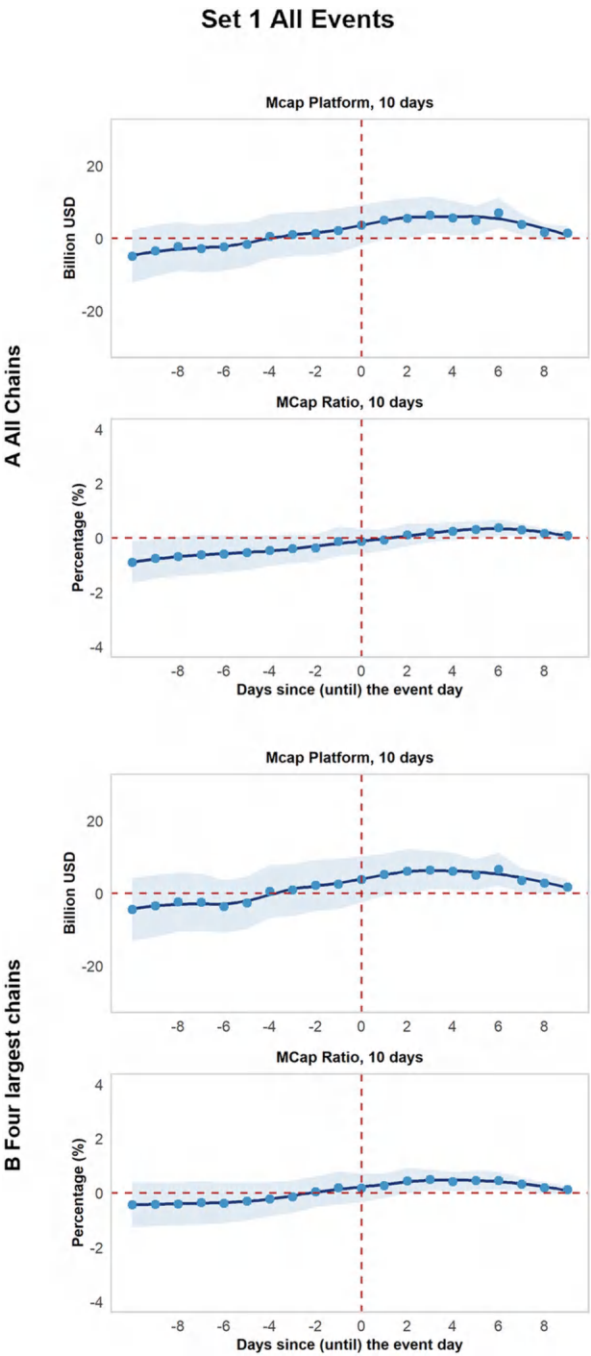
4.2 DID Event Study Regressions

The TWFE models provide insights into the average effects of DeFi exploits on blockchain platforms. To further explore the day-to-day effects within the estimation windows, we employ DID event study regressions, giving us a detailed view of valuation dynamics. Figures 3, 4, and 5 displays the results of 24 models with the same settings as the TWFE DID regressions. Prior to the treatment dates, all models consistently show insignificant results, supporting the parallel trend assumption, as there are no observable significant differences in market value changes prior to the attack event.

In the all-event sample regressions (see Fig. 3) with all control chains, positive effects are revealed in both 10- and 15-day windows on the market capitalization value and ratio. The positive effects diminish after eight days of the events toward the end of the estimation windows. Next, when using the four largest chains as control group, more significant impacts are revealed in comparison to TWFE models. The positive effects on the market capitalization value are found in the 10-day window between the third and eighth days post-event. For the market capitalization ratio, positive effects are found, but the effects periods are earlier than those for the all-chain control group.

The results for the large-event sample (see Fig. 4) mirror those from the corresponding TWFE models. All eight models indicate no significant effects on market capitalization value and ratio, confirming that the substantial financial losses from events do not impact the market capitalization value of the blockchain. Finally, Fig. 5 depicts the results from the ETH event specifications, of which the patterns align closely with the all-event sample regressions. However, while the event study regressions reveal positive effects following ETH events, these effects are more subtle than those observed in the TWFE models with ETH events.

Fig. 3 DID event study regressions for all-event sample with estimation windows of 10 days and 15 days



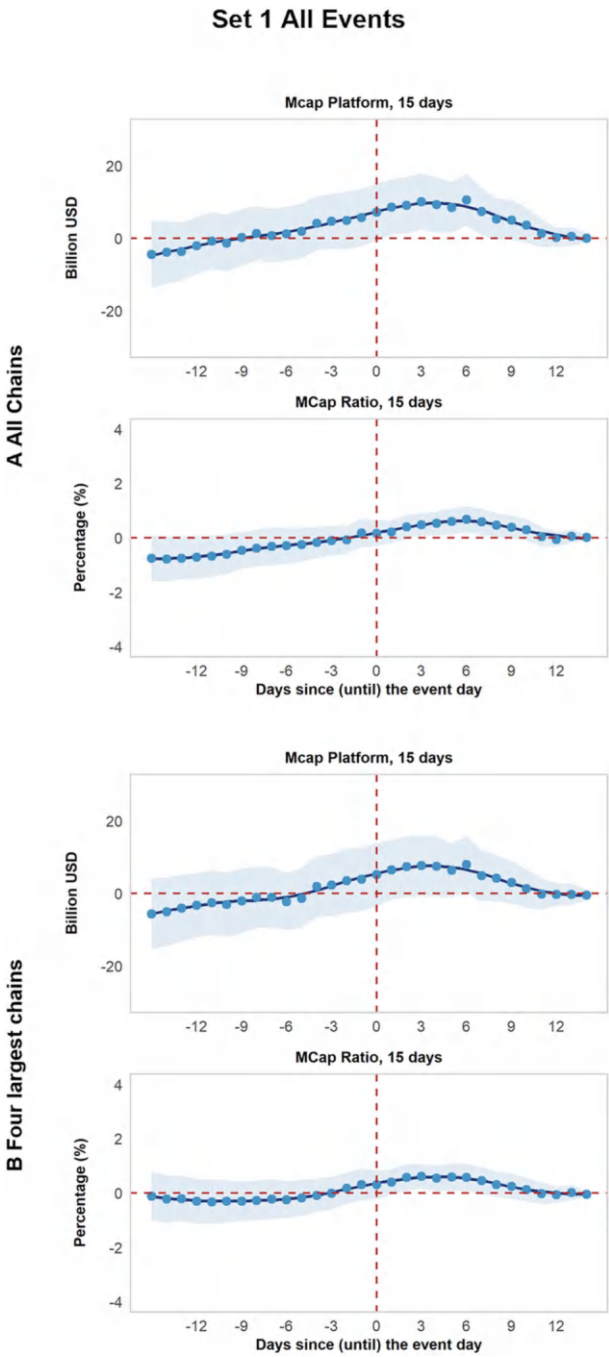
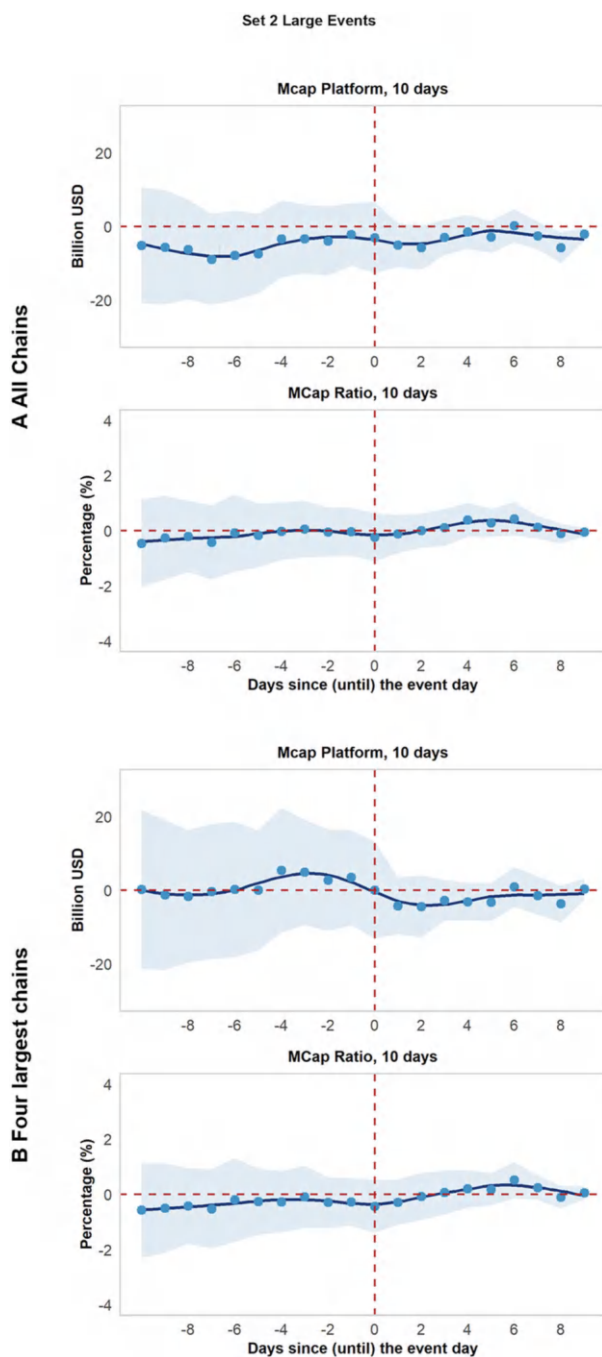


Fig. 3 (continued)



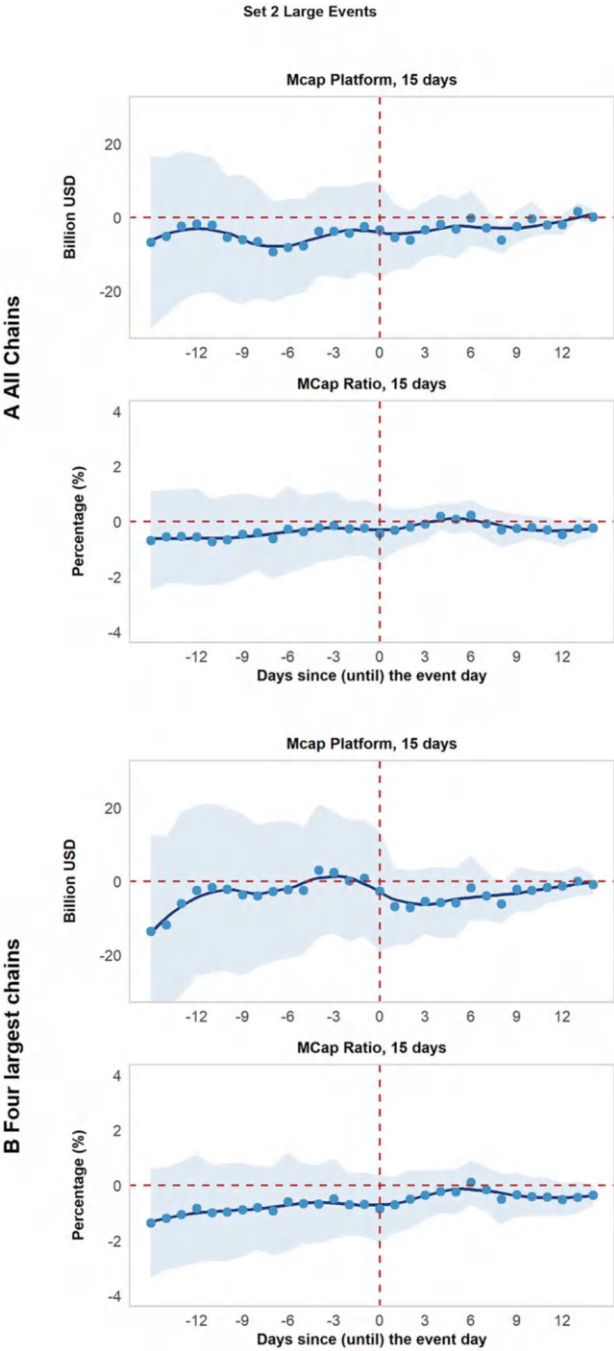


Fig. 4 (continued)

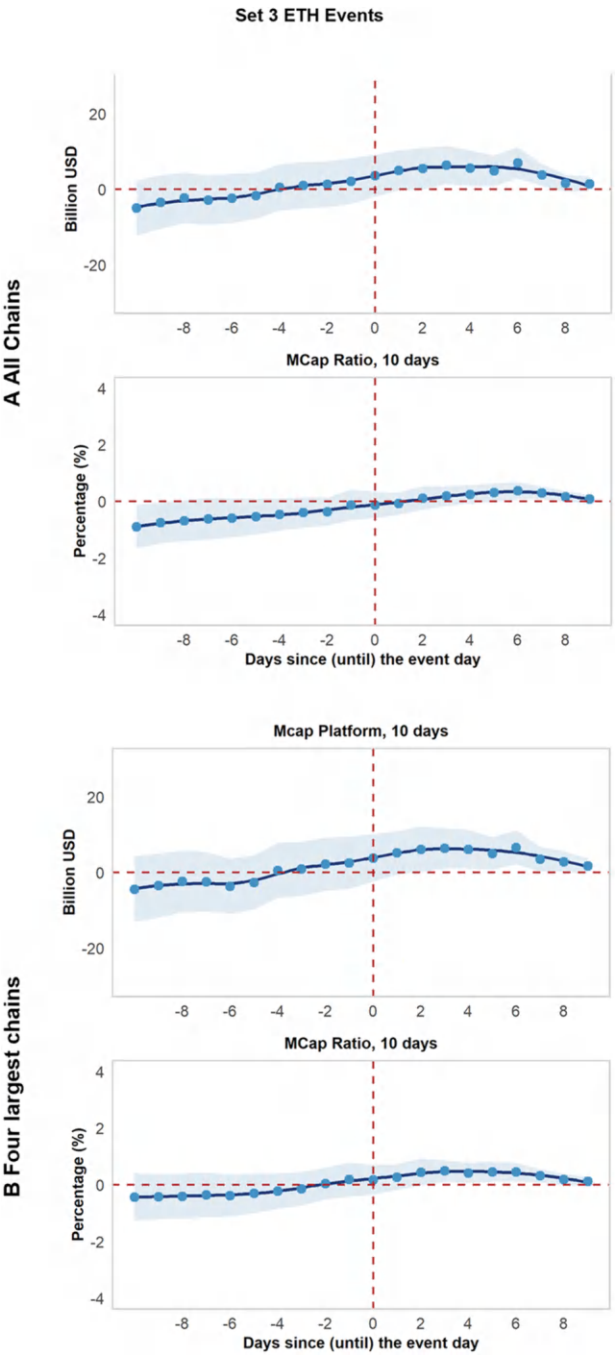


Fig. 5 DID event study regressions for ETH event samples with estimation windows of 10 days and 15 days

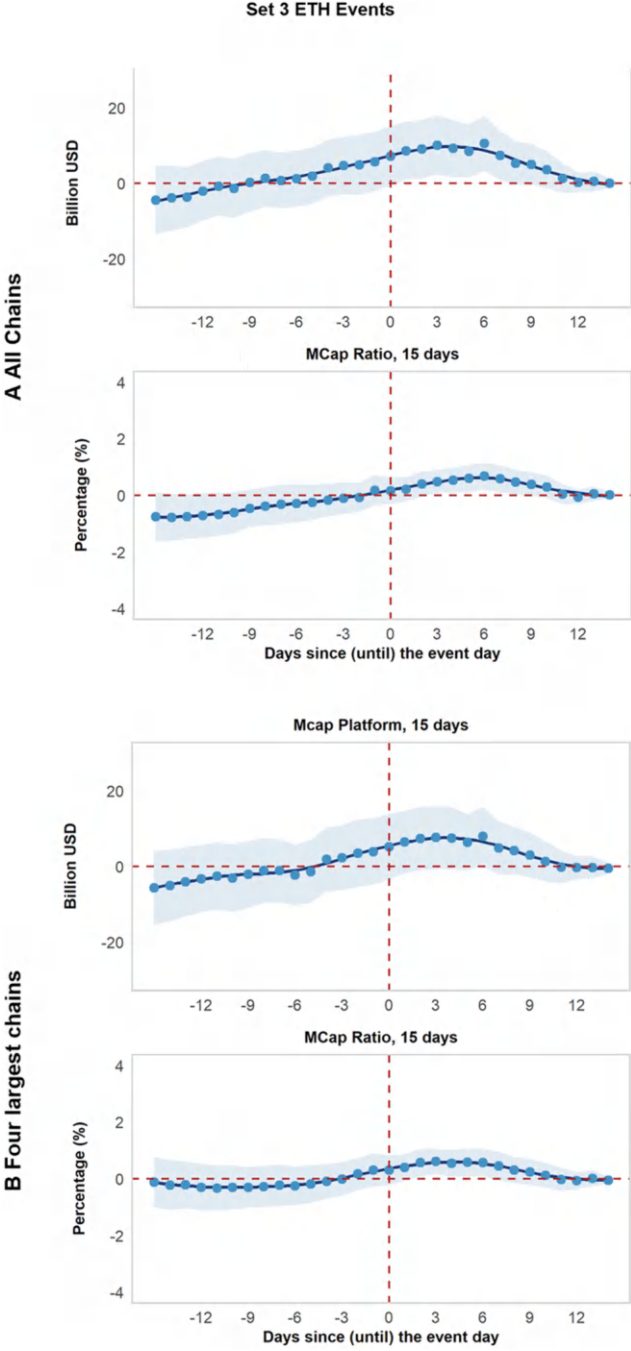


Fig. 5 (continued)

4.3 Discussion

The results of the regressions are consistent with the hypothesis that DeFi attacks positively impact the underlying native cryptoasset. An exploit event, although seemingly negative, can have a positive impact on the market capitalization of the native tokens of the affected blockchains. There are several reasons for this. First, the spillover effect of an event may cause the unaffected blockchain to be affected, resulting in negligible relative changes. Second, the news of such incidents may generate interest in researching the associated DeFi applications, which could stimulate demand for the blockchain. Third, in response to a DeFi attack or failure, the respective blockchain user communities and project developers can collaborate to address the issue and enhance the DeFi ecosystem's overall resilience by strengthening security measures.

In particular, the positive impact of ETH events is more pronounced. Ethereum is the most established and dominant blockchain in the DeFi space, accounting for over 56% of the total TVL [2]. It hosts a wide range of applications and protocols, providing diversity and connectivity in the ecosystem. This makes it a preferred choice for many users and investors, with a large network of developers. For instance, multiple online platforms have been gathering points for Ethereum developers and network participants to discuss the trends and technical developments. Ethereum channel on Reddit⁴ and Ethereum Magicians⁵ are two of the largest channels for such discussions. Therefore, the network effect can play a critical role in the event of an attack, as any improvements made can have a more profound impact than those made on other blockchains.

5 Conclusion

Our study analyzes a large set of DeFi attacks and subsequent valuation effects on the underlying blockchains' native asset market valuation. To determine the impact of such events on market values, we conduct both TWFE and event study difference-in-difference regressions, which allow us to observe the average and day-by-day treatment effects, showing the differences between market values changes of treated and control blockchains in response to attacks on the DeFi protocols they host. We establish various control groups and event types to examine the degree of the impacts of the exploit events in different settings. Contrary to the intuition that exploits and protocol attacks would have negative consequences for the underlying blockchain, we consistently find that market values of affected blockchains increase rather than decrease relative to control group blockchains.

⁴ <https://www.reddit.com/r/ethereum/>.

⁵ <https://ethereum-magicians.org/>.

Especially in the case of events on the Ethereum blockchain, we find amplified positive impacts resulting from DeFi exploit events. The findings suggest that the DeFi exploit events, particularly those that occur on established platforms like Ethereum, signal improved security measures and platform audits. This leads to increased investor and participant confidence, resulting in an increase in market value.

This chapter presents evidence in favor of treating DeFi protocol exploits as a temporary loss to the individual platform, but overall strengthening the resilience and suitability of the underlying blockchain to host such applications. It serves as a valuable foundation for future research in the area of DeFi attack and security, enabling researchers to delve deeper into the relationship between exploit incidents and the underlying blockchain infrastructure, as well as the attribution of their impact. Furthermore, this chapter offers insight into how investors and the wider DeFi community perceive these exploits, highlighting implications for practitioners and regulators in terms of a nuanced understanding of how such incidents can influence market perceptions and the actions of various stakeholders.

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Imagining a Democratic, Affordable Future of Foundation Models: A Decentralised Avenue



Fengxiang He , Lihao Nan , and Tongtian Zhu

Abstract Foundation models show astonishing performance for a variety of tasks while requiring extremely huge amounts of computing resources in both training and inference. Such costs are beyond the affordability of most users; consequently, foundation models are dominantly occupied by tech giants. To pursue an affordable and democratic future of foundation models, there is growing interest in examining decentralised learning approaches. This chapter provides a thorough review of the current decentralised solutions and offers insights into prospective strategies to overcome the existing barriers. We also describe our insights in facilitating decentralised learning by blockchain, as well as challenges and future work. In our vision, decentralised learning will energise the foundation model economy, but is still obstructed by major challenges such as establishing robust incentive mechanisms and developing training strategies suitable for heterogeneous environments.

1 Introduction

Recently, foundation models [8] (e.g., T5 [61], GPT-3 [13], PaLM [17], OPT [93], GPT-4 [1], Llama-2 [75], Mistral [35] and DALL-E 3 [5]) have made ground-breaking advancements in understanding and generating natural language and images, driven by substantial increases in model size and training data size [13]. Notably, GPT-3 consists of over 100 billion parameters and leverages immensely large datasets for training [13]. This expansion necessitates extremely high demand

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of CPU, memory, and GPU hardware. Furthermore, the operational costs associated with running these foundation models at such a massive scale have escalated substantially. For instance, OpenAI reportedly incurs a daily expenditure of \$700,000 to maintain ChatGPT [84], despite the fact that training the GPT-3 foundation model alone costs over \$5 million [84]. Consequently, only a small set of large corporations with sufficient data and computation resources control the access to the best AI models.

To democratise these advanced technologies for a broader user base, decentralised learning emerges as a promising strategy. The general idea of decentralised learning is to crowdsource the training of machine learning models with thousands of regular volunteers provided by decentralised volunteers via a peer-to-peer (P2P) network. Concretely, one could partition a large model (e.g., a neural network) into thousands of parallel segments and let each volunteer manage one of the segments. The advantages are mainly threefold:

- **Cost amortisation.** As more volunteers contribute their computational resources, the costs of computational tasks are spread over a larger number of participants. This reduces the individual cost burden, making participation in the network more affordable.
- **Autonomy.** Decentralisation creates an environment where autonomous and democratic participation is naturally encouraged.
- **Fault tolerance.** As the volunteer network grows, the resilience of the system to node and communication failures strengthens, enhancing its overall robustness.

Despite these advantages, training foundation models in a fully decentralised manner presents unique obstacles. The primary challenges include managing and potentially incentivising intricate coordination among a massive number of heterogeneous volunteers (i.e., with data, computational power, and model heterogeneity) with inconsistent network connectivity. This leads us to an important question:

? Question

How can we effectively coordinate the decentralised training of foundation models with heterogeneous volunteers under inconsistent network connectivity?

To answer this pivotal question, this chapter provides a thorough review of the existing solutions and envisions future strategies aimed at advancing towards a democratic and affordable future of foundation models. Section 2 introduces the basic concepts of deep learning and foundation models, while Sect. 3 discusses decentralised machine learning strategies and their cutting-edge extensions for training foundation models, highlighting the benefits of communication efficiency and cost-sharing mechanisms in decentralised approaches. Sections 3.1 and 3.2 detail the core principles, motivations, and algorithmic development of decentralised training methods. Section 3.3 explores the specific challenges associated with scaling decentralised techniques to support the training of foundation models and

summarises current advancements. The chapter further assesses the advantages of integrating Blockchain technology within decentralised learning systems in Sect. 4.

2 Deep Learning and Foundation Models

Deep learning emerges as a transformative force in artificial intelligence, fundamentally reshaping our understanding and potential within the field. Drawing inspiration from the neural networks of the human brain, deep learning models can learn from large datasets to identify underlying patterns and generalise, that is, to make accurate predictions on unseen data. They have proven to be exceptionally adept across various domains, including language processing [13, 22] to vision [31, 39], outperforming traditional machine learning models.

Foundation models, exemplified by ground-breaking language models like OpenAI's ChatGPT, represent another leap forward. These models are trained on extensive datasets, which lays the groundwork for their remarkable ability to adapt to specialised tasks through fine-tuning. ChatGPT, in particular, also extends beyond standard fine-tuning by incorporating reinforcement learning from human feedback (RLHF) [18], a promising way to align foundation models with human intents. The performance of foundation models can be further elevated by employing techniques such as prompt tuning [13] and in-context learning [82], which refine its ability to interpret and respond to prompts in a context-aware manner. Furthermore, techniques like LoRA [32] enable more resource-efficient fine-tuning by integrating low-rank layers into the original model, thereby avoiding the retraining of entire parameters.

Despite these technical achievements, foundation models like GPT-3.5 also present substantial challenges, particularly in terms of the economic investment required for their deployment and development. GPT-3.5, with its extensive ability to generate human-like text, answer complex questions, and craft creative content, comes at the cost of significant hardware and computational demands. These demands render the deployment and training of such models economically infeasible for many individuals and academic institutions. These barriers stand as significant impediments to democratising access to state-of-the-art models, potentially stifling scientific advancement.

3 Decentralised Machine Learning

This aforementioned challenge necessitates the development of sophisticated distributed learning paradigms. In this section, we introduce decentralised machine learning, which integrates the idea of volunteer computing into distributed machine learning. We start from the formal definition of distributed machine learning, then introduce the algorithmic aspects of decentralised learning, and move on to summarise the latest research on decentralised training of foundation models.

3.1 Distributed Machine Learning

Notations Let $\mathcal{X} \subseteq \mathbb{R}^{d_x}$ and $\mathcal{Y} \subseteq \mathbb{R}$ represent the input and output domains, respectively. We denote the training dataset as $\mu = \{z_1, \dots, z_N\}$, where each $z_\zeta = (x_\zeta, y_\zeta)$, for $\zeta = 1, \dots, N$, is independently and identically distributed (i.i.d.) according to an unknown distribution $\mathcal{D}$ over $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$. In supervised learning, the goal is to obtain a predictor (or hypothesis) $g(\theta; \cdot)$, parameterised by $\theta \in \mathbb{R}^d$ with d being an arbitrary finite dimension. This hypothesis aims to model the function mapping between inputs $x \in \mathcal{X}$ and outputs $y \in \mathcal{Y}$, based on the training set μ . We employ the cost function $c : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ to evaluate the quality of the hypothesis g . The loss associated with a hypothesis g with respect to the training data point $z_\zeta = (x_\zeta, y_\zeta)$ is thus given by $L(\theta; z_\zeta) = c(g(\theta; x_\zeta), y_\zeta)$ to measure the performance of the model parameterised by θ . Therefore, the empirical risk associated with θ , which is the target of optimisation, is defined as follows:

$$L_\theta^\mu = \frac{1}{N} \sum_{\zeta=1}^N L(\theta; z_\zeta). \quad (1)$$

Distributed Learning Traditional distributed learning considers optimising the empirical risk jointly with multiple workers [66]. In this framework, each worker, for $j = 1, \dots, m$, can access $|\mu_j|$ i.i.d. local training examples $\mu_j = \{z_{j,1}, \dots, z_{j,|\mu_j|}\}$. The global empirical risk of θ then becomes

$$L_\theta^\mu = \frac{1}{m} \sum_{j=1}^m L_\theta^{\mu_j} = \frac{1}{m} \sum_{j=1}^m \frac{1}{|\mu_j|} \sum_{\zeta=1}^{|\mu_j|} L(\theta; z_{j,\zeta}), \quad (2)$$

where $L_\theta^{\mu_j} = \frac{1}{|\mu_j|} \sum_{\zeta=1}^{|\mu_j|} L(\theta; z_{j,\zeta})$ denotes the local empirical risk on the j -th worker and $\mu_j = \{z_{j,\zeta}\}_{\zeta=1}^{|\mu_j|}$ represents the local training dataset. The optimisation of Eq. (2) is also a distributed consensus problem [12].

Due to the increasing scale of training data and the model complexity, various parallelisation strategies have been proposed for effective large-scale distributed learning. The primary strategies include data parallelism, pipeline parallelism, and tensor parallelism.

Data Parallelism In machine learning scenarios where fixed training datasets are available, the aforementioned distributed strategy corresponds to randomly partitioning the whole training samples among the multiple machines. Each machine is thus endowed with a subset of samples that are i.i.d. drawn from the same source distribution. Extending the principle, data parallelism [45] shares the workload by distributing a large mini-batch across multiple devices. Each device trains its own local model replica independently. Ensuring consistency across these local models necessitates synchronisation of gradients information via a parameter server

Algorithm 1 Parallel SGD [21, 43]**Worker** $j = 1, \dots, m$ (in parallel):

- 1: Receive $\theta^0 = 0$ from server
- 2: **for** step $t = 1$ to T **do**
- 3: Sample training batch $\{z_{j,i}\}_{i=1}^{|\mu_j^t|}$ from local training dataset
- 4: Compute gradient $g_j^t := \frac{1}{|\mu_j^t|} \sum_{i=1}^{|\mu_j^t|} \nabla L(\theta^t, z_{j,i})$ ▷ mini-batch gradient computation
- 5: Push g_j^t to server
- 6: Receive θ^{t+1} from server

Server:

- 7: **for** $t = 1$ to T **do**
- 8: Aggregate $g^t := \frac{1}{m} \sum_{j=1}^m g_j^t$ ▷ global gradient aggregation
- 9: Set learning rate as η^t , update $\theta^{t+1} := \theta^t - \eta^t g^t$ ▷ global weight update

Algorithm 2 FedAvg [53]**Client** $j = 1, \dots, m$ (in parallel):

- 1: Receive $\theta^0 = 0$ from server
- 2: **for** round $t = 1$ to T **do**
- 3: **for** local step $l = 1$ to E **do**
- 4: Sample local training batch $\{z_{j,i}\}_{i=1}^{|\mu_j^{t,l}|}$ from local training dataset
- 5: Compute gradient $g_j^{t,l} := \frac{1}{|\mu_j^{t,l}|} \sum_{i=1}^{|\mu_j^{t,l}|} \nabla L(\theta^{t,l}, z_{j,i})$ ▷ local gradient computation
- 6: Set local learning rate as $\eta_j^{t,l}$, compute $\theta_j^{t,l+1} := \theta_j^{t,l} - \eta_j^{t,l} g_j^{t,l}$ ▷ local weight update
- 7: Push $\theta_j^{t,E}$ to server
- 8: Receive θ^{t+1} from server

Server:

- 9: **for** $t = 1$ to T **do**
- 10: Aggregate $\theta^{t+1} := \frac{1}{m} \sum_{j=1}^m \theta_j^{t,E}$ ▷ global weight aggregation

[21, 43, 71] or through an AllReduce operation [45, 60]. Algorithm 1 is an example of implementing stochastic gradient descent with a parameter server. In each step, each worker computes their gradients locally, and then the server aggregates these gradients to update the shared parameter. Federated learning (FL) [9, 47, 53] is another famous example of data parallelism. FL considers the problem of collaborative learning with heterogeneous edge devices, where each device j keeps its own local data. A typical FL approach is FedAvg (see Algorithm 2), where clients update locally with E steps, and then the server averages these local updated models to refine the global model. The updated global model is then redistributed to the clients, facilitating subsequent local training cycles. This decentralised approach leverages the computational resources of the edge devices and also respects user privacy by keeping sensitive data locally. Despite the advantages offered by data parallelism, its scalability is mainly constrained by the following factors: the significant communication overhead required for model parameter synchronisation which could limit scalability [14, 42, 47], the reduction of GPU utilisation as the

per-GPU batch size becomes too small [57], and the memory constraints that render loading an entire large model onto a single GPU impractical.

Pipeline Parallelism Addressing the challenge of individual GPU memory, pipeline parallelism partitions a model into finer slices at the layer level, each processed on separate devices in a sequential fashion [24, 33, 55]. The downside of this approach is that the sequential layer processing creates dependencies that can limit scaling efficiency. These dependencies often result in potential idle time, known as “bubble time”, where some devices wait for others to complete their tasks before proceeding [56, 57].

Tensor Parallelism In tensor model parallelism [70], matrix multiplications within each individual layer are split over multiple devices. This form of parallelism is especially fitting for super-large models and requires access to high communication-bandwidth environments for efficiently handling the intensive data exchange [57]. However, in cases when intragroup communication is not fast enough, tensor model parallelism may exhibit subpar performance. Therefore, tensor parallelism is typically applied within a single physical server, in conjunction with complementary parallelisation strategies [34, 57].

3.2 *Decentralised Machine Learning*

To mitigate the communication bottleneck of server-based distributed machine learning, decentralised learning emerges as a powerful alternative. Employing a peer-to-peer approach, decentralised training harnesses the power of locally connected computing resources, effectively distributing the workload without the need for a central coordinating server [4, 11, 52, 81, 92] (Fig. 1).

The conceptual foundations of decentralised training algorithms are rooted in the early and influential work of [76, 77] and [59]. These studies provide the groundwork for the development of algorithms such as decentralised parallel stochastic gradient descent (D-SGD) [38, 48], which integrates the principles of decentralisation with gradient-based optimisation (see Algorithm 3). In the vanilla

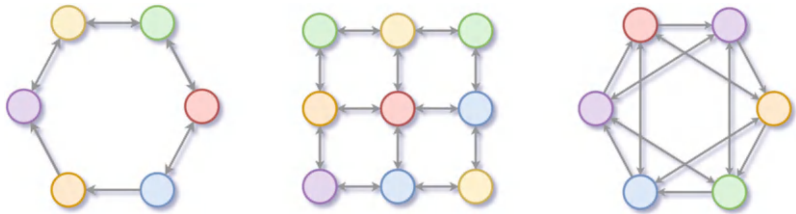


Fig. 1 An illustration of peer-to-peer communication topology in decentralised learning

Algorithm 3 Decentralised parallel SGD [38, 48]

Input: Given communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and mixing matrix $P = [P_{j,k}] \in \mathbb{R}^{m \times m}$
Worker $j = 1, \dots, m$ (in parallel):
1: Initialize $\theta_j^0 = 0$
2: **for** step $t = 1$ to T **do**
3: Sample training batch $\{z_{j,i}\}_{i=1}^{|\mu_j^t|}$ from local training dataset
4: **for all** neighbors $k : \{j, k\} \in \mathcal{E}$ **do**
5: Compute $\theta_j^{t+\frac{1}{2}} = \sum_{k=1}^m P_{j,k} \theta_k^t$ ▷ gossip weight aggregation
6: Compute gradient $g_j^t := \frac{1}{|\mu_j^t|} \sum_{i=1}^{|\mu_j^t|} \nabla L(\theta_j^t, z_{j,i})$ ▷ mini-batch gradient computation
7: Compute $\theta_j^{t+1} = \theta_j^{t+\frac{1}{2}} - g_j^t$ ▷ local weight update

Adapt-While-Communicate (AWC) version of D-PSGD [48, 59], each worker updates its own model locally and incorporating weights from peers. During weight exchange, a “sender” shares its locally trained model with its neighbours, and a “receiver” integrates these models into its local model. This peer-to-peer communication is through a gossip protocol orchestrated by a mixing matrix $P = [P_{j,k}] \in \mathbb{R}^{m \times m}$, which characterises the connectivity of the underlying communication topology $\mathcal{G}$ [96]. The central goal of D-PSGD is to establish a consensus model by optimising the empirical risk $\frac{1}{m} \sum_{j=1}^m \frac{1}{|\mu_j|} \sum_{\zeta=1}^{|\mu_j|} L(\theta; z_{j,\zeta})$ (see Eq. (2)) cooperatively through m locally connected workers.

Theoretical research has shown that large-scale models can effectively converge with D-PSGD [50, 68], with asymptotic linear speedup in convergence rate similar to centralised parallel SGD (C-SGD) [20, 44]. Recent studies [95] have further linked D-PSGD to a centralised generalisation-enhancing algorithm called sharpness-aware minimisation (SAM), suggesting that decentralised learning may offer additional generalisation benefits compared to server-based learning paradigms.

The development of decentralised algorithms has been characterised by their flexibility in adapting to complex environments. Notably, decentralised algorithms have been adapted to various contexts, including time-varying topologies [38, 51, 58, 91], asynchronous settings [10, 49, 54, 88], personalised settings [46], data heterogeneous scenarios [41, 72, 78], and Byzantine-robust versions [25, 90]. Decentralised optimisation problems have been further extended beyond standard single-level minimisation problems, including compositional [27], minimax [15, 87, 94], and bi-level [16, 26, 89] optimisation problems. Despite these advancements, existing decentralised training approaches predominantly focus on data parallelism, which alone could be inadequate for foundation models whose parameter sets are too large to be accommodated by a single device.

3.3 *Decentralised Training and Inference of Foundation Models*

Foundation models have reaped substantial rewards from the expansion of training data and model complexity, in accordance with the principles of scaling laws [37, 63]. However, this trend towards larger data size and models has outstripped the evolution of hardware, which trails behind the escalating requirements for computing power and memory. As a result, training and deploying modern foundation models not only requires advanced GPUs, but often necessitates specialised high-performance computing (HPC) clusters to handle their substantial computational demands. Sophisticated parallelisation strategies like data, pipeline, and tensor parallelism are widely used, yet they assume the availability of luxury data centres equipped with fast interconnects, which is beyond the budget of many individuals and academic institutions. The immensity of this challenge is exemplified by the requirements of foundation models like GPT-3, which requires 325GB of GPU memory [67] and 3.64K petaflop/s-days for training [13]. Such requirements starkly illustrate the daunting barriers faced by those with limited access to such computational resources.

Fully Decentralised Training of Foundation Models Thanks to the advantages in communication efficiency, cost sharing, and fault tolerance, decentralised approaches have emerged as a promising alternative to train foundation models such as large language models (LLMs, with a summary of key methods provided in Table 1). Leveraging the concept of volunteer computing [2, 3, 40, 69], Learning@home [65] and DeDLOC [23] spearhead the collaborative volunteer training of foundation models. Learning@home [65] proposes a promising decentralised mixture-of-experts (MoE) training paradigm to handle massive poorly connected participants with a decentralised hash table (DHT) used to route inputs to the appropriate expert. However, the training and evaluation of Learning@home is confined to relatively smaller datasets. DeDLOC [23] uses a decentralised adaptive averaging strategy that considers the diverse internet speeds and connectivity limitations of volunteers, but still relies on data parallelism. [92] explore the potential of training standard GPT-style foundation models with a new decentralised model parallelism over a heterogeneous and lower-bandwidth interconnected network. The major contribution is a scheduling algorithm allocating computational “tasklets”. Subsequent work by SWARM parallelism [64] leverages fault-tolerant pipelines and dynamically rebalances nodes across stages to train foundation models on heterogeneous devices under slower connectivity. In parallel, [73] propose FusionAI supporting dynamic join and quit policy for training large language models with underutilised consumer-grade GPUs. Based on swarm parallelism, Petal [11] develops a decentralised pipeline inference framework to amortise inference cost of LLMs. Petal facilitates a collaborative environment wherein users can donate heterogeneous computation resources to perform inference and small-scale fine-tuning collaboratively. A more recent work inference method called HexGen [36] can

Table 1 Review of methods, framework, and platforms for decentralised training and inference of foundation models

Methods	Description
Learning@home [65]	A decentralised mixture-of-experts (MoE) training paradigm for massive, poorly connected networks
DeDLOC [23]	A decentralised data-parallel framework using adaptive averaging strategy for collaborative training under diverse internet speeds and connectivity challenges
DT-FM [92]	A decentralised pipeline parallel method for training GPT-style foundation models, employing a specialised algorithm for “tasklet” allocation over heterogeneous and lower-bandwidth networks
SWARM parallelism [64]	A parallel training strategy for training billions of parameters across unreliable, heterogeneous devices with slow connectivity
FusionAI [73]	A distributed system supporting dynamic join and quit policy for training large language models with underutilised consumer-grade GPUs
Petal [11]	A decentralised collaborative inference service engine for cost sharing
HexGen [36]	A decentralised inference method supporting asymmetric partitioning of the inference computation by reformulating the scheduling problem as a constrained optimisation problem
AQ-SGD [80]	A decentralised activation compression algorithm for communication-efficient pipeline parallelism training over slow networks
CocktailSGD [79]	A communication-efficient algorithm combining decentralisation, sparsification, and quantisation
SAKSHI [6]	A decentralised platform for energy-efficient, trust-free, and incentive-compatible AI service hosting and delivery

further allocate the asymmetric inference tasklets among workers by reformulating the scheduling problem as a constrained optimisation problem. At the algorithmic level, CocktailSGD [79] elegantly combines decentralisation, sparsification, and quantisation for communication-efficient fine-tuning of foundation models on slow networks. AQ-SGD [80] introduces a decentralised activation compression algorithm for communication-efficient pipeline parallelism training over slow networks. Beyond the framework and algorithmic design, SAKSHI [6] emerges as a new decentralised platform for energy-efficient, trust-free, and incentive-compatible AI service hosting and delivery.

4 Decentralised Learning on Blockchain

As highlighted in the preceding section, decentralised learning stands out as an attractive strategy for training foundation models, offering notable benefits in terms

of communication efficiency and cost sharing. However, the absence of effective incentive mechanism and reliable security assurances remains a critical hurdle for such systems. Blockchain technology, characterised by its secure, auditable, immutable, incentive-based, and decentralised nature, presents a natural auxiliary to decentralised learning that encourages a collaborative environment [30]. In this section, we discuss the potential benefits of integrating Blockchain technology into decentralised learning, which are outlined in Table 2.

Automation By combining blockchain technology with smart contracts [74], users can execute verifiable and traceable transactions autonomously, aligning with the self-organising principles in decentralised learning systems.

Security and Integrity The security and integrity of transactions in a blockchain network are ensured through a verification process. Each account in the blockchain holds a public key and a private key, with the public key available to everyone and the private key only visible to the account owner. When a user, designated as the request node, submits a transaction, it uses its private key to create a digital signature for the transaction data. This digital signature is unique to the transaction and the private key of the request node. The verification process involves the following steps:

1. **Hash the transaction data:** The transaction data is hashed, creating a fixed-length string of characters that uniquely represents the transaction.
2. **Decrypt the digital signature:** The public key is used to decrypt the digital signature, revealing the original hash of the transaction data.
3. **Compare hashes:** The decrypted hash is compared to the hash of the transaction data. If they match, it means the transaction data has not been tampered with.
4. **Verify ownership:** The network also verifies that the public key used to create the digital signature belongs to the request node. If the signature is valid and the public key matches the public key of the request node on record, the transaction is considered legitimate.

Blockchain, further integrated with zero-knowledge proofs [7, 28, 29], could offer a robust framework for safeguarding information exchange in decentralised learning, where nodes can confirm the legitimacy of the contributions of others

Table 2 Advantages of integrating blockchain with decentralised learning

Aspect	Role of blockchain	Impact on decentralised learning
Automation	Smart contracts	Automatically execute decentralised training
Security & Integrity	Robust encryption mechanisms; tamper-resistant ledgers	Secures model/data exchange by ensuring only authorised access, maintaining the immutability of records
Incentivisation	Token-based reward mechanism	Encourages active and fair participation in model training

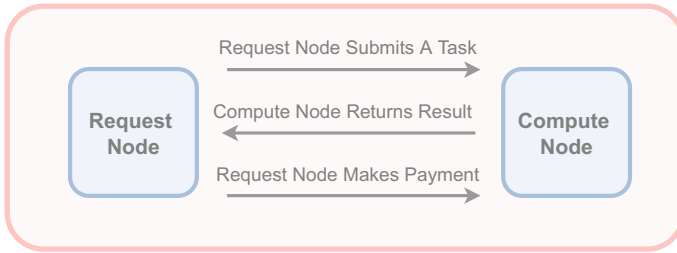


Fig. 2 Procedure of computation in a centralised cloud computing environment

without compromising privacy. In scenarios where there are many heterogeneous participants, the data owner and computation provider can be decoupled. Consider a request node falls short for substantial computational tasks. A straightforward solution is to transmit the code and data to an entity with stronger computational power, known as the compute node, as illustrated in Fig. 2. In the scenario of decentralised training, a natural question is:

? Questions

Execution Verification: How to confirm that the compute node has actually executed the instruction from the request node and not fabricated the results in a fully decentralised learning system?

In a collaborative learning system, the compute node can generate a zk-SNARK proof after completing a computation task. This proof validates the correctness of the computation without revealing the details. When a new computational task is submitted to the blockchain, miner nodes can verify this zk-SNARK proof in the same way they would verify a transaction, without access to the specific details of the computation. The strength of this system lies in the inherent immutability of blockchain; once transactions of model updates are recorded on the ledger, they are permanent and cannot be altered, effectively preventing unauthorised modifications. This characteristic is vital for creating a verifiable and trustable collaborative learning environment, particularly in sensitive sectors such as healthcare where data provenance and integrity are critical [81].

Incorporate Incentives One of the primary goals of a self-organised decentralised learning system is to foster sustainable collaboration among diverse participants, thus necessitating the design of a robust incentive mechanism to effectively motivate contributors, while also preventing unconstructive participation from receiving rewards. In a manner akin to how Filecoin [85] complements IPFS [86] by providing an incentive layer, blockchain could facilitate the establishment for such a mechanism in decentralised learning, distributing tokens or cryptocurrencies as

rewards for valuable contributions based on smart contracts. These incentives are crucial for encouraging participants to contribute computational and communication resources or even local data, foundational to establishing a transparent and cost-effective collaborative environment for training foundation models.

Despite the potential benefits, leveraging blockchain for training large-scale foundation models presents challenges.

High Communication Costs BAFFLE [62] and DeepChain [83] have leveraged blockchain to mitigate the security and privacy issues in federated learning. However, these works have not considered leveraging Blockchain to train large-scale deep learning models, such as foundation models. The core challenges here lie in the inefficiencies in parallelising the structure of current foundation models and the high communication costs associated with Blockchain [19]. Therefore, designing model-parallel, communication-efficient decentralised algorithms for training foundation models could be a promising future direction.

5 Conclusion

Foundation models are exceptionally effective for a broad range of tasks; however, they require substantial computational resources for both training and inference, making them financially and technically unattainable for most players. As a result, the control over foundation models is predominantly held by tech giants. The pursuit of a democratic and affordable future is in the interest of wide communities. In this chapter, we discuss of a possible avenue via decentralised learning. We provide a comprehensive review of current decentralised learning methods, the open problems and existing technical challenges, and prospective approaches to address them. We envision that decentralised methodologies could energise the economy based upon foundation models; however, progress is still hindered by challenges including establishing robust incentive mechanisms and developing training strategies suitable for heterogeneous environments. In this context, blockchain technologies can play a significant role in facilitating decentralised learning.

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The Development of Blockchain Platform for Maritime Supply Chain



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Abstract In recent years, supply chain processes have been improved through digitalization. However, digital data is sometimes unsynchronized and untrustworthy. Many blockchain applications have been developed in different supply chain industries to solve the issue based on the features of blockchain, such as tamper-proof, immutability, and traceability. The text focuses on a specific supply chain, the maritime supply chain, due to its significant position in global trade. The text aims to explore some blockchain applications in the field, compare them, and discuss them from different aspects. The text begins with the basic idea of blockchain and its evolution, followed by the basic knowledge of supply chain and maritime supply chain. By delving deeper into these topics, the text reinterprets the concept of blockchain 4.0, proposes a framework to explain the bottlenecks of digitalization in the field, and claims that blockchain can break the bottlenecks. Subsequently, the text discusses the applications from the aspect of blockchain 4.0. The advantages and disadvantages of each application's design are shown, and suggestions to reach blockchain 4.0 are provided. Researchers and application builders can refer to the text to discover research topics and develop successful blockchain applications. The ecosystem will then become more and more efficient.

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1 Introduction

Supply chain industries have undergone a significant transformation in recent years due to the influence of Industry 4.0, particularly digitalization. Digitalization has enhanced information transparency, data availability, process efficiency, and real-time data collection [1]. However, digital documents and data can be easily tampered with, resulting in unsynchronized and untrustworthy data. Trustworthiness, which is crucial for developing strong business connections in business-to-business and business-to-customer scenarios [2, 3], can be undermined. Leveraging blockchain, on the other hand, might solve these problems while enjoying the benefits of digitalization.

The text focuses on a specific supply chain, the maritime supply chain. The maritime supply chain has emerged as the predominant mode of international trade because of its cost-effective and efficient transportation system [4]. Over 70% of global trade by value is through ocean transport [5]. Since economic growth has been highly connected with shipping trade [6], the maritime supply chain is worthy of notice. Many novel applications leveraging blockchain technology in the maritime industry have recently emerged [7–10]. The text introduces several such applications. Some global applications are introduced, and one local application developed by iOne Information Technology (iOne) [11] is also explored. Although, currently, the local application is only used by organizations, primarily in Taiwan, it does have some unique characteristics that can be a reference for future development of maritime applications. The text aims to explore the applications, compare each of them, and interpret these applications from different aspects. The viewpoints discussed in this article can be guidance for future research and practical applications.

The remaining article is structured as follows. First, a brief explanation of blockchain is presented, followed by basic knowledge of supply chain and maritime supply chain. Then, some global blockchain applications in the maritime industry are exhibited. Subsequently, the iOne application is explored deeply; finally, the introduced applications are discussed and compared.

2 An Overview of Blockchain Evolution

A blockchain is a data structure. It is a decentralized, distributed, shareable, and immutable database comprising chronological series/chains of blocks. Through cryptography, the security of a blockchain is preserved [12]. A centralized database holds data in the central server; conversely, each network participant (node) has a duplicate of the data in a decentralized system. Therefore, the availability and transparency quality of a system reach since anyone in a blockchain network can access the data anytime and inspect the data credibility. A blockchain conducts a decentralized consensus mechanism to verify the data so it is free of unsynchronized cases and tamper-proof. Every blockchain may apply different types of consensus

mechanisms. The overall process of consensus mechanisms involves voting from distributed nodes according to each version of data of these nodes to compare and verify whether the new information added is correct [13]. There are different types of blockchains. Public blockchains (also called public permissionless blockchains or permissionless blockchains) are blockchains that anyone can join and be a node. Private blockchains (also called private permissioned blockchains or fully private blockchains) are blockchains in a central organization that controls a node's entrance. Permissioned blockchains (also called public permissioned blockchains, consortium blockchains, or federated blockchains) have hybrid characteristics between public and private blockchains. Access to the network and the consensus procedure in a permissioned blockchain is owned by some pre-chosen nodes. This taxonomy is sometimes ambiguous, and the definition of each type of blockchain may be slightly different in the literature. For example, in some articles, permissioned blockchains may refer to both the aforementioned private and permissioned ones [14–19]. In addition, blockchain, or blockchain technology, has broader meanings in practice. It can be referred to as the mentioned data structure, a consensus mechanism based on the structure, and applications established on the architectures [20, 21]. In this section, the text states the evolution of blockchain technology from blockchain 1.0 to blockchain 3.0. The characteristics of each of the advancement phases are highlighted. Also, the future vision of blockchain 4.0 is discussed. The text first differentiates between these phases, according to Mukherjee and Pradhan [22]. However, several other ways exist to classify the phases; the text also clarifies them later in this section.

2.1 Before Blockchain

Before the inception of blockchain technology, intermediaries dominate the financial world. Companies relied on fair, trustworthy, centralized organizations such as banks to complete transactions. However, the process not only tends to be lengthy and expensive but also poses potential credit risks if an intermediary fails [23].

2.2 Blockchain 1.0: Cryptocurrency

The advent of blockchain brings a game-changing innovation to the financial sector. Distributed ledger technology (DLT) [24, 25] encompasses the majority of blockchain 1.0, and its most prevalent application is cryptocurrency, specifically Bitcoin [26, 27]. The characteristics of decentralized systems include the persistence of network functionalities, even if specific nodes experience failures, distinguishing them from centralized systems. The absence of a need to evaluate the reliability of intermediaries or fellow network participants fosters increased trust [23].

2.3 Blockchain 2.0: Smart Contract

Blockchain 1.0 implementation faces a significant constraint in its incapability to support smart contracts and broaden its applications beyond the financial industry [22]. Smart contracts [28–30], pioneered by Ethereum [31, 32], marked the emergence of Blockchain 2.0 and expanded the applications of blockchain technology beyond the financial sector [33] (e.g., copyright [34, 35], healthcare [36, 37], energy [38], robotics [39], and supply chain [40–43]). Smart contracts are computer programs residing on a blockchain. They are guided by pre-set terms between two parties following the “if this, then that” logic. The programs are carried out automatically if the terms are reached. Once created, these contracts are secure from hacking or tampering [28]. Therefore, smart contracts markedly decrease verification, execution, and fraud prevention costs. This era signifies the genesis of blockchain technology in the maritime industry.

2.4 Blockchain 3.0: Improve the Overall Performance

Blockchain 1.0 and blockchain 2.0 primarily employed Proof of Work (POW) [44–46], a decentralized consensus mechanism mandating network participants (miners) to invest computational effort in deciphering an encrypted hexadecimal number to build new blocks [47, 48]. Through this high computational cost, a network can verify (prove) the integrity of miners because the miners will correctly build new blocks to avoid being rejected by the other nodes and wasting energy costs. Bitcoin, emblematic of blockchain 1.0, applies Hashcash [49], the POW algorithm; also, Ethereum, the epitome of blockchain 2.0, deployed POW before September 2022. Nonetheless, POW is energy-consuming [50, 51] and time-consuming (slow and low scalability) [52]. These drawbacks prompted the introduction of the third generation of blockchain.

Blockchain 3.0, the current development stage, aims to address the mentioned problem and improve the overall scalability, interoperability, privacy, and sustainability of blockchain. Thus, it can be applied to broader scenarios and works in an organization’s backend. Blockchain 3.0 can be interpreted from the perspective of uses as decentralized applications (dApps) [53, 54] or business blockchain platforms [22]. Researchers and professionals have recently been dedicated to developing new decentralized consensus mechanisms such as Proof-of-Stake (POS) and Proof-of-Authority (POA) to enhance the overall performances of blockchains in different applications. Instead of safeguarding a network via energy consumption, POS utilizes the ownership (stake) of some cryptocurrencies as well as the duration of owning cryptocurrencies (coin age) to achieve the security of a network [50, 51, 55–57]. On the other hand, POA aims to optimize POS in private or permissioned blockchains. A small group of nodes is selected to build new blocks based on

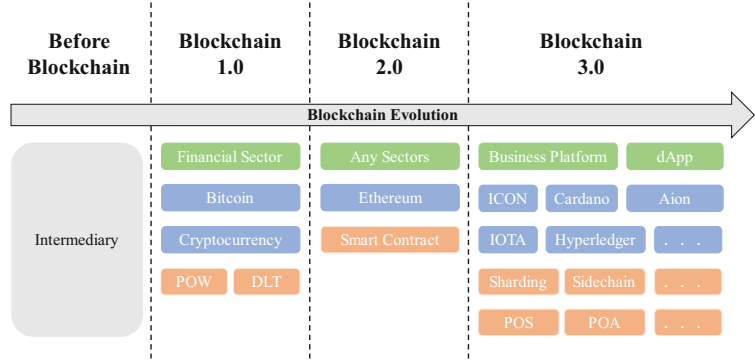


Fig. 1 Blockchain Evolution (Refer to Mukherjee and Pradhan (2021))

their identity and reputation rather than currency stakes or coin age [58–61]. Moreover, some techniques, such as sharding [62] and sidechain [63, 64], have been developed and applied to blockchains. Furthermore, there are some platforms and projects deemed as blockchain 3.0. Hyperledger Foundation, a part of the Linux Foundation [65], fosters interoperability and standardization via the open-source principles of the Linus Foundation; it aims to create a suitable environment for using decentralized technology across industries [66]. Many Hyperledger projects have been announced, such as Hyperledger Fabric, Hyperledger Aries, and Hyperledger Besu [67]. IOTA introduces the Tangle protocol [68], based on the directed acyclic graph to sharpen the network [69–73]. Other examples, such as the ICON project [74], Cardano [75], and Aion [76], also delve into the advancement. Figure 1 recaps the evolution.

2.5 Blockchain 4.0: The Future

Blockchain 4.0 emphasizes the integration of blockchain with other novel techniques, such as artificial intelligence, and improves further than the current state to form a business-usable platform (may be comprised of several interoperable platforms) in the entire industry or cross-industry. The platform(s) should be usable and valuable from the view of the entire industry or cross-industry; thus, it should be large scale. In addition, interoperability is also stressed here. Some examples Mukherjee and Pradhan regard as the precursors of blockchain 4.0 are Unibright [77] and Seele [78], which contribute to cross-blockchain and interoperability.

The idea of blockchain 4.0 here can be extended more. Currently, most blockchain applications applied in business aim to solve the pain point of inefficiency, and this means that without blockchain, the process would still work, but it would be inefficient. However, blockchain application builders and

supporters face the challenge of convincing enterprises in industries to apply the applications. The entrepreneurs conduct cost-benefit analyses, and some accept the proposal, but some reject it. The text would further interpret the concept of a business-usable platform in the entire industry or cross-industry as a platform or application that the enterprises in an industry or multi-industries must use to solve some inherent problems that cannot be handled before without blockchain, and the application will thrive quickly. Although the application may be small scale at that moment, it should become large scale in the future. For example, from the perspective of Internet security, the password-based security method is easily hacked. By leveraging blockchain technology as well as encryption techniques and InterPlanetary File System (IPFS) [79], the problem may be vastly improved to an unimaginable degree with classical Internet-based methods (for more information, refer to iOne Information Technology [80]).

2.6 Different Definitions of Blockchain Phases in Literature

Suzuki and Kawahara [81] interpret phases from the view of “web evolution” and “value.” Blockchain 1.0 is cryptocurrency and Bitcoin, representing value identification by money. Blockchain 2.0 points to smart contract and blockchain platforms applied to various fields, symbolizing value identification by platform. Blockchain 2.0 corresponds to Web 2.0, the era of meta platformer GAFa (Google, Apple, Facebook, and Amazon). Similarly, the idea of blockchain 3.0 is developed from Web 3.0. The core of Web 3.0 is self-sovereignty, which means individuals, instead of meta platformers, should own and manage their data, and blockchain technology is inevitable to complete this vision. Therefore, blockchain technology, which can incentivize Web 3.0, is called blockchain 3.0 and signifies value identification across applications (not just one platform). In summary, Suzuki and Kawahara’s blockchain 1.0 is the same as Mukherjee and Pradhan’s blockchain 1.0. However, the former’s blockchain 2.0 is a mixture of the latter’s blockchain 2.0 and blockchain 3.0. In addition, Suzuki and Kawahara’s blockchain 3.0 and Mukherjee and Pradhan’s blockchain 4.0 share a similar spirit although the viewpoints, scopes, and focuses are slightly different.

Di Francesco Maesa and Mori [82], as well as Khazanchi [83], define the phases in another way. The definition of blockchain 1.0 is the same as the other articles, and blockchain 2.0 is still smart contract. Nonetheless, blockchain 2.0 is restricted in the smart contract applications of cryptocurrency. The applications beyond money in the various sectors are regarded as blockchain 3.0. Therefore, Mukherjee and Pradhan’s blockchain 2.0 is a mixture of blockchain 2.0 and 3.0 here.

Because the definitions of blockchain 1.0–4.0 are different from the literature, for more accessible discussion, the text follows Mukherjee and Pradhan’s classification in the remaining sections.

3 Exploring Supply Chain and Maritime Landscape

3.1 *Supply Chain, Supply Chain Management, and Logistics*

Supply chain integrates various companies and their essential activities to develop, manufacture, transport, and use a product or service. Each enterprise aligns with one or several supply chains and makes a unique contribution. They rely on supply chains to develop sustainably and prosperously [84]. The term, supply chain, can be interpreted as a network of entities bringing products or services to market. It includes phases through upstream and downstream flows like production, procurement, and shipment, participated in by not only producers and suppliers but also shippers, warehouses, retailers, and customers or clients [85–89].

In addition, supply chain management (SCM) surfaced in the 1980s and was broadly embraced in the 1990s [84]. Mentzer et al. [89] have defined SCM as: “the systemic, strategic coordination of the traditional business functions and the tactics across these business functions within a particular company and across businesses within the supply chain, for the purposes of improving the long-term performance of the individual companies and the supply chain as a whole.”

Supply chain typically involves networks of collaborating organizations; logistics, on the other hand, commonly pertains to operations within the confines of a single organization. In addition, conventional logistics concentrates on functions such as procurement, distribution, maintenance, and inventory management. In contrast, supply chain encompasses not only the entire elements of traditional logistics but other activities such as marketing, new product development, finance, and customer service [84].

3.2 *Maritime Supply Chain*

The term, maritime supply chain, is also called “trans-ocean supply chain [90],” and in the case of container shipping, the more specific term, container shipping supply chain (CSSC) [5, 91], can be used. Based on the 2018 annual review of maritime transport by the United Nations Conference on Trade and Development, over 50% of global ocean trade by value was shipped by container ships [5]. Therefore, it is undoubted that container shipping is a crucial part of the maritime supply chain. In the remaining part of this section, the text explores the maritime supply chain in the context of container shipping.

In line with the explanation of supply chain mentioned in the previous part, maritime supply chain incorporates all the required activities and parties throughout the process such as distribution, collection, consolidation, de-consolidation, financial commitments, information transfer, and so on [90, 92]. Furthermore, Lam [6] has defined maritime supply chain as: “the connected series of activities pertaining to

shipping services which is concerned with planning, coordinating and controlling containerized cargoes from the point of origin to the point of destination.”

The main parties in maritime supply chains are shippers, freight forwarders, ocean carriers/shipping companies/shipping lines, port operators/terminal operators, inland carriers, and intermodal terminal operators/intermodal depot operators [93]. The explanation about these parties is displayed below [91]:

- **Shipper.** Transportation demand is generated when a shipper needs to move goods from the point of origin to the destination. Sometimes, the shipper is a consignor (exporter); however, other times, a consignee (importer) may be a shipper.
- **Freight forwarder.** Freight forwarders are responsible for coordinating and overseeing the transportation of goods between different locations or organizations. Traditionally, their role involved preparing the necessary documentation for customs clearance and insurance. Over time, they have expanded their services to include a broader range of logistics functions, such as inventory management, consolidation, warehousing, and inland transport. Acting as intermediaries between shippers and ocean carriers, freight forwarders are perceived as ocean carriers from the shippers’ view and shippers from the ocean carriers’ view.
- **Ocean carrier, shipping company, or shipping line.** They are tasked with transporting cargo from the port of origin to the port of destination. They operate ships and offer shipping services to shippers and freight forwarders. In addition, ocean carriers and container lessors must track the fleets, which includes managing empty container repositioning (ECR) to keep containers available.
- **Port operator or terminal operator.** They serve as intermediaries connecting seaborne transport with inland transport. They manage vessels, trucks, and trains, facilitating the seamless transfer of containers between modes of transportation.
- **Inland carrier.** The transportation of goods from one location to another within the hinterland is the job of inland carriers. Inland carriers can include road, rail, and barge carriers, who perform inland transportation for shippers, freight forwarders, and ocean carriers.
- **Intermodal terminal operator or intermodal depot operator.** Intermodal terminals or depots in the hinterland are places where containers can be temporarily put and accessed by trains and trucks. These facilities enable various logistics functions, including storage and warehousing of goods, inventory control, consolidation, movement of containers and goods, packing, cross-docking, unpacking, repair, and maintenance.

A shipper is a consignor when the consignor leads the transport of the goods. For example, the consignor and consignee have signed a periodic sales contract, and the consignor will send the goods to the consignee once a period. According to the contract, the consignor should lead the transportation and arrange all matters. Most of the time, the consignor commissions a freight forwarder to address complex matters during transportation. The freight forwarder helps the consignor schedule shipments with an ocean carrier. Also, it contacts inland carriers and intermodal

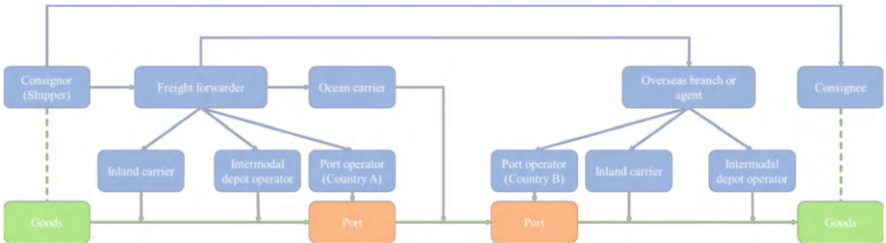


Fig. 2 Maritime supply chain framework (Consignor as Shipper)

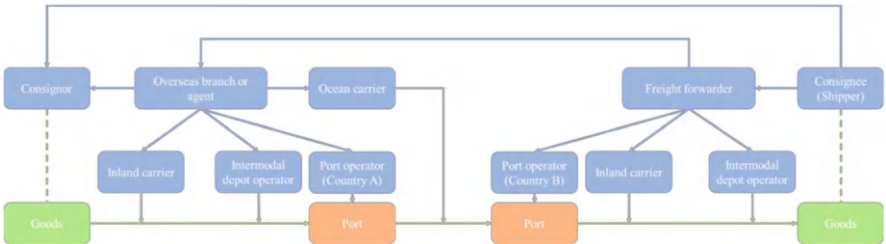


Fig. 3 Maritime supply chain framework (Consignee as Shipper)

depot operators to complete the inland transport from the consignor to the port. Then, the ocean carrier can transport the goods from the port in the consignor’s country to the port in the consignee’s country. Subsequently, the freight forwarder contacts its overseas branch or agent to arrange the affairs to finish the inland transport in the consignee’s country. The process involves organizing inland carriers and intermodal depot operators in the consignee’s country (see Fig. 2). On the other hand, a shipper is a consignee when the consignee leads the transport of the goods. In the same example, but this time, the consignee is responsible for all transportation matters. Similarly, the consignee usually designates a freight forwarder to manage complex matters during transportation, and the remainder of the process agrees with the case when a shipper is a consignor (see Fig. 3).

3.3 Challenges in the Maritime Supply Chain

As we can see in the last section, there are many players and complicated maritime supply chain processes. However, despite the emergence of digitalization in recent years, the maritime industry still depends primarily on humans. Since the maritime supply chain highly stresses information sharing, coordination, and communication among the parties, low levels of digitalization, possible human error, and untrustworthy data (led by human error) seriously reduce the supply chain efficiency and generate extra costs. Maritime supply chain challenges are mainly

attributed to trust, product traceability, and process optimization [4], and humans and low levels of digitalization largely lead to issues of these factors. The text lists some examples of the current challenges that the maritime supply chain confronts.

- **Complex documentation and paperwork.** The challenge can be described from all three factors: trust, product traceability, and process optimization. In 1999, Frankel [90] claimed that uncertainty and risk are the most uncontrollable and costly characteristics in maritime supply chains; therefore, a valid SCM should prioritize these. The risk mentioned above is not only operational risk, such as weather conditions, but also, more critically, human mistakes. SCM, mostly depending on physical paper documentation (humans), drives errors and delays. Unfortunately, the maritime supply chain process still involves intricate paperwork across the entities within the chain [5, 91], and the process contains a lot of personnel. For instance, the entire documentation process of an ocean shipment from East Asia to Europe encompasses approximately 30 relevant personnel. The digital documents are shared among parties; however, people in the different entities may alter the physical or digital copies of the documents without revising the original files. Hence, different versions of the same record lead to unsynchronized and untrustworthy data; finally, the entire process is delayed due to chaos and mistakes (trust issue). In addition, it would be unachievable or inconvenient to know immediately where the products are and when they can reach the next stage in the supply chain because of inconvenient maritime communication and limited digitalization. This situation causes problems. For example, assume the shipper is a consignor and will receive the proceeds after the delivery of the goods. The information and expectations about when the goods will reach the destination are essential to making decisions about procurement and finance (product traceability issue). Moreover, the unfavorable process induces a high percentage (15% in the above East Asia to Europe example) of total shipping costs owing to the inherent nature of human-based and paper-based operations (process optimization issue) [94].
- **Smuggling.** Despite the irreplaceable position of container shipping in global trade, it is exposed to the risk of smuggling [95]. Since the September 11 attacks occurred in 2001 in the United States, the US government has managed to avoid the menace of trafficking by terrorists in weapons of mass destruction into the state, and container security is one of the most focused [96]. At that time, the authorities screened containers for risk; nonetheless, the trade was also harmfully impacted due to the delay costs caused by the inspection. The responsible entities faced the trade-offs between the risk and the economics [97]. Accordingly, some research aims to optimize the process, including whether to adopt new equipment [98] or how many containers should be checked for valid inspection [96]. Also, the authorities built the Customs-Trade Partnership Against Terrorism (C-TPAT) and transferred some duties to commercial organizations in the maritime supply chain. C-TPAT requires its participants to follow some internal control standards to prevent smuggling; the participants can enjoy fewer and more uncomplicated inspections by the authorities [97]. Therefore, in the case of exporting to the

U.S., if one would like to minimize the delay, one should follow C-TPAT. Even if the destination is not in the U.S., the possible penalty is a noticeable reason why the parties must take care of this issue. Consequently, the maritime supply chain parties must have sufficient internal control to avoid delays and penalties; however, the objective is difficult to achieve cost-effectively under the human-centric and low levels of digitalization method (process optimization issue). Smuggling can also be interpreted from the aspects of trust and product traceability. For example, the physical record of the contents of a container may be changed from guns to hardware to import into a country that prohibits the unofficial importation of firearms. The authorities inspect containers to prevent this illegal behavior because of the untrustworthy record (trust issue) and the inability to trace the record of the containers (product traceability issue). The source of the untrustworthiness and the inability is the physical information rather than electronic files that involve the record of versions and are shared among the parties.

- **Inefficient customs clearance.** Supply chains usually waste considerable time and costs at intermediate stops [90]. In the maritime supply chain, customs are such the intermediate stops. First, the forwarder should send all correct documents to the importing country's customs office to ensure the consignee or its agent can complete customs clearance, and the goods can be released. The forwarder may omit some documents when physically sending or digitally emailing. Also, the physical or digital copy it sent may be an older version. For example, a container was planned to contain goods from four consignors. However, it ended up containing goods from five consignors (the original four consignors and the additional one) due to overestimating the space required by the original four pieces of goods. An employee then revised the physical or digital copy of the additional consignor's document, adjusting the container number without sharing the updated file with the other coworkers, and the older version may be sent. These cases lead to delays (trust and product traceability issues). Second, some customs are gradually digitalizing their process. If the forwarder still offers physical documentation, the customs clearance would be inefficient because the electronic record from customs cannot automatically match the paper-based data (process optimization issue). Finally, as we have discussed, the inspection to protect against smuggling poses delays as well (trust and product traceability issues). Figure 4 summarizes the idea.

As mentioned, humans and low levels of digitalization predominantly cause all these challenges and issues. As the digitalization level of the maritime supply chain has recently increased, the challenges and issues have sharpened [99–101]. However, efforts are required to solve the challenges and issues more thoroughly. People in the maritime field are realizing that the journey towards higher levels of digitalization is facing bottlenecks. In 2019, Jensen [102] identified three reasons related to digitalization that lead to challenges in the maritime supply chain. The challenges discussed in Jensen's article include high document processing costs, delivery time uncertainty, and container security monitoring costs, which can be

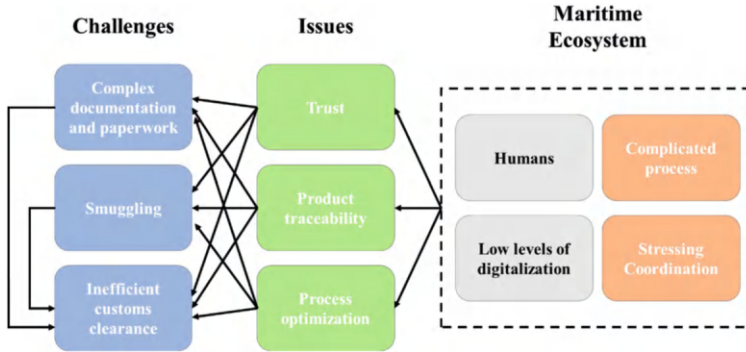


Fig. 4 Maritime supply chain challenges

linked to the challenges mentioned in the text (i.e., complex documentation and paperwork, smuggling, and inefficient customs clearance). The three reasons are explained as follows [102].

- **Incompatible or non-interoperable systems.** Each supply chain organization executes its preferred systems to digitalize the process, and then the efficiency of the internal processes (in each organization) improves. Nonetheless, the external processes (pertaining to multiple organizations) may not reap the benefits because some systems may be non-interoperable.
- **No standardized information or document format.** The International Organization for Standardization (ISO) has developed related standards, such as ISO 9735 [103]. Additional standards include the X12 Electronic Data Interchange (EDI) and Context Inspired Component Architecture (CICA) standards from the Accredited Standards Committee X12 (ASC X12) [104]. Despite the availability of these standards, only a few supply chain participants have utilized them. Moreover, some participants have sought to address integration issues by establishing large centralized information infrastructures and following standards. Unfortunately, these endeavors have seen limited success.
- **Regulations requiring paper documents.** Regulations frequently mandate original physical paper documents, complete with stamps and signatures.

The mix of these three reasons, along with the inherent limitations of traditional digitalization methods, is the cause of the bottlenecks in developing digitalization in the maritime sector. The documents can be shared among the relevant parties through email or some other system, and the unsynchronized and untrustworthy data issue seems to be addressed well. The customs clearance should then be efficient because it is free of error. The parties should be able to trace the products if the updated conditions of the products and containers are shared periodically. The process is optimized, and the human costs are reduced. However, in fact, digital files may be incorrect due to human errors (user or code errors) or even tampered with by malicious individuals. Therefore, the documents still cannot be truly

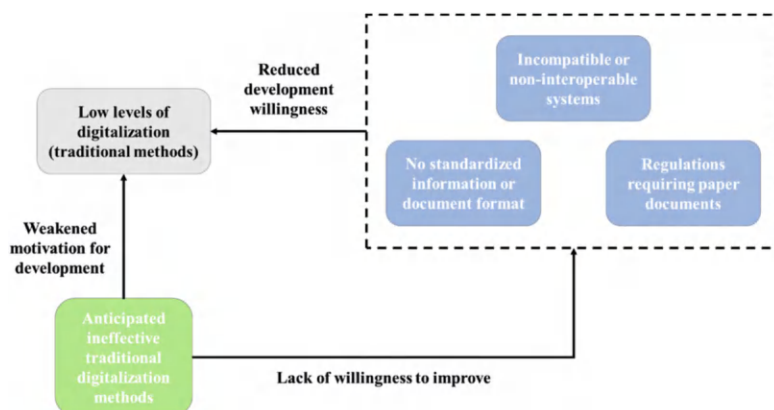


Fig. 5 Bottlenecks of traditional digitalization methods

trustworthy, and the expected outcome of the above inference cannot be achieved effectively. Furthermore, the three inconvenient facts, incompatible systems, the need for standardized information, and regulations requiring paper documents affect the willingness to use digitalization methods. The ineffective nature of digitalization methods also results in a lack of proactive support for system or data standards and regulation changes (see Fig. 5).

On the other hand, blockchain technology may provide an opportunity to break the bottlenecks and handle the challenges and issues better. Since blockchain is tamper-proof, the trust issue is eliminated. Customs clearance can be efficient and truly free of error. Additionally, the parties can trace the goods anytime on the blockchain. Moreover, tamper-proof blockchain can alleviate smuggling problems. Furthermore, there are no issues with system incompatibility or non-standard format in a blockchain application. Finally, regulations may be adjusted accordingly due to the immutable nature of blockchain methods and their capability for digital signature authentication, coupled with the effectiveness of this solution. The only concern is system incompatibility or non-standard format in various blockchain applications. According to the research conducted by Yang [9], establishing globally consistent standards for blockchain technology is crucial in the flourishing of blockchain applications. Therefore, builders of blockchain applications must pay attention to system and data interoperability. Then, the prevalence of digitalization with blockchain applications in the maritime field will gradually increase, and the challenges and issues discussed will be overcome (see Fig. 6).

In summary, suppose a blockchain application or several compatible applications can be designed carefully to leverage blockchain comprehensively (the perfect utilization of blockchain features) in the maritime field. In that case, the industry-usable application can be regarded as a blockchain 4.0 application, disruptive to the maritime industry. Furthermore, some problems in the industry can only be solved through blockchain technology, such as eliminating the way to smuggling

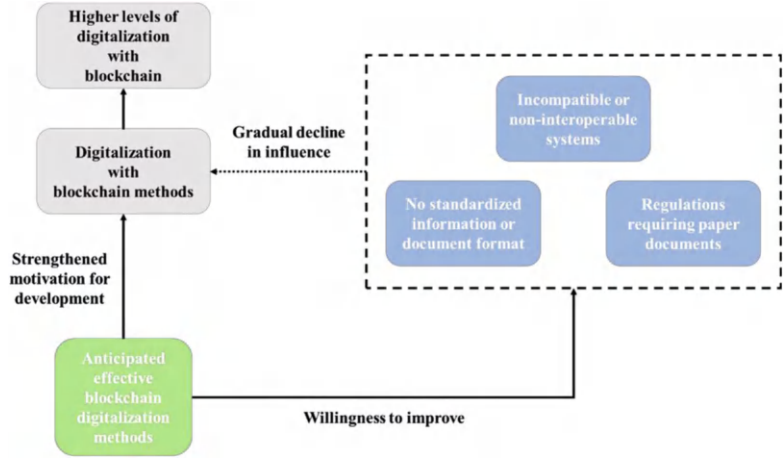


Fig. 6 Breaking the bottlenecks with blockchain digitalization methods

via tampering with the records. The applications that handle these cases also reach our broader definition of blockchain 4.0, and as mentioned, these applications will flourish easily. In the following sections, the text introduces several blockchain applications and discusses them from the aspect of reaching a blockchain 4.0 application. The advantages and disadvantages of the designs are discussed.

4 Global Maritime Consortiums and Blockchain Innovations

In this section, the text elaborates on three global consortiums and their four blockchain applications in the maritime supply chain. One consortium is the pioneer starting the journey of blockchain applications in the field, TradeLens (note that TradeLens is the name of one blockchain application; however, because its members form a specific ecosystem, people in the industry always regard it as a consortium). The other two consortiums are the Global Shipping Business Network (GSBN) [105] and the Digital Container Shipping Association (DCSA) [106]. One of the applications introduced is TradeLens, and the other three are based on the GSBN platform: Cargo Release, Trade Finance, and Electronic Bill of Lading (eBL).

4.1 TradeLens

History [102]. TradeLens, a pioneering solution in the digitizing maritime supply chain, emerged from a series of innovative initiatives led by Maersk and IBM:

Shipping Information Pipeline (SIP), Paperless Trade (PT), and Global Trade Digitization (GTD). These initiatives focus on information exchange, operational efficiency, and cost reduction within the container shipping industry (part of the maritime industry). Between 2014 and 2016, Maersk Line developed SIP prototypes to facilitate information exchange among maritime supply chain participants. SIP provided the digital infrastructure for gathering and sharing vital information about container events during the shipment, such as the completed stuffing, sealing, loading, and export and import documentation cleared. Notably, SIP did not apply blockchain technology then; it was just a digitalization application. Concurrently, IBM and Maersk aimed to apply blockchain technology in the container shipping industry. Initially, they concentrated on trade finance processes. However, due to the troublesome financial regulations (time-consuming), they finally proposed PT, a blockchain network that provided authorized participants access to encrypted electronic copies of documents. Then, SIP and PT were integrated into GTD and, after commercial availability, rebranded as TradeLens.

Overview [107]. TradeLens leverages IBM Cloud and IBM Blockchain Platform, based on Hyperledger Fabric [108]. Hyperledger Fabric is a permissioned blockchain platform that has access control capabilities. Each node is managed by TradeLens per se or a participant in the blockchain. The manager of the specific node can authorize other members (nodes) of the blockchain to access specified information. TradeLens provides open APIs and a web user interface to access the platform. The main functionalities of TradeLens are shipping milestone visibility and digitized documents, whose predecessors are SIP and PT, resp. Authorized participants can use the Shipment Manager user interface or APIs to track the real-time status of a shipment or container. Participants can share digital documents in TradeLens with security, version control, and privacy. The documents can be shared in either structured or unstructured data formats. The structured documents are ready to analyze and use. On the other hand, unstructured documents, such as PDF files, cannot be utilized directly. Blockchain technology is the core of TradeLens. It ensures tamper-proof information, verifiability, recoverability, provenance traceability, and privacy; thus, TradeLens can be a trustworthy platform that offers a digital solution without concern about poor decisions due to false information, leakage of confidential information, and the loss of essential documents. Regarding interoperability, the TradeLens data model and access control schema follow the United Nations Centre for Trade Facilitation and Electronic Business (UN/CEFACT) [109]. In addition, TradeLens employs non-proprietary, publicly available APIs that are straightforward to configure and implement.

Discontinuation. Unfortunately, on November 29, 2022, Maersk announced that TradeLens would be discontinued because it had not achieved the commercial viability required to sustain operations and fulfill the financial expectations as an independent enterprise. The platform would go offline by the end of quarter one, 2023 [110]. Before the discontinuation, over 100 parties in the industry had adopted this innovative platform, and TradeLens had held a substantial 61% share of the global shipping market.

Starting with SIP, Maersk conducted digitalization to enhance the maritime ecosystem in terms of product traceability and process optimization issues. However, SIP without blockchain faced the bottleneck, and then Maersk introduced blockchain into the maritime field to further sharpen the industry (see Figs. 5 and 6). With the development of PT, GTD, and TradeLens, Maersk and IBM have addressed the noted challenges and issues to some extent (see Fig. 4). Because the platform emphasized interoperability, it could be broadly used by different parties in the maritime field, and later, it succeeded in having over 100 members. TradeLens was a blockchain 3.0 application then. Theoretically, if all parties in the maritime supply chain worldwide applied TradeLens, it would be a business-usable platform in the entire industry and a blockchain 4.0 application. Because the parties in the maritime supply chain anticipate that the blockchain digitalization platform will be helpful to and benefit all parties in the supply chain, they should have the motivation to adopt the platform. Eventually, they would all attend the consortium, but it might take a long time and require endeavors of TradeLens to convince the parties. Therefore, the issue is not blockchain but business model. Whether a business model can support a platform to survive and operate stably finally is crucial. Unfortunately, before the vision was reached, TradeLens failed first. The text claims some possible reasons as follows.

First, TradeLens initially primarily covered the port-to-port process (during voyages) because Maersk is an ocean carrier, and TradeLens started with Maersk and the other ocean carriers. Subsequently, some other parties in the maritime supply chain gradually applied TradeLens. The usage of TradeLens was limited before the participation of sufficient parties. However, even though it was discontinued, it still did not reach that sufficient level (the reason is related to the following points).

Second, there are no incentives for freight forwarders to adopt TradeLens. From the freight forwarders' perspective, the only benefit of using TradeLens is that tracking and tracing goods for customers (shippers) on a single application rather than many applications is a bit more convenient. Freight forwarders could not improve other processes because other parties, such as inland carriers, intermodal depots, and ports locally, might still not utilize TradeLens. The incentive for freight forwarders to use TradeLens was not enough initially, and most did not adopt TradeLens. Noticeably, freight forwarders are the only parties with information about which shippers' goods are in a container. Therefore, before enough freight forwarders worldwide applied TradeLens, TradeLens only had container-level data from ocean carriers. It could only trace and track the container or shipment information instead of the goods information. The application is sufficient if an entire container only involves goods from one consignor, and the goods will be sent to one consignee (CY to CY/FCL to FCL). Nonetheless, the application is inadequate if a container contains goods from various consignors or the goods will be shipped to several consignees (CFS to CFS/LCL to LCL or CFS to CY/LCL to FCL or CY to CFS/FCL to LCL). When there are many consignees, the goods cannot be traced and tracked in the inland transport processes in the importing country since the goods for different consignees will be unloaded from the container at the port of the importing country and transported separately. Similarly, in the case

of various consignors, the goods can only be traced and tracked after loading them into the container in the exporting country. Unfortunately, most shipments are not CY to CY, and the usage of TradeLens is restricted.

Finally, there are no incentives for inland carriers and intermodal depots to apply TradeLens. These parties primarily interact with freight forwarders. If freight forwarders do not use TradeLens, it is hard to persuade these parties to use it. Then, the willingness of freight forwarders to employ the platform is also negatively affected (vicious circle).

In summary, the scope of TradeLens was limited, and some parties in the supply chain needed more motivation to utilize the platform. The blockchain features were only applied partially in the entire supply chain. Before the stable operation of the platform, it ended up failing first. Therefore, the business model and application development strategy are imperative for a successful blockchain platform.

4.2 GSBN

GSBN is a neutral, not-for-profit technology consortium that envisions a transformative journey for the maritime industry. With technology partners such as Oracle, Microsoft, AntChain, and Alibaba Cloud, it has developed its operating platform with Hyperledger Fabric (permissioned blockchain). Beyond the development and maintenance of the operating platform, GSBN offers APIs to empower developers to build applications on its platform. GSBN also contributed two applications: Cargo Release and Trade Finance. IQAX [111] designed another application: Electronic Bill of Lading (eBL). The text explains these three applications as follows [112].

Cargo Release [113]. When an ocean carrier arrives at a port, the cargo must complete a series of customs clearance processes before it can be released, and then the consignee or its agent can take the goods away. As mentioned, these processes are primarily based on humans with low levels of digitalization, and the overall customs clearance is inefficient. Subsequently, this leads to costs, port congestion, and heightened pollution levels. Therefore, it impacts not only the parties in the supply chain but also national and global economics and the environment. GSBN then announced Cargo Release to handle this problem. Cargo release is a digital system that connects all relevant entities at the port of import. The solution digitalized the customs clearance (cargo release) and brought trust and transparency with blockchain under the GSBN platform, which fundamentally solved the problem. Cargo Release dramatically reduces operational time and benefits several parties in the supply chain, economics, and the environment. For instance, it cuts the document preparation time from days to hours. Additionally, shippers can gain more profit owing to reduced turnaround times. Cargo Release's rollout commenced in China and Southeast Asia in August 2021. Subsequent global support from terminals facilitated the expansion of Cargo Release into Rotterdam, the Netherlands, by March 2022. Four months later, it debuted in Latin America, initiating operations in ports across Mexico and Panama. Since September 2022,

Cargo Release has been utilized by over 10,000 customers for real-time updates on more than 1 million shipments.

Trade Finance [114]. Financial institutions reject 40% of financing requests of small and medium enterprises (SMEs). However, most trade companies in the maritime supply chain are SMEs. Financing is inevitable for them to complete global trade since they usually cannot receive payment until delivery, and the process is lengthy. The rejection is due to the industry's highly uncertain nature and non-real-time trustworthy information led by the paperwork and low levels of digitalization processes. Thus, it is too risky for banks to accept the requests. GSBN started strategic partnerships with leading trade finance banks to address this financing gap. In September 2021, the Trade Finance Advisory Group was established with several banks. GSBN actively engaged with institutions within the Advisory Group to develop digital solutions to enhance banks' financing process. Later, GSBN introduced Trade Finance for consent collection. This innovation enables banks to source data from ocean carriers directly, ensuring the verification of information based on a single and reliable source. This functionality decreases banks' risk, making trade finance more accessible to SMEs. The most critical parts of this application are traceability and trustworthiness, which come from blockchain. GSBN successfully facilitated the collaboration with the relevant parties, achieving a milestone in September 2022 with the completion of a pilot transaction in under 20 minutes with Bank of China (Hong Kong), Hapag-Lloyd, and A & W Food Service (Hong Kong) Ltd.

Electronic Bill of Lading (eBL) [115]. The documents in the maritime supply chain are often transferred physically to complete shipment (paperwork, low levels of digitalization). One such document is the bill of lading (B/L). The B/L is a legal document between a shipper (or freight forwarder) and an ocean carrier describing the information about the goods, such as quantity, type, origin, and destination [102]. The ocean carrier issues this document to the counterparty (shipper or freight forwarder) and acknowledges cargo receipt from the counterparty. The consignee or agent in the importing country should receive the original B/L and present it, and the cargo can be released. Otherwise, the Telex Release can be an alternative. After a shipper or consignee applies for a Telex Release, the cargo can be released to the consignee or agent without the original B/L. This process requires some documents to be transferred to the ocean carrier or freight forwarder by email, and the request may only be completed once all related fees are paid to the ocean carrier and freight forwarder. The physical transport of B/L is inefficient and costly, and the Telex Release also mandates troublesome and inefficient preparation documents and checking processes. In addition, both of them have the risk of fraud. IQAX designed an Electronic Bill of Lading (eBL) application based on the GSBN platform to address this problem. All interested parties can access the real-time electronic information about B/L. The process is hugely enhanced by eliminating complicated information transport processes. The application leverages blockchain to ensure traceability and tamper-proof; problems like fraud can be improved. IQAX eBL

received approval from the International Group of P&I Clubs in June 2022 and has garnered adoption from three shipping companies.

Through the single operating platform and the ability to develop different applications on it, the GSBN platform is scalable and interoperable internally. Moreover, it is also ready to interoperate externally with APIs. However, the platform needs more internal or interoperable external applications to make it business-usable and become a blockchain 4.0 platform. Currently, each of the three applications on the platform addresses a single process in the maritime supply chain. Without an application or a series of applications to connect all these applications on the platform, the characteristics of blockchain cannot be thoroughly leveraged.

First, Cargo Release focuses on the customs clearance and release of goods process. It enhances the process with digitalization and blockchain. Blockchain, in a single process, may improve the trustworthiness of the record. However, the data cannot be genuinely tamper-proof and correct because it can be changed before the cargo release process. Also, it lacks traceability. The smuggling challenge cannot be sharpened. The process is still inefficient.

Second, Trade Finance makes an effort to improve the financing process. This application can be treated as our broader defined blockchain 4.0 application because it can improve the financing problem, which cannot be solved without blockchain. The risk of the bank to finance the shipment involves not only the uncertainty of arrival time but also the truth of the shipment. A company may conspire with other parties in the supply chain to forge the shipment. All documents, such as the container number, bill of lading, and sales contract, are available, but the shipment is fake (e.g., the case of New Site Industries Inc. fraud [116]). Therefore, although real-time digital records can give financial institutions more resources to estimate and control risk, the verity of shipment and relevant documents are crucial. Trade Finance highlights the single and reliable source of information from the ocean carrier that transports the goods, so it can vastly decrease the risk of fake shipment. Nevertheless, it can be sharpened by involving the whole process in the maritime supply chain, including door-to-port and port-to-door processes (i.e., inland processes). Then, the application will be more unassailable since the more participants attend, oversee, and operate with the blockchain application, the less likely fraud will occur.

Third, eBL aims to digitalize the B/L and enhance the process. Because digital data is easily tampered with, eBL applies blockchain to diminish forgery. Nonetheless, similar to the problem of Cargo Release and Trade Finance, the fabrication can only be solved partially when an application focuses on a single process instead of the entire process.

In summary, the framework of GSBN is practical owing to its ability to interoperate internally and externally. However, it relies on developers or platform builders worldwide to contribute more applications to or connect their applications with the GSBN platform.

4.3 DCSA [106]

Established in 2019, DCSA primarily collaborated with several of the largest international container shipping companies to drive digitalization and standardization in the industry. It aims to enhance the operational efficiency of the maritime field and enable more efficient information exchange among relevant parties. DCSA did not develop its blockchain platforms or applications but has significantly contributed to advancing blockchain in the maritime field. Standards are essential for the development of blockchain in a sector (see Figs. 5 and 6). DCSA has created several standards, such as the Bill of Lading [117] and Booking standards [118].

5 Forwarder Blockchain Platform

Freight forwarders must interact and collaborate with all parties in the supply chain to ensure on-time, efficient, and cost-effective transport for shippers. This nature motivates iOne to build a platform for freight forwarders named Forwarder Blockchain Platform [119]. By solving the pain points freight forwarders face and attracting their adoption, the relevant parties can also benefit, and the overall maritime supply chain can become more blockchain and improve. Nowadays, some freight forwarders in Taiwan and their supply chain collaborators have adopted Forwarder Blockchain Platform. In this section, the text starts with an exploration of freight forwarders' pain points. Then, an in-depth description of the platform is exhibited.

5.1 Exploring the Pain Points of Freight Forwarders

- **Integrity of Information across Internal Departments.** A freight forwarder involves various internal departments to complete complicated jobs. These internal departments usually keep their physical or digital copies of documents. If they need to revise documents, they change their copies without sharing. Because of this paperwork or low levels of digitalization, the unsynchronized and untrustworthy data issue may happen, and subsequently, mistakes and delays would occur.
- **Automation or Data-Based Processes and Decisions.** Most freight forwarders' processes are human-based rather than data-based or automated. The process spends a substantial amount of time on the pending phase. The staff cannot control their time reasonably due to the lack of expectations when the work will come. In addition, the paper-based or low levels of digitalization processes lead to difficulty applying AI or data-driven decisions.

- **Integrating Information from External Partners.** The human-based and paper-based (low levels of digitalization) collaboration among parties in the maritime supply chain leads to a tedious manual integration process by freight forwarders' employees. The process not only wastes lots of time but also is prone to mistakes.
- **Documents Management.** Freight forwarders and their external partners rely on email, messaging software, fax, or physical mail to transmit and receive relevant documentation. These parties may need to utilize different ways to transmit documents based on the requirements or agreements with distinct parties. The dispersed data across various locations without centralized management makes it challenging for freight forwarders and other parties to find and trace the original documentation.
- **Documents Revisions Synchronization.** Various factors often cause item and fee modifications during the exporting. The lack of real-time synchronization in updating the latest prices and related records among upstream and downstream supply chain entities brings errors and tribulations in subsequent document processing.
- **Tracking Process.** Forwarders and their customers must track cargo to control it and ensure timely deliveries. For example, they must ensure that cargo arrival at the exporting port complies with the deadline. Otherwise, they should use the next scheduled ship to transport the goods. Then, the delay would happen, and the customers experience substantial losses due to deadline overrun. The quality of forwarders' service and customer satisfaction were negatively impacted. However, owing to paperwork or low levels of digitalization, the tracking process is time-consuming and ineffective.

The pain points listed are associated with paperwork (low levels of digitalization) and human-based work, and they can be connected with the complex documentation and paperwork challenges we have discussed in Sect. 3. For the other challenges in the maritime ecosystem, if a freight forwarder can improve the smuggling and inefficient customs clearance challenges, its service would become more satisfactory and competitive.

5.2 Overview and System Architecture

In 2021, iOne developed Forwarder Blockchain Platform. It adopts Ethereum to build the private platforms centered around each collaborated freight forwarder. The standard application is well-designed and ready to use. The freight forwarders interested in using the application can purchase and install the standard platform or require a customization platform. The freight forwarder utilizing the platform can then share a user interface with its partners and customers.

Figure 7 shows the procedure of the platform. The process has 13 events (documents) corresponding to some tracking nodes. After logging in, relevant parties can

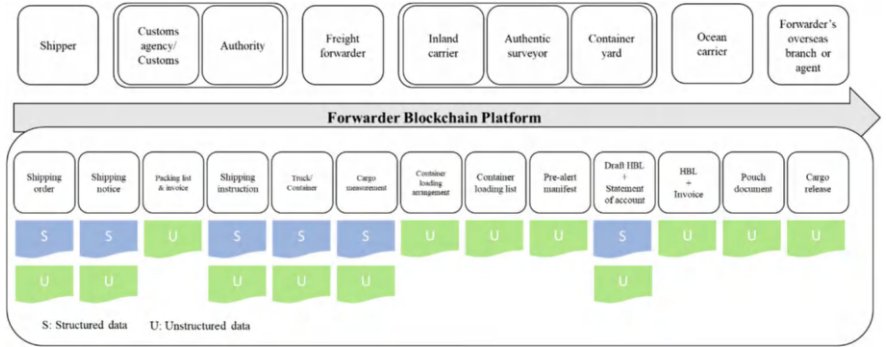


Fig. 7 Forwarder blockchain platform process

check the progress of the shipment on the mobile app or web page, and each node requires different related parties to upload the corresponding documents. The access control function manages which parties can search for and upload what documents. All upload work can be completed via the web, and some crucial documents can be uploaded by app. The different versions of the uploaded documents can be looked up on the web. The processes are primarily during the door-to-port (exporting country) phase since freight forwarders have many complex works during this stage. The door-to-port and port-to-door (importing country) phases are symmetrical (see Figs. 2 and 3). Therefore, the platform can be leveraged similarly. Although the platform does not involve the tracking process during the port-to-port (voyages) phase, the platform's nodes flow contains the pouch document function and cargo release process. The pouch document function provides the latest version of documents for customs declaration in the importing country. The customs in the importing country and consignee or customs agency can access the latest documents after the container is stuffed in the exporting country. The platform has two kinds of data forms: structured and unstructured. The structured data can be extracted and analyzed directly. The platform has leveraged these data to establish dashboards to make freight forwarders and relevant parties' tracking, control, and decision processes more efficient. The unstructured data are image or PDF files. It cannot be analyzed directly but checked and shared among relevant parties. When the relevant parties upload some documents with the structured feature, the platform organizes the structured data automatically and saves both unstructured (document image or PDF file) and structured data into the system. Otherwise, only unstructured data is saved if this feature does not support the documents.

The platform uses the private blockchain with the IPFS file management method. When documents are uploaded, the platform applies smart contracts to conduct pre-set instructions to complete the file-saving process. These instructions involve what information is extracted from the document for the structured feature. Blocks have file size limits on the blockchain. Therefore, the platform cannot upload images or PDF files directly. The platform has a local database and a blockchain. All

uploaded data is saved into the local database and then the blockchain. Regarding unstructured data, the local database has the original images or PDF files, and the blockchain has the corresponding hash value generated via IPFS. The hash value is unique to the underlying data, and it is content-based. Any documents with the same content produce the same hash value; a slight change makes the hash value thoroughly different [120]. Regarding structured data, both the local database and blockchain have the same information. Hence, if a document is supported by the structured feature, the corresponding block on the blockchain has an IPFS hash value (unstructured part), the basic information of the block (e.g., uploader, time), and structured data extracted by the platform, such as the origin and destination of the shipment (structured part). On the other hand, if the structured feature does not support a document, the corresponding block only has an IPFS hash value and the basic information of the block.

The platform leverages blockchain to ensure the data is tamper-proof, immutable, and traceable. The freight forwarder's processes are digitalized with these blockchain characteristics. In addition, the content-based IPFS hash makes the version control function easy because every revision automatically generates distinct records. The fragmented characteristic of IPFS also provides extra security. Besides, structured records and dashboards can satisfy the need for data-based decisions. Almost all pain points discussed in the last section are solved except the tracking process in the port-to-port phase, but freight forwarders care more about tracking in the door-to-port process since it is associated with the freight forwarders' job. Customers may inquire about their shipping status during the port-to-port phase, but it only requires freight forwarders to search on the ocean carriers' system, which is simple. Moreover, the pouch document function (realized easily under the IPFS file management) makes the customs declaration and clearance more efficient because it ensures the latest, correct documents. The shipper benefits, and the freight forwarders achieve higher customer satisfaction. Furthermore, the consignors (shippers) who are SMEs can more easily get loans from financial institutions by sharing the tamper-proof, immutable, real-time information on their platform with them (the concept is the same as the Trade Finance application). Similarly, customer satisfaction is enhanced. All these factors motivate freight forwarders to adopt the platform. Before the freight forwarders apply the platform, they persuade their inland partners to use it together. From the inland partners' view, the process is relatively similar, so they commonly accept the requirement. The ecosystem then becomes more blockchain and digitalized.

5.3 Future Vision

Currently, the freight forwarders use their independent Forwarder Blockchain Platform. The partners collaborating with them must use different user interfaces to log in, upload data, and look up data, which is inconvenient. Therefore, iOne plans to design a public blockchain platform. The supply chain partners can then log in to the

public platform and upload and access the specific forwarder’s private blockchain. Moreover, the public blockchain can also connect with global blockchain platforms such as the GSBN platform.

6 Discussion

Table 1 summarizes the global applications in Sect. 4 and the local application in Sect. 5. TradeLens started from the aspect of ocean carriers and focused on the tracking process of the entire supply chain as well as the digitalized documents. TradeLens contributed to the introduction of blockchain in the maritime field and the positive enhancement shown in Fig. 6. The functions are all imperative and valuable for some parties in the supply chain. However, the platform’s design failed to attract freight forwarders and inland parties. Finally, the usage was limited, blockchain characteristics could not be applied thoroughly, and TradeLens was discontinued. The GSBN platform is an interoperable ecosystem. It is scalable and positively impacts blockchain development in the maritime field. Nonetheless, each application on the GSBN platform addresses a single process (point) rather than the sequential processes (line), which TradeLens and Forwarder Blockchain Platform handle. Likewise, the characteristics of blockchain could not be applied thoroughly. Forwarder Blockchain Platform succeeds in appealing to freight forwarders, and their inland partners are also convinced to use the platform. Therefore, it solves the problem of TradeLens’s design. Nevertheless, its disadvantage is independence among private platforms, which is inconvenient for freight forwarders’ partners. This design may increase freight forwarders’ difficulty in persuading their inland partners. If most inland partners are unwilling to use the platform, the freight forwarders will not utilize it, and the platform may be prevalent more slowly. As a local innovative application, it is an excellent strategy to start in a simple manner.

Table 1 Summary of applications

Application	Builder	Framework	Focus
TradeLens	Maersk, IBM	Hyperledger Fabric	Tracking process, Digitalized documents
Cargo Release (GSBN platform)	GSBN	Hyperledger Fabric	Cargo release process
Trade Finance (GSBN platform)	GSBN	Hyperledger Fabric	Financing process
eBL (GSBN platform)	IQAX	Hyperledger Fabric	Electronic bill of lading (one digitalized document)
Forwarder Blockchain Platform	iOne	Ethereum	Tracking process, Digitalized documents

However, updating the software for the public blockchain function is essential to promote the platform faster.

In summary, all these applications' designs have advantages and disadvantages. Unfortunately, TradeLens was discontinued before the development of Forwarder Blockchain Platform. Otherwise, the collaboration between these platforms may bring us closer to blockchain 4.0. The GSBN platform and Forwarder Blockchain Platform are on the right path. As more applications are established on the GSBN platform, more freight forwarders and their inland partners apply Forwarder Blockchain Platform, and more local and global interoperable blockchain platforms and applications develop, blockchain 4.0 is on the horizon if all these systems are connected.

7 Conclusion

In conclusion, the text introduces the concept of blockchain 4.0. The text also shows that blockchain can break the bottlenecks of digitalization and make the entire maritime ecosystem more efficient, less fraudulent, less smuggling, less cost, and better in general. Thus, the text focuses on reaching blockchain 4.0, and the mentioned vision can be achieved. The introduced applications' designs have some merits and demerits. The text analyzes and discusses these designs and offers suggestions to reach blockchain 4.0. Application builders can refer to the concepts in this article when designing their applications. Researchers can expand the topics from the discussion. Accumulating knowledge about relevant topics and more successful applications in the field can lead us to blockchain 4.0 and the vision.

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